



Eidgenössische Technische Hochschule Zürich  
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# Regularity of Soap-Film-Like Minimal Surfaces

The More Accesible Case for Sets of Minimal Perimeter

Master Thesis

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### **Abstract**

This thesis investigates key aspects of geometric measure theory, beginning with an introduction to essential tools and concepts in Chapter 2. The objective is to familiarize the reader with the fundamental techniques required for understanding the following chapters. Chapter 3 introduces the concept of sets of finite perimeter, focusing on the properties functions of bounded variation. This chapter lays the foundation for the more detailed studies that follow. Furthermore, the regularity results of de Giorgi and Federer are proven.

Chapter 4, which illustrates the main result of this thesis, presents the existence and regularity results for sets of minimal perimeter. These results are central to understanding the geometric properties of such sets. The chapter provides comprehensive insights into the behavior and structure of minimal perimeter sets.

In Chapter 5, the discussion is extended to almost minimal sets, as defined by Almgren. This chapter places the results from Chapter 4 within a larger theoretical framework, demonstrating how these sets fit into the broader context of geometric measure theory. The introduction of almost-minimal sets offers a new perspective and further depth to the study of minimal surfaces.

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## Notation

|                            |                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{diam}(K)$           | For a subset $K \subset X$ of a metric space $(X, d)$ this denotes its diameter $\text{diam}(K) = \sup_{x, y \in K} d(x, y)$ .                                                                                                          |
| $\ x\ $                    | This will denote the standard norm of $x \in \mathbb{R}^N$ for $n \geq 2$ .                                                                                                                                                             |
| $ x $                      | This will denote the absolute value/standard norm of $x \in \mathbb{R}$ .                                                                                                                                                               |
| $d(x, A)$                  | For a subset $A \subset X$ and an element $x \in X$ of metric space $X$ this will denote the distance $d(x, A) = \inf_{a \in A} d(x, a)$ between $A$ and $x$ .                                                                          |
| $\mathcal{B}(X)$           | For a topological space $X$ this will denote the Borel $\sigma$ -algebra on $X$ .                                                                                                                                                       |
| $C(X)$                     | For a topological space $X$ this denotes the set of continuous functions $X \rightarrow \mathbb{R}$ .                                                                                                                                   |
| $C^k(\Omega)$              | For a open set $\Omega \subset \mathbb{R}^N$ this denotes the set of $k$ -times continuously differentiable functions $\Omega \rightarrow \mathbb{R}$ .                                                                                 |
| $C_c(X)$                   | For a topological space $X$ this denotes the set of continuous functions $X \rightarrow \mathbb{R}$ with compact support.                                                                                                               |
| $C_c^k(\Omega)$            | For a open set $\Omega \subset \mathbb{R}^N$ this denotes the set of $k$ -times continuously differentiable functions $\Omega \rightarrow \mathbb{R}$ .                                                                                 |
| $L^p(X, \mu)$              | For a measure space $X$ with measure $\mu$ and $p \geq 1$ this denotes the space of all $\mu$ -measurable functions $f : X \rightarrow \mathbb{R}$ such that $\int_X  f(x) ^p d\mu(x) < \infty$ .                                       |
| $L_{\text{loc}}^p(X, \mu)$ | For a measure space $X$ with measure $\mu$ and $p \geq 1$ this denotes the space of all $\mu$ -measurable functions $f : X \rightarrow \mathbb{R}$ such that $\int_K  f(x) ^p d\mu(x) < \infty$ for all compact subsets $K \subset X$ . |
| $\det$                     | This denotes the determinant function on $n \times n$ matrices with entries in $\mathbb{R}$ for $n \geq 1$ .                                                                                                                            |
| $A^*$                      | For $n \times n$ matrix $A$ with entries in $\mathbb{R}$ this denotes its transpose.                                                                                                                                                    |
| $B(x, r)$                  | For an element $x \in X$ of a metric space $(X, d)$ and $r > 0$ this will denote the open ball about $x$ of radius $r$ , i.e. $B(x, r) = \{y \in X \mid d(x, y) < r\}$ .                                                                |
| $\bar{B}(x, r)$            | For an element $x \in X$ of a metric space $(X, d)$ and $r > 0$ this will denote the closed ball about $x$ of radius $r$ , i.e. $\bar{B}(x, r) = \{y \in X \mid d(x, y) \leq r\}$ .                                                     |
| $\partial B(x, r)$         | For an element $x \in X$ of a metric space $(X, d)$ and $r > 0$ this will denote the boundary of the open ball about $x$ of radius $r$ , i.e. $\partial B(x, r) = \{y \in X \mid d(x, y) = r\}$ .                                       |
| $S^{N-1}$                  | This will denote the unit sphere in $\mathbb{R}^N$ for $N \geq 2$ .                                                                                                                                                                     |
| $\mu - a.e.$               | For a measure space $X$ with measure $\mu$ this will be an abbreviation for $\mu$ -almost every(where).                                                                                                                                 |

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|                      |                                                                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\nu \perp \mu$      | For a measure space $X$ and measures $\mu, \nu$ on $X$ this will denote that there are disjoint subsets $E, F \subset X$ such that for all measurable sets $A \subset X$ we have that $A \cap E, A \cap F$ are measurable and $\mu(A \cap E) = \mu(E), \nu(A \cap F) = \nu(A)$ . |
| $\mu \lfloor A$      | For a measure space $X$ and measure $\mu$ on $X$ this will denote the restriction of $\mu$ to a subset $A \subset X$ , i.e. $\mu \lfloor A(B) = \mu(A \cap B)$ .                                                                                                                 |
| $A(x, r_1, r_2)$     | This denotes the (open) annulus $A(x, r_1, r_2) = B(x, r_2) \setminus B(x, r_1)$ about an element $x \in X$ of a metric space $(X, d)$ with inner radius $r_1 > 0$ and outer radius $r_2 > 0$ .                                                                                  |
| $o(f(x))$            | For a function $f : X \rightarrow \mathbb{R}$ and $x_0 \in X$ this denotes that $\lim_{x \rightarrow x_0} \left  \frac{f(x)}{g(x)} \right  = 0$ . Most of the times we have $X \subset [-\infty, \infty]$ and $x_0 = 0$ or $x_0 = \pm\infty$ .                                   |
| $A \subset\subset U$ | For subset $U$ of a topological space $X$ , we write $A \subset\subset U$ if $\overline{A} \subset U^\circ$ is and $\overline{A}$ is compact.                                                                                                                                    |



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## Chapter 1

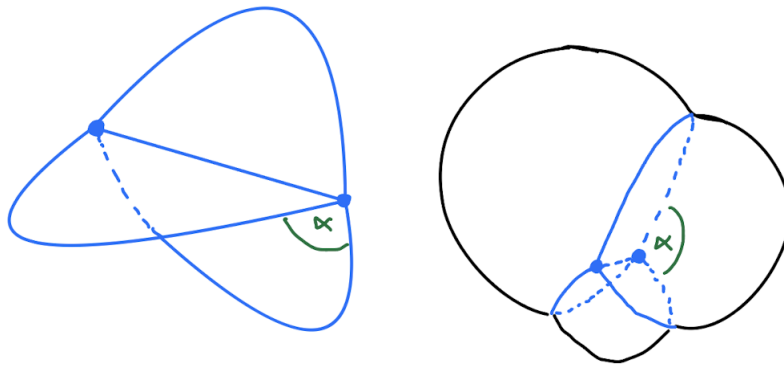
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# Introduction

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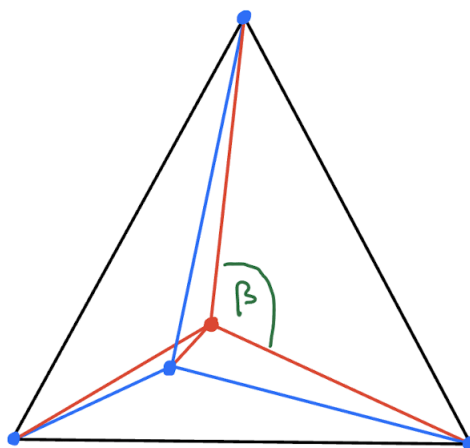
In 1873, the Belgian physicist and mathematician Joseph Plateau published his observations from experiments with soap films, describing their structures as seen in foams. He summarized his observations in the following four laws:

- (i) **Unbroken Surfaces:** Soap films are unbroken surfaces, meaning they are connected.
- (ii) **Constant Mean Curvature:** At any point on a soap film, the mean curvature is constant, indicating that the surface tension of soap films is uniform.
- (iii) **Triple Junction Angle:** If three soap films meet on an edge, they always do so at an angle of  $\alpha = \arccos\left(-\frac{1}{2}\right) = 120^\circ$ . This is captured in Figure 1.1.



**Figure 1.1:** Three soap films meeting at an edge.

- (iv) **Quadruple Junction Angle:** Edges as in (iii) meet in fours at the tetrahedral angle of  $\beta = \arccos\left(-\frac{1}{3}\right) \approx 109.47^\circ$ . This is illustrated in Figure 1.2.



**Figure 1.2:** Tetrahedral angle formed by soap films.

Plateau observed that configurations of soap films that do not obey these four laws are inherently unstable and will rearrange to conform to these laws.

These early observations by Plateau at the end of the 19<sup>th</sup> century arguably sparked the beginning of the mathematical discipline of geometric measure theory in the 20<sup>th</sup> century. This thesis will demonstrate how this discipline has its origins in so-called Plateau-type problems. In particular, Taylor rigorously proved in [1] in 1976 that minimal<sup>1</sup> surfaces, whose nature will be explained later in this thesis, obey these laws using tools from geometric measure theory. This is the historical context in which my thesis is situated.

I will restrict the focus of this thesis to the simpler case of sets of finite perimeter, in contrast to Jean Taylor's seminal work on minimal surfaces, particularly soap-film-like structures. Sets of finite perimeter, which are also known as Caccioppoli sets, provide a mathematical framework for modeling shapes with boundaries that exhibit some degree of irregularity but are generally well behaved in a measure-theoretic sense. These sets are effective in capturing certain aspects of minimal surfaces, such as those that can be modeled by a single interface. In the context of soap films, the perimeter of such a set can represent the surface area of a soap film formed by dipping a

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<sup>1</sup>The nature of said minimality will be explained in the course of this thesis.

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wire frame in soapy water. This approach is particularly useful in analyzing Plateau's first law, which asserts that the surface of a soap film is locally flat, minimizing the area subject to boundary conditions.

However, while sets of finite perimeter provide a powerful tool for understanding these simpler minimal surfaces, they fall short in capturing the full complexity of soap-film structures, especially when it comes to singularities and the junctions where multiple surfaces meet. These limitations are significant because real-world soap films are rarely as simple as single-interface models imply. Soap films often form intricate networks of surfaces that meet at junctions, creating singularities where the smooth structure of the film breaks down. These singularities are characterized by specific geometric configurations, such as triple junctions where three surfaces meet at 120-degree angles or more complex configurations where four or more surfaces converge.

Jean Taylor's work, particularly her 1976 paper "The Structure of Singularities in Soap-Bubble-Like and Soap-Film-Like Minimal Surfaces," addresses these complexities using more sophisticated mathematical tools. Taylor employs varifold theory and advanced techniques from geometric measure theory to rigorously describe the behavior of minimal surfaces near these singularities. Her work provides a comprehensive classification of the local structure of singularities in minimal surfaces, showing that the singular set consists of curves along which three sheets of the surface meet smoothly at equal angles, as well as isolated points where four such curves meet. This classification is crucial for understanding how minimal surfaces behave in the presence of complex boundary conditions, such as those found in soap films adhering to Plateau's third and fourth laws, which describe the angles between surfaces and the conditions under which multiple surfaces meet at a common curve or point.

Despite the constraints of focusing on the simpler model of sets of finite perimeter, this thesis will capture minimal surface theory within this framework and the related methodologies, while recognizing that more advanced tools, like those developed by Taylor, are necessary to fully capture the intricate geometry and topology of soap-film-like minimal surfaces.

This thesis is structured as follows: The first chapter establishes the foundational concepts and tools of geometric measure theory. This is essential to ensure that all the readers understand the setting in which we will work. This chapter provides an overview of geometric measure theory and its classical tools, with a focus on introducing vector-valued measures and their properties. The concept of rectifiability, which will allow us to generalize tangent spaces in less regular spaces, is of utmost importance. This generalization enables us to perform calculus on sets that are more irregular than manifolds in differential geometry. This foundational knowledge will

be useful in subsequent chapters, as the sets of finite perimeter discussed will exhibit a certain degree of irregularity. Essentially, the purpose of geometric measure theory is to generalize concepts from differential geometry, transitioning from smooth to unsmooth structures.

The following chapter is entirely dedicated to the study of functions of bounded variation. This will provide a measure of the regularity of functions and, subsequently, of sets. This is important as it serves as the basis for defining sets of finite perimeter. These functions and sets are the natural framework for applying tools from geometric measure theory and performing calculus. As mentioned, they are useful in modeling physical phenomena such as soap films, as well as in phase transitions, fracture mechanics, and fluid dynamics. Closely related to the soap films considered in this thesis is the modeling of interfaces between different fluid phases. In this context, sets of finite perimeter provide the mathematical framework to represent these interfaces. The perimeter, whose definition will be provided in this chapter, corresponds to the interface's total variation, capturing its possible irregularity. Soap films can also be viewed as an interface between air on both sides if we think of air in fluid-dynamical terms. Mathematically, functions of bounded variation are essential tools for defining less regular solutions to various problems in the calculus of variations and PDE theory, with Plateau-type problems being just one of many. An interesting example, aside from the problem considered in this thesis, is the inviscid Burger's equation:

$$u_t + f(u)_x = 0. \quad (1.1)$$

Even for smooth initial data  $u_0$ , the partial differential equation 1.1 can develop discontinuities in finite time, resulting in the so-called shock solutions. Physically, one can think of a drop of water hitting a small puddle, which then produces shock solutions. With this in mind, the keyword of this chapter should be "discontinuity", because the goal of this thesis is to investigate the intricacies of possibly very irregular sets. In the setting of functions of bounded variation, one gains some control over such discontinuities in solutions.

Finally, after these preliminary but equally important chapters, we can delve into the main objective of this thesis: presenting a proof of a regularity result for sets of finite perimeter that minimize said perimeter. This is crucial for modeling sets of finite perimeter as soap-film-type membranes in accordance with the first of Plateau's laws, i.e., the smoothness condition. I will demonstrate that sets minimizing perimeter are indeed regular, aside from a negligible set of irregularities. This is illustrated in Figures 1.1 and 1.2, where soap films are smooth everywhere except at the edges and vertices where they meet.

From a mathematical standpoint, this chapter illustrates the methodology

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used when considering such minimization problems. Starting with the existence of solutions via the direct method of the calculus of variations, we will then establish a monotonicity formula, which captures the behavior of minimizers at different scales. This can be thought of as zooming in and out around a point of minimal sets. Zooming in provides insight into the local structure, which is crucial when considering singular points. In contrast, zooming out helps to understand the global structure of the geometric shapes considered in this thesis.

The monotonicity formula is our main tool for studying the behavior of a quantity, in this case the perimeter. When this behavior is predictable, it indicates the likely regularity of the minimizer in the zoomed-in region. For a soap film spanned by a wire frame, the monotonicity formula tells us how the area of the soap film contained within a small ball changes as we shrink or extend the ball. If this area does not increase too rapidly, exhibiting the desired predictable behavior, it suggests that the soap film is regular in that region. This intuitive thought leads to the culmination of this chapter: de Giorgi's regularity theorem, which captures Plateau's first law, the smoothness condition.

Lastly, we will turn our attention to the more general setting considered by Taylor [1] and David [2]. The importance of this chapter lies in establishing more general definitions of minimality of surfaces suited for higher dimensions and proving a monotonicity formula. The most interesting part for me is that this chapter serves as a jumping-off point into the deeper study of soap films in higher dimensions. Although not as visualizable as in three dimensions, the difficulties and intricacies of Plateau-type problems remain and become even more pronounced in higher dimensions. For this reason, I have restricted this chapter to definitions and the monotonicity formula.

The difficulty of soap films in higher dimensions lies, in my opinion, in two main points. Firstly, in higher dimensions, the types and structures of singularities that minimal surfaces can develop become more complex. While in two dimensions, a minimal surface might have isolated singular points, in higher dimensions, singularities form along higher-dimensional sets, such as lines and surfaces. This complexity is already evident in three dimensions, which is the focus of Taylor's more general study. The findings of Taylor's theorem in three dimensions highlight the second difficulty when considering soap films in higher dimensions: the problem of classification. In three dimensions, Taylor was able to classify all types of singularities of minimal soap films in accordance with the last two laws of Plateau. However, this classification becomes increasingly challenging for dimensions greater than three. In fact, this is an open problem, as starting from dimension  $N = 4$ , we do not have a complete list of singularities exhibited by soap films. This highlights an area for further research.

In conclusion, the main point of this thesis is to establish the mathematical framework for modeling soap films using sets of finite perimeter and to investigate the intricacies of such sets as solutions to a certain Plateau-type problem, bringing them mathematically into accordance with Plateau's first law. The final chapter serves as a pointer to further research possibilities for both the reader and me. Along the way, I hope the reader gains a deeper understanding of the methodology and tools provided in the context of the problem considered in this thesis, as well as in the preliminary chapters on geometric measure theory. Personally, after spending hours, days, and months on this thesis, I have learned many new useful tools and methodologies, and I am in awe of the immense work done in this field of research with many still unsettled open problems. I thank my advisor, Prof. Dr. Joaquim Serra, for the opportunity to write this thesis and explore the inherently interesting problem of soap films under his supervision, as well as for all the guidance he provided in writing this thesis.

## Chapter 2

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# Geometric measure theory

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This chapter is entirely dedicated to introducing the reader to geometric measurement theory, which is used extensively in this thesis. It is based on the exposition given in Chapters 1 through 3 of [3].

### 2.1 Abstract measure theory

This section briefly repeats the necessary definitions and results of classical measure theory. Proofs will be mostly omitted, but references will be provided.

**Definition 2.1** *Let  $X$  be a set and  $\mathcal{E}$  a family of subsets of  $X$ . Then we define the following notions:*

- (i)  $\mathcal{E}$  is called an **algebra** if  $\emptyset \in \mathcal{E}$ ,  $E_1 \cup E_2 \in \mathcal{E}$  and  $X \setminus E \in \mathcal{E}$  whenever  $E_1, E_2, E \in \mathcal{E}$ .
- (ii)  $\mathcal{E}$  is called a  **$\sigma$ -algebra** if it is an algebra and for any sequence  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  we have  $\bigcup_{h \in \mathbb{N}} E_h \in \mathcal{E}$ .
- (iii) For any family  $\mathcal{G}$  of subsets of  $X$ , the  **$\sigma$ -algebra generated by  $\mathcal{G}$**  is the smallest  $\sigma$ -algebra containing  $\mathcal{G}$  and denoted by  $\sigma(\mathcal{G})$ , i.e.

$$\sigma(\mathcal{G}) = \bigcap_{\mathcal{F} \supset \mathcal{G} \text{ is } \sigma\text{-algebra}} \mathcal{F}. \quad (2.1)$$

Moreover, if  $(X, \tau)$  is a topological space, we denote by  $\mathcal{B}(X)$  the  $\sigma$ -algebra generated by the topology  $\tau$  and call it **Borel  $\sigma$ -algebra in  $X$**  and its elements **Borel sets**.

- (iv) If  $\mathcal{E}$  is a  $\sigma$ -algebra in  $X$ , then  $(X, \mathcal{E})$  is called a **measurable space**.

**Definition 2.2** *Let  $(X, \mathcal{E})$  be a measurable space, and  $\mu : \mathcal{E} \rightarrow [0, \infty)$  be a set function.*

(i)  $\mu$  is **additive**, if  $\mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2)$  whenever  $E_1, E_2 \subset \mathcal{E}$  and  $E_1 \cap E_2 = \emptyset$ .

(ii)  $\mu$  is  **$\sigma$ -subadditive**, if for any sequence  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  and  $E \subset \bigcup_{h \in \mathbb{N}} E_h$  it holds that

$$\mu(E) \leq \sum_{h \in \mathbb{N}} \mu(E_h). \quad (2.2)$$

(iii)  $\mu$  is a **positive measure on  $X$**  if  $\mu(\emptyset) = 0$  and  **$\sigma$ -additive**, i.e. for any sequence  $(E_h)_{h \in \mathbb{N}}$  of pairwise disjoint elements of  $\mathcal{E}$  it holds that

$$\mu\left(\bigcup_{h \in \mathbb{N}} E_h\right) = \sum_{h \in \mathbb{N}} \mu(E_h). \quad (2.3)$$

(iv) We say that a positive measure  $\mu$  on  $X$  is **finite** if  $\mu(X) < \infty$ .

(v) We say that a subset  $E \subset X$  is  **$\sigma$ -finite** with respect to a positive measure  $\mu$  on  $X$ , if  $E = \bigcup_{h \in \mathbb{N}} E_h$  for  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  with  $E_h \subset E_{h+1}$  and  $\mu(E_h) < \infty$  for all  $h \in \mathbb{N}$ . If  $X$  is  $\sigma$ -finite, then we call  $\mu$   **$\sigma$ -finite**.

(vi) If  $\mu$  is a positive measure on  $(X, \mathcal{E})$ , we call  $(X, \mathcal{E}, \mu)$  a **measure space**.

(vii) In a measure space  $(X, \mathcal{E}, \mu)$ , we call a subset  $N \subset X$  a  **$\mu$ -null set** if there is  $E \in \mathcal{E}$  such that  $\mu(E) = 0$  and  $N \subset E$ . We set  $\mathcal{E}_\mu$  to be the collection of subsets of  $X$  of the form  $F = E \cup N$  for  $E \in \mathcal{E}$  and  $N$  a  $\mu$ -null set. Then  $\mathcal{E}_\mu$  is a  $\sigma$ -algebra and we call it the  **$\mu$ -completion of  $\mathcal{E}$**  and say  $F \in \mathcal{E}_\mu$  is  **$\mu$ -measurable**.

(viii) We say that a property  $P(x)$  depending on a point  $x \in X$  holds  **$\mu$ -a.e.** in  $X$  if the set of points, where  $P(x)$  does not hold, is a  $\mu$ -null set.

**Definition 2.3** Let  $(X, \mathcal{E})$  be a measurable space, and  $m \in \mathbb{N}$ .

(i) We say that a set function  $\mu : \mathcal{E} \rightarrow \mathbb{R}^m$  is a  **$\mathbb{R}^m$ -valued measure on  $X$**  if  $\mu(\emptyset) = 0$  and for any sequence  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  of pairwise disjoint elements we have that

$$\mu\left(\bigcup_{h \in \mathbb{N}} E_h\right) = \sum_{h \in \mathbb{N}} \mu(E_h). \quad (2.4)$$

For  $m = 1$ , we call  $\mu$  a **real measure** on  $X$ .

(ii) If  $\mu$  is a  $\mathbb{R}^m$ -valued measure on  $X$ , we define its **total variation**  $\|\mu\|$  for every  $E \in \mathcal{E}$  by

$$\|\mu\|(E) := \sup \left\{ \sum_{h \in \mathbb{N}} \|\mu(E_h)\| \mid \begin{array}{l} E_h \in \mathcal{E} \text{ pairwise disjoint,} \\ E \subset \bigcup_{h \in \mathbb{N}} E_h. \end{array} \right\} \quad (2.5)$$

For  $m = 1$  we denote it by  $|\mu|$ .

(iii) If  $\mu$  is a real measure, then we define its **positive** and **negative parts** by

$$\mu^+ = \frac{|\mu| + \mu}{2} \text{ and } \mu^- = \frac{|\mu| - \mu}{2}. \quad (2.6)$$

**Remark 2.4** If  $\mu$  is a  $\mathbb{R}^m$ -valued measure on a measurable space  $(X, \mathcal{E})$ , then its total variation  $\|\mu\|$  is a finite positive measure on  $(X, \mathcal{E})$ . For a proof of this, see Theorem 1.6 in [3].

**Definition 2.5** Let  $(X, \mathcal{E}, \mu)$  be a measure space and  $(Y, \tau)$  a topological space.

- (i) A function  $f : X \rightarrow Y$  is called  **$\mathcal{E}$ -measurable** if  $f^{-1}(A) \in \mathcal{E}$  for every  $A \in \tau$ .
- (ii) If  $\mu$  is a positive measure on  $X$ , we call a function  $f : X \rightarrow Y$   **$\mu$ -measurable** if it is  $\mathcal{E}_\mu$ -measurable.
- (iii) We call a function  $f : X \rightarrow \mathbb{R}$  a **simple function** if its image is finite, that is,  $f$  is an element of the vector space spanned by the characteristic functions  $\chi_E$  of subsets  $E \subset X$ .
- (iv) For a simple  $\mu$ -measurable function  $u : X \rightarrow [0, \infty]$ , we define its integral with respect to  $\mu$  by  $\int_X u d\mu = \sum_{y \in u(X)} y \mu(u^{-1}(y))$ . If  $u : X \rightarrow [0, \infty]$  is  $\mu$ -measurable its integral is defined as

$$\int_X u d\mu = \sup_{v \text{ } \mu\text{-measurable, simple}} \int_X v d\mu. \quad (2.7)$$

- (v) We call a  $\mu$ -measurable function  $u : X \rightarrow \overline{\mathbb{R}}$   **$\mu$ -summable** if  $\int_X |u| d\mu < \infty$ . We call  $u$   **$\mu$ -integrable** if for  $u^\pm := \max(\pm u, 0)$ , we have  $\int_X u^+ d\mu < \infty$  or  $\int_X u^- d\mu < \infty$ . For  $u : X \rightarrow \overline{\mathbb{R}}$   $\mu$ -integrable we define:

$$\int_X u d\mu := \int_X u^+ d\mu - \int_X u^- d\mu. \quad (2.8)$$

- (vi) If  $\mu$  is an  $\mathbb{R}^m$ -valued measure and  $u : X \rightarrow \overline{\mathbb{R}}$  is  $\|\mu\|$ -summable, we define:

$$\int_X u d\mu := \left( \int_X u d\mu_1, \dots, \int_X u d\mu_m \right). \quad (2.9)$$

In addition, if  $u = (u_1, \dots, u_m) : X \rightarrow \mathbb{R}^m$  is such that each component  $u_i$  is  $\|\mu\|$ -summable, we define:

$$\int_X u \cdot d\mu := \sum_{i=1}^m \int_X u_i d\mu_i. \quad (2.10)$$

- (vii) If  $\mu$  is a real measure and  $u = (u_1, \dots, u_k) : X \rightarrow \mathbb{R}^k$  is  $|\mu|$ -summable (i.e. every component  $u_i$  is  $\mu$ -summable), then we set:

$$\int_X u d\mu := \left( \int_X u_1 d\mu, \dots, \int_X u_k d\mu \right). \quad (2.11)$$

(viii) For  $E \subset X$  and  $u : X \rightarrow \overline{\mathbb{R}}$  a function such that  $\mu\chi_E$  is  $\mu$ -integrable, we set  $\int_E u d\mu := \int_X u\chi_E d\mu$ .

**Remark 2.6** For every  $u : X \rightarrow \overline{\mathbb{R}}$   $\|\mu\|$ -summable with respect to a  $\mathbb{R}^m$ -valued measure  $\mu$  on  $X$ , it holds that:

$$\left| \int_X u d\mu \right| \leq \int_X |u| d\|\mu\|. \quad (2.12)$$

**Definition 2.7** Let  $(X, \mathcal{E}, \mu)$  be a measure space. We define for  $\mu$ -measurable function  $u : X \rightarrow \overline{\mathbb{R}}$ :

$$\|u\|_p := \left( \int_X |u|^p d\mu \right)^{\frac{1}{p}}, \quad (2.13)$$

if  $1 \leq p < \infty$  and

$$\|u\|_\infty = \inf\{C \in [0, \infty] \mid |u(x)| \leq C \text{ for } \mu\text{-a.e. } x \in X\}. \quad (2.14)$$

For  $p \in [0, \infty]$ , we say that  $u \in \mathcal{L}^p(X, \mu)$  if  $\|u\|_p < \infty$ . The quotient  $L^p(X, \mu)$  of  $\mathcal{L}^p(X, \mu)$  by the equivalence relation of  $\mu$ -a.e.-equality is a real vector space and  $\|\cdot\|_p$  is a norm on  $L^p(X, \mu)$ . Moreover, if  $(X, \tau)$  is a topological space, we define the vector spaces  $L^p_{\text{loc}}(X, \mu)$  as the space of all  $\mu$ -measurable functions  $u : X \rightarrow \overline{\mathbb{R}}$  such that  $u|_K \in L^1(K, \mu)$  for all  $K \subset X$  compact.

We now recall that for  $\sigma$ -finite measure  $\mu$  on  $X$  the dual of  $L^1(X, \mu)$  is  $L^\infty(X, \mu)$  and for  $p \in (1, \infty)$  the dual of  $L^p(X, \mu)$  is  $L^q(X, \mu)$ , where  $\frac{1}{p} + \frac{1}{q} = 1$  (see, e.g., Chapter 4 in [4]). This leads us to the following definition:

**Definition 2.8** Let  $p \in [1, \infty]$ ,  $(X, \mathcal{E})$  be a measurable space and  $\mu$  a  $\sigma$ -finite measure on  $X$ . We say that a sequence  $(f_h)_{h \in \mathbb{N}} \subset L^p(X, \mu)$  **converges weakly** to  $f \in L^p(X, \mu)$  if  $\lim_{h \rightarrow \infty} \int_X f_h g d\mu = \int_X f g d\mu$  for all  $g \in L^q(X, \mu)$ , where  $\frac{1}{p} + \frac{1}{q} = 1$  if  $p \in (1, \infty)$  and  $q = \infty$  if  $p = 1$ .

We also mention here *convergence in measure* and *almost everywhere convergence*.

**Definition 2.9** Let  $(X, \mathcal{E}, \mu)$  be a measure space. We say a sequence  $(f_h)_{h \in \mathbb{N}}$  of  $\mu$ -measurable functions **converges in measure** to a  $\mu$ -measurable function  $f$  if

$$\forall \varepsilon > 0 : \lim_{h \rightarrow \infty} \mu(\{x \in X \mid |f_h(x) - f(x)| > \varepsilon\}) = 0. \quad (2.15)$$

We say  $(f_h)_{h \in \mathbb{N}}$  **converges almost everywhere** to  $f$  if

$$\mu(X \setminus \{x \in X \mid \lim_{h \in \mathbb{N}} f_h(x) \neq f(x)\}) = 0. \quad (2.16)$$

We also have the following notion of *equiintegrability*.

**Definition 2.10** Let  $(X, \mathcal{E}, \mu)$  be a measure space and  $\mathcal{F} \subset L^1(X, \mu)$ . Then we say that  $\mathcal{F}$  is an **equiintegrable family** if the following conditions are satisfied:

- (i) For any  $\varepsilon > 0$  there exists a  $\mu$ -measurable set  $A \subset X$  with  $\mu(A) < \infty$  such that  $\int_{X \setminus A} |f| d\mu \leq \varepsilon$  for all  $f \in \mathcal{F}$ .
- (ii) For any  $\varepsilon > 0$  there is  $\delta > 0$  such that for every  $\mu$ -measurable set  $E \subset X$  the following implication holds:

$$\mu(E) < \delta \implies \int_E |f| d\mu < \varepsilon \forall f \in \mathcal{F}. \quad (2.17)$$

We have the following characterizations of equiintegrability.

**Proposition 2.11** Let  $(X, \mathcal{E}, \mu)$  be a measure space and  $\mathcal{F} \subset L^1(X, \mu)$ . Then the following holds:

- (i)  $\mathcal{F}$  is equiintegrable if and only if for every  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  such that  $E_h \rightarrow \emptyset$  it holds that  $\lim_{h \rightarrow \infty} \sup_{f \in \mathcal{F}} \int_{E_h} |f| d\mu = 0$ .
- (ii) If  $\mu$  is a finite measure and  $\mathcal{F}$  is bounded, then  $\mathcal{F}$  is equiintegrable if and only if there exists some increasing continuous function  $\varphi : [0, \infty) \rightarrow [0, \infty]$  with  $\frac{\varphi(t)}{t} \xrightarrow{t \rightarrow \infty} \infty$  such that

$$\mathcal{F} \subset \left\{ f \in L^1(X, \mu) \left| \int_X \varphi(|f|) d\mu \leq 1 \right. \right\}. \quad (2.18)$$

- (iii) If  $\mu$  is a finite measure and  $\mathcal{F}$  is bounded, then  $\mathcal{F}$  is equiintegrable if and only if

$$\lim_{t \rightarrow \infty} \sup_{f \in \mathcal{F}} \int_{\{|f| > t\}} |f| d\mu = 0. \quad (2.19)$$

- (iv) If  $\mu$  is finite,  $\mathcal{F}$  is countable, that is,  $\mathcal{F} = (f_h)_{h \in \mathbb{N}}$  is a sequence in  $L^1(X, \mu)$ , and  $f_h$  converges in  $L^1(X, \mu)$  to some  $f \in L^1(X, \mu)$ , then  $\mathcal{F}$  is equiintegrable.

**Proof** The proof of (i)-(iii) is given in Proposition 1.27 in [3]. For (iv) we can apply (iii) to obtain for every  $t \geq 0$  that

$$\sup_{h \in \mathbb{N}} \int_{\{|f_h| > t\}} |f_h| d\mu \leq \sup_{h \in \mathbb{N}} \int_{\{|f_h| > t\}} |f_h - f| d\mu + \sup_{h \in \mathbb{N}} \int_{\{|f_h| > t\}} |f| d\mu. \quad (2.20)$$

Since  $f_h \rightarrow f$  in  $L^1(X, \mu)$  and we can choose for every  $\varepsilon > 0$  and  $h \in \mathbb{N}$  a  $t_n \geq 0$  such that  $\mu(\{|f_h| > t_n\}) < \varepsilon$ , we can conclude that

$$\lim_{t \rightarrow \infty} \sup_{h \in \mathbb{N}} \int_{\{|f_h| > t\}} |f_h| d\mu = 0, \quad (2.21)$$

and so  $\mathcal{F}$  is equiintegrable by (iii).  $\square$

The next proposition is dedicated to all the convergence-related results for integrals.

**Proposition 2.12** *Let  $(X, \mathcal{E}, \mu)$  be a measure space.*

- (i) **(Monotone Convergence Theorem (MCT))** *For an increasing sequence  $(u_h)_{h \in \mathbb{N}}$  of  $\mu$ -measurable functions  $u_h : X \rightarrow \overline{\mathbb{R}}$  with  $u_h \geq g$  with  $g \in L^1(X, \mu)$  for any  $h \in \mathbb{N}$ , it holds that:*

$$\lim_{h \rightarrow \infty} \int_X u_h d\mu = \int_X \lim_{h \rightarrow \infty} u_h d\mu. \quad (2.22)$$

- (ii) **(Fatous's Lemma)** *Let  $(u_h)_{h \in \mathbb{N}}$  be a sequence of  $\mu$ -measurable functions  $u_h : X \rightarrow \overline{\mathbb{R}}$  and  $g \in L^1(X, \mu)$ . Then it holds that*

$$\int_X \liminf_{h \rightarrow \infty} u_h d\mu \leq \liminf_{h \rightarrow \infty} \int_X u_h d\mu, \quad (2.23)$$

*if  $u_h \geq g$  for any  $h \in \mathbb{N}$ , and*

$$\int_X \limsup_{h \rightarrow \infty} u_h d\mu \geq \limsup_{h \rightarrow \infty} \int_X u_h d\mu, \quad (2.24)$$

*if  $u_h \leq g$  for any  $h \in \mathbb{N}$ .*

- (iii) **(Dominated Convergence Theorem (DCT))** *Let  $u, u_h : X \rightarrow \overline{\mathbb{R}}$  be  $\mu$ -measurable functions and assume that  $\lim_{h \in \mathbb{N}} u_h(x) = u(x)$  for  $\mu$ -a.e.  $x \in X$ . If  $\int_X \sup_{h \in \mathbb{N}} |u_h| d\mu < \infty$ , then*

$$\lim_{h \rightarrow \infty} \int_X u_h d\mu = \int_X u d\mu. \quad (2.25)$$

- (iv) **(Vitali theorem)** *Consider  $u_h, u : X \rightarrow \mathbb{R}$  with  $u_h$   $\mu$ -measurable. If  $(u_h)_{h \in \mathbb{N}}$  converges to  $u$  in measure and there is an equiintegrable sequence  $(v_h)_{h \in \mathbb{N}}$  of positive functions such that  $|u_h(x)| \leq v_h(x)$  for  $\mu$ -a.e.  $x \in X$ , then  $\lim_{h \rightarrow \infty} \int_X u_h d\mu = \int_X u d\mu$ .*

**Proof** For a proof of these results, see, e.g. [4]. □

The next definition mainly concerns the terminology around the Radon-Nikodym theorem.

**Definition 2.13** *Let  $(X, \mathcal{E})$  be a measurable space.*

- (i) *Let  $\mu$  be a positive measure and  $\nu$  a  $\mathbb{R}^m$ -valued measure on  $X$ . We say that  $\nu$  is **absolutely continuous with respect to  $\mu$** , denoted by  $\nu \ll \mu$ , if for every  $B \in \mathcal{E}$  it holds that*

$$\mu(B) = 0 \implies \|\nu\|(B) = 0. \quad (2.26)$$

- (ii) Let  $\mu, \nu$  be positive measures. We say  $\mu$  and  $\nu$  are **mutually singular**, denoted by  $\mu \perp \nu$ , if there exists  $E \in \mathcal{E}$  such that  $\mu(E) = 0$  and  $\nu(X \setminus E) = 0$ . If  $\mu, \nu$  are  $\mathbb{R}^m$ -valued measures, we say they are **mutually singular**, if  $\|\mu\| \perp \|\nu\|$ .

**Remark 2.14** Let  $\mu$  be a positive measure and  $\nu$  a  $\mathbb{R}^m$ -valued measure on a measurable space  $(X, \mathcal{E})$ . Then  $\nu \ll \mu$  is equivalent to the following condition:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall E \in \mathcal{E} : \mu(E) < \delta \implies \|\nu\|(E) < \varepsilon. \quad (2.27)$$

If 2.27 holds, then for every  $\varepsilon > 0$ , the assumption  $\mu(E) = 0$  for any  $E \in \mathcal{E}$ , implies  $\|\nu\|(E) < \varepsilon$ . That is,  $\|\nu\|(E) = 0$ . So  $\nu \ll \mu$ . Conversely, assume  $\nu \ll \mu$ . Suppose, for the sake of contradiction, that there is  $\varepsilon > 0$  such that there is a sequence  $(E_h)_{h \in \mathbb{N}} \subset \mathcal{E}$  such that for any  $h \in \mathbb{N}$ ,  $\mu(E_h) < 2^{-h}$  and  $\|\nu\|(E_h) \geq \varepsilon$  hold. Since  $\mathcal{E}$  is a  $\sigma$ -algebra, we know that  $E := \bigcap_{h \in \mathbb{N}} \bigcup_{k \geq h} E_k \in \mathcal{E}$ . We note that for every  $h \in \mathbb{N}$  we have by  $\sigma$ -subadditivity we have:

$$\mu(E) \leq \sum_{k \geq h} \mu(E_k) \leq \sum_{k \geq h} 2^{-k}. \quad (2.28)$$

By convergence of the geometric series we get, as  $h \rightarrow \infty$ , that  $\mu(E) = 0$ . Since  $\|\nu\|(E_k) \geq \varepsilon$  for all  $k \in \mathbb{N}$ , we find that  $\|\nu\|(E) \geq \varepsilon > 0$ . This is a contradiction to  $\nu \ll \mu$ .

The next theorem is the famous Radon-Nikodym theorem.

**Theorem 2.15 (Radon-Nikodym)** Let  $(X, \mathcal{E})$  be a measurable space,  $\mu$  be a  $\sigma$ -finite positive measure, and  $\nu$  a  $\mathbb{R}^m$ -valued measure on  $X$ . Then there exists a unique pair of  $\mathbb{R}^m$ -valued measures  $\nu^a$  and  $\nu^s$  on  $X$  such that  $\nu^a \ll \mu$ ,  $\nu^s \perp \mu$  and  $\nu = \nu^a + \nu^s$ . Moreover, there is a unique function  $f \in [L^1(X, \mu)]^m$  such that  $\nu^a = f\mu$ .

**Proof** For a proof, see, e.g. [4]. □

**Remark 2.16** We call the measures  $\nu^a$  and  $\nu^s$  in Theorem 2.15 the **absolutely continuous**, respectively, **singular part** of  $\nu$ .

The expression  $\nu^a = f\mu$  in Theorem 2.15 is understood to mean:

$$\nu^a(E) = \int_E f d\mu = \left( \int_E f_1 d\mu, \dots, \int_E f_m d\mu \right), \quad E \in \mathcal{E}. \quad (2.29)$$

We refer to the function  $f \in [L^1(X, \mu)]^m$  in Theorem 2.15 as the **density of  $\nu$  with respect to  $\mu$** . We denote it by  $f = \frac{d\nu}{d\mu}$  in analogy to the notation of differential calculus.

The following is the **polar decomposition** of a measure and is an obvious corollary to Radon-Nikodym.

**Corollary 2.17 (Polar Decomposition)** *Let  $(X, \mathcal{E})$  be a measurable space and  $\mu$  be a  $\mathbb{R}^m$ -valued measure on  $X$ . Then  $f = \frac{d\mu}{d\|\mu\|} \in [L^1(X, \mu)]^m$  has values in  $S^{m-1}$  and satisfies  $\mu = f \|\mu\|$ .*

**Proof** For a proof, see, e.g. [4]. □

### 2.1.1 Radon measures and operations on measures

In this short section, we recall the important notion of *Radon measures* and some operations on measures.

**Definition 2.18** *Let  $X$  be a locally compact and separable (lcs) metric space,  $\mathcal{B}(X)$  its Borel  $\sigma$ -algebra.*

- (i) *A positive measure on  $(X, \mathcal{B}(X))$  is called a **Borel measure**. If a Borel measure is finite on compacts, it is called a **positive Radon measure**.*
- (ii) *A  $\mathbb{R}^m$ -valued set function defined on the relatively compact Borel sets of  $X$ , that is, a measure on  $(K, \mathcal{B}(K))$  for every compact set  $K \subset X$ , is called a  **$\mathbb{R}^m$ -valued Radon measure**. We denote by  $[\mathcal{M}_{\text{loc}}(X)]^m$  the space of  $\mathbb{R}^m$ -valued Radon measures.*
- (iii) *If  $\mu : \mathcal{B}(X) \rightarrow \mathbb{R}^m$  is a  $\mathbb{R}^m$ -valued measure with  $\|\mu\|(X) < \infty$ , we say that  $\mu$  is **finite  $\mathbb{R}^m$ -valued Radon measure**. We denote  $[\mathcal{M}(X)]^m$  the space of finite  $\mathbb{R}^m$ -valued Radon measures.*

We have the following neat property of Borel measures.

**Proposition 2.19** *Let  $X$  be an lcs metric space and  $\mu$  a Borel measure on  $X$ . Let  $E$  be a  $\mu$ -measurable set. Then it holds that:*

- (i) *If  $\mu$  is  $\sigma$ -finite then*

$$\mu(E) = \sup\{\mu(K) \mid K \subset E, K \text{ compact}\}. \quad (2.30)$$

- (ii) *If there is a sequence  $(X_h)_{h \in \mathbb{N}}$  of open subsets of  $X$  such that  $\mu(X_h) < \infty$  for any  $h \in \mathbb{N}$  and  $X = \bigcup_{h \in \mathbb{N}} X_h$ , then*

$$\mu(E) = \inf\{\mu(A) \mid E \subset A, A \text{ open}\}. \quad (2.31)$$

**Proof** For a proof, see Theorem 2.10 in [5]. □

**Remark 2.20** (ii) in Proposition 2.19 is in particular true in the case of  $X = \mathbb{R}^N$  and  $X_h = B(0, h)$  for any  $h \in \mathbb{N}$ .

It also makes sense to introduce here the *weak\* convergence* for measures here.

**Definition 2.21** *Let  $X$  be an lcs metric space and  $\mu \in [\mathcal{M}_{\text{loc}}(X)]^m$  and  $(\mu_h)_{h \in \mathbb{N}} \subset [\mathcal{M}_{\text{loc}}(X)]^m$ . Then, we say that  $\mu_h$  **locally weakly\* converges** to  $\mu$  if*

$$\lim_{h \rightarrow \infty} \int_X u d\mu_h = \int_X u d\mu, \quad \forall u \in C_c(X). \quad (2.32)$$

We have the following weak compactness property for sequences of finite Radon measures.

**Theorem 2.22** *If  $(\mu_h)_{h \in \mathbb{N}}$  is a sequence of finite positive or  $\mathbb{R}^m$ -valued Radon measures in an lcs metric space  $X$  with  $\sup\{\|\mu\|(X) \mid h \in \mathbb{N}\} < \infty$ , then it has a locally weakly\* converging subsequence. Moreover, the map  $\mu \mapsto \|\mu\|(X)$  is lower semicontinuous with respect to the local weak\* convergence.*

**Proof** For a proof, see Theorem 1.59 in [3].  $\square$

The next proposition characterizes the total variation of measures  $\mu \in [\mathcal{M}(X)]^m$ .

**Proposition 2.23** *Let  $X$  be an lcs metric space and  $\mu \in [\mathcal{M}(X)]^m$ . Then for every open set  $A \subset X$  it holds that:*

$$\|\mu\|(A) = \sup \left\{ \int_X u \cdot d\mu \mid u \in [C_c(A)]^m, \|u\|_\infty \leq 1 \right\}. \quad (2.33)$$

**Remark 2.24** *In Proposition 2.23 we mean by  $\|u\|_\infty$  the following norm on  $C_c(A)$ :*

$$\|u\|_\infty = \sup_{x \in A} \|u(x)\|. \quad (2.34)$$

**Proof** By the polar decomposition of  $\mu$  from Corollary 2.17, we can find a unique function  $f : X \rightarrow S^{m-1}$  in  $[L^1(X, \|\mu\|)]^m$  such that  $\mu = f \|\mu\|$ . So for any open set  $A \subset X$  and for any  $u \in [C_c(A)]^m$  with  $\|u\|_\infty \leq 1$  we have that:

$$\int_A u \cdot d\mu = \int_A u \cdot f d\|\mu\| \leq \|\mu\|(A). \quad (2.35)$$

This proves the inequality  $\geq$  in 2.33. We find that  $[C_c(A)]^m$  is dense in  $[L^1(A, \|\mu\|)]^m$ . So, there exists a sequence  $(u_h)_{h \in \mathbb{N}} \subset [C_c(A)]^m$  such that  $\|(u_h)_i - f_i\|_1 \rightarrow 0$  as  $h \rightarrow \infty$  for all  $i = 1, \dots, m$ . Up to replacing  $u_h$  by  $\frac{u_h}{\|u_h\|_\infty}$  we may assume that  $\|u_h\| = 1$ :

$$\lim_{h \rightarrow \infty} \int_A u_h \cdot d\mu = \lim_{h \rightarrow \infty} \int_A u_h \cdot f d\|\mu\| = \|\mu\|(A). \quad (2.36)$$

This proves the inequality  $\leq$  in 2.33.  $\square$

The following result due to Riesz characterizes Radon measures in terms of bounded linear functionals.

**Theorem 2.25 (Riesz)** *Let  $X$  be a lcs metric space and  $L : [C_c(X)]^m \rightarrow \mathbb{R}$  be a bounded linear functional, i.e.*

$$\|L\| = \sup\{L(u) \mid u \in [C_c(X)]^m, \|u\|_\infty \leq 1\} < \infty. \quad (2.37)$$

*Then, there exists a unique  $\mathbb{R}^m$ -valued finite Radon measure  $\mu$  on  $X$ , such that:*

$$L(u) = \int_X u \cdot d\mu, \quad \forall u \in [C_c(X)]^m, \quad (2.38)$$

*and  $\|L\| = \|\mu\|(X)$ .*

**Proof** For a proof, see Theorem 2.14 in [6]. □

Next, we will introduce the *support of a measure*.

**Definition 2.26** Let  $X$  be an lcs metric space and  $\mu$  a positive measure on  $X$ . We define the **support of  $\mu$**  by

$$\text{supp}(\mu) := \left\{ x \in X \mid \forall U \underset{\text{open}}{\subset} X : (x \in U \implies \mu(U) > 0) \right\}. \quad (2.39)$$

If  $\mu$  is a  $\mathbb{R}^m$ -valued measure we define its **support** by  $\text{supp}(\mu) := \text{supp}(\|\mu\|)$ . More generally, we have the following notion. We say that a  $\mathbb{R}^m$ -valued measure on a measurable space  $(X, \mathcal{E})$  is **concentrated on a subset**  $S \subset X$  if  $S \in \mathcal{E}$  and  $\|\mu\|(X \setminus S) = 0$ .

**Proposition 2.27** Let  $X$  be a lcs metric space and  $\mu$  a positive measure on  $X$ . Then,  $\text{supp}(\mu)$  is a closed subset of  $X$  and in particular we have:

$$\text{supp}(\mu) = \bigcap \{ A \subset X \mid A \text{ is closed and } \mu(X \setminus A) = 0 \}. \quad (2.40)$$

**Proof** It suffices to prove 2.40. Indeed, let  $x \in \text{supp}(\mu)$  and  $\mu$  be concentrated on a closed set  $A \subset X$ . Suppose  $x \notin A$ , that is,  $x \in X \setminus A$ . Then  $U = X \setminus A$  is an open neighborhood of  $x$  and since  $x \in \text{supp}(\mu)$  we must have  $\mu(U) > 0$ . But this is a contradiction, since  $\mu(U) = 0$  as  $\mu$  is concentrated on  $A$ . So we must have  $x \in A$  for all closed sets  $A \subset X$  with  $\mu(X \setminus A) = 0$ . Conversely, if  $x \in A$  for all closed sets  $A \subset X$  with  $\mu(X \setminus A) = 0$ , and  $U \subset X$  is an open neighborhood of  $x$ , suppose  $\mu(U) = 0$ . Then  $A = X \setminus U$  is closed and  $\mu$  is concentrated on  $A$ . So we must have  $x \in A$ . This contradicts  $x \in U = X \setminus A$ . So we must have  $\mu(U) > 0$ . This shows the equality 2.40. □

Now let us define the *restriction of a measure*.

**Definition 2.28** Let  $\mu$  be a positive or  $\mathbb{R}^m$ -valued measure on a measurable space  $(X, \mathcal{E})$ . If  $E \in \mathcal{E}$  we define the **restriction of  $\mu$  to  $E$**  by  $\mu \llcorner E(F) = \mu(E \cap F)$  for every  $F \in \mathcal{E}$ .

The next proposition establishes the circumstances under which the restriction of a Radon measure stays a Radon measure.

**Proposition 2.29** Let  $X$  be a lcs metric space and  $\mu$  a positive or  $\mathbb{R}^m$ -valued Radon measure on  $X$ . If  $E \in \mathcal{B}(X)$ , then  $\mu \llcorner E$  is a positive resp.  $\mathbb{R}^m$ -valued Radon measure.

**Proof** Suppose first that  $\mu$  is a positive Radon measure on  $X$ . Then for any  $A \subset X$  and Borel set  $B \in \mathcal{B}(X)$  we have that:

$$\begin{aligned} \mu \llcorner E(A) &= \mu(E \cap A) \\ &\stackrel{\mu \text{ Borel}}{=} \mu(E \cap A \cap B) + \mu((E \cap A) \setminus B) \\ &= \mu \llcorner E(A \cap B) + \mu \llcorner E(A \setminus B). \end{aligned} \quad (2.41)$$

And since  $\mu$  is finite on compacts, also  $\mu \llcorner E$  is finite on compacts. This shows that  $\mu$  is a Radon measure on  $X$ .

If  $\mu$  is a  $\mathbb{R}^m$ -valued Radon measure on  $X$ . The clearly  $\mu \llcorner E$  is still a set function defined on the relatively compact subsets of  $X$  and for every  $K \subset X$  compact, we have  $\mu \llcorner E$  still a measure on  $(K, \mathcal{B}(K))$ .  $\square$

Now, let us define the *pushforward of a measure*.

**Definition 2.30** Let  $(X, \mathcal{E})$  and  $(Y, \mathcal{F})$  be two measurable spaces and let  $f : X \rightarrow Y$  be such that  $f^{-1}(F) \in \mathcal{E}$  whenever  $F \in \mathcal{F}$ . For any positive or  $\mathbb{R}^m$ -valued measure  $\mu$  in  $(X, \mathcal{E})$  we define the measure  $f_{\#}\mu$  on  $(Y, \mathcal{F})$  by

$$f_{\#}\mu(F) := \mu(f^{-1}(F)), \quad \forall F \in \mathcal{F}. \quad (2.42)$$

**Proposition 2.31** Let  $X, Y$  be two lcs metric spaces and  $f : X \rightarrow Y$  be a continuous and proper function. If  $\mu$  is a Radon measure on  $X$  then  $f_{\#}\mu$  is a Radon measure on  $Y$ .

**Proof** By continuity we have  $f^{-1}(B) \in \mathcal{B}(X)$  whenever  $B \in \mathcal{B}(Y)$ . So  $f_{\#}\mu$  is a Borel measure on  $Y$ . Now let  $K \subset Y$ , then since  $f$  is proper we have  $f^{-1}(K) \subset X$  still compact and so  $f_{\#}\mu(K) = \mu(f^{-1}(K)) < \infty$ , as  $\mu$  is a Radon measure on  $X$ . So  $f_{\#}\mu$  is a Radon measure on  $Y$ .  $\square$

We close this section about abstract measure theory with the statement of the *Fubini theorem*.

**Theorem 2.32** Let  $(X_1, \mathcal{E}_1)$  and  $(X_2, \mathcal{E}_2)$  be measurable spaces and  $\mu_1$  and  $\mu_2$  be  $\sigma$ -finite measures on  $X_1$  resp.  $X_2$ . Then, there is a unique  $\sigma$ -finite measure  $\mu$  on  $(X_1 \times X_2, \mathcal{E}_1 \times \mathcal{E}_2)$  such that

$$\mu(E_1 \times E_2) = \mu_1(E_1)\mu_2(E_2), \quad \forall E_1 \in \mathcal{E}_1, E_2 \in \mathcal{E}_2. \quad (2.43)$$

Moreover, if  $u : X_1 \times X_2 \rightarrow \mathbb{R}$  is  $\mu$ -summable, we have that

$$x \mapsto \int_{X_2} u(x, y) d\mu_2(y) \text{ and } y \mapsto \int_{X_1} u(x, y) d\mu_1(x), \quad (2.44)$$

are  $\mu_1$ - resp.  $\mu_2$ -summable and

$$\begin{aligned} \int_{X_1 \times X_2} u d\mu &= \int_{X_1} \left( \int_{X_2} u(x, y) d\mu_2(y) \right) d\mu_1(x) \\ &= \int_{X_2} \left( \int_{X_1} u(x, y) d\mu_1(x) \right) d\mu_2(y). \end{aligned} \quad (2.45)$$

**Proof** For a proof, see, Theorem 2.32 in [4].  $\square$

## 2.2 Geometric measure theory

This section is dedicated to familiarizing the reader with the basics of geometric measure theory. We will go over the convolution of measures, some Sobolev space results, the Lebesgue-Besicovitch theorem, tangent measures, Hausdorff measures, rectifiability, area formula, approximate tangent spaces, and the coarea formula. In contrast to the previous section, the ambient space will always be Euclidean space  $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N))$  instead of an abstract measurable space  $(X, \mathcal{E})$ .

### 2.2.1 Convolution

This subsection concerns itself with the definition and properties of the convolution of functions and measures. The following proposition contains all results related to the convolution of functions. We also fix the notation  $\mathcal{L}^N$  for the **standard Lebesgue measure on  $\mathbb{R}^N$**  for  $N \in \mathbb{N}$  and write  $dx$  or  $d\mathcal{L}^N(x)$  for the integration with respect to  $\mathcal{L}^N$ .

**Proposition 2.33** *For any two functions  $f, g : \mathbb{R}^N \rightarrow \mathbb{R}$ , we define their **convolution** by*

$$f * g(x) = \int_{\mathbb{R}^N} f(y)g(x-y)dy, \quad x \in \mathbb{R}^N. \quad (2.46)$$

*Then the following properties are satisfied:*

- (i) *If  $f \in L^1(\mathbb{R}^N, \mathcal{L}^N)$  and  $g \in L^p(\mathbb{R}^N, \mathcal{L}^N)$  for  $p \in [1, \infty]$ , then  $f * g \in L^p(\mathbb{R}^N, \mathcal{L}^N)$  and  $\|f * g\|_p \leq \|f\|_1 \|g\|_p$ .*
- (ii) *If  $f \in L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  and  $g \in C_c(\mathbb{R}^N)$ , then  $f * g \in C(\mathbb{R}^N)$ .*
- (iii) *For the supports,  $\text{supp}(f * g) \subset \text{supp}(f) + \text{supp}(g)$ .*
- (iv) *If  $f \in L^1(\mathbb{R}^N, \mathcal{L}^N)$  and  $g \in C^\infty(\mathbb{R}^N)$  or  $f \in L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  and  $g \in C^\infty_c(\mathbb{R}^N)$ , then it holds that  $f * g \in C^\infty(\mathbb{R}^N)$  and*

$$\nabla^\alpha(f * g)(x) = f * (\nabla^\alpha g)(x), \quad (2.47)$$

*for every multiindex  $\alpha \in \mathbb{N}^N$ .*

- (v) *Let  $\rho \in C^\infty_c(\mathbb{R}^N)$  be a **mollifier**, i.e.  $\rho \geq 0$ ,  $\rho(-x) = \rho(x)$ ,  $\text{supp}(\rho) \subset B(0, 1)$  and  $\int_{\mathbb{R}^N} \rho(x)dx = 1$ . We call  $\rho_\varepsilon(x) = \varepsilon^{-N} \rho(\frac{x}{\varepsilon})$  for any  $\varepsilon > 0$  a **family of mollifiers**. If now  $f \in L^p_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$ , then  $f * \rho_\varepsilon \in C^\infty(\mathbb{R}^N)$  and*

$$\lim_{\varepsilon \rightarrow 0^+} \|f * \rho_\varepsilon - f\|_{L^p(A, \mathcal{L}^N)} = 0, \quad \forall A \subset\subset \mathbb{R}^N, \quad (2.48)$$

*for every  $p \in [1, \infty)$ .*

---

<sup>1</sup>If  $(X, \tau)$  is a topological space, we write  $A \subset\subset X$  if  $A \subset X$  is relatively compact, i.e.  $\overline{A}$  is compact in  $X$ .

(vi) If  $f \in C(\mathbb{R}^N)$ , then convergence  $f * \rho_\varepsilon \xrightarrow{\varepsilon \rightarrow 0^+} f$  is uniform on compacts in  $\mathbb{R}^N$ .

**Proof** For a proof of this, see the chapter on convolutions in [6].  $\square$

Knowing this, we now define the convolution of continuous functions with measures.

**Definition 2.34** Let  $\mu$  be an  $\mathbb{R}^m$ -valued Radon measure on an open set  $\Omega \subset \mathbb{R}^N$ . If  $f \in C(\Omega)$  we define the **convolution of  $f$  and  $\mu$**  by

$$\mu * f(x) := \int_{\Omega} f(x - y) d\mu(y), \quad (2.49)$$

whenever it makes sense to do so, i.e. when the integral in 2.49 is well-defined.

We have the following regularity and convergence results:

**Proposition 2.35** Let  $\Omega \subset \mathbb{R}^N$  be an open set,  $\mu = (\mu_1, \dots, \mu_n) \in [\mathcal{M}_{\text{loc}}(\Omega)]^m$ ,  $(\mu_h)_{h \in \mathbb{N}} \subset [\mathcal{M}_{\text{loc}}(\Omega)]^m$  and  $(\rho_\varepsilon)_{\varepsilon \in \mathbb{R}_{>0}}$  a family of mollifiers.

- (i) If  $\mu_h$  locally weak\* converges to  $\mu$  and  $f \in C_c(\Omega)$ , then  $\mu_h * f \xrightarrow{h \rightarrow \infty} \mu * f$  uniformly on compact subsets of  $\Omega$ .
- (ii) The functions  $\mu * \rho_\varepsilon$  belong to  $[C^\infty(\Omega_\varepsilon)]^m$  and  $\nabla^\alpha(\mu * \rho_\varepsilon) = \mu * \nabla^\alpha \rho_\varepsilon$  for any  $\alpha \in \mathbb{N}^N$ .<sup>2</sup>
- (iii) The measures  $\mu_\varepsilon := \mu * \rho_\varepsilon \mathcal{L}^N$  locally weakly\* converge in  $\Omega$  to  $\mu$  as  $\varepsilon \rightarrow 0$  and the estimate

$$\int_E \|\mu * \rho_\varepsilon\| dx \leq \|\mu\| (U_\varepsilon(E)), \quad (2.50)$$

holds whenever  $E \subset \Omega_\varepsilon$  is a Borel set and where  $U_\varepsilon(E) := \{x \in E \mid d(x, E^c) < \varepsilon\}$ .

- (iv) The measures  $\|\mu_\varepsilon\|$  locally weak\* converge in  $\Omega$  to  $\|\mu\|$  as  $\varepsilon \rightarrow 0^+$ .

**Remark 2.36** We note that if  $v \in L^1(\Omega, \mathcal{L}^N)$  and either  $\mu$  is concentrated on  $\Omega_\varepsilon$  or  $v = 0$   $\mathcal{L}^N$ -a.e. outside of  $\Omega_\varepsilon$  we have the following by application of Fubini's theorem 2.32 and the fact that  $\rho_\varepsilon(-x) = \rho_\varepsilon(x)$ :

$$\begin{aligned} \int_{\Omega} \mu * \rho_\varepsilon(x) v(x) dx &= \int_{\Omega} \int_{\Omega_\varepsilon} \rho_\varepsilon(x - y) d\mu(y) v(x) dx \\ &= \int_{\Omega_\varepsilon} \int_{\Omega} \rho_\varepsilon(y - x) v(x) dx d\mu(y) \\ &= \int_{\Omega} v * \rho_\varepsilon(y) d\mu(y). \end{aligned} \quad (2.51)$$

<sup>2</sup>Here  $\Omega_\varepsilon$  denotes the set of all  $x \in \Omega$  with  $d(x, \partial\Omega) > \varepsilon$ .

**Proof** We give the proof of (i)-(iii) and for (iv) direct the reader to Theorem 2.2 in [3].

- (i) This follows immediately from the local weak\* convergence of  $\mu_h$  to  $\mu$  as for any  $K \subset \Omega$  compact and for every  $x \in K$  we have the following:

$$\|\mu_h * f(x) - \mu * f(x)\| = \left\| \int_{\Omega} f(x-y) d\mu_h - \int_{\Omega} f(x-y) d\mu \right\| \rightarrow 0, \quad (2.52)$$

as  $h \rightarrow \infty$ . This proves uniform convergence on compacts.

- (ii) We fix standard orthonormal basis  $\{e_i\}_{i=1}^N$  of  $\mathbb{R}^N$  and consider for arbitrary  $x \in \Omega_\varepsilon$  and any  $i = 1, \dots, N$  the difference quotient for any positive sequence  $s_k \xrightarrow{k \rightarrow \infty} 0$  we have that

$$\frac{\mu * \rho_\varepsilon(x + s_k e_i) - \mu * \rho_\varepsilon(x)}{s_k} = \int_{\Omega} \frac{\rho_\varepsilon(x - y + s_k e_i) - \rho_\varepsilon(x - y)}{s_k} d\mu(y). \quad (2.53)$$

Now by the Dominated Convergence Theorem 2.25 we find that for all  $i = 1, \dots, N$ :

$$\begin{aligned} \frac{\partial}{\partial x_i}(\mu * \rho_\varepsilon)(x) &= \lim_{k \rightarrow \infty} \int_{\Omega} \frac{\rho_\varepsilon(x - y + s_k e_i) - \rho_\varepsilon(x - y)}{s_k} d\mu(y) \\ &= \int_{\Omega} \frac{\partial}{\partial x_i} \rho_\varepsilon(x - y) d\mu(y) \\ &= (\mu * \frac{\partial}{\partial x_i} \rho_\varepsilon)(x). \end{aligned} \quad (2.54)$$

This proves (ii) for all  $\alpha \in \mathbb{N}^N$  with  $\|\alpha\|_1 := \alpha_1 + \dots + \alpha_N = 1$ . Assume that (ii) holds for all  $d \geq 1$  and all  $\alpha \in \mathbb{N}^N$  with  $\|\alpha\|_1 \leq d$ . Now, replace  $\alpha \in \mathbb{N}^N$  with  $\|\alpha\|_1 = d + 1$ . Then we can choose  $k = 1, \dots, N$  such that  $\alpha_k \neq 0$ . If  $\alpha_k = d + 1$  we have by induction hypothesis that

$$\begin{aligned} \nabla^\alpha(\mu * \rho_\varepsilon) &= \frac{\partial}{\partial x_k} \frac{\partial^d}{\partial x_i^d} \\ &= \frac{\partial}{\partial x_k} (\mu * \frac{\partial^d}{\partial x_k^d} \rho_\varepsilon) \\ &= \mu * \nabla^\alpha \rho_\varepsilon. \end{aligned} \quad (2.55)$$

Otherwise we can define  $\tilde{\alpha} = \alpha - \alpha_k e_k$  with  $\|\tilde{\alpha}\|_1 \leq d$  and so by induc-

tion hypothesis:

$$\begin{aligned}
 \nabla^\alpha(\mu * \rho_\varepsilon) &= \frac{\partial^{\alpha_k}}{\partial x_k^{\alpha_k}} \nabla^{\tilde{\alpha}}(\mu * \rho_\varepsilon) \\
 &= \frac{\partial^{\alpha_k}}{\partial x_k^{\alpha_k}} (\mu * \nabla^{\tilde{\alpha}} \rho_\varepsilon) \\
 &= \mu * \frac{\partial^{\alpha_k}}{\partial x_k^{\alpha_k}} \nabla^{\tilde{\alpha}} \rho_\varepsilon \\
 &= \mu * \nabla^\alpha \rho_\varepsilon.
 \end{aligned} \tag{2.56}$$

This shows (ii) for all  $\alpha \in \mathbb{N}^N$  and in particular  $\mu * \rho_\varepsilon \in C^\infty(\Omega_\varepsilon)$ .

(iii) By Remark 2.36 we deduce that for all  $\varepsilon > 0$  and for all  $u \in C_c(\Omega_\varepsilon)$  that:

$$\int_{\Omega} \mu * \rho_\varepsilon(x) u(x) d\mathcal{L}^N(x) = \int_{\Omega} u * \rho_\varepsilon(y) d\mu(y). \tag{2.57}$$

By Proposition 2.33 (iv) and the continuity of  $u$  we find as  $\varepsilon \rightarrow 0$  that 2.57 implies the local weak\* convergence of  $\mu_\varepsilon = \mu * \rho_\varepsilon \mathcal{L}^N$  to  $\mu$ . Moreover, if  $E \subset \Omega_\varepsilon$  is a Borel set we see by Fubini's theorem:

$$\begin{aligned}
 \int_E \|\mu * \rho_\varepsilon\| (x) dx &= \int_E \left\| \int_{\Omega} \rho_\varepsilon(x-y) d\mu(y) \right\| dx \\
 &\leq \int_E \int_{\Omega} \rho_\varepsilon(x-y) d\|\mu\| dx \\
 &= \int_{\Omega} \int_E \rho_\varepsilon(x-y) dx d\|\mu\| \\
 &\stackrel{(*)}{=} \int_{U_\varepsilon(E)} \int_E \rho_\varepsilon(x-y) dx d\|\mu\| \\
 &\leq \|\mu\| (U_\varepsilon(E)).
 \end{aligned} \tag{2.58}$$

At (\*) we used that for  $x \in E$  we have  $\rho_\varepsilon(x-y) \neq 0$  if and only if  $y \in U_\varepsilon(E)$ .  $\square$

### 2.2.2 Sobolev spaces and Lipschitz functions

In this section, we discuss the definitions and properties pertaining to Sobolev spaces and Lipschitz functions. Of interest is in particular *the Rademacher theorem* establishing almost everywhere differentiability of Lipschitz functions. For most proofs, we direct the reader to Sections 2.2 and 2.3 of [3]. We start with the definition of Sobolev spaces and weak derivatives.

**Definition 2.37** Let  $\Omega \subset \mathbb{R}^N$  be an open set,  $i = 1, \dots, N$  and  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . If there is  $g \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  such that

$$\int_{\Omega} u \frac{\partial \varphi}{\partial x_i} dx = - \int_{\Omega} \varphi g dx, \quad \forall \varphi \in C_c^\infty(\Omega), \tag{2.59}$$

then we say that  $u$  has **weak  $i$ -th derivative** given by  $g$ . If the  $i$ -th weak derivative exists, it is unique and is denoted by  $\nabla_i u$  or  $\frac{\partial u}{\partial x_i}$ .

For  $p \in [0, \infty]$  we say that  $u \in W^{1,p}(\Omega)$  if  $u \in L^p(\Omega, \mathcal{L}^N)$  and has all weak derivatives in  $L^p(\Omega, \mathcal{L}^N)$  for every  $i = 1, \dots, N$ . We write for  $u \in W^{1,p}(\Omega)$ :

$$\nabla u := (\nabla_1 u, \dots, \nabla_N u). \quad (2.60)$$

We call the space  $W^{1,p}(\Omega)$  **Sobolev** and it becomes a Banach space with the **Sobolev norm**

$$\|u\|_{W^{1,p}(\Omega)}^p = \|u\|_p^p + \sum_{i=1}^N \|\nabla_i u\|_p^p, \quad (2.61)$$

if  $1 \leq p < \infty$ , and

$$\|u\|_{W^{1,\infty}(\Omega)} = \|u\|_\infty + \sum_{i=1}^N \|\nabla_i u\|_\infty. \quad (2.62)$$

The next proposition summarizes all the properties of Sobolev spaces.

**Proposition 2.38** Let  $\Omega \subset \mathbb{R}^N$  and  $(u_h)_{h \in \mathbb{N}} \subset W^{1,p}(\Omega)$  converge in  $L^p(\Omega, \mathcal{L}^N)$  to some  $u \in L^p(\Omega, \mathcal{L}^N)$ . The following statements are true:

- (i) If  $p \in [0, \infty]$  and for every  $i = 1, \dots, N$  there is  $g_i \in L^p(\Omega, \mathcal{L}^N)$  such that  $\|\nabla_i u_h - g_i\| \xrightarrow{h \rightarrow \infty} 0$ , then  $u \in W^{1,p}(\Omega)$  and  $\nabla_i u = g_i$ .
- (ii) If  $p \in (1, \infty]$  and the sequences  $(\nabla_i u_h)_{h \in \mathbb{N}}$  are bounded for any  $i = 1, \dots, N$ , then  $u \in W^{1,p}(\Omega)$  and  $\nabla_i u_h \rightarrow \nabla_i u$  weakly for any  $i = 1, \dots, N$ .
- (iii) If  $u, g \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  and  $i = 1, \dots, N$  we have the following equivalence:  $g = \nabla_i u$  if and only if there is a sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(\Omega)$  such that  $u_h \rightarrow u$  and  $\nabla_i u_h \rightarrow g$  in  $L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . Moreover, if  $p \in [1, \infty)$  the space  $C^\infty(\Omega) \cap W^{1,p}(\Omega)$  is dense in  $W^{1,p}(\Omega)$ .

**Proof** For proofs of this, see Chapter 1 in [7]. □

Now we turn to more tangible functions in the realm of *Lipschitz functions*.

**Definition 2.39** Let  $E \subset \mathbb{R}^N$ . We say a function  $f : E \rightarrow \mathbb{R}^m$  is a **Lipschitz function** in  $E$  and write  $f \in [\text{Lip}(E)]^m$  if

$$\text{Lip}(f, E) := \sup \left\{ \frac{\|f(x) - f(y)\|}{\|x - y\|} \mid x \neq y, x, y \in E \right\} < \infty. \quad (2.63)$$

We call  $\text{Lip}(f, E)$  the **Lipschitz constant** of  $f$ .

**Remark 2.40** By definition we have  $\text{Lip}(f, E)$  the least constant  $L \in [0, \infty]$  such that for all  $x, y \in E$  it holds that:

$$\|f(x) - f(y)\| \leq L \|x - y\|. \quad (2.64)$$

The next proposition establishes the relation of Lipschitz functions to Sobolev functions.

**Proposition 2.41** *Let  $\Omega$  be a bounded, convex, and open set, and  $u : \Omega \rightarrow \mathbb{R}$  a function. Then we have  $u \in W^{1,\infty}(\Omega)$  if and only if  $\text{Lip}(u, \Omega) < \infty$ . In particular, if  $u \in W^{1,\infty}(\Omega)$  then  $\|\nabla u\|_{L^\infty(\Omega, \mathcal{L}^N)} = \text{Lip}(u, \Omega)$ .*

The next theorem due to Rademacher establishes the almost-everywhere differentiability of  $W^{1,\infty}(\Omega)$  function.

**Theorem 2.42** *Any function  $u \in W^{1,\infty}(\Omega)$  is differentiable  $\mathcal{L}^N$ -a.e. in  $\Omega$  and the differential coincides  $\mathcal{L}^N$ -a.e. with the weak derivative  $\nabla u$  in  $L^\infty(\Omega, \mathcal{L}^N)$ .*

### 2.2.3 Lebesgue-Besicovitch differentiation theorem

This section gives an overview of the results of Lebesgue and Besicovitch on the differentiation of measures. In addition, it introduces as a consequence the concept of *Lebesgue points*.

**Theorem 2.43** *Let  $\mu$  be a positive Radon measure on an open set  $\Omega \subset \mathbb{R}^N$  and  $\nu$  a  $\mathbb{R}^m$ -valued Radon measure. Then, for  $\mu$ -a.e.  $x \in \text{supp}(\mu)$  the limit*

$$f(x) := \lim_{r \rightarrow 0^+} \frac{\nu(B(x, r))}{\mu(B(x, r))} \in \mathbb{R}^m, \quad (2.65)$$

*exists and defines a Borel measurable function  $f \in [L^1_{\text{loc}}(\Omega, \mathcal{L}^N)]^m$ . Moreover, the decomposition of  $\nu$  given by the Radon-Nikodym theorem 2.15 is given by  $\nu = f\mu + \nu^s$ , where  $\nu^s = \nu \llcorner E$  and  $E$  is the  $\mu$ -null set given by*

$$E = (\Omega \setminus \text{supp}(\mu)) \cup \left\{ x \in \text{supp}(\mu) \mid \lim_{r \rightarrow 0^+} \frac{\|\nu\|(B(x, r))}{\mu(B(x, r))} \right\}. \quad (2.66)$$

**Proof** For a proof, see Theorem 2.22 in [3]. □

**Remark 2.44** *By Theorem 2.43 we have established that the density of  $\nu$  with respect to  $\mu$  is given  $\mu$ -a.e. by*

$$\frac{d\nu}{d\mu}(x) = \lim_{r \rightarrow 0^+} \frac{\nu(B(x, r))}{\mu(B(x, r))} \in \mathbb{R}^m. \quad (2.67)$$

The following corollary characterizes the so-called *Lebesgue points*.

**Corollary 2.45** *Let  $\mu$  be a positive Radon measure on an open set  $\Omega \subset \mathbb{R}^N$  and  $f \in L^1(\Omega, \mu)$ . Then for  $\mu$ -a.e.  $x \in \Omega$  the following equality holds:*

$$\lim_{r \rightarrow 0^+} \frac{1}{\mu(B(x, r))} \int_{B(x, r)} |f(y) - f(x)| d\mu(y) = 0. \quad (2.68)$$

**Proof** Consider for any  $q \in \mathbb{Q}$  the  $\mathbb{R}$ -valued Radon measure  $\nu_q := |f - q| \mu$ . By Theorem 2.43 we have for  $\mu$ -a.e.  $x \in \Omega$  the following limit exists:

$$\lim_{r \rightarrow 0^+} \frac{\nu_q(B(x, r))}{\mu(B(x, r))} = |f(x) - q|. \quad (2.69)$$

The last equality holds  $\mu$ -a.e.. By the density of  $\mathbb{Q}$  in  $\mathbb{R}$  the equation 2.68 follows.  $\square$

**Remark 2.46** We call the points  $x \in \Omega$ , where 2.68 holds, the **Lebesgue points** of  $f$ .

### 2.2.4 Tangent measures

This section gives an overview of the definition of *tangent measures* and how they relate to Radon measures.

**Definition 2.47** Let  $\mu$  be an  $\mathbb{R}^m$ -valued Radon measure on an open set  $\Omega \subset \mathbb{R}^N$ . We define for all  $x \in \mathbb{R}^N$  and  $r > 0$  the rescaled measure:

$$\mu_{x,r}(B) := \mu(x + rB). \quad (2.70)$$

Then we define the set of  $\text{Tan}(\mu, x)$  of all  $\mathbb{R}^m$ -valued finite Radon measures  $\nu$  on  $B(0, 1)$ , which are weak\* limits of

$$\frac{1}{c_{x,r_i}} \mu_{x,r_i} \text{ for } c_{x,r} := \|\mu\|(B(x, r)), \quad (2.71)$$

where  $(r_i)_{i \in \mathbb{N}} \subset (0, \infty)$  is a sequence with  $\lim_{i \rightarrow \infty} r_i = 0$ . We call  $\nu \in \text{Tan}(\mu, x)$  a **tangent measure to  $\mu$  at  $x$** .

We now consider an example of tangent measures.

**Example 2.48** Let  $\mu = f \mathcal{L}^N$  for some  $f \in L^1(\mathbb{R}^N, \mathcal{L}^N)$ . Then we note first that for all  $r \in (0, \infty)$  and  $x \in \mathbb{R}^N$  a Lebesgue point of  $f$  such that  $f(x) \neq 0$  it holds

$$\frac{d\mu_{x,r}}{d\mathcal{L}^N}(y) = r^N f(x + ry), \quad (2.72)$$

and as  $r \rightarrow 0$

$$c_{x,r} = (1 + o(1)) |f(x)| \omega_N r^N, \quad (2.73)$$

where  $\omega_N = \mathcal{L}^N(B(0, 1))$ . Then we find for any  $\varphi \in C(B(0, 1))$  that:

$$\begin{aligned} \lim_{r \rightarrow 0} \frac{1}{c_{x,r}} \int_{B(0,1)} \varphi(y) d\mu_{x,r}(y) &= \lim_{r \rightarrow 0} \frac{1}{(1 + o(1)) |f(x)| \omega_N r^N} \\ &\quad \int_{B(0,1)} r^N f(x + ry) d\mathcal{L}^N(y) \\ &= \frac{\text{sign}(f(x))}{\omega_N} \int_{B(0,1)} \varphi(y) d\mathcal{L}^N(y). \end{aligned} \quad (2.74)$$

This shows that any tangent measure to  $\mu$  at a Lebesgue point  $x$  of  $f$ , is given by  $\frac{\text{sign}(f(x))}{\omega_N} \mathcal{L}^N$ .

Now we give a complete characterisation of tangent measures in terms of the total variation measure.

**Theorem 2.49** *Let  $\mu$  be a  $\mathbb{R}^m$ -valued Radon measure on a open set  $\Omega \subset \mathbb{R}^N$ . Let  $\mu = f \|\mu\|$  be the polar decomposition of  $\mu$  (see Corollary 2.17). Then for any Lebesgue point  $x \in \Omega$  of  $f$  it holds that:*

$$v \in \text{Tan}(\mu, x) \iff \|v\| \in \text{Tan}(\|\mu\|, x) \quad (2.75)$$

and the polar decomposition of  $v$  is given by  $v = f(x) \|v\|$ . In particular, we have  $\text{Tan}(\mu, x) = f(x) \text{Tan}(\|\mu\|, x)$ .

**Proof** For a proof, consult Theorem 2.44 in [3].  $\square$

### 2.2.5 Hausdorff measures

This next part introduces all the theory used in this thesis pertaining *Hausdorff measures*. For this we generalise the Lebesgue measure  $\omega_N = \mathcal{L}^N(B(0, 1))$  of the  $N$ -dimensional unit ball to the following constant for any  $k \in [0, \infty)$ :

$$\omega_k = \frac{\pi^{\frac{k}{2}}}{\Gamma\left(1 + \frac{k}{2}\right)}, \quad (2.76)$$

where  $\Gamma(t) := \int_0^\infty s^{t-1} e^{-s} ds$  is the **Euler Gamma function**.

**Definition 2.50** *Let  $k \in [0, \infty)$  and  $E \subset \mathbb{R}^N$ . The  $k$ -dimensional Hausdorff measure of  $E$  is given by:*

$$\mathcal{H}^k(E) := \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^k(E) = \sup_{\delta > 0} \mathcal{H}_\delta^k(E), \quad (2.77)$$

where for  $\delta \in (0, \infty]$ ,  $\mathcal{H}_\delta^k(E)$  is the  $k$ -dimensional Hausdorff measure of  $E$  with step size  $\delta$  and given by

$$\mathcal{H}_\delta^k(E) := \inf \left\{ \sum_{i \in I} \omega_k \left[ \frac{\text{diam}(E_i)}{2} \right]^k \mid \text{diam}(E_i) < \delta, E \subset \bigcup_{i \in I} E_i \right\}. \quad (2.78)$$

The index sets  $I$  in 2.78 are finite and we have the convention  $\text{diam}(\emptyset) = 0$ .

The next proposition summarizes the most important properties of Hausdorff measures.

**Proposition 2.51** (i) *For all  $k \in [0, \infty)$  the measures  $\mathcal{H}^k$  on  $\mathbb{R}^N$  are Borel measures. Moreover, if  $E \in \mathcal{B}(\mathbb{R}^N)$  with  $\mathcal{H}^k(E) < \infty$  the restriction  $\mathcal{H}^k \llcorner E$  is a Radon measure on  $\mathbb{R}^N$ .*

(ii) For all  $k \in [0, \infty)$ ,  $E \subset \mathbb{R}^N$ ,  $z \in \mathbb{R}^N$  and  $\lambda > 0$  it holds that:

$$\mathcal{H}^k(E + z) = \mathcal{H}^k(E) \quad (2.79)$$

$$\mathcal{H}^k(\lambda E) = \lambda^k \mathcal{H}^k(E) \quad (2.80)$$

$$\mathcal{H}^k(E) = 0 \quad \text{if } k > N. \quad (2.81)$$

(iii) For all  $0 \leq \ell < k$  it holds that:  $\mathcal{H}^k(E) > 0 \implies \mathcal{H}^\ell(E) = \infty$ .

(iv) For all  $k \in [0, \infty)$ , open set  $\Omega \subset \mathbb{R}^N$  and for any Lipschitz function  $f : \Omega \rightarrow \mathbb{R}^m$  it holds that

$$\mathcal{H}^k(f(E)) \leq \text{Lip}(f, \Omega)^k \mathcal{H}^k(E), \quad \forall E \subset \Omega. \quad (2.82)$$

(v) **(Isodiametric Inequality)**  $\mathcal{L}^N(B) \leq \omega_N \left( \frac{\text{diam}(B)}{2} \right)^N$  for all  $\mathcal{L}^N$ -measurable sets  $B \subset \mathbb{R}^N$ .

(vi)  $(\mathcal{L}^N = \mathcal{H}^N)$  For all Borel sets  $B \subset \mathbb{R}^N$  and every  $\delta \in (0, \infty]$  it holds that

$$\mathcal{L}^N(B) = \mathcal{H}_\delta^N(B) = \mathcal{H}^N(B). \quad (2.83)$$

**Proof** (i) For a proof of this, see Proposition 2.49 (i) in [3].

(ii) Note that the function  $E \mapsto \text{diam}(E)^k$  satisfies both 2.79 and 2.80, therefore  $\mathcal{H}^k$  satisfies 2.79 and 2.80. Now let  $k > N$ . Note that any unit cube  $Q \subset \mathbb{R}^N$  can be covered by  $n^N$  unit cubes with side length  $\frac{1}{n}$ . This gives us the estimate

$$\mathcal{H}_\delta^k(Q) \leq \omega_k \left( \frac{\sqrt{N}}{2} \right)^k n^{N-k}, \quad (2.84)$$

for all  $\delta > \frac{\sqrt{N}}{n}$ . Then for  $n \rightarrow \infty$  (and thus  $\delta \rightarrow 0$ ) we find  $\mathcal{H}^k(Q) = 0$ . Since we can cover  $\mathbb{R}^N$  with countably many unit cubes, we get by subadditivity that  $\mathcal{H}^k(E) \leq \mathcal{H}^k(\mathbb{R}^N) = 0$  for all  $E \subset \mathbb{R}^N$ .

(iii) See Proposition 2.49 (iii) in [3] for a proof.

(iv) This is immediate from the inequality  $\text{diam}(f(E)) \leq \text{Lip}(f, \Omega) \text{diam}(E)$ .

(v) See Proposition 2.52 in [3] for a proof.

(vi) See Theorem 2.53 in [3] for a proof.  $\square$

We have also the following nice connection between Radon measures and Hausdorff measures.

**Definition 2.52** Let  $\mu$  be a positive or  $\mathbb{R}^m$ -valued Radon measure on an open set  $\Omega \subset \mathbb{R}^N$  and  $k \geq 0$ . The **upper** and **lower  $k$ -dimensional densities of  $\mu$  at  $x$**  are respectively given by

$$\Theta_k^*(\mu, x) := \limsup_{r \rightarrow 0^+} \frac{\mu(B(x, r))}{\omega_k r^k} \text{ resp. } \Theta_{*k}(\mu, x) := \liminf_{r \rightarrow 0^+} \frac{\mu(B(x, r))}{\omega_k r^k}. \quad (2.85)$$

If  $\Theta_k^*(\mu, x) = \Theta_{*k}(\mu, x)$ , we denote the  **$k$ -dimensional density of  $\mu$  at  $x$**  by  $\Theta_k(\mu, x)$ . Furthermore, we define for  $E \subset \Omega$  the **upper** and **lower  $k$ -dimensional densities of  $E$  at  $x$**  by

$$\Theta_k^*(E, x) := \limsup_{r \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap B(x, r))}{\omega_k r^k} \quad (2.86)$$

$$\Theta_{*k}(E, x) := \liminf_{r \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap B(x, r))}{\omega_k r^k}. \quad (2.87)$$

If  $\Theta_k^*(E, x) = \Theta_{*k}(E, x)$ , we denote the  **$k$ -dimensional density of  $E$  at  $x$**  by  $\Theta_k(E, x)$ .

We have the following implication of densities of measures and sets.

**Proposition 2.53** Let  $\Omega \subset \mathbb{R}^N$  be an open set and  $\mu$  be a positive Radon measure  $\mu$  in  $\Omega$ . Then for any  $t \in (0, \infty)$  and any Borel set  $B \subset \Omega$  the following implications hold:

$$\Theta_k^*(\mu, x) \geq t \quad \forall x \in B \implies \mu \geq t \mathcal{H}^k \llcorner B \quad (2.88)$$

$$\Theta_k^*(\mu, x) \leq t \quad \forall x \in B \implies \mu \leq 2^k t \mathcal{H}^k \llcorner B. \quad (2.89)$$

**Proof** For a proof, see Theorem 2.56 in [3].  $\square$

**Theorem 2.54** If  $\mu$  is Borel regular measure on open set  $\Omega \subset \mathbb{R}^N$ ,  $A \subset \Omega$  a Borel set with  $\mu(A) < \infty$ , then

$$\Theta_N^*(\mu \llcorner A, x) = 0 \quad (2.90)$$

for  $\mathcal{H}^N$ -a.e.  $x \in \Omega \setminus A$ .

**Proof** See for a proof Theorem 3.5 in [8].  $\square$

### 2.2.6 Rectifiable sets

In this section, we give a short overview of *rectifiable sets*. In this section,  $k \leq N$  will always denote an integer. The theory here comes from [3] and [5].

**Definition 2.55** Let  $1 \leq k < N$  and  $E \subset \mathbb{R}^N$  a  $\mathcal{H}^k$ -measurable set. Then define the following:

- (i)  $E$  is **countably  $\mathcal{H}^k$ -rectifiable** if there is a countable collection  $\{f_i\}_{i \in \mathbb{N}}$  of Lipschitz maps  $f_i : \mathbb{R}^k \rightarrow \mathbb{R}^N$  such that

$$\mathcal{H}^k \left( E \setminus \bigcup_{i \in \mathbb{N}} f_i(\mathbb{R}^k) \right) = 0. \quad (2.91)$$

- (ii)  $E$  is  **$\mathcal{H}^k$ -rectifiable** if  $E$  is countably  $\mathcal{H}^k$ -rectifiable and  $\mathcal{H}^k(E) < \infty$ .  
 (iii)  $E$  is **locally  $\mathcal{H}^k$ -rectifiable** if  $E \cap K$  is  $\mathcal{H}^k$ -rectifiable for all compact subsets  $K \subset \mathbb{R}^n$ .

We also define what a *rectifiable measure* is.

**Definition 2.56** Let  $\mu$  be a  $\mathbb{R}^m$ -valued Radon measure on  $\mathbb{R}^N$ . We say that  $\mu$  is  **$k$ -rectifiable** if there is a countably  $\mathcal{H}^k$ -measurable set  $S$  and a Borel function  $\theta : S \rightarrow \mathbb{R}^m$  such that  $\mu = \theta \mathcal{H}^k \llcorner S$ .

Now we give criteria of rectifiability of sets.

**Proposition 2.57** Let  $E \subset \mathbb{R}^N$  be  $\mathcal{H}^k$ -measurable.

- (i) If  $E$  is a Borel set and  $\mathcal{H}^k(E) < \infty$ , then  $E$  is  $\mathcal{H}^k$ -rectifiable if and only if  $\Theta_k(E, x) = 1$  for  $\mathcal{H}^k$ -a.e.  $x \in E$ .  
 (ii)  $E$  is countably  $\mathcal{H}^k$ -rectifiable if and only if there exists a sequence  $\{\Gamma_i\}_{i \in \mathbb{N}}$  of  $k$ -dimensional  $C^1$  submanifolds of  $\mathbb{R}^N$  such that:

$$\mathcal{H}^k \left( E \setminus \bigcup_{i \in \mathbb{N}} \Gamma_i \right) = 0. \quad (2.92)$$

- (iii)  $E$  is countably  $\mathcal{H}^k$ -rectifiable if and only if there exists a sequence  $(\Gamma_i)_{i \in \mathbb{N}}$  graphs  $\Gamma_i \subset \mathbb{R}^N$  of Lipschitz functions  $f_i : \mathbb{R}^k \rightarrow \mathbb{R}^{N-k}$  such that

$$\mathcal{H}^k \left( E \setminus \bigcup_{i \in \mathbb{N}} \Gamma_i \right) = 0. \quad (2.93)$$

Moreover, for any  $\varepsilon > 0$ , the Lipschitz functions  $f_i$  can be chosen such that  $\text{Lip}(f_i, \mathbb{R}^k) < \varepsilon$ .

**Proof** See for a proof, see, e.g. Proposition 2.76 in [3]. □

### 2.2.7 Approximate tangent space and tangential differentiability

We are now ready to define *approximate tangent spaces* for both measures and sets. We denote in the following by  $\mathbf{G}(k, N)$  the space of  $k$ -dimensional subspaces  $\pi \subset \mathbb{R}^N$ . Also, we denote by  $\pi^\perp$  the orthogonal complement of  $\pi \in \mathbf{G}(k, N)$ . If it is needed we identify with  $\pi \in \mathbf{G}(k, N)$  the projection  $\mathbb{R}^N \rightarrow \pi$  onto  $\pi$ , also denoted by  $\pi$ . This will be evident from the context.

**Definition 2.58** Let  $\mu$  be an  $\mathbb{R}^m$ -valued Radon measure on an open set  $\Omega \subset \mathbb{R}^N$  and  $x \in \Omega$ . Then  $\mu$  has **approximate tangent space**  $\pi \in \mathbf{G}(k, N)$  with **multiplicity**  $\theta \in \mathbb{R}^m$  at  $x$ , if  $(r^{-k}\mu_{x,r})_{r>0}$  locally weak\* converges to  $\theta\mathcal{H}^k \llcorner \pi$  in  $\mathbb{R}^N$  as  $r \rightarrow 0^+$ . If this is the case, we write  $\text{Tan}^k(\mu, x) = \theta\mathcal{H}^k \llcorner \pi$ .

**Remark 2.59** Definition 2.58 can be reformulated as follows:  $\mu$  has approximate tangent space  $\pi \in \mathbf{G}(k, N)$  with multiplicity  $\theta \in \mathbb{R}^m$  at  $x$  if it holds that:

$$\lim_{r \rightarrow 0^+} r^{-k} \int_{\Omega} \varphi \left( \frac{y-x}{r} \right) d\mu(y) = \theta \int_{\pi} \varphi(y) d\mathcal{H}^k(y), \quad \forall \varphi \in C_c(\mathbb{R}^N). \quad (2.94)$$

Moreover, if such an approximate tangent space  $\pi$  to  $\mu$  at  $x$  exists, then it is unique. We also note that if  $\mu = f \|\mu\|$  and  $x$  is a Lebesgue point of  $f$  with respect to  $\|\mu\|$ , then we have that  $v = \text{Tan}^k(\mu, x)$  if and only if  $\|v\| = \text{Tan}^k(\|\mu\|, x)$ .

We now state a rectifiability criterion for measures.

**Proposition 2.60** Let  $\mu$  be a positive Radon measure on an open set  $\Omega \subset \mathbb{R}^N$ . Then the following holds:

- (i) If  $\mu = \theta\mathcal{H}^k \llcorner S$  and  $S$  is countably  $\mathcal{H}^k$ -rectifiable, then  $\mu$  admits an approximate tangent space with multiplicity  $\theta(x)$  for  $\mathcal{H}^k$ -a.e.  $x \in S$ . In particular,  $\theta(x) = \Theta_k(\mu, x)$  for  $\mathcal{H}^k$ -a.e.  $x \in S$ .
- (ii) If  $\mu$  is concentrated on a Borel set  $S$  and admits an approximate tangent space with multiplicity  $\theta(x) > 0$  for  $\mu$ -a.e.  $x \in S$ , then  $S$  is countably  $\mathcal{H}^k$ -rectifiable and  $\mu = \theta\mathcal{H}^k \llcorner S$ .

**Proof** See for a proof, see, e.g. Theorem 2.83 in [3]. □

**Definition 2.61** Let  $E \subset \mathbb{R}^N$  be locally  $\mathcal{H}^k$ -rectifiable and  $x \in E$ . If it exists, we define the **approximate tangent space to  $E$  at  $x$**  as  $\text{Tan}^k(E, x) := \text{Tan}^k(\mathcal{H}^k \llcorner E, x)$ , i.e. it must be satisfied that:

$$\lim_{r \rightarrow 0^+} \int_{E_{x,r}} \varphi(y) d\mathcal{H}^k(y) = \theta \int_{\pi} \varphi(y) d\mathcal{H}^k(y), \quad \forall \varphi \in C_c(\mathbb{R}^N), \quad (2.95)$$

where  $E_{x,r} = \frac{E-x}{r}$  for all  $r > 0$  and  $\pi \in \mathbf{G}(k, N)$  is the approximate tangent space to  $E$  at  $x$ . In particular, by definition,  $\pi$  has multiplicity 1.

The following proposition establishes the almost everywhere existence of approximate tangent spaces to locally rectifiable sets.

**Proposition 2.62** If  $E \subset \mathbb{R}^N$  is locally  $\mathcal{H}^k$ -rectifiable, then for  $\mathcal{H}^k$ -almost every  $x \in E$  there exists a unique approximate tangent space  $\text{Tan}^k(E, x)$  to  $E$  at  $x$ .

**Proof** See for a proof, see, e.g. Theorem 10.2 in [5]. □

**Definition 2.63** Let  $E \subset \mathbb{R}^N$  be locally  $\mathcal{H}^k$ -rectifiable and  $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$  a Lipschitz function. We say that  $f$  is **tangentially differentiable** at  $x \in E$  if the restriction of  $f$  to  $x + \text{Tan}^k(E, x) \subset \mathbb{R}^N$  is differentiable at  $x$ . We denote by  $d^E f_x : \text{Tan}^k(E, x) \rightarrow \mathbb{R}^m$  the **tangential differential** of  $f$ .

As in Theorem 2.42, we have the following existence result of the tangential differential.

**Proposition 2.64** If  $E \subset \mathbb{R}^N$  is locally  $\mathcal{H}^k$ -rectifiable and  $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$  is a Lipschitz continuous function, then  $f$  is tangentially differentiable at  $\mathcal{H}^k$ -almost every  $x \in E$ .

**Proof** See for a proof, see, e.g. Theorem 11.4 [5]. □

### 2.2.8 Area formula

The definitions from the previous section now give us the tools to state the *area formula* and its generalization to rectifiable sets. We also fix the following notation for the *Jacobian*.

**Definition 2.65** Let  $V, W$  be Hilbert spaces with  $\dim(V) = k \leq N = \dim(W)$  and  $L : V \rightarrow W$  a linear map. The  **$k$ -dimensional Jacobian** of  $L$  is defined as:

$$J_k L := \sqrt{\det(L^* \circ L)}, \quad (2.96)$$

where  $L^* : W^* \rightarrow V^*$  is the transpose of  $L$ .

**Theorem 2.66** Let  $f : \mathbb{R}^k \rightarrow \mathbb{R}^N$  be a Lipschitz function with  $k \leq N$ . Then, for any  $\mathcal{L}^k$ -measurable set  $E \subset \mathbb{R}^k$  the function  $y \in \mathbb{R}^N \mapsto \mathcal{H}^0(E \cap f^{-1}(y))$  is  $\mathcal{H}^k$ -measurable in  $\mathbb{R}^N$  and

$$\int_{\mathbb{R}^N} \mathcal{H}^0(E \cap f^{-1}(y)) d\mathcal{H}^k(y) = \int_E J_k df_x dx. \quad (2.97)$$

If  $f$  is injective, we even have that:

$$\mathcal{H}^k(f(E)) = \int_E J_k df_x dx. \quad (2.98)$$

**Proof** For a proof, see Theorem 2.71 in [3]. □

Using the tangential differential, we can extend this even to locally rectifiable sets.

**Theorem 2.67** Let  $f : \mathbb{R}^M \rightarrow \mathbb{R}^N$  be a Lipschitz function and  $E \subset \mathbb{R}^M$  be locally  $\mathcal{H}^k$ -rectifiable. Then the function  $y \in \mathbb{R}^N \mapsto \mathcal{H}^0(E \cap f^{-1}(y))$  is  $\mathcal{H}^k$ -measurable in  $\mathbb{R}^N$  and

$$\int_{\mathbb{R}^N} \mathcal{H}^0(E \cap f^{-1}(y)) d\mathcal{H}^k(y) = \int_E J_k d^E f_x d\mathcal{H}^k(x). \quad (2.99)$$

If  $f$  is injective, we even have that:

$$\mathcal{H}^k(f(E)) = \int_E \mathbf{J}_k \mathrm{d}f_x \mathrm{d}\mathcal{H}^k(x). \quad (2.100)$$

**Proof** For a proof, see Theorem 2.91 in [3].  $\square$

### 2.2.9 Coarea formula

The purpose of this final section is to establish the *coarea formula*. First, we introduce the *coarea factor*

**Definition 2.68** Let  $V, W$  be Hilbert spaces with  $\dim(V) = N \geq k = \dim(W)$  and  $L : V \rightarrow W$  a linear map. The  **$k$ -dimensional coarea factor of  $L$**  is defined as:

$$\mathbf{C}_k L := \sqrt{\det(L \circ L^*)}, \quad (2.101)$$

where  $L^* : W^* \rightarrow V^*$  is the transpose of  $L$ .

The *coarea formula* states the following.

**Theorem 2.69** Let  $f : \mathbb{R}^M \rightarrow \mathbb{R}^k$  be a Lipschitz function and let  $E \subset \mathbb{R}^M$  be locally  $\mathcal{H}^N$ -rectifiable. Then the function  $t \mapsto \mathcal{H}^{N-k}(E \cap f^{-1}(t))$  is  $\mathcal{L}^N$ -measurable in  $\mathbb{R}^k$ ,  $E \cap f^{-1}(t)$  is locally  $\mathcal{H}^{N-k}$ -rectifiable for  $\mathcal{L}^k$ -a.e.  $t \in \mathbb{R}^k$  and

$$\int_E \mathbf{C}_k \mathrm{d}^E f_x \mathcal{H}^N(x) = \int_{\mathbb{R}^k} \mathcal{H}^{N-k}(E \cap f^{-1}(t)) \mathrm{d}t. \quad (2.102)$$

**Proof** For a proof, see Theorem 2.93 in [3].  $\square$



## Chapter 3

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# Functions of Bounded Variation and Sets of Finite Perimeter

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This chapter introduces the main concepts from the theory of *functions of bounded variation* and *sets of finite perimeter*, which will be fundamental when talking about *sets of minimal perimeter*. We will follow the exposition given in [3] and [5]. In this chapter, we fix the ambient space  $\mathbb{R}^N$  for some  $N \in \mathbb{N}$ .

### 3.1 The space of functions of bounded variation

We will start by defining and exploring the properties of the space of *functions of bounded variation*.

**Definition 3.1** Let  $\Omega \subset \mathbb{R}^N$  be an open set, and  $u \in L^1(\Omega, \mathcal{L}^N)$ .  $u$  is called a **function of bounded variation in  $\Omega$**  if there exists a finite  $\mathbb{R}^N$ -valued Radon measure  $Du = (D_1 u, \dots, D_N u)$  on  $\Omega$ , such that the following satisfied:

$$\int_{\Omega} u \operatorname{div}(\varphi) \, dx = - \int_{\Omega} \varphi \cdot dDu, \quad \forall \varphi \in [C_c^\infty(\Omega)]^N. \quad (3.1)$$

The vector space of all functions of bounded variation in  $\Omega$  is denoted by  $BV(\Omega)$ . Hence, we will refer to  $Du$  as the **distributional derivative of  $u$** .

**Remark 3.2** Let  $u \in BV(\Omega)$  and let  $\varphi = (\varphi_1, \dots, \varphi_N) \in [C_c^1(\Omega)]^N$  or  $\varphi : \Omega \rightarrow \mathbb{R}^N$  be a Lipschitz function. Then, we can consider the function  $\varphi_\varepsilon = (\varphi_1 * \rho_\varepsilon, \dots, \varphi_N * \rho_\varepsilon)$  for a family  $(\rho_\varepsilon)_{\varepsilon \in (0, \infty)}$  of mollifiers. We note that  $\varphi_\varepsilon \in [C_c^\infty(\Omega)]^N$  and by the continuity of  $\varphi$ , we have  $\varphi_\varepsilon \rightarrow \varphi$  uniformly on compact subsets of  $\Omega$  as  $\varepsilon \rightarrow 0^+$  (see Proposition 2.33 (v)). Then by 3.1 we have the following:

$$\int_{\Omega} u \operatorname{div}(\varphi_\varepsilon) \, dx = - \int_{\Omega} \varphi \cdot dDu, \quad \forall \varphi_\varepsilon \in [C_c^\infty(\Omega)]^N. \quad (3.2)$$

By Proposition 2.33 (iv),  $\operatorname{div}(\varphi_\varepsilon) = \operatorname{div}(\varphi) * \rho_\varepsilon$  and  $\operatorname{div}(\varphi)$  is  $\mathcal{L}^N$ -a.e. defined (by Rademacher). Moreover, by the continuity of  $\operatorname{div}(\varphi)$  we have that  $\operatorname{div}(\varphi_\varepsilon) \rightarrow$

$\operatorname{div}(\varphi)$  uniformly on compact subsets of  $\Omega$  as  $\varepsilon \rightarrow 0^+$ . Thus, passing 3.2 to the limit as  $\varepsilon \rightarrow 0^+$ , we find that 3.1 is satisfied for all  $\varphi \in [C_c^1(\Omega)]^N$  and for all Lipschitz functions  $\varphi : \Omega \rightarrow \mathbb{R}^N$ .

**Remark 3.3** We note that 3.1 is equivalent to that it holds for all  $i = 1, \dots, N$ :

$$\int_{\Omega} u \frac{\partial \varphi}{\partial x_i} dx = - \int_{\Omega} \varphi dD_i u, \quad \forall \varphi \in C_c^\infty(\Omega). \quad (3.3)$$

The next proposition establishes some basic properties of the distributional derivative of a function of bounded variation.

**Proposition 3.4** Let  $u \in BV(\Omega)$ . Then following properties hold:

- (i) If  $Du = 0$ , then  $u$  is constant on any connected component of  $\Omega$ .
- (ii) For any Lipschitz function  $\psi : \Omega \rightarrow \mathbb{R}$  we have that  $u\psi \in BV(\Omega)$  and:

$$D(u\psi) = \psi Du + (u \otimes \nabla \psi) \mathcal{L}^N. \quad (3.4)$$

- (iii) If  $(\rho_\varepsilon)_{\varepsilon \in (0, \infty)}$  is a family of mollifiers, then it holds for every  $\varepsilon > 0$  and in  $\Omega_\varepsilon$ , that

$$\nabla(u * \rho_\varepsilon) = Du * \rho_\varepsilon. \quad (3.5)$$

**Proof** (i) Assuming (iii) in Proposition 3.4 is true, we can consider the smooth functions  $u_\varepsilon = u * \rho_\varepsilon$ . We know by Proposition 2.33, that  $u_\varepsilon \rightarrow u$  in  $L^1(\Omega, \mathcal{L}^N)$  as  $\varepsilon \rightarrow 0^+$ . Moreover, we see by (iii) that  $\nabla u_\varepsilon = 0$  under the assumption  $Du = 0$ . By passing to the limit, we find that on connected components of  $\Omega$ , we must have that  $u$  is equal in  $L^1(\Omega, \mathcal{L}^N)$  to a constant.

- (ii) We use 3.4 to verify 3.1 for  $u\psi$ . Let  $\varphi \in [C_c^\infty(\Omega)]^N$ . Then we compute the following:

$$\begin{aligned} \int_{\Omega} \varphi \cdot dDu\psi &= \int_{\Omega} \varphi \cdot d(\psi Du + (u \otimes \nabla \psi) \mathcal{L}^N) \\ &= \int_{\Omega} \psi \varphi \cdot dDu + \int_{\Omega} u \varphi \cdot \nabla \psi dx \\ &\stackrel{(*)}{=} - \int_{\Omega} u \operatorname{div}(\psi \varphi) dx + \int_{\Omega} u \varphi \cdot \nabla \psi dx \\ &\stackrel{(**)}{=} - \int_{\Omega} u \psi \operatorname{div}(\varphi) dx - \int_{\Omega} u \varphi \cdot \nabla \psi dx + \int_{\Omega} u \varphi \cdot \nabla \psi dx \\ &= - \int_{\Omega} u \psi \operatorname{div}(\varphi) dx. \end{aligned} \quad (3.6)$$

At (\*) we used Remark 3.2 applied to the Lipschitz function  $\psi\varphi : \Omega \rightarrow \mathbb{R}^N$ ; and at (\*,\*) we used that  $\psi$  is almost everywhere differentiable and so the identity  $\operatorname{div}(\psi\varphi) = \psi\operatorname{div}(\varphi) + \varphi \cdot \nabla\psi$  holds almost everywhere. This proves that  $u\psi \in BV(\Omega)$  and the identity 3.4.

- (iii) We use 3.3 to verify 3.5. For this we note that  $\rho_\varepsilon \in C_c^\infty(\Omega)$  and is supported on  $\Omega_\varepsilon$ . So for any  $x \in \Omega_\varepsilon$  we find that for every  $i = 1, \dots, N$ :

$$\begin{aligned} (D_i u * \rho_\varepsilon)(x) &= \int_{\Omega} \rho_\varepsilon(x-y) dD_i u(y) \\ &\stackrel{3.3}{=} - \int_{\Omega} u(y) \frac{\partial \rho_\varepsilon}{\partial x_i}(x-y) dy \\ &= \int_{\Omega} u(x-y) \frac{\partial \rho_\varepsilon}{\partial x_i}(y) dy \\ &= \left( u * \frac{\partial \rho_\varepsilon}{\partial x_i} \right)(x) \\ &= \frac{\partial}{\partial x_i} (u * \rho_\varepsilon)(x). \end{aligned} \tag{3.7}$$

This shows that  $Du * \rho_\varepsilon = \nabla(u * \rho_\varepsilon)$  on  $\Omega_\varepsilon$ .

Now we introduce a different way of thinking about functions of bounded variation via the so-called *variation* of a function.

**Definition 3.5** Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . The *variation of  $u$  in  $\Omega$* , denoted by  $V(u, \Omega)$ , is defined by

$$V(u, \Omega) := \sup \left\{ \int_{\Omega} u \operatorname{div}(\varphi) dx \mid \varphi \in [C_c^1(\Omega)]^N, \|\varphi\|_\infty \leq 1 \right\}. \tag{3.8}$$

**Remark 3.6** We note that for  $V(u, \Omega) \geq 0$ . In fact, since any constant vector field  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  satisfies  $0 = \int_{\Omega} u \operatorname{div}(\varphi) dx \leq V(u, \Omega)$ .

**Remark 3.7** Consider  $u \in C^1(\Omega)$ . Then for every  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty$  we find by the identity  $\operatorname{div}(u\varphi) = u \operatorname{div}(\varphi) + \varphi \cdot \nabla u$ :

$$\begin{aligned} \int_{\Omega} u \operatorname{div}(\varphi) dx &= \int_{\Omega} \operatorname{div}(u\varphi) dx - \int_{\Omega} \varphi \cdot \nabla u dx \\ &\stackrel{(*)}{=} \int_{\partial\Omega} u\varphi \cdot \nu_{\partial\Omega} dx - \int_{\Omega} \varphi \cdot \nabla u dx \\ &\stackrel{(**)}{=} - \int_{\Omega} \varphi \cdot \nabla u dx \\ &\leq \int_{\Omega} \|\nabla u\| dx. \end{aligned} \tag{3.9}$$

At (\*) we used the Gauss' divergence theorem and at (\*\*) we used that  $\varphi$  is compactly supported in  $\Omega$ .

This shows that  $V(u, \Omega) \leq \int_{\Omega} \|\nabla u\| \, dx$ . But also note that if we consider  $\varphi = -\frac{\nabla u}{\|\nabla u\|}$  we get the equality 3.9 and so  $V(u, \Omega) = \int_{\Omega} \|\nabla u\| \, dx$ .

**Remark 3.8** We see that  $V(u, A)$  is defined for every open subset  $A \subset \Omega$ . Then we can extend  $\tilde{V}(u, B) := \inf\{V(u, A) \mid A \supset B \text{ open}\}$  to a Borel measure  $V(u, \cdot)$  on  $\Omega$ .

The next proposition establishes the lower semicontinuity of  $V(\cdot, \Omega)$ .

**Proposition 3.9** Let  $(u_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  be a sequence that converges to  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  in  $L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . Then it holds that:

$$V(u, \Omega) \leq \liminf_{h \rightarrow \infty} V(u_h, \Omega). \quad (3.10)$$

**Proof** Let  $\varphi \in [C_c^1(\Omega)]$  with  $\|\varphi\|_{\infty} \leq 1$ . Then by Fatou's lemma 2.23 and the  $L^1_{\text{loc}}$ -convergence  $u_h \rightarrow u$  we find that:

$$\begin{aligned} \int_{\Omega} u \operatorname{div}(\varphi) \, dx &= \int_{\Omega} \lim_{h \rightarrow \infty} u_h \operatorname{div}(\varphi) \, dx \\ &\stackrel{\text{Fatou}}{\leq} \liminf_{h \rightarrow \infty} \int_{\Omega} u_h \operatorname{div}(\varphi) \, dx \\ &\leq \liminf_{h \rightarrow \infty} V(u_h, \Omega). \end{aligned} \quad (3.11)$$

Considering the supremum over all such  $\varphi \in [C_c^1(\Omega)]$ , we can conclude our proof.  $\square$

Now we characterize the space  $BV(\Omega)$  in terms of  $V(\cdot, \Omega)$ .

**Proposition 3.10** Let  $u \in L^1(\Omega, \mathcal{L}^N)$ . Then it holds that:

$$u \in BV(\Omega) \iff V(u, \Omega) < \infty. \quad (3.12)$$

Moreover,  $V(u, \Omega) = \|Du\|(\Omega)$  for all  $u \in BV(\Omega)$ .

**Proof** Assume that  $u \in BV(\Omega)$ . Then, by Remark 3.2, for any  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_{\infty} \leq 1$  it holds that:

$$\begin{aligned} \int_{\Omega} u \operatorname{div}(\varphi) \, dx &= - \int_{\Omega} \varphi \cdot dDu \\ &\leq \left| \int_{\Omega} \varphi \cdot dDu \right| \\ &\leq \int_{\Omega} \|\varphi\|_{\infty} \, d\|Du\| \\ &\stackrel{\|u\|_{\infty} \leq 1}{\leq} \|Du\|(\Omega) < \infty, \end{aligned} \quad (3.13)$$

where we used in the last step that  $Du$  is a finite  $\mathbb{R}^N$ -valued Radon measure.

Conversely, assume that  $V(u, \Omega) < \infty$ . We first note that the linear functional

$$L_0 : \varphi \in [C_c^1(\Omega)]^N \mapsto \int_{\Omega} u \operatorname{div}(\varphi) \, dx \quad (3.14)$$

is a bounded linear functional, as  $\|L_0\| \leq V(u, \Omega) < \infty$  by definition of  $V(u, \Omega)$ . By the Hahn-Banach theorem, there exists a bounded linear functional  $L : [C_c^1(\Omega)]^N \rightarrow \mathbb{R}$  such that  $L(\varphi) = L_0(\varphi)$  for all  $\varphi \in [C_c^1(\Omega)]^N$ , and  $\|L\| \leq V(u, \Omega)$ . By Riesz' theorem 2.25, there exists a unique finite  $\mathbb{R}^N$ -valued Radon measure  $\mu$  on  $\Omega$ , such that for all  $\varphi \in [C_c^1(\Omega)]^N$  we have:

$$\int_{\Omega} u \operatorname{div}(\varphi) \, dx = \int_{\Omega} \varphi \cdot d\mu, \quad (3.15)$$

and  $\|\mu\|(\Omega) = \|L\| \leq V(u, \Omega)$ . Hence,  $u \in BV(\Omega)$  with  $Du = -\mu$ . Moreover, we have  $\|Du\|(\Omega) = \|\mu\|(\Omega) \leq V(u, \Omega)$ .  $\square$

The space  $BV(\Omega)$  of functions of bounded variations in  $\Omega$  is in fact a Banach space endowed with the norm

$$\|u\|_{BV(\Omega)} := \int_{\Omega} |u| \, dx + \|Du\|(\Omega). \quad (3.16)$$

This is justified by the next proposition.

**Proposition 3.11** *Let  $\Omega \subset \mathbb{R}^N$  be an open set. Then  $(BV(\Omega), \|\cdot\|_{BV(\Omega)})$  is a Banach space.*

**Proof** We only need to check the completeness. Let  $(u_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  a Cauchy sequence with respect to the norm  $\|\cdot\|_{BV(\Omega)}$ . By completeness of  $L^1(\Omega, \mathcal{L}^N)$ , there is a function  $u \in L^1(\Omega, \mathcal{L}^N)$  such that  $\int_{\Omega} |u_h - u| \, dx \xrightarrow{h \rightarrow \infty} 0$ . Now by lower semicontinuity of  $V(\cdot, \Omega)$  with respect to  $L^1_{\text{loc}}$ -convergence, and hence with respect to  $L^1$ -convergence, and Proposition 3.10 we find that for any  $h \in \mathbb{N}$

$$\begin{aligned} \|D(u_h - u)\|(\Omega) &= V(u_h - u, \Omega) \\ &\leq \liminf_{k \rightarrow \infty} V(u_h - u_k). \end{aligned} \quad (3.17)$$

As  $h \rightarrow \infty$ , the last term in the above computation goes to 0, as  $(u_h)_{h \in \mathbb{N}}$  is Cauchy in  $BV(\Omega)$ . This shows that  $\|u_h - u\|_{BV(\Omega)} \xrightarrow{h \rightarrow \infty} 0$ .  $\square$

The next theorem will be vital in the theory of *sets of finite perimeter*, as it allows approximations by smooth functions.

**Theorem 3.12** Let  $u \in L^1(\Omega, \mathcal{L}^N)$ . Then,  $u \in BV(\Omega)$  if and only if there is a sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(\Omega)$  such that  $\|u_h - u\|_1 \xrightarrow{h \rightarrow \infty} 0$  and

$$L := \lim_{h \rightarrow \infty} \int_{\Omega} \|\nabla u_h\| \, dx < \infty. \quad (3.18)$$

Moreover, if  $u \in BV(\Omega)$ , then  $L = \|Du\|(\Omega)$ .

**Proof** Assume first that we find a sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(\Omega)$  satisfying 3.18 such that  $u_h \rightarrow u$  in  $L^1(\Omega, \mathcal{L}^N)$  as  $h \rightarrow \infty$ . Note that for all  $h \in \mathbb{N}$  we have that:

$$\|\nabla u_h \mathcal{L}^N\|(\Omega) = \int_{\Omega} \|\nabla u_h\| \, dx < \infty, \quad (3.19)$$

as  $u_h$  are compactly supported. Then, up to extracting a subsequence, by Theorem 2.22, we can assume that  $\nabla u_h \mathcal{L}^N$  locally weak\* converges to some finite  $\mathbb{R}^N$ -valued Radon measure  $\mu$  such that  $\|\mu\|(\Omega) \leq L$ . By integration by parts using for each  $h \in \mathbb{N}$  and every  $\varphi \in [C_c^\infty(\Omega)]^N$  the identity  $\operatorname{div}(u_h \varphi) = u_h \operatorname{div}(\varphi) + \varphi \cdot \nabla u_h$  we obtain the following:

$$\begin{aligned} \int_{\Omega} u_h \operatorname{div}(\varphi) \, dx &= \int_{\Omega} \operatorname{div}(u_h \varphi) \, dx - \int_{\Omega} \varphi \cdot \nabla u_h \, dx \\ &\stackrel{(*)}{=} - \int_{\Omega} \varphi \cdot \nabla u_h \, dx, \end{aligned} \quad (3.20)$$

where at  $(*)$  we used the divergence theorem and that  $u_h \varphi$  must vanish at the boundary by compact support. Now passing 3.20 to the limit as  $h \rightarrow \infty$ , we find that for all  $\varphi \in [C_c^\infty(\Omega)]^N$ :

$$\int_{\Omega} u \operatorname{div}(\varphi) \, dx = - \int_{\Omega} \varphi \cdot d\mu. \quad (3.21)$$

This shows that  $u \in BV(\Omega)$  and  $\|Du\|(\Omega) = \|\mu\|(\Omega) \leq L$ . Conversely, assume that  $u \in BV(\Omega)$ . For any  $\delta > 0$  we can construct a function  $v_\delta \in C^\infty(\Omega)$  such that

$$\int_{\Omega} |u - v_\delta| \, dx < \delta \quad (3.22)$$

$$\int_{\Omega} \|\nabla v_\delta\| \, dx \leq \|Du\|(\Omega) + \delta. \quad (3.23)$$

Indeed, let  $\delta > 0$ . Note that we can find a countable family  $\{\Omega_h\}_{h \in \mathbb{N}}$  such that  $\Omega = \bigcup_{h \in \mathbb{N}} \Omega_h$ , for every  $x \in \Omega$  we have  $|\{h \in \mathbb{N} \mid x \in \Omega_h\}| \leq 4$  and  $\Omega_h \subset \Omega$  has compact closure. To this end, consider

$$\Omega_{k,1} := \left\{ x \in \Omega \cap (B(0, k+1) \setminus \overline{B}(0, k-1)) \mid d(x, \partial\Omega) > \frac{1}{2} \right\} \quad (3.24)$$

$$\Omega_{k,p} := \left\{ x \in \Omega \cap (B(0, k+1) \setminus \overline{B}(0, k-1)) \mid \frac{1}{p-1} > d(x, \partial\Omega) > \frac{1}{p+1} \right\}, \quad (3.25)$$

where  $k \geq 1$  and  $p > 1$  are integers and  $B(0,0) := 0$ . This defines such an open covering of  $\Omega$ . So choose now a partition of unity  $(\varphi_h)_{h \in \mathbb{N}}$  subordinate to the cover  $\{\Omega_h\}_{h \in \mathbb{N}}$  of  $\Omega$ . Let  $(\rho_\varepsilon)_{\varepsilon \in (0,\infty)}$ . For any  $h \in \mathbb{N}$  we can find  $\varepsilon_h > 0$  such that  $\text{supp}((u\varphi_h) * \rho_{\varepsilon_h}) \subset \Omega_h$  and

$$\int_{\Omega} |(u\varphi_h) * \rho_{\varepsilon_h} - u\varphi_h| + |(u \otimes \nabla \varphi_h) * \rho_{\varepsilon_h} - u \otimes \nabla \varphi_h| dx < 2^{-h}\delta. \quad (3.26)$$

Then we set  $v_\delta := \sum_{h \in \mathbb{N}} (u\varphi_h * \rho_{\varepsilon_h})$ . This function is smooth and, by choice of  $\varepsilon_h$  it follows that  $\int_{\Omega} |u - v_\delta| dx < \delta$ . Now by Proposition 3.4 (ii) and (iii) we can evaluate  $\|Dv_\delta\|(\Omega)$ :

$$\begin{aligned} \nabla v_\delta &= \sum_{h \in \mathbb{N}} \nabla((u\varphi_h) * \rho_{\varepsilon_h}) \\ &= \sum_{h \in \mathbb{N}} D(u\varphi_h) * \rho_{\varepsilon_h} \\ &= \sum_{h \in \mathbb{N}} (\varphi_h Du) * \rho_{\varepsilon_h} + \sum_{h \in \mathbb{N}} (u \otimes \nabla \varphi_h) * \rho_{\varepsilon_h} \\ &= \sum_{h \in \mathbb{N}} (\varphi_h Du) * \rho_{\varepsilon_h} + \sum_{h \in \mathbb{N}} [(u \otimes \nabla \varphi_h) * \rho_{\varepsilon_h} - (u \otimes \nabla \varphi_h)]. \end{aligned} \quad (3.27)$$

Using 3.26 and 3.27 we find that:

$$\begin{aligned} \|Dv_\delta\|(\Omega) &= \int_{\Omega} \|\nabla v_\delta\| dx \\ &\leq \delta + \sum_{h \in \mathbb{N}} \int_{\Omega} \varphi_h \|Du\| \\ &= \delta + \|Du\|(\Omega). \end{aligned} \quad (3.28)$$

This proves the existence of our desired smooth functions  $v_\delta$  for every  $\delta > 0$ . Consider now  $\delta_h = 2^{-h}$  and  $u_h = v_{\delta_h}$ . Then we find that by 3.24 and 3.25:

$$\lim_{h \rightarrow \infty} \int_{\Omega} |u_h - u| dx = 0 \quad (3.29)$$

$$\limsup_{h \rightarrow \infty} \int_{\Omega} \|\nabla u_h\| dx \leq \|Du\|(\Omega). \quad (3.30)$$

By the lower semi continuity of variation, we can infer that  $\|Du_h\|(\Omega)$  converges to  $\|Du\|(\Omega)$ . Thus,  $\|Du\|(\Omega) = L$ .  $\square$

**Remark 3.13** Theorem 3.12 in fact shows that  $BV(\Omega)$  is a Banach algebra with pointwise multiplication. Indeed, if  $u, v \in BV(\Omega)$ , by Theorem 3.12 we can find sequences  $(u_h)_{h \in \mathbb{N}}, (v_h)_{h \in \mathbb{N}} \subset C_c^\infty(\Omega)$  such that  $\|u_h - u\|_{BV(\Omega)} \xrightarrow{h \rightarrow \infty} 0$  and  $\|v_h - v\|_{BV(\Omega)} \xrightarrow{h \rightarrow \infty} 0$ . We then see that  $(u_h v_h)_{h \in \mathbb{N}}$  is sequence of smooth functions and we have that:

$$\begin{aligned} \|u(v_h - v)\|_{BV(\Omega)} &\xrightarrow{h \rightarrow \infty} 0 \\ \|(u_h - u)v\|_{BV(\Omega)} &\xrightarrow{h \rightarrow \infty} 0. \end{aligned} \quad (3.31)$$

Hence, we find that:

$$\begin{aligned} \|u_h v_h - uv\|_{BV(\Omega)} &= \|u_h v_h - uv_h + uv_h - uv\|_{BV(\Omega)} \\ &\leq \|u(v_h - v)\|_{BV(\Omega)} + \|(u_h - u)v\|_{BV(\Omega)} \\ &\xrightarrow{h \rightarrow \infty} 0. \end{aligned} \quad (3.32)$$

So by Theorem 3.12 we have  $uv \in BV(\Omega)$ .

Furthermore, we have by Theorem 3.12 and Hölder's inequality<sup>1</sup>:

$$\begin{aligned} \|uv\|_{BV(\Omega)} &= \|uv\|_1 + \|D(uv)\|(\Omega) \\ &\leq \|u\|_1 \|v\|_1 + \lim_{h \rightarrow \infty} \int_{\Omega} \|\nabla(u_h v_h)\| \, dx \\ &\leq \|u\|_1 \|v\|_1 + \lim_{h \rightarrow \infty} \int_{\Omega} |u_h| \|\nabla v_h\| \, dx + \int_{\Omega} |v_h| \|\nabla u_h\| \, dx \\ &\leq \|u\|_1 \|v\|_1 + \|u\|_1 \lim_{h \rightarrow \infty} \int_{\Omega} \|\nabla v_h\| \, dx + \|v\|_1 \lim_{h \rightarrow \infty} \int_{\Omega} \|\nabla u_h\| \, dx \\ &= \|u\|_1 \|v\|_1 + \|u\|_1 \|Dv\|(\Omega) + \|v\|_1 \|Du\|(\Omega) \\ &\leq (\|u\|_1 + \|Du\|(\Omega))(\|v\|_1 + \|Dv\|(\Omega)) \\ &= \|u\|_{BV(\Omega)} \|v\|_{BV(\Omega)}. \end{aligned} \quad (3.33)$$

This shows that  $BV(\Omega)$  is a Banach algebra.

### 3.1.1 Extension domain

Sometimes, it is useful to extend  $u \in BV(\Omega)$  to the whole of  $\mathbb{R}^N$ . This is possible in the so-called *extension domains*.

**Definition 3.14** A open set  $\Omega \subset \mathbb{R}^N$  is an **extension domain** if  $\partial\Omega$  is bounded and for any open subset  $A \supset \overline{\Omega}$  if there exists a bounded linear operator  $T : BV(\Omega) \rightarrow BV(\mathbb{R}^N)$  such that

- (i)  $T(u) = 0$   $\mathcal{L}^N$ -a.e. on  $\mathbb{R}^N \setminus A$  for all  $u \in BV(\Omega)$ .
- (ii)  $\|DT(u)\|(\partial\Omega) = 0$  for all  $u \in BV(\Omega)$ .
- (iii) For any  $p \in [1, \infty]$ , the restriction of  $T$  to  $W^{1,p}(\Omega)$  is a bounded linear map  $W^{1,p}(\Omega) \rightarrow W^{1,p}(\mathbb{R}^N)$ .

---

<sup>1</sup>If  $\frac{1}{p} + \frac{1}{q} = 1$  for  $p, q \in [0, \infty]$  and  $f \in L^p(X, \mu)$  and  $g \in L^q(X, \mu)$ , then

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

**Remark 3.15** It should be noted that in (iii) we used the fact that  $W^{1,p}(\Omega)$  is a subspace of  $BV(\Omega)$ . Indeed, if  $u \in W^{1,p}(\Omega)$  we know that there exists  $\nabla u = (\nabla_1 u, \dots, \nabla_N u) \in [L^p(\Omega, \mathcal{L}^N)]^N$  such that for all  $\varphi \in C_c^\infty(\Omega)$  and any  $i = 1, \dots, N$  it holds that

$$\int_{\Omega} u \frac{\partial \varphi}{\partial x_i} dx = - \int_{\Omega} \varphi \nabla_i u dx. \quad (3.34)$$

Hence, we see that  $\nabla u \mathcal{L}^N$  is a finite  $\mathbb{R}^N$ -valued Radon measure satisfying 3.1. Thus,  $u \in BV(\Omega)$  with  $Du = \nabla u \mathcal{L}^N$ .

There is a class of open sets in  $\mathbb{R}^N$ , which are all extension domains, and are reasonably regular: open sets with compact Lipschitz boundary. In particular, these will include all open balls in  $\mathbb{R}^N$ .

**Definition 3.16** Let  $\Omega \subset \mathbb{R}^N$  be an open set. We say that  $\Omega$  has **Lipschitz boundary** if for each  $p \in \partial\Omega$  there exists  $r_p > 0$  and a Lipschitz function  $\psi_p : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$  such that up to orthogonal transformation it holds that:

$$\begin{aligned} \partial\Omega \cap B(p, r_p) &= \{(y_1, \dots, y_N) \mid \psi_p(y_1, \dots, y_{N-1}) = y_N\} \cap B(p, r_p) \\ \Omega \cap B(p, r_p) &= \{(y_1, \dots, y_N) \mid \psi_p(y_1, \dots, y_{N-1}) < y_N\} \cap B(p, r_p). \end{aligned} \quad (3.35)$$

**Example 3.17** Let  $R > 0$ . Then,  $B(0, R)$  has the Lipschitz boundary. Indeed we can write  $B(0, R) = B^+(0, R) \cup B^-(0, R)$ , where

$$\begin{aligned} B^+(0, R) &= \{(y_1, \dots, y_N) \in B(0, R) \mid y_N \geq 0\} \\ B^-(0, R) &= \{(y_1, \dots, y_N) \in B(0, R) \mid y_N \leq 0\}. \end{aligned} \quad (3.36)$$

Note that then  $\partial B(0, R) = \partial B^+(0, R) \cup \partial B^-(0, R)$ . We define the Lipschitz functions  $\psi_{\pm} : \mathbb{R}^{N-1} \rightarrow \mathbb{R}$  by  $\psi_{\pm}(y) = \pm \sqrt{R^2 - \|y\|^2}$ . Then we can choose for every  $p \in \partial B^+(0, R)$  a radius  $r_p > 0$  such that for all  $y = (y_1, \dots, y_N) \in B(p, r_p)$  we have  $y_N \geq 0$ . Now we note that then:

$$\begin{aligned} \partial B(0, R) \cap B(p, r_p) &= \{(y_1, \dots, y_N) \mid \psi_+(y_1, \dots, y_{N-1}) = y_N\} \cap B(p, r_p) \\ B(0, R) \cap B(p, r_p) &= \{(y_1, \dots, y_N) \mid \psi_+(y_1, \dots, y_{N-1}) < y_N\} \cap B(p, r_p). \end{aligned} \quad (3.37)$$

Analogously, we get the same for  $p \in \partial B^-(0, R)$  with  $\psi_-$ . This shows that  $B(0, R)$  has Lipschitz boundary. Moreover, by translation and scaling (both Lipschitz transformations), this also shows that  $B(x, R)$  has Lipschitz boundary for any  $x \in \mathbb{R}^N$  and  $R > 0$ .

The next proposition says that any open set with compact Lipschitz boundary is an extension domain.

**Proposition 3.18** Any open set  $\Omega \subset \mathbb{R}^N$  with compact Lipschitz boundary is an extension domain.

**Proof** For a proof of this, see Proposition 3.21 in [3].  $\square$

To finish this introduction to the space of functions of bounded variation, we state and prove here a compactness theorem. For this we denote by  $BV_{\text{loc}}(\Omega)$  the **space of functions of locally bounded variation**, that is,  $u \in BV_{\text{loc}}(\Omega)$  if  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  and for every open subset  $A \subset\subset \Omega$  it holds that  $u|_A \in BV(A)$ .

**Theorem 3.19** Any sequence  $(u_h)_{h \in \mathbb{N}} \subset BV_{\text{loc}}(\Omega)$  with

$$\sup \left\{ \|u_h\|_{BV(A)} \mid h \in \mathbb{N} \right\} < \infty \quad (3.38)$$

for every open  $A \subset\subset \Omega$ , admits a subsequence  $(u_{h_k})_{k \in \mathbb{N}}$  converging in  $L^p_{\text{loc}}(\Omega, \mathcal{L}^N)$  to some  $u \in BV_{\text{loc}}(\Omega)$ . Moreover, if  $\Omega$  is a bounded extension domain and  $(u_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  any bounded sequence, then the subsequence  $(u_{h_k})_{k \in \mathbb{N}}$  converges in  $L^1(\Omega, \mathcal{L}^N)$  to  $u \in BV(\Omega)$  and  $(Du_{h_k})_{k \in \mathbb{N}}$  weak\* converges to  $Du$ .

**Proof** See the proof of Theorem 3.23 in [3].  $\square$

## 3.2 Sets of finite perimeter

In this section, we introduce the so-called *sets of finite perimeter*, using of course all the properties of functions of bounded variation. We follow here the exposition of [3] and [5].

**Definition 3.20** Let  $E \subset \mathbb{R}^N$  be  $\mathcal{L}^N$ -measurable. For any open set  $\Omega \subset \mathbb{R}^N$  the **perimeter of  $E$  in  $\Omega$**  is defined as  $P(E, \Omega) := V(\chi_E, \Omega)$ , i.e.

$$P(E, \Omega) := \sup \left\{ \int_E \operatorname{div}(\varphi) \, dx \mid \varphi \in [C_c^1(\Omega)]^N, \|\varphi\|_\infty \leq 1 \right\}. \quad (3.39)$$

We say that  $E$  is a **set of finite perimeter in  $\Omega$**  if  $P(E, \Omega) < \infty$ . We say that  $E$  is a **set of locally finite perimeter in  $\Omega$**  if  $E$  has finite perimeter in every bounded open set  $A \subset \Omega$ .

The next proposition will illustrate the connection between sets of finite perimeter and functions of bounded variation even further.

**Proposition 3.21** Let  $\Omega \subset \mathbb{R}^N$  be an open set. Then, for every set  $E$  of finite perimeter in  $\Omega$  the distributional derivative  $D\chi_E$  is a finite  $\mathbb{R}^N$ -valued Radon measure on  $\Omega$ . Moreover, if  $D\chi_E = \nu_E \|D\chi_E\|$  is the polar decomposition of  $D\chi_E$ , then  $P(E, \Omega) = \|D\chi_E\|(\Omega)$  and the **generalised Gauss-Green formula** holds:

$$\int_E \operatorname{div}(\varphi) \, dx = - \int_\Omega \varphi \cdot \nu_E \, d\|D\chi_E\|, \quad \forall \varphi \in [C_c^1(\Omega)]^N. \quad (3.40)$$

**Remark 3.22** Note that if  $E$  has  $C^1$  boundary within  $\Omega$  such that  $\mathcal{H}^{N-1}(\Omega \cap \partial E) < \infty$ , then  $E$  has finite perimeter in  $\Omega$ . Indeed, by the divergence theorem we have for all  $\varphi \in [C_c^1(\Omega)]^N$  that

$$\int_E \operatorname{div}(\varphi) \, dx = - \int_{\Omega \cap \partial E} \varphi \cdot \nu_E \, d\mathcal{H}^{N-1}, \quad (3.41)$$

where  $\nu_E$  is the inner unit normal to  $E$  inside  $\Omega$ . Clearly, we have  $-\nu_E \in [C_c^1(\Omega)]^N$  with  $\|\nu_E\|_\infty = 1$  and therefore we find by taking the supremum over all  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  in 3.41, we find that

$$P(E, \Omega) = \int_{\Omega \cap \partial E} \nu_E \cdot \nu_E \, d\mathcal{H}^{N-1} = \mathcal{H}^{N-1}(\Omega \cap \partial E). \quad (3.42)$$

**Proof (Proposition 3.21)** We note that  $\chi_E \in L_{\text{loc}}^1(\Omega, \mathcal{L}^N)$ , so we must have  $\chi_E \in BV_{\text{loc}}(\Omega)$ . So for every open subset  $A \subset\subset \Omega$  we have that:

$$\|\mathbf{D}\chi_E\|(A) = V(\chi_E, A) \leq V(\chi_E, \Omega) = P(E, \Omega). \quad (3.43)$$

The generalized Gauss-Green formula is just a reformulation of  $\chi_E \in BV_{\text{loc}}(\Omega)$  in terms of polar decomposition.  $\square$

We now define the modes of convergence of sets of finite perimeter.

**Definition 3.23** Let  $(E_h)_{h \in \mathbb{N}}$  be a sequence of  $\mathcal{L}^N$ -measurable subsets in  $\mathbb{R}^N$  and  $E \subset \mathbb{R}^N$   $\mathcal{L}^N$ -measurable.

- (i) We say  $E_h$  converges to  $E$  in  $\Omega$  if  $\chi_{E_h}$  converges to  $\chi_E$  in  $L^1(\Omega, \mathcal{L}^N)$ .
- (ii) We say that  $E_h$  locally converges to  $E$  in  $\Omega$  if  $\chi_{E_h}$  converges to  $\chi_E$  in  $L_{\text{loc}}^1(\Omega, \mathcal{L}^N)$ .

The next proposition summarizes the properties of the perimeter.

**Proposition 3.24** (i) The function  $E \mapsto P(E, \Omega)$  is lower semicontinuous with respect to local convergence in  $\Omega$ . Moreover, the map  $\Omega \mapsto P(E, \Omega)$  is the restriction to open sets of a Borel measure on  $\mathbb{R}^N$ .

(ii) If  $\mathcal{L}^N(\Omega \cap (E \Delta F)) = 0$ , then  $P(E, \Omega) = P(F, \Omega)$ .

(iii)  $P(E, \Omega) = P(\mathbb{R}^N \setminus E, \Omega)$  and

$$P(E \cup F, \Omega) + P(E \cap F, \Omega) \leq P(E, \Omega) + P(F, \Omega). \quad (3.44)$$

**Proof** (i) If  $(E_h)_{h \in \mathbb{N}}$  converges locally to  $E$  in  $\Omega$ , we have that  $\chi_{E_h} \xrightarrow{h \rightarrow \infty} \chi_E$  in  $L_{\text{loc}}^1(\Omega, \mathcal{L}^N)$ . So by Proposition 3.9 we have that

$$P(E, \Omega) = V(\chi_E, \Omega) \leq \liminf_{h \rightarrow \infty} V(\chi_{E_h}, \Omega) = \liminf_{h \rightarrow \infty} P(E_h, \Omega). \quad (3.45)$$

- (ii) Since  $\mathcal{L}^N(\Omega \cap (E \Delta F)) = 0$ , it follows that  $\mathcal{L}^N(\Omega \cap (E \setminus F)) = \mathcal{L}^N(\Omega \cap (F \setminus E)) = 0$ . Then, for every  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  we find that:

$$\begin{aligned} \int_E \operatorname{div}(\varphi) \, dx &= \int_{E \setminus F} \operatorname{div}(\varphi) \, dx + \int_{E \cap F} \operatorname{div}(\varphi) \, dx \\ &= \int_{F \setminus E} \operatorname{div}(\varphi) \, dx + \int_{E \cap F} \operatorname{div}(\varphi) \, dx \\ &= \int_F \operatorname{div}(\varphi) \, dx. \end{aligned} \quad (3.46)$$

Hence, taking the supremum over all such  $\varphi$ , we find  $P(E, \Omega) = P(F, \Omega)$ .

- (iii) By Theorem 3.12 we can find sequences  $(u_h)_{h \in \mathbb{N}}, (v_h)_{h \in \mathbb{N}} \subset C^\infty(\Omega)$  such that  $u_h \rightarrow \chi_E$  and  $v_h \rightarrow \chi_F$  in  $L_{\text{loc}}^1(\Omega, \mathcal{L}^N)$  as  $h \rightarrow \infty$  and

$$\begin{aligned} \lim_{h \rightarrow \infty} \int_\Omega \|\nabla u_h\| \, dx &= P(E, \Omega) \\ \lim_{h \rightarrow \infty} \int_\Omega \|\nabla v_h\| \, dx &= P(F, \Omega). \end{aligned} \quad (3.47)$$

Then, we note that  $(u_h v_h)_{h \in \mathbb{N}}$  converges to  $\chi_{E \cap F}$  and  $(u_h + v_h - u_h v_h)_{h \in \mathbb{N}}$  converges to  $\chi_{E \cup F}$  in  $L_{\text{loc}}^1(\Omega, \mathcal{L}^N)$  and we have that

$$\begin{aligned} \int_\Omega \|\nabla u_h v_h\| \, dx + \int_\Omega \|\nabla u_h + v_h - u_h v_h\| \, dx \\ \leq \int_\Omega \|\nabla u_h\| \, dx + \int_\Omega \|\nabla v_h\| \, dx. \end{aligned} \quad (3.48)$$

Now, passing to the limit as  $h \rightarrow \infty$ , we conclude 3.44.

For the other statement we note that  $\chi_{\mathbb{R}^N \setminus E} = 1 - \chi_E$  and hence for every  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  we find that:

$$\begin{aligned} \int_{\mathbb{R}^N \setminus \Omega} \operatorname{div}(\varphi) \, dx &= \int_\Omega \operatorname{div}(\varphi) \, dx + \int_E \operatorname{div}(\varphi) \, dx \\ &= \int_{\partial\Omega} \varphi \cdot \nu_{\partial\Omega} \, dx + \int_E \operatorname{div}(\varphi) \, dx \\ &= \int_E \operatorname{div}(\varphi) \, dx. \end{aligned} \quad (3.49)$$

So, by taking the supremum over all  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  we get  $P(\mathbb{R}^N \setminus E, \Omega) = P(E, \Omega)$ .

As a direct consequence of Theorem 2.18 we have the following compactness result of sets of locally finite perimeter.

**Theorem 3.25** Any sequence  $(E_h)_{h \in \mathbb{N}}$  of  $\mathcal{L}^N$ -measurable sets in  $\mathbb{R}^N$  such that

$$\sup_{h \in \mathbb{N}} P(E_h, A) < \infty, \quad \forall A \subset \subset \Omega \text{ open}, \quad (3.50)$$

admits a subsequence  $(E_{h_k})_{k \in \mathbb{N}}$  locally converging in  $\Omega$ . If  $\mathcal{L}^N(\Omega) < \infty$ , then we have the convergence in  $\Omega$ .

Next, we tackle the coarea formula for functions of bounded variation. For this we adopt the following notation for function  $u : \Omega \rightarrow \mathbb{R}$  reminiscent of classical probability theory

$$\begin{aligned} \{u > t\} &:= \{x \in \mathbb{R}^N \mid u(x) > t\} \\ \{u < t\} &:= \{x \in \mathbb{R}^N \mid u(x) < t\} \\ \{u \geq t\} &:= \{x \in \mathbb{R}^N \mid u(x) \geq t\} \\ \{u \leq t\} &:= \{x \in \mathbb{R}^N \mid u(x) \leq t\} \\ \{u = t\} &:= \{x \in \mathbb{R}^N \mid u(x) = t\}. \end{aligned} \quad (3.51)$$

**Theorem 3.26** For any open set  $\Omega \subset \mathbb{R}^N$  and  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  it holds that

$$V(u, \Omega) = \int_{-\infty}^{\infty} P(\{u > t\}, \Omega) dt. \quad (3.52)$$

In particular, if  $u \in BV(\Omega)$ , the set  $\{u > t\}$  has finite perimeter in  $\Omega$  for  $\mathcal{L}^1$ -a.e.  $t \in \mathbb{R}$  and for every Borel set  $B \subset \Omega$

$$\begin{aligned} \|Du\|(B) &= \int_{-\infty}^{\infty} \|D\chi_{\{u > t\}}\| dt \\ Du(B) &= \int_{-\infty}^{\infty} D\chi_{\{u > t\}} dt. \end{aligned} \quad (3.53)$$

**Proof** We first note that since the integrals in Theorem 3.26 are invariant under modifications of  $u$  on  $\mathcal{L}^N$ -null sets, we can assume  $u$  is Borel measurable. Then, the map  $(x, t) \mapsto \chi_{\{u > t\}}(x)$  is Borel measurable.

Now suppose  $u \in C^1(\Omega)$ . Then we have by Sard's theorem that  $\mathcal{H}^{N-1}(E \cap \{u = t\}) = 0$  for  $\mathcal{L}^1$ -a.e.  $t \in \mathbb{R}$ . Hence we can infer that

$$P(\{u > t\}, \Omega) = \mathcal{H}^{N-1}(\{u = t\}), \quad \forall \mathcal{L}^1\text{-a.e. } t \in \mathbb{R}. \quad (3.54)$$

By the coarea formula for Lipschitz functions (Theorem 2.69) we then deduce that

$$\begin{aligned} \int_{-\infty}^{\infty} P(\{u > t\}, \Omega) dt &= \int_{-\infty}^{\infty} \mathcal{H}^{N-1}(\{u = t\}) dt \\ &= \int_{\Omega} \mathbf{C}_1 du_x dx \\ &= \int_{\Omega} \|\nabla u\| dx \\ &= V(u, \Omega). \end{aligned} \quad (3.55)$$

### 3. FUNCTIONS OF BOUNDED VARIATION AND SETS OF FINITE PERIMETER

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This proves 3.52 for  $u \in C^1(\Omega)$ . Now we show it for  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  Borel measurable. First we note that it holds for all  $x \in \Omega$ :

$$u(x) = \int_0^\infty \chi_{\{u>t\}}(x) dt - \int_{-\infty}^0 1 - \chi_{\{u>t\}}(x) dt. \quad (3.56)$$

So for any  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$  we have that since the integral of  $\text{div}(\varphi)$  vanishes on  $\Omega$ :

$$\begin{aligned} \int_\Omega u(x) \text{div}(\varphi)(x) dx &= \int_\Omega \left( \int_{-\infty}^\infty \chi_{\{u>t\}}(x) dt \right) \text{div}(\varphi)(x) dx \\ &= \int_{-\infty}^\infty \int_\Omega \chi_{\{u>t\}}(x) \text{div}(\varphi)(x) dx dt \\ &\leq \int_{-\infty}^\infty P(\{u > t\}, \Omega) dt. \end{aligned} \quad (3.57)$$

So, taking the supremum over all  $\varphi \in [C_c^1(\Omega)]^N$  with  $\|\varphi\|_\infty \leq 1$ , we find that  $V(u, \Omega) \leq \int_{-\infty}^\infty P(\{u > t\}, \Omega) dt$ .

Now, we may assume that  $V(u, \Omega) < \infty$ , as otherwise the equality in 3.52 is trivial. We may also assume that  $u \in \mathcal{L}^1(\Omega)$ , otherwise we can argue by covering  $\Omega$  with sets, whose closure is compact in  $\Omega$ . Under these assumptions, we have  $u \in BV(\Omega)$ . By Theorem 3.12, we find a sequence  $(u_h)_{h \in \mathbb{N}}$  converging to  $u$  in  $BV(\Omega)$ . This means that  $\{u_h > t\}$  converges to  $\{u > t\}$  in  $\Omega$ . By the lower semicontinuity of perimeter, we find that

$$\begin{aligned} \int_{-\infty}^\infty P(\{u > t\}, \Omega) dt &\leq \int_{-\infty}^\infty \liminf_{h \rightarrow \infty} P(\{u_h > t\}, \Omega) dt \\ &\leq \liminf_{h \rightarrow \infty} \int_{-\infty}^\infty P(\{u_h > t\}, \Omega) dt \\ &= \liminf_{h \rightarrow \infty} \|Du_h\|(\Omega) \\ &= \|Du\| \\ &\stackrel{u \in BV(\Omega)}{=} V(u, \Omega). \end{aligned} \quad (3.58)$$

This shows that  $\int_{-\infty}^\infty P(\{u > t\}, \Omega) dt \leq V(u, \Omega)$ .

Now, let  $B \subset \Omega$  be a Borel set and  $u \in BV(\Omega)$ . Then we have by the above:

$$\begin{aligned} \|Du\|(B) &= \int_{-\infty}^\infty P(\{u > t\}, B) dt \\ &\stackrel{u \in BV(\Omega)}{=} \int_{-\infty}^\infty \|D\chi_{\{u>t\}}\|(B) dt. \end{aligned} \quad (3.59)$$

Next, we find for all  $\varphi \in [C_c^\infty(\Omega)]^N$ , that:

$$\begin{aligned}
 \int_{\Omega} \varphi \cdot dDu &= - \int_{\Omega} u(x) \operatorname{div}(\varphi)(x) dx \\
 &= - \int_{\Omega} \left( \int_{-\infty}^{\infty} \chi_{\{u>t\}}(x) dt \right) \operatorname{div}(\varphi)(x) dx \\
 &= - \int_{-\infty}^{\infty} \int_{\Omega} \chi_{\{u>t\}}(x) \operatorname{div}(\varphi)(x) dx dt \\
 &= \int_{-\infty}^{\infty} \int_{\Omega} \varphi \cdot dD\chi_{\{u>t\}} dt.
 \end{aligned} \tag{3.60}$$

This shows that  $Du(B) = \int_{-\infty}^{\infty} D\chi_{\{u>t\}}(B) dt$ .  $\square$

### 3.2.1 Isoperimetric inequalities

In this short excursion, we will go over the *isoperimetric inequalities* and the *Poincaré inequalities* of functions of bounded variation. For this purpose we will define the **mean value**  $u_{\Omega}$  of a function  $u \in L^1(\Omega, \mathcal{L}^N)$  by

$$u_{\Omega} := \int_{\Omega} u(x) dx = \frac{1}{\mathcal{L}^N(\Omega)} \int_{\Omega} u(x) dx. \tag{3.61}$$

The following first theorem controls the difference between a function of bounded variation and their mean value, at least in  $L^1(\Omega, \mathcal{L}^N)$ .

**Theorem 3.27** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded connected extension domain. Then, there exists  $C = C(\Omega) > 0$  only depending on  $\Omega$  such that*

$$\int_{\Omega} |u - u_{\Omega}| dx \leq C \|Du\|(\Omega), \quad \forall u \in BV(\Omega). \tag{3.62}$$

**Proof** Assume for the sake of contradiction that there exists a sequence  $(u_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  such that  $\int_{\Omega} |u_h - (u_h)_{\Omega}| dx \geq h \|Du_h\|(\Omega)$  for all  $h \in \mathbb{N}$ . We note that both sides of the inequalities are homogeneous of degree 1. Set now for every  $h \in \mathbb{N}$ :

$$v_h := \frac{u_h - (u_h)_{\Omega}}{\int_{\Omega} |u_h - (u_h)_{\Omega}| dx} \in BV(\Omega). \tag{3.63}$$

Then the functions  $v_h \in BV(\Omega)$  satisfy the following:

$$\int_{\Omega} |v_h| dx = \frac{1}{\int_{\Omega} |u_h - (u_h)_{\Omega}| dx} \int_{\Omega} |u_h - (u_h)_{\Omega}| dx = 1 \tag{3.64}$$

$$\int_{\Omega} v_h dx = \frac{1}{\int_{\Omega} |u_h - (u_h)_{\Omega}| dx} \left( \int_{\Omega} u_h dx - \mathcal{L}^N(\Omega)(u_h)_{\Omega} \right) = 0 \tag{3.65}$$

$$h \|Dv_h\|(\Omega) = \frac{h}{\int_{\Omega} |u_h - (u_h)_{\Omega}| dx} \|Du_h\|(\Omega) \leq 1. \tag{3.66}$$

This shows that for all  $h \in \mathbb{N}$  that  $\|v_h\|_{BV(\Omega)} \leq 1 + \frac{1}{h}$ , that is  $\sup_{h \in \mathbb{N}} \|v_h\|_{BV(\Omega)} \leq 1$  and so  $(v_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  is a bounded sequence. Since  $\Omega$  is a bounded extension domain, we can conclude by Theorem 3.19, that up to restricting to a subsequence,  $(v_h)_{h \in \mathbb{N}}$  converges to some  $v \in BV(\Omega)$  in  $L^1(\Omega, \mathcal{L}^N)$  with  $\int_{\Omega} |v| dx = 1$  and  $\int_{\Omega} v dx = 0$ . But, since  $\|Dv_h\|(\Omega) \leq \frac{1}{h}$  for all  $h \in \mathbb{N}$ . By lower semicontinuity of variation and Proposition 3.10 it must follow that  $\|Dv\|(\Omega) = 0$ . By Proposition 3.4 (i) and since  $\Omega$  is connected, it follows that  $v$  is constant on  $\Omega$ . As  $\int_{\Omega} |v| dx = 1$  we must have  $v \neq 0$ . But then  $\int_{\Omega} v dx = v_0 \mathcal{L}^N(\Omega) \neq 0$  for some constant  $v_0 \neq 0$ . This is a contradiction, as by our construction  $\int_{\Omega} v dx = 0$ . This proves the existence of a constant  $C = C(\Omega) > 0$  as claimed in Theorem 3.27.  $\square$

A useful corollary of Theorem 3.27 is its application to open balls in  $\mathbb{R}^N$ .

**Corollary 3.28** *For every  $x \in \mathbb{R}^N$  and  $r > 0$  it holds that:*

$$\int_{B(x,r)} |u - u_{B(x,r)}| dx \leq \gamma_1 r \|Du\|(B(x,r)), \quad \forall u \in BV(B(x,r)). \quad (3.67)$$

where  $\gamma_1 > 0$  is a constant only depending on the unit ball  $B(0,1)$ . In particular, for any set  $E \subset \mathbb{R}^N$  of locally finite perimeter in  $\mathbb{R}^N$ , it holds that for all  $x \in \mathbb{R}^N$  and  $r > 0$  that:

$$\min \left\{ (\chi_E)_{B(x,r)}, 1 - (\chi_E)_{B(x,r)} \right\} \leq \gamma_1 \frac{P(E, B(x,r))}{\omega_N r^{N-1}}. \quad (3.68)$$

**Proof** Note first that with any  $u \in BV(B(x,r))$  we can identify the function  $u_0 \in BV(B(0,1))$  defined by  $u_0(y) := u(x + ry)$ . This function satisfies (using at  $(*)$  the substitution  $y \mapsto x + ry$ )

$$\begin{aligned} (u_0)_{B(0,1)} &\stackrel{(*)}{=} \frac{1}{\omega_N} \int_{B(0,1)} u_0(y) dy \\ &= \frac{1}{\omega_N} \int_{B(0,1)} u(x + ry) dy \\ &= \frac{1}{\omega_N r^N} \int_{B(x,r)} u(y) dy \\ &= u_{B(x,r)}, \end{aligned} \quad (3.69)$$

and there is a constant  $\gamma_1 > 0$  only depending on  $B(0,1)$  such that

$$\int_{B(0,1)} |u_0 - (u_0)_{B(0,1)}| dy \leq \gamma_1 \|Du_0\|(B(0,1)), \quad \forall u \in BV(B(x,r)). \quad (3.70)$$

Now, first suppose  $u \in C^1(B(x,r))$ . We note that by substitution at  $(*)$  and chain rule at  $(**)$  we have

$$\int_{B(0,1)} |u_0(y) - (u_0)_{B(0,1)}| dy \stackrel{(*)}{=} \frac{1}{r^N} \int_{B(x,r)} |u(y) - u_{B(x,r)}| dy, \quad (3.71)$$

and

$$\begin{aligned} \int_{B(0,1)} \|\nabla u_0(y)\| \, dy &\stackrel{(**)}{=} r \int_{B(0,1)} \|\nabla u(x+ry)\| \, dy \\ &\stackrel{(*)}{=} \frac{1}{r^{N-1}} \int_{B(x,r)} \|\nabla u(y)\| \, dy. \end{aligned} \quad (3.72)$$

Using 3.71, 3.72 and Remark 3.7 in 3.70 we find that for all  $u \in C^1(B(x,r))$  it holds that:

$$\begin{aligned} \int_{B(x,r)} |u(y) - u_{B(x,r)}| \, dy &= r^N \int_{B(0,1)} |u_0(y) - (u_0)_{B(0,1)}| \, dy \\ &\leq \gamma_1 r^N \|Du_0\| (B(0,1)) \\ &= \gamma_1 r^N \int_{B(0,1)} \|\nabla u_0(y)\| \, dy \\ &= \gamma_1 r^N \frac{1}{r^{N-1}} \int_{B(x,r)} \|\nabla u(y)\| \, dy \\ &= \gamma_1 r \|Du\| (B(x,r)). \end{aligned} \quad (3.73)$$

This proves 3.67 for all  $u \in C^1(B(x,r))$ . Now, let  $u \in BV(B(x,r))$ . Then, there is a sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(B(x,r))$  such that  $u_h \xrightarrow{h \rightarrow \infty} u$  in  $L^1(B(x,r), \mathcal{L}^N)$  and  $\lim_{h \rightarrow \infty} \|Du_h\| (B(x,r)) = \|Du\| (B(x,r))$ . Then it follows that

$$\begin{aligned} \int_{B(x,r)} |u - u_{B(x,r)}| \, dy &= \lim_{h \rightarrow \infty} \int_{B(x,r)} |u_h - (u_h)_{B(x,r)}| \, dy \\ &\leq \lim_{h \rightarrow \infty} \gamma_1 r \|Du_h\| (B(x,r)) \\ &= \gamma_1 r \|Du\| (B(x,r)). \end{aligned} \quad (3.74)$$

Let now  $E \subset \mathbb{R}^N$  be a set of locally finite perimeter. We note that we have  $\chi_E \in BV(B(x,r))$  for all  $x \in \mathbb{R}^N$  and  $r > 0$ . Then  $(\chi_E)_{B(x,r)} = \frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))}$  and  $(\chi_{\mathbb{R}^N \setminus E})_{B(x,r)} = 1 - (\chi_E)_{B(x,r)}$ . We note that  $(\chi_E)_{B(x,r)} \leq \frac{1}{2}$  or  $1 - (\chi_E)_{B(x,r)} \leq \frac{1}{2}$  depending on which of  $E \cap B(x,r)$  and  $B(x,r) \setminus E$  has smaller  $\mathcal{L}^N$ -measure. So it follows that:

$$\begin{aligned} 2(\chi_E)_{B(x,r)}(1 - (\chi_E)_{B(x,r)})\mathcal{L}^N(B(x,r)) &\leq \int_{B(x,r)} |\chi_E - (\chi_E)_{B(x,r)}| \, dy \\ &\stackrel{3.67}{=} \gamma_1 r \|D\chi_E\| (B(x,r)) \\ &\stackrel{3.21}{=} \gamma_1 r P(E, B(x,r)). \end{aligned} \quad (3.75)$$

Now, we note that for  $t \in [0, 1]$  it holds that  $\min\{t, 1-t\} \leq 2t(1-t)$  and as  $t = (\chi_E)_{B(x,r)} \in [0, 1]$  it follows that:

$$\min\{(\chi_E)_{B(x,r)}, 1 - (\chi_E)_{B(x,r)}\} \leq \frac{\gamma_1}{\omega_N r^{N-1}} P(E, B(x,r)). \quad (3.76)$$

This ends the proof of the corollary.  $\square$

**Remark 3.29** The corollary 3.28 also holds for open balls in any other norm  $\|\cdot\|$  on  $\mathbb{R}^N$  up to changing the constants  $\gamma_1$  and  $\omega_N$  to the corresponding constants depending on the unit ball in norm  $\|\cdot\|$ .

Now, with this, we are able to prove the *isoperimetric inequality*.

**Theorem 3.30** Let  $N \in \mathbb{N}$  and  $N > 1$ . For any set  $E$  of finite perimeter in  $\mathbb{R}^N$  it holds that either  $E$  or  $\mathbb{R}^N \setminus E$  has finite  $\mathcal{L}^N$ -measure and the *isoperimetric inequality* holds:

$$\min \left\{ \mathcal{L}^N(E), \mathcal{L}^N(\mathbb{R}^N \setminus E) \right\} \leq \gamma_2 P(E, \mathbb{R}^N)^{\frac{N}{N-1}}, \quad (3.77)$$

where  $\gamma_2 = \gamma_2(N) > 0$  only depends on  $N$ .

**Proof** Denote for any  $x \in \mathbb{R}^N$  and  $r > 0$  by  $Q(x, r)$  the open cube with center  $x$  and side length  $2r$ , that is, the open ball about  $x$  of radius  $r$  with respect to the norm  $\|y\|_\infty = \max_{i=1, \dots, N} |y_i|$ . By Remark 3.29 and Corollary 3.28 there is a dimensional constant  $c > 0$  (depending only on  $Q(0, 1)$ ) such that

$$\min \left\{ (\chi_E)_{Q(x, r)}, 1 - (\chi_E)_{Q(x, r)} \right\} \leq \frac{c}{r^{N-1}} P(E, B(x, r)). \quad (3.78)$$

We fix now  $r = [3cP(E, \mathbb{R}^N)]^{\frac{1}{N-1}}$ . Note that  $(\chi_E)_{Q(x, r)} \leq \frac{1}{2}$ , if  $\mathcal{L}^N(E \cap B(x, r)) \leq \mathcal{L}^N(E \setminus B(x, r))$ , and  $(\chi_E)_{Q(x, r)} \geq \frac{1}{2}$ , if  $\mathcal{L}^N(E \cap B(x, r)) \geq \mathcal{L}^N(E \setminus B(x, r))$ . If  $\mathcal{L}^N(E \cap B(x, r)) \leq \mathcal{L}^N(E \setminus B(x, r))$ , then we find that by 3.78 we get

$$\frac{\mathcal{L}^N(E \cap Q(x, r))}{(2r)^N} = (\chi_E)_{Q(x, r)} \leq \frac{c}{r^{N-1}} P(E, Q(x, r)). \quad (3.79)$$

Now we may cover  $\mathcal{L}^N$ -almost all of  $\mathbb{R}^N$  by countably many disjoint open cubes  $\{Q(x_h, r)\}_{h \in \mathbb{Z}^N}$ . By 3.79 we find that

$$\begin{aligned} \mathcal{L}^N(E) &= \sum_{h \in \mathbb{Z}^N} \mathcal{L}^N(E \cap Q(x_h, r)) \\ &\leq 2^N c r \sum_{h \in \mathbb{Z}^N} P(E, Q(x_h, r)) \\ &\leq 2^N c r P(E, \mathbb{R}^N) \\ &= 2^N c [3cP(E, \mathbb{R}^N)]^{\frac{1}{N-1}} P(E, \mathbb{R}^N) \\ &= 2^N 3^{\frac{1}{N-1}} c^{\frac{N}{N-1}} P(E, \mathbb{R}^N)^{\frac{N}{N-1}}. \end{aligned} \quad (3.80)$$

If, on the other hand,  $\mathcal{L}^N(E \cap B(x, r)) \geq \mathcal{L}^N(E \setminus B(x, r))$  we can apply the same argument to  $\mathbb{R}^N \setminus E$  and it will yield the same result. So we can choose  $\gamma_2 := 2^N 3^{\frac{1}{N-1}} c^{\frac{N}{N-1}}$  for the constant postulated in Theorem 3.30.  $\square$

**Theorem 3.31** *Let  $N > 1$  and  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  with  $V(u, \mathbb{R}^N) < \infty$ . Then there exists  $m \in \mathbb{R}$  such that*

$$\|u - m\|_{\frac{N}{N-1}} \leq \gamma_3 V(u, \mathbb{R}^N), \quad (3.81)$$

*for some constant  $\gamma_3 = \gamma_3(N)$  depending only on  $N$ . If  $u \in L^1(\mathbb{R}^N, \mathcal{L}^N)$ , then  $m = 0$ ,  $u \in BV(\mathbb{R}^N)$  and  $\|u\|_{\frac{N}{N-1}} \leq \gamma_3 \|Du\|(\mathbb{R}^N)$ . In particular, the embedding  $BV(\mathbb{R}^N) \hookrightarrow L^{\frac{N}{N-1}}(\mathbb{R}^N, \mathcal{L}^N)$  is a bounded linear operator.*

We need the following lemma to prove Theorem 3.31.

**Lemma 3.32** *For every  $p \in [0, \infty)$  and decreasing function  $g : (0, \infty) \rightarrow [0, \infty)$  it holds for all  $T > 0$  that*

$$p \int_0^T g(s) s^{p-1} ds \leq \left( \int_0^T g(s)^{\frac{1}{p}} ds \right)^p. \quad (3.82)$$

**Proof** Note that if 3.82 holds for a bounded decreasing  $g$ , it must hold for any decreasing  $g$ . Indeed, if that is the case, we can take decreasing  $g$  and approximate it from below by  $g_k := \min\{g, k\}$  and apply 3.82 to  $g_k$  and conclude 3.82 for  $g$  by passing to the limit and the monotone convergence theorem 2.12 (i). So assume  $g$  is decreasing and bounded. Define now for any  $t \in [0, \infty)$

$$\begin{aligned} \psi(t) &= p \int_0^t g(s) s^{p-1} ds \\ \varphi(t) &= \left( \int_0^t g(s)^{\frac{1}{p}} ds \right)^p. \end{aligned} \quad (3.83)$$

Then, we have  $\psi(0) = \varphi(0) = 0$ ,  $\psi, \varphi$  absolutely continuous in any bounded subinterval of  $[0, \infty)$  and at points continuity we find:

$$\begin{aligned} \psi'(t) &= pg(t)t^{p-1} \\ \varphi'(t) &= pg(t)^{\frac{1}{p}} \left( \int_0^t g(s)^{\frac{1}{p}} ds \right)^{p-1}. \end{aligned} \quad (3.84)$$

As  $\psi'(t) = pg(t)^{\frac{1}{p}} \left( tg^{\frac{1}{p}} \right)^{p-1}$ , by the monotonicity of  $g$  it follows that  $\psi'(t) \leq \varphi'(t)$  for  $\mathcal{L}^1$ -a.e.  $t \in (0, \infty)$ . Hence, integrating from 0 to  $T$  we conclude 3.82.  $\square$

**Proof (Theorem 3.31)** Let  $T = \{t \in \mathbb{R} \mid P(\{u > t\}, \mathbb{R}^N) < \infty\}$ . By the coarea formula 3.26 we find that  $T$  must be dense in  $\mathbb{R}$  (as  $P(\{u > t\}, \mathbb{R}^N) < \infty$  for  $\mathcal{L}^1$ -a.e.  $t \in \mathbb{R}$ ). By Theorem 3.30, we see that for every  $t \in T$  either  $\mathcal{L}^N(\{u > t\}) < \infty$  or  $\mathcal{L}^N(\{u \leq t\}) < \infty$ . Define now

$$m = \inf\{t \in T \mid \mathcal{L}^N(\{u > t\}) < \infty\}. \quad (3.85)$$

If  $t > m$ , by density, we find  $\tau \in T \cap (m, t)$  such that  $\mathcal{L}^N(\{u > \tau\}) < \infty$  and hence  $\mathcal{L}^N(\{u > t\}) < \infty$ . Using the same argument for every  $t < m$  we have  $\mathcal{L}^N(\{u \leq t\}) < \infty$ . We now show that  $m \in \mathbb{R}$ . Indeed, suppose for the sake of contradiction that  $m = -\infty$ . Then for every  $t \in \mathbb{R}$  we have  $\mathcal{L}^N(\{u > t\}) < \infty$  and by the coarea formula for any  $h \in \mathbb{N}$  we find

$$\int_{-(h+1)}^{-h} P(\{u > t\}, \mathbb{R}^N) dt \leq V(u, \mathbb{R}^N) < \infty. \quad (3.86)$$

By density of  $T$  in  $\mathbb{R}$  we can find  $t_h \in (-(h+1), -h)$  such that  $\sup_{h \in \mathbb{N}} P(\{u > t_h\}, \mathbb{R}^N) < \infty$ . Then, by Theorem 3.30 we have  $\sup_{h \in \mathbb{N}} \mathcal{L}^N(\{u > t_h\}) < \infty$ , but we have  $\bigcup_{h \in \mathbb{N}} \{u > t_h\} = \mathbb{R}^N$  which gives us the contradiction. The same argument for  $m = \infty$  and the sets  $\{u \leq t\}$  yields the same contradiction. Hence,  $m \in \mathbb{R}$ . Up to replacing  $u$  by  $u - m$ , we may assume that  $m = 0$ . Let us now  $v := \max\{u, 0\}$ . Now by Lemma 3.32 for  $p = \frac{N}{N-1}$ ,  $g(s) := \mathcal{L}^N(\{u > s\})s^{\frac{1}{N-1}}$  and Fubini's theorem we get the following:

$$\begin{aligned} \int_{\mathbb{R}^N} v^{\frac{N}{N-1}} dx &= \int_0^\infty \mathcal{L}^N(\{v^{\frac{N}{N-1}} > t\}) dt \\ &= \frac{N}{N-1} \int_0^\infty \mathcal{L}^N(\{u > s\}) s^{\frac{1}{N-1}} ds \\ &\leq \left( \int_0^\infty \mathcal{L}^N(\{u > s\})^{\frac{N-1}{N}} ds \right)^{\frac{N}{N-1}}. \end{aligned} \quad (3.87)$$

As  $\mathcal{L}^N(\{u > s\}) < \infty$  for every  $s \in (0, \infty)$  (as  $m = 0$ ), we infer by Theorem 3.30 that there is some constant  $c > 0$  only depending on  $N$  such that

$$\mathcal{L}^N(\{u > s\})^{\frac{N-1}{N}} \leq c P(\{u > s\}, \mathbb{R}^N), \quad \forall s > 0. \quad (3.88)$$

So by 3.100 we find by the coarea formula that

$$\begin{aligned} \|v\|_{\frac{N}{N-1}}^{\frac{N}{N-1}} &= \left( \int_{\mathbb{R}^N} v^{\frac{N}{N-1}} dx \right)^{\frac{N-1}{N}} \\ &\leq c \int_0^\infty P(\{u > s\}, \mathbb{R}^N) ds \\ &\leq c \|Du\|(\mathbb{R}^N). \end{aligned} \quad (3.89)$$

Now consider  $w := \max\{-u, 0\}$ . By Lemma 3.32 for  $p = \frac{N}{N-1}$ ,  $g(s) := \mathcal{L}^N(\{u \leq s\})s^{\frac{1}{N-1}}$  and Fubini's theorem we get that

$$\begin{aligned} \int_{\mathbb{R}^N} w^{\frac{N}{N-1}} dx &= \int_0^\infty \mathcal{L}^N(\{w^{\frac{N}{N-1}} > t\}) dt \\ &= \frac{N}{N-1} \int_{-\infty}^0 \mathcal{L}^N(\{u \leq s\}) s^{\frac{1}{N-1}} ds \\ &\leq \left( \int_{-\infty}^0 \mathcal{L}^N(\{u \leq s\})^{\frac{N-1}{N}} ds \right)^{\frac{N}{N-1}}. \end{aligned} \quad (3.90)$$

As  $\mathcal{L}^N(\{u > s\}) < \infty$  for every  $s \in (-\infty, 0)$  (as  $m = 0$ ), we infer by Theorem 3.30 that there is some constant  $c > 0$  only depending on  $N$  such that

$$\mathcal{L}^N(\{u \leq s\})^{\frac{N-1}{N}} \leq cP(\{u > s\}, \mathbb{R}^N), \quad \forall s < 0. \quad (3.91)$$

So by 3.90 we find by the coarea formula that

$$\begin{aligned} \|w\|_{\frac{N}{N-1}} &= \left( \int_{\mathbb{R}^N} w^{\frac{N}{N-1}} dx \right)^{\frac{N-1}{N}} \\ &\leq c \int_{-\infty}^0 P(\{u \leq s\}, \mathbb{R}^N) ds \\ &\leq c \|Du\|(\mathbb{R}^N). \end{aligned} \quad (3.92)$$

This shows that we find a dimensional constant  $\gamma_3 > 0$  such that:

$$\begin{aligned} \|u\|_{\frac{N}{N-1}} &= \|v - w\|_{\frac{N}{N-1}} \\ &\leq \|v\|_{\frac{N}{N-1}} + \|w\|_{\frac{N}{N-1}} \\ &\leq \gamma_3 \|Du\|(\mathbb{R}^N). \end{aligned} \quad (3.93)$$

Let now  $u \in L^1(\mathbb{R}^N, \mathcal{L}^N)$ . Then by Chebyshev's inequality<sup>2</sup> we find that  $\mathcal{L}^N(\{u > t\}) \leq \frac{1}{t} \|u\|_1 < \infty$  and  $\mathcal{L}^N(\{u > t\}) = \infty$  for all  $t < 0$ . Therefore  $m = 0$ .  $\square$

**Corollary 3.33** *Let  $N > 1$  and  $\Omega \subset \mathbb{R}^N$  a bounded extension domain. Then, the embedding  $BV(\Omega) \hookrightarrow L^{\frac{N}{N-1}}(\Omega, \mathcal{L}^N)$  is a bounded linear operator, and for every  $p \in [1, \frac{N}{N-1})$  the embedding  $BV(\Omega) \hookrightarrow L^p(\Omega, \mathcal{L}^N)$  is a compact operator.*

**Proof** Since  $\Omega$  is an extension domain, we find a bounded linear operator  $T : BV(\Omega) \rightarrow BV(\mathbb{R}^N)$  that meets (i)-(iii) of Definition 3.14. Note that we have an embedding  $L^p(\Omega, \mathcal{L}^N) \hookrightarrow L^p(\mathbb{R}^N, \mathcal{L}^N)$  sending  $u \in L^p(\Omega, \mathcal{L}^N)$  to  $\tilde{u} \in L^p(\mathbb{R}^N, \mathcal{L}^N)$  defined by  $\tilde{u}|_{\Omega} = u$  and  $\tilde{u}|_{\mathbb{R}^N \setminus \Omega} = 0$ . Then we have that the following diagram commutes for every  $p \in [1, \frac{N}{N-1}]$ :

$$\begin{array}{ccc} BV(\Omega) & \xrightarrow{T} & BV(\mathbb{R}^N) \\ \iota_{\Omega} \downarrow & & \downarrow \iota \\ L^p(\Omega, \mathcal{L}^N) & \xhookrightarrow{\iota_p} & L^p(\mathbb{R}^N, \mathcal{L}^N). \end{array} \quad (3.94)$$

<sup>2</sup>If  $(X, \mathcal{E}, \mu)$  is a measure space and  $u : X \rightarrow \mathbb{R}$  is measurable it follows that for every  $t > 0$  and  $1 \leq p < \infty$ :

$$\mu(\{u > t\}) \leq \frac{1}{t^p} \int_{\{u > t\}} |u|^p d\mu.$$

By Theorem 3.31, the embedding  $\iota$  is a bounded linear operator. Also the embedding  $\iota_p$  is a bounded linear operator as we only extend by zero, it is even an isometry. Therefore, we can find an inverse function  $\psi_p$  to  $\iota : L^p(\Omega, \mathcal{L}^N) \rightarrow \iota(L^p(\Omega, \mathcal{L}^N))$ . Hence, we see that  $\iota_\Omega = \psi_p \circ \iota \circ T$  is a bounded linear operator for every  $p \in [1, \frac{N}{N-1}]$ . By Theorem 3.19, we see that the embedding  $BV(\Omega) \hookrightarrow L^1(\Omega, \mathcal{L}^N)$  is a compact operator. By Hölder's inequality, the compactness of  $BV(\Omega) \hookrightarrow L^p(\Omega, \mathcal{L}^N)$  for  $p \in [1, \frac{N}{N-1}]$  follows.  $\square$

To finish this section, we go over the **Poincaré inequalities** for functions of bounded variation and sets of finite perimeter.

**Theorem 3.34** *Let  $N > 1$ .*

- (i) *If  $\Omega \subset \mathbb{R}^N$  is a bounded connected extension domain, then there exists a constant  $C = C(\Omega) > 0$  only depending on  $\Omega$ , such that it holds for all  $p \in [1, \frac{N}{N-1}]$ :*

$$\|u - u_\Omega\|_p \leq C \|Du\|(\Omega), \quad \forall u \in BV(\Omega). \quad (3.95)$$

- (ii) *For all  $x \in \mathbb{R}^N$  and  $r > 0$ , there is a constant  $\gamma_4 > 0$  only depending on the unit ball in  $\mathbb{R}^N$  such that for all  $p \in [1, \frac{N}{N-1}]$ :*

$$\left\|u - u_{B(x,r)}\right\|_p \leq \frac{\gamma_4 r^{\frac{N}{p}}}{r^{N-1}} \|Du\|(B(x,r)), \quad \forall u \in BV(B(x,r)). \quad (3.96)$$

- (iii) *Let  $E \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Then there exists a constant  $\gamma_5 > 0$  only depending on  $N$ , such that it holds for all  $x \in \mathbb{R}^N$  and  $r > 0$ :*

$$\min \left\{ \mathcal{L}^N(E \cap B(x,r)), \mathcal{L}^N(E \setminus B(x,r)) \right\}^{\frac{N-1}{N}} \leq \gamma_5 P(E, B(x,r)). \quad (3.97)$$

The proof of this theorem is quite similar to the proof of Corollary 3.28. Nonetheless, we will give the proof here.

**Proof** (i) By Theorem 3.27 there exists a constant  $C_1 > 0$  only depending on  $\Omega$  such that  $\|u - u_\Omega\|_1 \leq C_1 \|Du\|(\Omega)$  for all  $u \in BV(\Omega)$ . By Corollary 3.33 we know that the embedding  $BV(\Omega) \hookrightarrow L^p(\Omega, \mathcal{L}^N)$  is a bounded linear operator for every  $p \in [1, \frac{N}{N-1}]$ . This, there exists a constant  $C_2 > 0$  only depending on  $\Omega$  such that  $\|u\|_p \leq C_2 \|u\|_{BV(\Omega)}$  for all  $u \in BV(\Omega)$ . So we find for all  $u \in BV(\Omega)$  that:

$$\begin{aligned} \|u - u_\Omega\|_p &\leq C_2 \|u - u_\Omega\|_{BV(\Omega)} \\ &= C_2 (\|u - u_\Omega\|_1 + \|Du\|(\Omega)) \\ &\leq (C_1 C_2 + 1) \|Du\|(\Omega). \end{aligned} \quad (3.98)$$

So  $C = C_1 C_2 + 1 > 0$  is the constant that is needed, which depends only on  $\Omega$ .

- (ii) For each  $u \in BV(B(x, r))$  we consider again the corresponding  $u_0 \in BV(B(0, 1))$  given by  $u_0(y) = u(x + ry)$ . As in Corollary 3.28 we note that  $u_{B(x, r)} = (u_0)_{B(0, 1)}$  and  $\|u - u_{B(x, r)}\|_p = r^{\frac{N}{p}} \|u_0 - (u_0)_{B(0, 1)}\|_p$  for each  $p \in [1, \frac{N}{N-1}]$  and  $u \in BV(B(x, r))$ . By (i), there is a constant  $\gamma_4 > 0$  that depends only on  $B(0, 1)$  such that  $\|u_0 - (u_0)_{B(0, 1)}\|_p \leq \gamma_4 \|Du_0\|(B(0, 1))$  for all  $u \in BV(B(x, r))$ . Now consider  $u \in C^1(B(x, r))$ . By Remark 3.7 at (\*), chain rule and substitution at (\*\*) we find that:

$$\begin{aligned}
 \|u - u_{B(x, r)}\|_p &= r^{\frac{N}{p}} \|u_0 - (u_0)_{B(0, 1)}\|_p \\
 &\leq \gamma_4 r^{\frac{N}{p}} \|Du_0\|(B(0, 1)) \\
 &\stackrel{(*)}{=} \gamma_4 r^{\frac{N}{p}} \int_{B(0, 1)} \|\nabla u_0(y)\| \, dy \\
 &\stackrel{(**)}{=} \frac{\gamma_4 r^{\frac{N}{p}}}{r^{N-1}} \int_{B(x, r)} \|\nabla u(y)\| \, dy \\
 &\stackrel{(*)}{=} \frac{\gamma_4 r^{\frac{N}{p}}}{r^{N-1}} \|Du\|(B(x, r)).
 \end{aligned} \tag{3.99}$$

If  $u \in BV(B(x, r))$ , we can choose by Theorem 3.12 as sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(B(x, r))$  converging to  $u$  in  $BV(B(x, r))$  such that

$$\lim_{h \rightarrow \infty} \|Du_h\|(B(x, r)) = \|Du\|(B(x, r)) \tag{3.100}$$

By continuity of the embedding  $BV(\Omega) \hookrightarrow L^p(\Omega, \mathcal{L}^N)$ , we find that  $u_h$  converges to  $u$  in  $L^p(\Omega, \mathcal{L}^N)$ . So by 3.99 we find that:

$$\begin{aligned}
 \|u - u_{B(x, r)}\|_p &= \lim_{h \rightarrow \infty} \|u_h - (u_h)_{B(x, r)}\|_p \\
 &\leq \frac{\gamma_4 r^{\frac{N}{p}}}{r^{N-1}} \lim_{h \rightarrow \infty} \|Du_h\|(B(x, r)) \\
 &= \frac{\gamma_4 r^{\frac{N}{p}}}{r^{N-1}} \|Du\|(B(x, r)).
 \end{aligned} \tag{3.101}$$

This proves (ii).

- (iii) Applying (ii) to the characteristic function  $\chi_E$  and noting that (ii) holds for an open ball in any norm on  $\mathbb{R}^N$  and then applying the same argument as in Theorem 3.30 gives the result in (iii).

This ends this section on the isoperimetric inequalities and sets us up to investigate the structure of sets of finite perimeter.

### 3.2.2 The reduced boundary and the structure of sets of finite perimeter

In this section, we will define the measure-theoretic concepts of boundary, that is, we will introduce the *reduced boundary*. The goal is then to prove De Giorgi's result about the structure of sets of finite perimeter.

**Definition 3.35** Let  $E \subset \mathbb{R}^N$  be a  $\mathcal{L}^N$ -measurable set and  $\Omega \subset \mathbb{R}^N$  the largest open set such that  $E$  has locally finite perimeter in  $\Omega$ . We define the **reduced boundary**  $\partial^*E$  of  $E$  as the set of all  $x \in \text{supp } \|D\chi_E\| \cap \Omega$  such that the following limit exist and satisfies that

$$v_E(x) := \lim_{r \rightarrow 0^+} \frac{D\chi_E(B(x, r))}{\|D\chi_E\|(B(x, r))} \in S^{N-1}. \quad (3.102)$$

We call the function  $v_E : \partial^*E \rightarrow S^{N-1}$  the **generalised inner unit normal to  $E$** .

**Remark 3.36** By the Lebesgue-Besicovitch differentiation theorem (Theorem 2.43) we know that  $v_E$  in Definition 3.35 is a Borel measurable function and  $\partial^*E$  is a Borel set.

First we make general comments about points of the reduced boundary.

**Lemma 3.37** Let  $E \subset \mathbb{R}^N$  be a  $\mathcal{L}^N$ -measurable set. Any point  $x \in \partial^*E$  is a Lebesgue point<sup>3</sup> of  $v_E$  with respect to  $\|D\chi_E\|$ .

**Proof** We note that for  $r > 0$  small enough we have:

$$\frac{1}{2\|D\chi_E\|(B(x, r))} \int_{B(x, r)} \|v_E(y) - v_E(x)\|^2 d\|D\chi_E\|(y) \quad (3.103)$$

$$= 1 - v_E(x) \cdot \frac{D\chi_E(B(x, r))}{\|D\chi_E\|(B(x, r))}. \quad (3.104)$$

So for  $r \rightarrow 0^+$  the above expression goes to 0 if  $x \in \partial^*E$ . This shows that any  $x \in \partial^*E$  is a Lebesgue point of  $v_E$  with respect to  $\|D\chi_E\|$ .  $\square$

Next, we note that we can bound the perimeter of intersections with open balls as follows.

**Proposition 3.38** Let  $E \subset \mathbb{R}^N$  be a set of finite perimeter in an open set  $\Omega \subset \mathbb{R}^N$ ,  $x_0 \in \Omega$  and set  $\delta = d(x_0, \partial\Omega)$ . Then it holds for every  $r \in (0, \delta)$ , that

$$P(E \cap B(x_0, r), \mathbb{R}^N) \leq P(E, \bar{B}(x_0, r)) + \ell'_+(r), \quad (3.105)$$

where  $\ell(r) = \mathcal{L}^N(E \cap B(x_0, r))$  and  $\ell'_+$  is the right derivative of  $\ell$ <sup>4</sup>.

<sup>3</sup>See Corollary 2.45.

<sup>4</sup>For a function  $f : \mathbb{R} \rightarrow \mathbb{R}$  we denote by  $f'_+(r)$  (if it exists) its **right derivative at  $r \in \mathbb{R}$**  defined by

$$f'_+(r) = \lim_{t \rightarrow r^+} \frac{f(t) - f(r)}{t - r}.$$

**Proof** We show the following more general inequality for any  $u \in BV(\Omega)$ ,  $u_r := u\chi_{B(x_0, r)}$  and  $r \in (0, \delta)$

$$\|Du_r\|(\mathbb{R}^N) \leq \|Du\|(\overline{B}(x_0, r)) + \left( \int_{B(x_0, r)} |u(x)| dx \right)'_{+}. \quad (3.106)$$

Note that 3.105 follows from 3.106 for  $u = \chi_E$ . Let  $u \in BV(\Omega)$ . For any  $\rho \in (0, \delta - r)$  we define  $u^\rho \in BV(\mathbb{R}^N)$  by  $u^\rho(x) := u(x)\gamma_\rho(\|x\|)$ , where

$$\gamma_\rho(t) := \begin{cases} 1, & t \leq r \\ 1 + \frac{r-t}{\rho}, & r \leq t \leq r + \rho \\ 0, & t \geq r + \rho. \end{cases} \quad (3.107)$$

Since  $\gamma_\rho$  defines a Lipschitz map, it follows by Proposition 3.4 that  $u^\rho \in BV(\mathbb{R}^N)$  with support in  $\overline{B}(x_0, r + \rho)$  and coinciding with  $u$  in  $B(x_0, r)$ . Moreover, we have that

$$Du^\rho = \gamma_\rho(\|x\|)Du + u(x)\gamma'_\rho(x)\frac{x}{\|x\|}\mathcal{L}^N, \quad (3.108)$$

and hence

$$\begin{aligned} \|Du^\rho\|(\mathbb{R}^N) &\leq \int_{B(x_0, r+\rho)} \gamma_\rho(\|x\|) d\|Du\|(x) + \int_{B(x_0, r+\rho)} |u(x)| \gamma'_\rho(x) dx \\ &\leq \|Du\|(B(x_0, r + \rho)) + \rho^{-1} \int_{B(x_0, r+\rho) \setminus B(x_0, r)} |u(x)| dx. \end{aligned} \quad (3.109)$$

By the convergence of  $u^\rho$  to  $u_r$  in  $L^1(\mathbb{R}^N, \mathcal{L}^N)$  as  $\rho \rightarrow 0^+$  and the lower semicontinuity of variation, the inequality 3.108 follows from 3.109.  $\square$

**Remark 3.39** If  $P(E, \partial B(x_0, r)) = 0$  in Proposition 3.38 we can subtract in 3.105 on both sides  $P(E, B(x_0, r))$  to obtain

$$P(E \cap B(x_0, r), \partial B(x_0, r)) \leq \ell'_+(r). \quad (3.110)$$

Moreover, by noting that  $\mathcal{L}^N(E \cap B(x_0, r)) = \int_0^r \mathcal{L}^N(E \cap \partial B(x_0, t)) dt$  we have  $\ell'_+(r) = \mathcal{L}^N(E \cap \partial B(x_0, r))$  for  $\mathcal{L}^1$ -a.e.  $r \in (0, \delta)$  and hence by 3.110 we have shown that:

$$P(E \cap B(x_0, r), \partial B(x_0, r)) \leq \mathcal{L}^N(E \cap \partial B(x_0, r)). \quad (3.111)$$

That is, if the perimeter of  $E$  vanishes in the boundaries of open balls, then its perimeter in open balls is controlled by its Lebesgue measure in the boundary of open balls.

Now, we give bounds on the perimeter and Lebesgue measure in neighborhoods of the reduced boundary. This result will establish that, in a sense, the reduced boundary and the perimeter are very much  $(N - 1)$ -dimensional entities.

**Lemma 3.40** *Let  $N > 1$  and  $\Omega \subset \mathbb{R}^N$  be an open set. Let  $E \subset \mathbb{R}^N$  be a set of finite perimeter in  $\Omega$  and  $x_0 \in \partial^* E \cap \Omega$ . Then, there exists  $r_0 \in (0, d(x_0, \Omega))$  and constant  $\alpha, \beta > 0$  only depending on  $N$  such that*

$$P(E, B(x_0, r)) \leq \alpha r^{N-1} \quad \forall r \in (0, r_0) \quad (3.112)$$

$$\min\{\mathcal{L}^N(E \cap B(x_0, r)), \mathcal{L}^N(B(x_0, r) \setminus E)\} \geq \beta r^N \quad \forall r \in (0, r_0). \quad (3.113)$$

**Proof** We may choose  $r_0 \in (0, \frac{d(x_0, \Omega)}{2})$  such that

$$P(E, B(x_0, r)) = \|D\chi_E\|(B(x_0, r)) \leq \|D\chi_E(B(x_0, r))\|, \quad \forall r \in (0, 2r_0). \quad (3.114)$$

Let now  $T \subset (0, 2r_0)$  be the subset of all  $t \in (0, 2r_0)$  such that  $P(E, \partial B(x_0, t)) = 0$ . Set  $E_t = E \cap B(x_0, t)$  for  $t \in T$ . By Remark 3.39 and since  $D\chi_{E_t}(\mathbb{R}^N) = 0$  we infer that for every  $t \in T$  it holds that:

$$\begin{aligned} \|D\chi_E(B(x_0, t))\| &= \|D\chi_{E_t}(B(x_0, t))\| \\ &= \|D\chi_{E_t}(\partial B(x_0, t))\| \\ &\leq \ell'_+(t). \end{aligned} \quad (3.115)$$

Also, we note that  $T \subset (0, 2r_0)$  is dense, and so for our choice of  $r_0$  we find that for  $\mathcal{L}^1$ -a.e.  $r \in (0, 2r_0)$  it holds that:

$$P(E, B(x_0, r)) \leq 2\ell'(r). \quad (3.116)$$

So given any  $r \in (0, r_0)$  integration from  $r$  to  $2r$  yields:

$$\begin{aligned} P(E, B(x_0, r)) &\leq r^{-1} \int_r^{2r} P(E, B(x_0, t)) dt \\ &\leq \frac{2\ell(2r)}{r} \\ &\leq 2^{N+1} \omega_N r^{N-1}. \end{aligned} \quad (3.117)$$

So with  $\alpha = 2^{N+1} \omega_N$ , 3.112 follows.

Now note that by Proposition 3.38 and computation 3.116 we have  $P(E_t, \mathbb{R}^N) \leq 3\ell'(t)$  for  $\mathcal{L}^1$ -a.e.  $t \in (0, r_0)$ . By Theorem 3.30 and recalling that  $\ell(t) = \mathcal{L}^N(E_t)$  we find for  $\mathcal{L}^1$ -a.e.  $t \in (0, r_0)$  that

$$\begin{aligned} \left(\ell^{\frac{1}{N}}\right)'(t) &= \frac{1}{N} \ell^{\frac{1-N}{N}}(t) \ell'(t) \\ &\geq \frac{1}{3N} \ell^{\frac{1-N}{N}}(t) P(E_t, \mathbb{R}^N) \\ &\geq \frac{1}{3N} \gamma_2^{\frac{1-N}{N}}. \end{aligned} \quad (3.118)$$

Now by integration in  $t$  from 0 to  $r$  we get  $\ell(r) \geq \beta r^N$  for all  $r \in (0, r_0)$ , where  $\beta := \frac{1}{3N} \gamma_2^{\frac{1-N}{N}}$ . Using the same argument for  $\mathbb{R}^N \setminus E$ , we find that for this  $\beta$  inequality 3.113 holds.  $\square$

Now we are in the position to state and prove De Giorgi's result about the structure of sets of finite perimeter.

**Theorem 3.41** *Let  $N > 1$  and  $E \subset \mathbb{R}^N$  a  $\mathcal{L}^N$ -measurable set. Then the reduced boundary  $\partial^* E$  is a countably  $(N-1)$ -rectifiable set and  $\|D\chi_E\| = \mathcal{H}^{N-1} \llcorner \partial^* E$ . Additionally, for any  $x_0 \in \partial^* E$  the following properties hold:*

- (i) *The sets  $\frac{E-x_0}{r}$  locally converge in  $\mathbb{R}^N$  as  $r \rightarrow 0^+$  to the half space  $H$  orthogonal to  $\nu_E(x_0)$  containing  $\nu_E(x_0)$ , i.e.  $\nu_E(x_0) + \nu_E(x_0)^\perp$ .*
- (ii) *The approximate tangent measure to  $\mathcal{H}^{N-1} \llcorner \partial^* E$  at  $x_0$  is given by  $\text{Tan}^{N-1}(\mathcal{H}^{N-1} \llcorner \partial^* E, x_0) = \mathcal{H}^{N-1} \llcorner \nu_E(x_0)^\perp$  and in particular it holds that*

$$\lim_{r \rightarrow 0^+} \frac{\mathcal{H}^{N-1}(\partial^* E \cap B(x_0, r))}{\omega_{N-1} r^{N-1}} = 1. \quad (3.119)$$

Before the proof of this theorem, we state another theorem about the structure of one-dimensional sets of finite perimeter:

**Theorem 3.42** *If  $E$  has finite perimeter in an open interval  $(a, b) \subset \mathbb{R}$  and  $\mathcal{L}^1(E \cap (a, b)) > 0$ , then there exists an integer  $k \geq 1$  and  $k$  pairwise disjoint intervals  $J_i = [a_{2i-1}, a_{2i}] \subset \mathbb{R}$  such that  $E \cap (a, b)$  is equivalent to the union of the  $J_i$  and*

$$P(E, (a, b)) = |\{i \in \{1, \dots, 2k\} \mid a_i \in (a, b)\}|. \quad (3.120)$$

**Proof** A proof this can be seen in Theorem 3.52 in [3].  $\square$

**Proof (Theorem 3.41)** Let  $x_0 \in \partial^* E$  and set  $\nu := \nu_E(x_0)$ . Fix now  $r_0 > 0$  as in Lemma 3.40 and set  $E_r := \frac{E-x_0}{r}$ . We start by proving (i), that is, the convergence of  $E_r$  to  $H$ . We note that  $P(E_r, B(0, R)) = \frac{1}{r^{N-1}} P(E, B(x_0, rR))$ , this follows easily as the perimeter defines a Borel measure. Then by Lemma 3.40 it follows that for every  $r \in (0, r_0)$  and every  $R > 0$  that:

$$P(E_r, B(0, R)) \leq \alpha. \quad (3.121)$$

Hence, by Theorem 3.25, we can assume that  $E_r$  converges locally in  $\mathbb{R}^N$ . Hence, it suffices to show that for any sequence  $r_h \rightarrow 0^+$  the limit  $F$  of  $E_{r_h}$  coincides with  $H$ . Since  $D\chi_{E_{r_h}}$  locally weak\* converges in  $\mathbb{R}^N$  to  $D\chi_F$  as  $h \rightarrow \infty$ , and since  $x_0$  is a Lebesgue point by Lemma 3.37, it follows that  $\|D\chi_{E_{r_h}}\|$  weak\* converges to  $\|D\chi_F\|$  and the polar decomposition of  $D\chi_F$  is given by  $D\chi_F = \nu \|D\chi_F\|$ . This shows, in particular, that  $D\chi_F$  has no

component orthogonal to  $\nu$  and  $D\chi_F \cdot \nu = \|\nu\|^2 \|D\chi_F\| \geq 0$ . Now for a family  $(\rho_\varepsilon)_{\varepsilon \in (0, \infty)}$  of mollifiers we note that:

$$\nabla(\chi_F * \rho_\varepsilon) = (D\chi_F) * \rho_\varepsilon = (\|D\chi_F\| * \rho_\varepsilon)\nu. \quad (3.122)$$

□

This gives us  $\chi_F * \rho_\varepsilon = \gamma_\varepsilon(x \cdot \nu)$  for a suitable increasing family of functions  $\gamma_\varepsilon : \mathbb{R} \rightarrow [0, 1]$ . So for  $\gamma := \lim_{\varepsilon \rightarrow 0^+} \gamma_\varepsilon$  with  $D\gamma \geq 0$  we find that  $\chi_F(x) = \gamma(x \cdot \nu)$  for  $\mathcal{L}^N$ -a.e.  $x \in \mathbb{R}^N$ . Since  $\chi_F \in \{0, 1\}$   $\mathcal{L}^N$ -a.e., it must be that  $\gamma$  is equal in  $L^1(\mathbb{R}^1, \mathcal{L}^1)$  to a characteristic function  $\chi_L$ . By Theorem 3.42  $L$  must be a half-line  $(c, \infty)$  for some  $c \in \mathbb{R}$ . If  $c > 0$ , then set  $d = \min\{c, 1\}$  and note that

$$\begin{aligned} \lim_{h \rightarrow \infty} \frac{\mathcal{L}^N(E \cap B(x_0, dr_h))}{r_h^N} &= \lim_{h \rightarrow \infty} \mathcal{L}^N(E_{r_h} \cap B(0, d)) \\ &= \mathcal{L}^N(F \cap B(0, d)) \\ &= 0. \end{aligned} \quad (3.123)$$

This contradicts  $x_0 \in \partial^* E$ , where density is strictly positive. Symmetric arguments for  $c < 0$  give the same contradiction. Hence,  $c = 0$  and hence  $\chi_F(x) = 1$  if and only if  $x \cdot \nu > 0$ , that is,  $F = H$ .

Since  $E_r$  converges to  $H$ , we have that  $D\chi_{E_r}$  weak\* converges to  $D\chi_H$  as  $r \rightarrow 0^+$ . This implies that  $\text{Tan}(D\chi_E, x_0) = \nu \mathcal{H}^{N-1} \llcorner \partial H$ . By Remark 2.59, it follows that:

$$\text{Tan}^{N-1}(\|D\chi_E\|, x_0) = \mathcal{H}^{N-1} \llcorner \partial H. \quad (3.124)$$

Now by Proposition 2.60 (ii), it follows that since  $\|D\chi_E\|$  is concentrated on the Borel set  $\partial^* E$  and by 3.124 admits an approximate tangent space with multiplicity 1 at  $x_0$ , it must follow that  $\|D\chi_E\|$  is countably  $\mathcal{H}^{N-1}$ -rectifiable and  $\|D\chi_E\| = \mathcal{H}^{N-1} \llcorner \partial^* E$ . □

Next, we introduce another concept of boundary: the *essential boundary*.

**Definition 3.43** For every  $t \in [0, 1]$  and every  $\mathcal{L}^N$ -measurable set  $E \subset \mathbb{R}^N$  we define the *set of points of density*  $t$  by

$$E^t := \left\{ x \in \mathbb{R}^N \mid \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} = t \right\}. \quad (3.125)$$

We define the *essential boundary* of  $E$  by  $\partial^e E := \mathbb{R}^N \setminus (E^0 \cup E^1)$ .

**Remark 3.44** We note that if  $E \subset \mathbb{R}^N$  is a  $\mathcal{L}^N$ -measurable set,  $x \in \mathbb{R}^N$  and  $\{C_r\}_{r \in (0, r_0)}$  is a collection of Borel sets, such that there are constants  $0 < \alpha < \beta$  with  $B(x, \alpha r) \subset C_r \subset B(x, \beta r)$  for  $r \in (0, r_0)$ , then for  $t = 0, 1$  we have  $x \in E^t$  if and only if

$$\lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(E \cap C_r)}{\mathcal{L}^N(C_r)} = t. \quad (3.126)$$

This follows simply from the following chain of inequalities:

$$\frac{\mathcal{L}^N(E \cap B(x, \alpha r))}{\omega_N(\beta r)^N} \leq \frac{\mathcal{L}^N(E \cap C_r)}{\mathcal{L}^N(C_r)} \leq \frac{\mathcal{L}^N(E \cap B(x, \beta r))}{\omega_N(\alpha r)^N}. \quad (3.127)$$

If  $x \in E^0$ , then the last term in 3.127 tends to 0 as  $r \rightarrow 0^+$ . If  $x \in E^1$ , then the first term in 3.127 tends to 1 as  $r \rightarrow 0^+$ . This proves 3.126. Conversely, if 3.126 holds for  $t = 0$ , then the first term in 3.127 tends to 0 if  $r \rightarrow 0^+$  and  $x \in E^0$ . If 3.126 holds for  $t = 1$ , then the last term in 3.127 tends to 1 if  $r \rightarrow 0^+$  and  $x \in E^1$ .

We now state and prove the following result about the essential boundary of sets of finite perimeter by Federer.

**Theorem 3.45** *Let  $E \subset \mathbb{R}^N$  be a set of finite perimeter in an open set  $\Omega \subset \mathbb{R}^N$ . Then the following holds:*

- (i)  $\partial^* E \cap \Omega \subset E^{\frac{1}{2}} \subset \partial^e E$ ;
- (ii)  $\mathcal{H}^{N-1}(\Omega \setminus (E^0 \cup \partial^* E \cup E^1))$  and in particular for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Omega$   $E$  has density 0,  $\frac{1}{2}$  or 1 and  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial^e E \cap \Omega$  belongs to  $\partial^* E$ .

**Proof** (i) Let  $x \in \partial^* E \cap \Omega$ . By Theorem 3.41  $\frac{E-x}{r}$  locally converges in  $\mathbb{R}^N$  to the half space  $H$  containing  $v_E(x)$ . We clearly have that  $H$  bisects the unit ball  $B(0, 1)$  into two equal parts, i.e.  $\mathcal{L}^N(H \cap B(0, 1)) = \frac{\omega_N}{2}$ . Then by the local convergence it follows that:

$$\begin{aligned} \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} &= \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N\left(\frac{E-x}{r} \cap B(0, 1)\right)}{\omega_N} \\ &= \frac{\mathcal{L}^N(H \cap B(0, 1))}{\omega_N} \\ &= \frac{1}{2}. \end{aligned} \quad (3.128)$$

This shows that  $\partial^* E \cap \Omega \subset E^{\frac{1}{2}}$ .

- (ii) Consider  $\alpha(r) = \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N}$ . By 3.97 we know that  $\min\{\alpha(r), 1 - \alpha(r)\} \rightarrow 0$  as  $r \rightarrow 0^+$ . So we must have  $\alpha(r) \rightarrow 0$  or  $\alpha(r) \rightarrow 1$  as  $r \rightarrow 0^+$ . Now, since  $P(E, B(x, r)) = \mathcal{H}^{N-1}(\partial^* E \cap B(x, r))$ , we must have that  $\partial^e E \cap \Omega$  is contained in the set of strictly positive  $(N-1)$ -dimensional upper density of  $\mathcal{H}^{N-1} \llcorner \partial^* E$ . But also  $\mathcal{H}^{N-1}$ -a.e. point in  $\partial^e E \cap \Omega$  belongs to  $\partial^* E$ .

**Remark 3.46** *By Theorem 3.41 and Theorem 3.45, for any set of finite perimeter  $E \subset \mathbb{R}^N$  in an open set  $\Omega \subset \mathbb{R}^N$ :*

$$P(E, \Omega) = \mathcal{H}^{N-1}(\partial^* E \cap \Omega) = \mathcal{H}^{N-1}(\partial^e E \cap \Omega) = \mathcal{H}^{N-1}\left(E^{\frac{1}{2}} \cap \Omega\right). \quad (3.129)$$

In particular, for every  $\varphi \in [C_c^1(\Omega)]^N$  the following versions of the Gauss-Green formular hold

$$\int_E \operatorname{div}(\varphi) \, dx = - \int_{\partial^* E} \varphi \cdot \nu_E \, d\mathcal{H}^{N-1} \quad (3.130)$$

$$\int_E \operatorname{div}(\varphi) \, dx = - \int_{\partial^e E} \varphi \cdot \nu_E \, d\mathcal{H}^{N-1} \quad (3.131)$$

$$\int_E \operatorname{div}(\varphi) \, dx = - \int_{E^{\frac{1}{2}}} \varphi \cdot \nu_E \, d\mathcal{H}^{N-1}. \quad (3.132)$$

Finally, we make the connection between the topological and measure-theoretic boundaries.

**Proposition 3.47** *Any open set  $\Omega \subset \mathbb{R}^N$  with  $\mathcal{H}^{N-1}(\partial\Omega) < \infty$  has finite perimeter in  $\mathbb{R}^N$  and  $\|D\chi_\Omega\| \leq \mathcal{H}^{N-1} \llcorner \partial\Omega$ . Equality holds if  $\Omega$  has Lipschitz boundary.*

**Proof** Without loss of generality, we may assume that  $\Omega$  is bounded. Then  $\partial\Omega$  is compact and for any  $\delta > 0$  there is a finite family  $\{C_i\}_{i \in I}$  of closed sets with diameters  $2r_i < \delta$  such that  $\partial\Omega \subset \bigcup_{i \in I} C_i$  and

$$\sum_{i \in I} \omega_{N-1} r_i^{N-1} < \mathcal{H}^{N-1}(\partial\Omega) + \delta. \quad (3.133)$$

Let now  $\{B_i\}_{i \in I}$  be a family of closed balls with radii  $2r_i$  centered at arbitrary points  $x_i \in \Omega$ . Then we still have  $\partial\Omega \subset \bigcup_{i \in I} B_i$ . Set now  $\Omega_\delta = \Omega \cup \bigcup_{i \in I} B_i$ . As  $\partial\Omega_\delta \subset \bigcup_{i \in I} \partial B_i$ , we can infer by Remark 3.22 and 3.133 that:

$$\begin{aligned} P(\Omega_\delta, \mathbb{R}^N) &\leq \sum_{i \in I} P(B_i, \mathbb{R}^N) \\ &= \sum_{i \in I} \mathcal{H}^{N-1}(\partial B_i) \\ &\leq \frac{N\omega_N 2^{N-1}}{\omega_{N-1}} (\mathcal{H}^{N-1}(\partial\Omega) + \delta). \end{aligned} \quad (3.134)$$

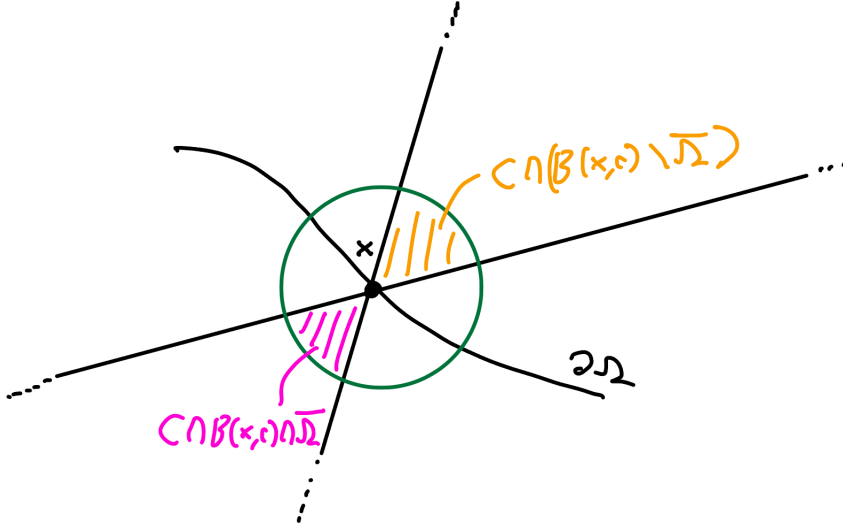
Since  $\mathcal{L}^N(\Omega_\delta \setminus \Omega) \rightarrow 0$  as  $\delta \rightarrow 0^+$ , we find that

$$\begin{aligned} P(\Omega, \mathbb{R}^N) &\leq \liminf_{\delta \rightarrow 0^+} P(\Omega_\delta, \mathbb{R}^N) \\ &\leq \frac{N\omega_N 2^{N-1}}{\omega_{N-1}} \mathcal{H}^{N-1}(\partial\Omega) < \infty. \end{aligned} \quad (3.135)$$

Since  $\|D\chi_\Omega\| = \mathcal{H}^{N-1} \llcorner \partial^* \Omega$  by Theorem 3.41 and as  $\partial^* \Omega \subset \partial\Omega$ , it follows that  $\|D\chi_\Omega\| \leq \mathcal{H}^{N-1} \llcorner \partial\Omega$ .

Suppose now that  $\Omega$  has Lipschitz boundary. Let  $x \in \partial\Omega$ . Then we can find  $r > 0$  and a solid cone  $C$  (see Definition 5.17) centered at  $x$  such that one side of  $C \cap B(x, r)$  is contained in  $\Omega$  and the other side in  $\mathbb{R}^N \setminus \overline{\Omega}$ .

This implies that  $x \in \partial^* \Omega$  and so  $\partial\Omega \subset \partial^* \Omega$  and hence  $\|D\chi_\Omega\| = \mathcal{H}^{N-1} \llcorner \partial\Omega$ .  $\square$


 Figure 3.1: Cone  $C$  centered at  $x \in \partial\Omega$ .

### 3.3 Approximate continuity and differentiability

We want now to extend the properties of sets of finite perimeter set forth in the preceding section to all functions of bounded variation. For this, we introduce the concept of *approximate limits*.

**Definition 3.48** Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . We say that  $u$  has **approximate limit** at  $x \in \Omega$  if there exists  $z \in \mathbb{R}$  such that

$$\lim_{r \rightarrow 0^+} \int_{B(x,r)} |u(y) - z| \, dy = 0. \quad (3.136)$$

We denote by  $S_u$  the set of points  $x \in \Omega$ , where 3.136 is **not** satisfied, and call it **the appropriate discontinuity set**. For any  $x \in \Omega \setminus S_u$ , we denote  $\tilde{u}(x) \in \mathbb{R}$  the uniquely determined point satisfying 3.136.

We say  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  is **approximately continuous** at  $x \in \Omega$  if  $x \notin S_u$  and  $\tilde{u}(x) = u(x)$  (that is,  $x$  is a Lebesgue point of  $u$ ).

The following proposition summarizes the properties of approximate limits.

**Proposition 3.49** Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ .

- (i)  $S_u$  is an  $\mathcal{L}^N$ -null set and a Borel set. Moreover, the function  $\tilde{u} : \Omega \setminus S_u \rightarrow \mathbb{R}$  is a Borel measurable function coinciding  $\mathcal{L}^N$ -a.e. on  $\Omega \setminus S_u$  with  $u$ .
- (ii) If  $x \in \Omega \setminus S_u$  and  $(\rho_\epsilon)_{\epsilon \in (0, \infty)}$  is a family of mollifiers, then  $u * \rho_\epsilon(x)$  converges to  $\tilde{u}(x)$  as  $\epsilon \rightarrow 0^+$ .

(iii) If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a Lipschitz function and  $v = f \circ u$ , then  $S_v \subset S_u$  and  $\tilde{v} = f \circ \tilde{u}$  on  $\Omega \setminus S_u$ .

**Proof** (i) Since the complement of the set of Lebesgue points of  $u$  is an  $\mathcal{L}^N$ -null set, it follows that  $S_u$  is and  $\mathcal{L}^N$ -null set.

Now we note that:

$$\Omega \setminus S_u = \bigcap_{n=1}^{\infty} \bigcup_{q \in \mathbb{Q}} \left\{ x \in \Omega \left| \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(x) - q| \, dy < \frac{1}{n} \right. \right\}. \quad (3.137)$$

In fact, let  $x \in \Omega \setminus S_u$  and  $n \in \mathbb{N}$ . By density of  $\mathbb{Q}$  in  $\mathbb{R}$  there exist  $q \in \mathbb{Q}$  such that  $|\tilde{u}(x) - q| < \frac{1}{n}$  and hence:

$$\begin{aligned} \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(y) - q| \, dy &= \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |\tilde{u}(x) - q| \, dy \\ &= |\tilde{u}(x) - q| \\ &\leq \frac{1}{n}. \end{aligned} \quad (3.138)$$

This shows the inclusion  $\subset$  in 3.137. If  $x$  is element of the set on the right side of 3.137 we can find a sequence  $(q_n)_{n \in \mathbb{N}}$  such that:

$$\limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(x) - q| \, dy < \frac{1}{n}, \quad \forall n \in \mathbb{N}. \quad (3.139)$$

For all  $n, m \in \mathbb{N}$  we find that:

$$\begin{aligned} |q_n - q_m| &= \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |q_n - q_m| \, dy \\ &\leq \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(x) - q_n| \, dy + \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(x) - q_m| \, dy \\ &\leq \frac{1}{n} + \frac{1}{m}. \end{aligned} \quad (3.140)$$

Letting  $n, m \rightarrow \infty$ , we see that by 3.140  $|q_n - q_m| \rightarrow 0$ . So  $(q_n)_{n \in \mathbb{N}}$  is a Cauchy sequence, and we see that its limit  $z \in \mathbb{R}$  satisfies 3.136. This shows the inclusion  $\supset$  in 3.137. This shows that  $S_u$  is a Borel set. By 3.136 for any  $x \in \Omega \setminus S_u$  the mean values  $u_{B(x,r)}$  of  $u$  converge on  $B(x, r)$  to  $\tilde{u}(x)$  as  $r \rightarrow 0^+$ . Hence,  $\tilde{u}$  as the pointwise limit of  $x \mapsto u_{B(x,r)}$  as  $r \rightarrow 0^+$ , must be a Borel measurable function.

(ii) By the substitution  $x - \varepsilon z = y$  we find that:

$$\begin{aligned} |u * \rho_\varepsilon(x) - \tilde{u}(x)| &\leq \int_{\mathbb{R}^N} |u(x - \varepsilon z) - \tilde{u}(x)| \rho(z) \, dz \\ &\leq \|\rho\|_\infty \omega_N \int_{B(x, \varepsilon)} |u(y) - \tilde{u}(x)| \, dy. \end{aligned} \quad (3.141)$$

So, as  $\varepsilon \rightarrow 0^+$ , we have  $|u * \rho_\varepsilon(x) - \tilde{u}(x)| \rightarrow 0$ .

(iii) By the estimate  $|v(y) - f(\tilde{u}(x))| \leq \text{Lip}(f, \mathbb{R}) |u(y) - \tilde{u}(x)|$  and integration we find the result.

We may also characterize approximate limits  $z \in \mathbb{R}$  in terms of the sets

$$E_\varepsilon := \{y \in \Omega \mid |u(y) - z| > \varepsilon\}. \quad (3.142)$$

**Proposition 3.50** *If  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ ,  $x \in \Omega \setminus S_u$  and  $z = \tilde{u}(x)$ , then for every  $\varepsilon > 0$  the set  $E_\varepsilon$  has density 0 at  $x$ . On the other hand, if  $u \in L^\infty_{\text{loc}}(\Omega, \mathcal{L}^N)$  and all  $E_\varepsilon$ 's have density 0 at  $x$ , then  $u$  has approximate limit at  $x$  and  $\tilde{u}(x) = z$ .*

**Proof** Note that if  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ ,  $x \in \Omega \setminus S_u$  and  $z = \tilde{u}(x)$ , we have by Chebyshev inequality gives us for all  $\varepsilon > 0$ :

$$\varepsilon \frac{1}{\omega_N r^N} \mathcal{L}^N(E_\varepsilon \cap B(x, r)) \leq \int_{B(x, r)} |u(y) - z| \, dy. \quad (3.143)$$

So, if  $r \rightarrow 0^+$  we find that  $E_\varepsilon$  has density 0 at  $x$  for every  $\varepsilon > 0$ .

Conversely, if  $u \in L^\infty_{\text{loc}}(\Omega, \mathcal{L}^N)$  and all  $E_\varepsilon$ 's have density 0 at  $x$ , we find  $M > 0$  and  $r_0 > 0$  such that  $|u(x)| \leq M$  for all  $x \in B(x, r_0)$ . So for all  $r \in (0, r_0)$  we find that:

$$\int_{B(x, r)} |u(y) - z| \, dy \leq (M + |z|) \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} + \varepsilon \frac{\mathcal{L}^N(B(x, r) \setminus E_\varepsilon)}{\omega_N r^N}. \quad (3.144)$$

As  $r \rightarrow 0^+$ , we find that  $\limsup_{r \rightarrow 0^+} \int_{B(x, r)} |u(y) - z| \, dy \leq \varepsilon$  for every  $\varepsilon > 0$ . Hence,  $z = \tilde{u}(x)$ .  $\square$

We now introduce some notation for half balls oriented by some  $\nu \in S^{N-1}$  and so-called jump discontinuities. For  $u : \mathbb{R}^N \rightarrow \mathbb{R}$ ,  $x \in \mathbb{R}^N$ ,  $r > 0$  and  $a, b \in \mathbb{R}$  we define:

$$B^+(x, r, \nu) := \{y \in B(x, r) \mid (y - x) \cdot \nu > 0\} \quad (3.145)$$

$$B^-(x, r, \nu) := \{y \in B(x, r) \mid (y - x) \cdot \nu < 0\} \quad (3.146)$$

$$u_{a,b,\nu}(y) := \begin{cases} a, & y \cdot \nu > 0 \\ b, & y \cdot \nu < 0. \end{cases} \quad (3.147)$$

**Definition 3.51** *Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  and  $x \in \Omega$ . We say that  $x$  is an **approximate jump point** of  $u$  if there exist  $a, b \in \mathbb{R}$  and  $\nu \in S^{N-1}$  such that  $a \neq b$  and*

$$\begin{aligned} \lim_{r \rightarrow 0^+} \int_{B^+(x, r, \nu)} |u(y) - a| \, dy &= 0 \\ \lim_{r \rightarrow 0^+} \int_{B^-(x, r, \nu)} |u(y) - b| \, dy &= 0. \end{aligned} \quad (3.148)$$

Up to permutation of  $(a, b)$  and sign of  $v$ , the triplet  $(a, b, v)$  is uniquely determined, and we denote it by  $(u^+(x), u^-(x), v_u(x))$ . The set of approximate jump points is denoted by  $J_u$ .

To make the connection to the reduced boundary and sets of finite perimeter, we consider the following example.

**Example 3.52** If  $E \subset \mathbb{R}^N$  is  $\mathcal{L}^N$ -measurable, we may consider  $u = \chi_E \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . Then, by definition, we note that  $S_u = \partial^e E$  and we have  $J_u \subset E^{\frac{1}{2}}$  and  $\{u^+(x), u^-(x)\} = \{0, 1\}$  for any  $x \in J_u$ . If  $E$  is a set of finite perimeter, then  $\partial^* E \subset J_u$  by Theorem 3.41 and the following sets have all the same  $\mathcal{H}^{N-1}$ -measure:

$$\partial^* E \cap \Omega, J_{\chi_E} \cap \Omega, E^{\frac{1}{2}} \cap \Omega, \partial^e E \cap \Omega. \quad (3.149)$$

The following proposition summarizes the properties of the set of approximate jump points.

**Proposition 3.53** Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . Then it holds that:

(i) The set  $J_u \subset S_u$  is a Borel set and there exists Borel functions

$$(u^+, u^-, v_u) : J_u \rightarrow \mathbb{R} \times \mathbb{R} \times S^{N-1} \quad (3.150)$$

such that  $(u^+(x), u^-(x), v_u(x))$  satisfies 3.148 for every  $x \in J_u$ .

(ii) If  $x \in J_u$  and  $(\rho_\varepsilon)_{\varepsilon \in (0, \infty)}$  is a family of mollifiers, then the functions  $u * \rho_\varepsilon$  converge to  $\frac{u^+(x) + u^-(x)}{2}$  as  $\varepsilon \rightarrow 0^+$ .

(iii) If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a Lipschitz function,  $v = f \circ u$  and  $x \in J_u$ , then  $x \in J_v$  if and only if  $f(u^+(x)) \neq f(u^-(x))$  and in that case

$$(v^+(x), v^-(x), v_v(x)) = (f(u^+(x)), f(u^-(x)), v_u(x)). \quad (3.151)$$

Otherwise,  $x \notin S_v$  and  $\tilde{v}(x) = f(u^+(x)) = f(u^-(x))$ .

**Proof** The proof follows the same arguments as the proof of Proposition 3.49.  $\square$

Lastly, we introduce the concept of approximate differentiability.

**Definition 3.54** Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  and  $x \in \Omega \setminus S_u$ . Then,  $u$  is **approximately differentiable at  $x$**  if there exists a vector  $L \in \mathbb{R}^N$  such that

$$\lim_{r \rightarrow 0^+} \int_{B(x, r)} \frac{|u(y) - \tilde{u}(x) - L \cdot (y - x)|}{r} dy = 0 \quad (3.152)$$

If  $u$  is approximately differentiable at  $x$ , then  $L$  is uniquely determined and we call it the **approximate differential of  $u$  at  $x$**  and denote it by  $\nabla u(x)$ . The set of approximate differentiability points is denoted by  $\mathcal{D}_u$ .

The following proposition summarizes the property of the set of approximate differentiability points.

**Proposition 3.55** *Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . Then it holds that:*

- (i) *The set  $\mathcal{D}_u \subset \Omega \setminus S_u$  is a Borel set in  $\Omega$  and  $\nabla u : \mathcal{D}_u \rightarrow \mathbb{R}^N$  is a Borel function and  $\nabla u(x)$  satisfies 3.152 for all  $x \in \mathcal{D}_u$ .*
- (ii) *If  $x \in \mathcal{D}_u$  and  $f \in C^1(\mathbb{R})$ , then  $v = f \circ u$  is approximately differentiable at  $x$  and*

$$\nabla v(x) = \nabla f(\tilde{u}(x)) \nabla u(x). \quad (3.153)$$

**Proof** (i) Similar arguments as in Proposition 3.49 (i) by choosing a countable dense subset of  $\mathbb{R}^N$  in place of  $\mathbb{Q}$ , shows that  $\mathcal{D}_u$  is a Borel set. Same ideas from Proposition 3.49 (i) hold also for the Borel measurability of  $\nabla u$ .

- (ii) See for a proof, the proof of Proposition 3.71 in [3].

We may rephrase all these approximate concepts as follows.

**Remark 3.56** *Let  $u \in L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$  and define for all  $x \in \Omega$  and  $r > 0$  the rescaled function  $u^{x,r}(y) = u(x + ry)$ . Then we have the following equivalences:*

- (i)  *$u$  has approximate limit  $z$  at  $x$  if and only if  $u^{x,r}$  converges to  $z$  in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  as  $r \rightarrow 0^+$ .*
- (ii)  *$x \in J_u$  and  $(u^+(x), u^-(x), \nu_u(x)) = (a, b, \nu)$  if and only if  $u^{x,r}$  converges to  $u_{a,b,\nu}$  in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  as  $r \rightarrow 0^+$ .*
- (iii)  *$x \in \mathcal{D}_u$  and  $\nabla u(x) = L$  if and only if  $v_r(y) = \frac{u^{x,r}(y) - \tilde{u}(x)}{r}$  converges to  $y \mapsto L \cdot y$  in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  as  $r \rightarrow 0^+$ .*

### 3.4 The trace of functions of bounded variation

This section is dedicated to the fine properties of functions of bounded variation and should end with the introduction of the *trace of a function of bounded variation*. This section is in fact an extension of the section about the reduced boundary of sets of finite perimeter. We begin this section with some important preliminary results.

**Lemma 3.57** *Let  $(E_h)_{h \in \mathbb{N}}$  be a sequence of Borel subsets of an open set  $\Omega \subset \mathbb{R}^N$ . If  $P(E_h, \Omega) \rightarrow 0$  and  $\mathcal{L}^N(E_h) \rightarrow 0$  as  $h \rightarrow \infty$ , then it holds for all  $\alpha > 0$  that*

$$\mathcal{H}^{N-1} \left( \bigcap_{h=1}^{\infty} \{x \in \Omega \mid \Theta_N^*(E_h, x) \geq \alpha\} \right) = 0. \quad (3.154)$$

**Proof** We show for  $\Omega' \subset\subset \Omega$ ,  $\sigma = d(\Omega', \partial\Omega)$ ,  $\alpha \in (0, 1)$ ,  $p > \frac{2}{\alpha}$  and  $\delta = \left(\frac{p\mathcal{L}^N(E)}{\omega_N}\right)^{\frac{1}{N}}$ , where  $E \subset \Omega$  is Borel, that there are constants  $\gamma_1, \xi > 0$  such that if  $\delta < \sigma$ :

$$\mathcal{H}_{2\delta}^{N-1}(E^\alpha \cap \Omega') \leq \frac{2\gamma_1 \xi \omega_{N-1}}{\omega_N \alpha} P(E, \Omega). \quad (3.155)$$

In 3.155 we denote  $E^\alpha = \{x \in \mathbb{R}^N \mid \Theta_N^*(E, x) \geq \alpha\}$ . We will use the Besicovitch covering theorem<sup>5</sup>. Note that for every  $x \in E^\alpha \cap \Omega'$  the ratio  $\frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} \geq \frac{\alpha}{2}$  for all  $r > 0$  by definition of upper density and  $E^\alpha$ . Then by choice of  $\delta$  we have:

$$\frac{\mathcal{L}^N(E \cap B(x, \delta))}{\omega_N \delta^N} \leq \frac{\mathcal{L}^N(E)}{\omega_N \delta^N} = \frac{1}{p} < \frac{\alpha}{2}. \quad (3.156)$$

By Theorem 3.34 we find  $r \in (0, \delta)$  such that  $\mathcal{L}^N(E \cap B(x, r)) = \frac{\alpha \omega_N r^N}{2}$  and  $2\gamma_1 P(E, B(x, r)) \geq \omega_N \alpha r^{N-1}$ . By the Besicovitch covering theorem, we can cover  $E^\alpha \cap \Omega'$  by a countable family of such balls  $B(r_i, x_i)$  such that at most  $\xi$  of them overlap. Now we can estimate:

$$\begin{aligned} \mathcal{H}_{2\delta}^{N-1}(E^\alpha \cap \Omega') &\leq \omega_{N-1} \sum_i r_i^{N-1} \\ &\leq \frac{2\gamma_1 \omega_{N-1}}{\omega_N \alpha} \sum_i P(E, B(x_i, r_i)) \\ &\leq \frac{2\gamma_1 \xi \omega_{N-1}}{\omega_N \alpha} P(E, \Omega). \end{aligned} \quad (3.157)$$

So for  $h$  large enough we can apply 3.155 to  $E_h$  and obtain for  $\delta_h = \left(\frac{p\mathcal{L}^N(E_h)}{\omega_N}\right)^{\frac{1}{N}}$  that

$$\begin{aligned} \mathcal{H}_{2\delta_h}^{N-1}\left(\bigcap_{i=0}^{\infty} E_i^\alpha \cap \Omega'\right) &\leq \mathcal{H}_{2\delta_h}^{N-1}(E_h^\alpha \cap \Omega') \\ &\leq \frac{2\gamma_1 \xi \omega_{N-1}}{\omega_N \alpha} P(E_h, \Omega). \end{aligned} \quad (3.158)$$

Hence, for  $h \rightarrow \infty$  and  $\Omega' \uparrow \Omega$ , the lemma follows.  $\square$

By this lemma, we find that  $\mathcal{H}^{N-1}$ -a.e. The mean value of the absolute value of functions of bounded variations on balls as the radius goes to 0 is uniformly bounded.

<sup>5</sup>The **Besicovitch covering theorem** states: If  $A \subset \mathbb{R}^N$  is a bounded set and  $\rho : A \rightarrow (0, \infty)$  is a function, then there is a countable subset  $S \subset A$  such that  $A \subset \bigcup_{x \in S} B(x, \rho(x))$  and every point of  $\mathbb{R}^N$  belongs to at most  $\xi$  balls  $B(x, \rho(x))$  for  $x \in S$ . For a proof of this result, see Theorem 2.18 in [3].

**Lemma 3.58** For any  $u \in BV(\Omega)$  the set

$$L := \left\{ x \in \Omega \left| \limsup_{r \rightarrow 0^+} \int_{B(x,r)} |u(y)|^{\frac{N}{N-1}} dy = \infty \right. \right\} \quad (3.159)$$

is an  $\mathcal{H}^{N-1}$ -null set.

**Proof** Up to replacing  $u$  by  $|u|$  we may assume that  $u \geq 0$ . Consider the set

$$D := \left\{ x \in \Omega \left| \limsup_{r \rightarrow 0^+} \frac{\|Du\|(B(x,r))}{r^{N-1}} = \infty \right. \right\}. \quad (3.160)$$

By Proposition 2.53 we know that  $\Theta_{N-1}^*(\|Du\|, x) < \infty$  for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Omega$  and so  $D$  is a  $\mathcal{H}^{N-1}$ -null set. Now choose for every  $h \in \mathbb{N}$  a  $t_h \in (h, h+1)$  such that:

$$P(\{u > t_h\}, \Omega) \leq \int_h^{h+1} P(\{u > t\}, \Omega) dt. \quad (3.161)$$

Since  $u$  is integrable, we have  $\mathcal{L}^N(\{u > t_h\}) \rightarrow 0$  as  $h \rightarrow \infty$ . By the coarea formula 3.52 we know that

$$\sum_{h \in \mathbb{N}} P(\{u > t_h\}, \Omega) \leq \int_0^\infty P(\{u > t\}, \Omega) dt = \|Du\|(\Omega) < \infty. \quad (3.162)$$

For any  $h \in \mathbb{N}$ , let now  $E_h$  be the set of all points  $x \in \Omega$  such that  $\Theta_N^*(\{u > t_h\}, x) = 1$ . By Lemma 3.57, we know that  $\mathcal{H}^{N-1}(\bigcap_{h \in \mathbb{N}} E_h) = 0$ .

We claim now that  $L \subset D \cup \bigcap_{h \in \mathbb{N}} E_h$ . Suppose  $x \notin D \cup \bigcap_{h \in \mathbb{N}} E_h$  and set  $c_r := u_{B(x,r)}$  for  $r > 0$ . By Theorem 3.34 (ii) we know that there is a dimensional constant  $\gamma_4 > 0$  such that

$$r^{-N} \int_{B(x,r)} |u(y) - c_r|^{\frac{N}{N-1}} dy \leq \gamma_4^{\frac{N}{N-1}} \left( \frac{\|DB(x,r)\|}{r^{N-1}} \right)^{\frac{N}{N-1}}. \quad (3.163)$$

To show that  $x \notin L$ , it suffices now to show that  $(c_r)$  is bounded as  $r \rightarrow 0^+$ . For the sake of contradiction, suppose that there is a sequence  $(r_i)_{i \in \mathbb{N}}$  such that  $r_i \xrightarrow{i \rightarrow \infty} 0$  and  $c_{r_i} \xrightarrow{i \rightarrow \infty} \infty$ . Consider now  $v_i = u^{x,r_i} - c_{r_i}$ . Since  $x \notin D$ , we know that  $\left( \frac{\|Du\|(B(x,r_i))}{r_i^{N-1}} \right)_{i \in \mathbb{N}}$  is a bounded sequence, and therefore for some constant  $c > 0$  we find

$$\int_{B(0,1)} |v_i(y)| dy \leq c \|Dv_i\|(B(0,1)) = c \frac{\|Du\|(B(x,r_i))}{r_i^{N-1}}. \quad (3.164)$$

Hence, the sequence  $(v_i)_{i \in \mathbb{N}}$  is relatively compact in  $L^1(B(0,1), \mathcal{L}^N)$ . So, up to restriction to a subsequence, we can assume that  $(v_i)_{i \in \mathbb{N}}$  converges  $\mathcal{L}^N$ -a.e. to some  $w \in L^1(B(0,1), \mathcal{L}^N)$ . So for  $\mathcal{L}^N$ -a.e.  $y \in B(0,1)$  we have

that  $u(x + r_i y) = c_{r_i} + v_i(y)$  converges to  $\infty$  as  $i \rightarrow \infty$ . Now we note that  $\omega_N = \lim_{i \rightarrow \infty} \frac{\mathcal{H}^N(\{u > t_h\} \cap B(x, r_i))}{r_i^N}$  and so  $\Theta_N^*(\{u > t_h\}, x) = 1$  and so  $x \in E_h$ , contradicting that  $x \notin \bigcap_{h \in \mathbb{N}} E_h$ .  $\square$

**Lemma 3.59** *Let  $u \in BV(\Omega)$ . Then it holds that:*

$$(i) \quad \|Du\| \geq |u^+ - u^-| \mathcal{H}^{N-1} \llcorner J_u.$$

(ii) *For every Borel set  $B \subset \Omega$  it holds that:*

$$\mathcal{H}^{N-1}(B) \implies \|Du\|(B) = 0 \quad (3.165)$$

$$\mathcal{H}^{N-1}(B) < \infty, B \cap S_u = \emptyset \implies \|Du\|(B) = 0. \quad (3.166)$$

**Proof** (i) Let  $x \in J_u$ . By definition of  $J_u$ , the sequence of functions  $u^{x,r}(y) = u(x + ry)$  converges in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  to

$$w_x(y) = \begin{cases} u^+(x), & y \cdot \nu_u(x) > 0 \\ u^-(x), & y \cdot \nu_u(x) < 0. \end{cases} \quad (3.167)$$

By lower semicontinuity of variation we have that:

$$\begin{aligned} |u^+(x) - u^-(x)| &= \frac{\|Dw_x\|(B(0,1))}{\omega_{N-1}} \\ &= \liminf_{r \rightarrow 0^+} \frac{\|Du^{x,r}\|(B(0,1))}{\omega_{N-1}} \\ &= \liminf_{r \rightarrow 0^+} \frac{\|Du\|(B(x,r))}{\omega_{N-1} r^{N-1}} \\ &= \Theta_{N-1}(\|Du\|, x). \end{aligned} \quad (3.168)$$

So we have found that for every  $x \in J_u$  it holds that  $\Theta_{N-1}(\|Du\|, x) \geq |u^+(x) - u^-(x)|$ . Thus, since  $J_u$  is a Borel set, by Proposition 2.53 it follows that  $\|Du\| \geq |u^+ - u^-| \mathcal{H}^{N-1} \llcorner J_u$ .

(ii) By Theorem 3.41 we know that if  $u = \chi_E$  for set  $E$  of finite perimeter in  $\Omega$ , that  $\|Du\| = \mathcal{H}^{N-1} \llcorner \partial^* E$  and if  $\mathcal{H}^{N-1}(B) = 0$  it follows that  $\|Du\|(B) = \mathcal{H}^{N-1}(B \cap \partial^* E) = 0$ . Consider now a general function  $u \in BV(\Omega)$ , then if  $\mathcal{H}^{N-1}(B) = 0$  we have, by the coarea formula, that:

$$\|Du\|(B) = \int_{-\infty}^{\infty} \|D\chi_{\{u > t\}}\|(B) dt = 0. \quad (3.169)$$

(iii) For  $u \in L^1(\Omega, \mathcal{L}^N)$ , we have at any point  $x \in \Omega$  such that an approximate limit  $\tilde{u}(x)$  exists, that  $x \in \partial^e \{u > t\}$  for at most one  $t \in \mathbb{R}$ , namely  $t = \tilde{u}(x)$ . Indeed, if  $t > \tilde{u}(x)$ :

$$\mathcal{L}^N(\{u > t\} \cap B(x, r)) \leq \frac{1}{t - \tilde{u}(x)} \int_{B(x, r)} |u(y) - \tilde{u}(x)| dy \xrightarrow{r \rightarrow 0^+} 0. \quad (3.170)$$

Then  $\Theta_N(\{u > t\}, x) = 0$  and so  $x \notin \partial^e \{u > t\}$ . Similarly, if  $t < \tilde{u}(x)$  we have  $\Theta_N(\{u > t\}, x) = 1$  and so  $x \notin \partial^e \{u > t\}$ . Suppose now that  $\mathcal{H}^{N-1}(B) < \infty$  and  $B \cap S_u = \emptyset$ . Then by Fubini we find that

$$\begin{aligned} \|Du\|(B) &= \int_{-\infty}^{\infty} \mathcal{H}^{N-1} \llcorner B(\partial^e \{u > t\}) dt \\ &= \int_{-\infty}^{\infty} \int_{B \cap \{u > t\}} d\mathcal{H}^{N-1} dt \\ &= \int_B \mathcal{L}^1(\{t \in \mathbb{R} \mid x \in \partial^e \{u > t\}\}) d\mathcal{H}^{N-1}(x) \\ &= 0. \end{aligned} \tag{3.171}$$

□

**Definition 3.60** Let  $\Gamma \subset \mathbb{R}^N$  be a countably  $\mathcal{H}^{N-1}$ -rectifiable set. We say that  $\Gamma$  is **oriented by a Borel measurable function**  $\nu : \Gamma \rightarrow S^{N-1}$  if for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Gamma$  we have  $\text{Tan}^{N-1}(\Gamma, x) = \nu(x)^\perp$ .

**Example 3.61** We note that by Theorem 3.41 we see that  $\partial^* E$  is oriented by the generalized inner unit normal  $\nu_E$ .

Now we can state and prove our first trace theorem.

**Theorem 3.62** Let  $u \in BV(\Omega)$  and  $\Gamma \subset \Omega$  be a countably  $\mathcal{H}^{N-1}$ -rectifiable set oriented by  $\nu$ . Then, for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Omega$  there exist  $u_\Gamma^\pm(x) \in \mathbb{R}$  such that

$$\lim_{r \rightarrow 0^+} \int_{B^+(x, r, \nu)} |u(y) - u_\Gamma^+(x)| dy = 0 \tag{3.172}$$

$$\lim_{r \rightarrow 0^+} \int_{B^-(x, r, \nu)} |u(y) - u_\Gamma^-(x)| dy = 0. \tag{3.173}$$

Furthermore, we have the following representation of  $Du \llcorner \Gamma$ :

$$Du \llcorner \Gamma = (u_\Gamma^+ - u_\Gamma^-) \otimes \nu \mathcal{H}^{N-1} \llcorner \Gamma. \tag{3.174}$$

**Proof** We prove this result in several steps.

**Step 1:** First, we show that 3.172 and 3.173 hold for characteristic functions  $u = \chi_E$  of sets of finite perimeter  $E \subset \Omega$ . By Theorem 3.45 (ii) we know that  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Omega$  belongs to  $E^0 \cup \partial^* E \cup E^1$ . If  $x \in E^0$ , then we can choose  $u_\Gamma^\pm(x) = 0$ ; and if  $x \in E^1$ , then we can choose  $u_\Gamma^\pm(x) = 1$ . Both options satisfy 3.172 and 3.173. Now suppose  $x \in \partial^* E$ . By Theorem 3.41 (i), the rescaled sets  $E_r = \frac{E-x}{r}$  converge locally in  $\mathbb{R}^N$  to the half-space  $H$  orthogonal to  $\nu_E(x)$  and containing  $\nu_E(x)$ . Hence, we have for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial^* E \cap \Gamma$  we have  $\nu_E(x) = \pm \nu(x)$ . If  $\nu_E(x) = \nu(x)$ , we can choose  $u_\Gamma^+(x) = 1$  and  $u_\Gamma^-(x) = 0$  to satisfy 3.172 and 3.173. If  $\nu_E(x) = -\nu(x)$ , we can choose  $u_\Gamma^+(x) = 0$  and  $u_\Gamma^-(x) = 1$  to satisfy 3.172 and 3.173.

**Step 2:** By linearity  $u_\Gamma^\pm$  there still exist  $\mathcal{H}^{N-1}$ -a.e. on  $\Gamma$  for a simple function  $u \in BV(\Omega)$ , that is, if  $u = \sum_i z_i \chi_{E_i}$  is such that  $z_i \neq 0$  for only finitely many  $i$  and  $E_i$  are sets of finite perimeter in  $\Omega$ .

**Step 3:** Now if  $u \in BV(\Omega)$  is bounded, then there exists a sequence  $(u_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  of simple functions uniformly converging to  $u$ <sup>6</sup>. Then we can define  $\mathcal{H}^{N-1}$ -a.e. on  $\Gamma$   $u_\Gamma^\pm(x) = \lim_{h \rightarrow \infty} (u_h)_\Gamma^\pm(x)$ .

**Step 4:** Consider now a general function  $u \in BV(\Omega)$ . Then we can consider the bounded functions  $u_h = \min\{h, \max\{u, -h\}\} \in BV(\Omega)$ . Consider now the  $\mathcal{H}^{N-1}$ -null set  $L$  from Lemma 3.58 and  $M_h \subset \Gamma$  the  $\mathcal{H}^{N-1}$ -null set, where  $(u_h)_\Gamma^\pm$  are not defined. We claim that  $u_\Gamma^\pm(x)$  exists for all  $x \in \Gamma \setminus (L \cup \bigcup_{h \in \mathbb{N}} M_h)$ . Indeed, the sequences  $((u_h)_\Gamma^\pm)_{h \in \mathbb{N}}$  are bounded since  $x \notin L$  and  $|u_h| \leq |u|$ . Then, by Bolzano-Weierstrass, there are subsequences  $((u_{h_k})_\Gamma^\pm)_{k \in \mathbb{N}}$  converging to real numbers  $z^\pm \in \mathbb{R}$ . Now we can estimate:

$$\begin{aligned} \int_{B^\pm(x, r, \nu(x))} |u(y) - z^\pm| \, dy &= \int_{B^\pm(x, r, \nu(x))} |u(y) - u_{h_k}(y)| \, dy \\ &\quad + \int_{B^\pm(x, r, \nu(x))} |u_{h_k}(y) - (u_{h_k})_\Gamma^\pm(x)| \, dy \\ &\quad + \frac{\omega_N r^N}{2} |(u_{h_k})_\Gamma^\pm(x) - z^\pm|. \end{aligned} \quad (3.175)$$

So by division by  $\omega_N r^N$  and passing to the limit as  $r \rightarrow 0^+$  and  $k \rightarrow \infty$  we find that  $u_\Gamma^\pm(x) = z^\pm$  satisfies 3.172 and 3.173.

**Step 5:** Now we show the representation 3.174 of  $Du \llcorner \Gamma$ . Without loss of generality, we may assume that  $\mathcal{H}^{N-1}(\Gamma) < \infty$  and  $u_\Gamma^\pm(x)$  exist for all  $x \in \Gamma$ . Denote now  $D_1 u = Du \llcorner \Gamma$  and  $D_2 u Du - D_1 u$ . By Lemma 3.59 (i), we know that  $D_1 u \ll \mathcal{H}^{N-1} \llcorner \Gamma$ . So by Theorem 2.43 we only need to show that

$$\lim_{r \rightarrow 0^+} \frac{D_1 u(B(x, r))}{\mathcal{H}^{N-1}(\Gamma \cap B(x, r))} = (u_\Gamma^+(x) - u_\Gamma^-(x)) \nu(x). \quad (3.176)$$

We know that  $\|D_2 u\|(B(x, r)) = o(r^{N-1})$  and  $\lim_{r \rightarrow 0^+} \frac{\mathcal{H}^{N-1}(\Gamma \cap B(x, r))}{\omega_{N-1} r^{N-1}} = 1$  for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Gamma$ . So we will only need to show that there is a sequence  $r_i > 0$  such that  $r_i \rightarrow 0$  as  $i \rightarrow \infty$  and

$$\lim_{i \rightarrow \infty} \frac{D_1 u(B(x, r_i))}{\mathcal{H}^{N-1}(\Gamma \cap B(x, r_i))} = (u_\Gamma^+(x) - u_\Gamma^-(x)) \nu(x), \quad (3.177)$$

<sup>6</sup>We can consider  $u_h(y) = \frac{i-1}{2^h}$  if  $\frac{i-1}{2^h} \leq u(x) < \frac{i}{2^h}$ .

for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Omega$ .

Let now  $x \in \Gamma$  be such that  $\frac{\|Du\|_{(B(x,r))}}{r^{N-1}}$  is bounded as  $r \rightarrow 0^+$ . We will show that the set of such  $x$ , where 3.177 is an  $\mathcal{H}^{N-1}$ -null set. By definition of  $u_{\Gamma}^{\pm}(x)$  we have that  $u^{x,r}(y) = u(x + ry)$  converges in  $L_{\text{loc}}^1(\mathbb{R}^N, \mathcal{L}^N)$  to

$$w_x(y) = \begin{cases} u_{\Gamma}^{+}(x), & y \cdot \nu(x) > 0 \\ u_{\Gamma}^{-}(x), & y \cdot \nu(x) < 0. \end{cases} \quad (3.178)$$

By the boundedness of  $\|Du^{x,r}\| = \frac{\|Du\|_{(B(x,r))}}{r^{N-1}}$  and by Theorem 3.19 we find that  $Du^{x,r}$  weak\* converges in  $B(0,1)$  to  $Dw_x$  as  $r \rightarrow 0^+$ . Let now  $s_i > 0$  be a sequence such that  $s_i \rightarrow 0$  and  $\|Du^{x,s_i}\|$  weak\* converges in  $B(0,1)$  to some measure  $\mu$  in  $B(0,1)$  as  $i \rightarrow \infty$ . Let  $t \in (0,1)$  be such that  $\mu(\partial B(0,t)) = 0$  and  $t_i = ts_i$ . Then it holds that

$$\begin{aligned} \lim_{i \rightarrow \infty} \frac{Du(B(x, t_i))}{\omega_{N-1} t_i^{N-1}} &= \lim_{i \rightarrow \infty} \frac{Du^{x,s_i}(B(0, t))}{\omega_{N-1} t^{N-1}} \\ &= \frac{Dw_x(B(0, t))}{\omega_{N-1} t^{N-1}}. \end{aligned} \quad (3.179)$$

Since  $Dw_x = (u_{\Gamma}^{+} - u_{\Gamma}^{-})\nu(x)\mathcal{H}^{N-1} \llcorner \nu(x)^{\perp}$ , we have shown 3.177 for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \Gamma$ .

**Definition 3.63** We will call for  $u \in BV(\Omega)$  the (by Theorem 3.62  $\mathcal{H}^{N-1}$ -a.e. on  $\Gamma$  defined) functions  $u_{\Gamma}^{\pm} : \Gamma \rightarrow \mathbb{R}$  the *interior* resp. *exterior trace* of  $u$ .

### 3.4.1 Boundary trace theorem

Now we are in a position to tackle the *boundary trace theorem*. This will give us the tools to state an integration by parts formula for  $u \in BV(\Omega)$  and  $\Omega$  with Lipschitz boundary of the form

$$\int_{\Omega} u \operatorname{div}(\varphi) \, dx = - \int_{\Omega} \varphi \cdot dDu - \int_{\partial\Omega} u^{\Omega} \nu_{\Omega} \cdot \varphi \, d\mathcal{H}^{N-1}, \quad (3.180)$$

where  $\varphi \in C_c^1(\Omega)$  and  $u^{\Omega}$  is the so-called *trace* on  $\partial\Omega$ , which we will construct in the following.

First, we consider the following summability condition for *glueing* two functions  $u, v \in BV(\Omega)$  at an interface given by a set  $E$  of finite perimeter in  $\Omega$  to obtain another function  $w = u\chi_E + v\chi_{\Omega \setminus E} \in BV(\Omega)$ .

**Theorem 3.64** Let  $u, v \in BV(\Omega)$  and  $E$  be a set of finite perimeter in  $\Omega$ . Assume that  $\partial^* E \cap \Omega$  is oriented by  $\nu_E$  and let  $u_{\partial^* E}^{+}$  and  $v_{\partial^* E}^{-}$  be the interior trace of  $u$  and the interior trace of  $v$ . Then, the following holds:

$$(i) \quad w := u\chi_E + v\chi_{\Omega \setminus E} \in BV(\Omega) \iff \int_{\partial^* E \cap \Omega} |u_{\partial^* E}^{+} - v_{\partial^* E}^{-}| \, d\mathcal{H}^{N-1} < \infty.$$

(ii) If  $w \in BV(\Omega)$ , then we have the following representation of  $Dw$ :

$$Dw = Du \llcorner E^1 + (u_{\partial^* E}^+ - v_{\partial^* E}^-) \otimes \nu_E \mathcal{H}^{N-1} \llcorner (\partial^* E \cap \Omega) + Dv \llcorner E^0. \quad (3.181)$$

**Proof** To make the notation a bit simpler, we may assume, up to replacing  $E$  by  $E \cap \Omega$ , that  $\partial^* E \subset \Omega$ .

(i) Assume first that  $w \in BV(\Omega)$ . Then we know by Theorem 3.62 for  $\Gamma = \partial^* E$  that

$$Dw \llcorner \partial^* E = (w_{\partial^* E}^+ - w_{\partial^* E}^-) \otimes \nu_E \mathcal{H}^{N-1} \llcorner \partial^* E, \quad (3.182)$$

where  $w_{\partial^* E}^\pm$  satisfy 3.172 and 3.173. Now, let  $x \in \partial^* E$ . Then we have the following estimate as  $r \rightarrow 0^+$ :

$$\begin{aligned} & \int_{B^+(x, r, \nu_E(x))} |u(y) - w_{\partial^* E}^+(x)| \, dy \leq \int_{E \cap B^+(x, r, \nu_E(x))} |w(y) - w_{\partial^* E}^+(x)| \, dy \\ & + \int_{B^+(x, r, \nu_E(x)) \setminus E} |u(y) - w_{\partial^* E}^+(x)| \, dy = o(r^N) + r^N \int_{F_{x,r}} |u^{x,r}(y) - w_{\partial^* E}^+(x)| \, dy, \end{aligned} \quad (3.183)$$

where  $u^{x,r}(y) = u(x + ry)$  and  $F_{x,r} = B^+(0, 1, \nu_E(x)) \setminus (\frac{E-x}{r})$ . By Theorem 3.41  $\frac{E-x}{r}$  converges to the half-space orthogonal to  $\nu_E(x)$  and containing  $\nu_E(x)$  as  $r \rightarrow 0^+$ , and so  $F_{x,r}$  converges to  $\emptyset$  as  $r \rightarrow 0^+$ . Set now  $g^{x,r} = |u^{x,r} - w_{\partial^* E}^+(x)|$ . We note that  $g^{x,r}$  converges in  $L^1(B(0, 1), \mathcal{L}^N)$  as  $r \rightarrow 0^+$ , so by Proposition 2.11 (iv)  $(g^{x,r})_{r \geq 0}$  is equiintegrable. By the Vitali theorem (Proposition 2.12 (iv)) we find then that  $(g^{x,r} \chi_{F_{x,r}})_{r \geq 0}$  converges to 0 in  $L^1(B(0, 1), \mathcal{L}^N)$ , and taking  $r \rightarrow 0^+$  in 3.183 we can conclude by the uniqueness of  $u_{\partial^* E}^+$  that  $u_{\partial^* E}^+(x) = w_{\partial^* E}^+(x)$  for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial^* E$ . By the same argument, we find that  $v_{\partial^* E}^-(x) = w_{\partial^* E}^-(x)$  for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial^* E$ . This then implies that

$$\begin{aligned} \int_{\partial^* E} |u_{\partial^* E}^+(x) - v_{\partial^* E}^-(x)| \, d\mathcal{H}^{N-1}(x) &= \int_{\partial^* E} |w_{\partial^* E}^+(x) - w_{\partial^* E}^-(x)| \, d\mathcal{H}^{N-1}(x) \\ &= \|Dw\|(\partial^* E) < \infty. \end{aligned} \quad (3.184)$$

Conversely, assume that  $\int_{\partial^* E} |u_{\partial^* E}^+(x) - v_{\partial^* E}^-(x)| \, d\mathcal{H}^{N-1}(x) < \infty$ . We consider the sequences  $(u_h)_{h \in \mathbb{N}}, (v_h)_{h \in \mathbb{N}} \subset BV(\Omega)$  given by

$$u_h = \min\{h, \max\{u, -h\}\} \text{ and } v_h = \min\{h, \max\{v, -h\}\}. \quad (3.185)$$

We note that  $u_h, v_h$  converges to  $u, v$  in  $BV(\Omega)$ . Now note that for  $w_h = u_h \chi_E + v_h \chi_{\Omega \setminus E}$  it holds since  $BV(\Omega)$  is a Banach algebra

$$\|Dw_h\|(\Omega) \leq \|Du_h\|(\Omega)P(E, \Omega) + \|Dv_h\|(\Omega)P(\Omega \setminus E, \Omega) < \infty. \quad (3.186)$$

Hence,  $w_h \in BV(\Omega)$ . We also note that  $(u_h)_{\partial^* E}^+ = \min\{h, \max\{u_{\partial^* E}^+, -h\}\}$  and  $(v_h)_{\partial^* E}^- = \min\{h, \max\{v_{\partial^* E}^-, -h\}\}$  and therefore we have the uniform bound by (ii) (proved below):

$$\begin{aligned} \|Dw_h\|(\Omega) &\leq \|Du_h\|(E^1) + \int_{\partial^* E} |(u_h)_{\partial^* E}^+ - (v_h)_{\partial^* E}^-| d\mathcal{H}^{N-1} + \|Dv_h\|(E^0) \\ &\leq \|Du\|(E^1) + \int_{\partial^* E} |u_{\partial^* E}^+ - v_{\partial^* E}^-| d\mathcal{H}^{N-1} + \|Dv\|(E^0) < \infty. \end{aligned} \quad (3.187)$$

Since  $w_h \rightarrow w$  in  $L^1(\Omega, \mathcal{L}^N)$ , by the lower semicontinuity of variation and the uniform bound 3.187 that  $w \in BV(\Omega)$ .

- (ii) Now we prove (ii). Since  $E^0 \cup \partial^* E \cup E^1$  covers  $\mathcal{H}^{N-1}$ -almost all of  $\Omega$  and since  $Dw$  vanishes on a  $\mathcal{H}^{N-1}$ -null set, we only need to prove that  $Dw \llcorner E^1 = Du \llcorner E^1$  and  $Dw \llcorner E^0 = Dv \llcorner E^0$ , since by (i) we have already shown that if  $w \in BV(\Omega)$  we have

$$Dw \llcorner \partial^* E = (u_{\partial^* E}^+ - v_{\partial^* E}^-) \otimes \nu_E \mathcal{H}^{N-1} \llcorner \partial^* E. \quad (3.188)$$

To this end, we will show that  $Df \llcorner E^1 = 0$  and  $Df \llcorner E^0 = 0$  whenever  $f = 0$   $\mathcal{L}^N$ -a.e. in  $E$  resp. in  $\Omega \setminus E$ . Then, we get the result for  $f = w - u$  resp.  $f = w - v$ . We note that at any  $x \in E^1$  the function  $f^{x,r}(y) = f(x + ry)$  converges in measure in  $B(0, 1)$  to 0. Hence, we have  $x \in \partial^* \{f > 0\}$  and so by the coarea formula for  $f$  and the Borel set  $E^1$  we obtain  $\|Df\|(E^1) = 0$ . Applying the same argument to  $\Omega \setminus E$  and noting that  $(\Omega \setminus E)^1 = E^0$ , yields the other result. Indeed, we have  $(\Omega \setminus E)^1 = E^0$  as

$$\begin{aligned} x \in (\Omega \setminus E)^1 &\iff \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(B(x, r) \setminus E)}{\omega_N r^N} = 1 \\ &\iff 1 - \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} = 1 \\ &\iff \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} = 0 \\ &\iff x \in E^0. \end{aligned} \quad (3.189)$$

**Remark 3.65** *It should be noted that the trace  $u_{\partial^* E}^\pm$  can also be characterized by convergence of  $\frac{E-x}{r}$  to the half plane orthogonal to  $v_E(x)$  containing  $v_E(x)$  as  $r \rightarrow 0^+$  by the formulas:*

$$\lim_{r \rightarrow 0^+} \int_{E \cap B(x, r)} |u(y) - u_{\partial^* E}^+(x)| \, dy = 0 \quad (3.190)$$

$$\lim_{r \rightarrow 0^+} \int_{B(x, r) \setminus E} |u(y) - u_{\partial^* E}^-(x)| \, dy = 0. \quad (3.191)$$

We now show that under certain conditions that  $u_{\partial^* E}^\pm$  are summable.

**Theorem 3.66** *Let  $u \in BV(\Omega)$  and  $\Gamma \subset\subset \Omega$  a Borel set such that*

$$\mathcal{H}^{N-1}(\Gamma \cap B(x, r)) \leq \Lambda r^{N-1}, \quad \forall x \in \Gamma, r \in (0, r_0), \quad (3.192)$$

*for some constant  $r_0 \in (0, d(\Gamma, \partial\Omega))$  and  $\Lambda \geq 0$ . Then it holds for  $u^\pm$  and the approximate limit  $\tilde{u}$  of  $u$  satisfy*

$$\int_{\Gamma \setminus S_u} |\tilde{u}| \, d\mathcal{H}^{N-1} + \int_{\Gamma \cap J_u} |u^+| + |u^-| \, d\mathcal{H}^{N-1} \leq C \|u\|_{BV(\Omega)}, \quad (3.193)$$

*for some constant  $C > 0$  depending only on  $\mathcal{H}^{N-1}(\Gamma)$ ,  $r_0$  and  $\Lambda$ .*

**Proof** Up to replacing  $u$  by  $|u|$ , we may assume that  $u \geq 0$ . Consider a set  $E$  of finite perimeter in  $\Omega$  such that  $3\mathcal{L}^N(E) < \omega_N r_0^N$ , e.g. any open ball of radius  $r < r_0$ . Consider now  $E^{\frac{1}{2}} = \{\Theta_N^*(x, E) \geq \frac{1}{2}\}$ . Let  $\gamma_1$  be the constant of the inequality 3.68. Now for any  $x \in E^{\frac{1}{2}} \cap \Gamma$  we see that for any  $0 < r < r_0$  we have

$$\frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N} \geq \frac{1}{3}. \quad (3.194)$$

On the other hand, we must have

$$\frac{\mathcal{L}^N(E) \cap B(x, r_0)}{\omega_N r_0^N} \leq \frac{\mathcal{L}^N(E)}{\omega_N r_0^N} < \frac{1}{3}. \quad (3.195)$$

Now consider the continuous function  $f : [0, r_0] \rightarrow [0, \infty)$  given by  $f(r) = \frac{\mathcal{L}^N(E \cap B(x, r))}{\omega_N r^N}$  with the convention  $f(0) = 0$ . By the intermediate value theorem since  $f(r_0) < \frac{1}{3}$  and  $f(r) \geq \frac{1}{3}$  for all  $r \in (0, r_0)$ , we conclude that there is some  $r \in (0, r_0)$  such that  $f(r) = \frac{1}{3}$ . Hence, by 3.68

$$\omega_N r^{N-1} \leq 3\gamma_1 P(E, B(x, r)). \quad (3.196)$$

By the Besicovitch covering theorem we can cover  $E^{\frac{1}{2}} \cap \Gamma$  by countably many such balls  $\{B(x_i, r_i)\}_{i \in I}$  such that at most  $\zeta$  of them intersect. By this cover,

we get the estimate:

$$\begin{aligned}
 \mathcal{H}^{N-1}(E^{\frac{1}{2}} \cap \Gamma) &\leq \sum_{i \in I} \mathcal{H}^{N-1}(\Gamma \cap B(x_i, r_i)) \\
 &\stackrel{3.192}{\leq} \Lambda \sum_{i \in I} r_i^{N-1} \\
 &\stackrel{3.196}{\leq} \frac{3\gamma_1 \Lambda}{\omega_N} \sum_{i \in I} P(E, B(x_i, r_i)) \\
 &= \frac{3\gamma_1 \xi \Lambda}{\omega_N} P(E, \Omega).
 \end{aligned} \tag{3.197}$$

For every  $x \in J_u$  we may define

$$w_x(y) := \begin{cases} u^+(x), & y \cdot \nu_u(x) > 0 \\ u^-(x), & y \cdot \nu_u(x) < 0. \end{cases} \tag{3.198}$$

We know that the functions  $u^{x,r}(y) = u(x + ry)$  converge in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  to  $w_x$  as  $r \rightarrow 0^+$  for any  $x \in J_u$ . This implies that for every  $t > 0$ :

$$\{x \in J_u \mid u^+(x) + u^-(x) > t\} \subset \left\{u > \frac{1}{2}\right\}^{\frac{1}{2}}. \tag{3.199}$$

If we assume that  $t > t_0 := \frac{6\|u\|_1}{\omega_N r_0^N}$  we find by Chebyshev inequality that  $E_t := \{u > \frac{t}{2}\}$  satisfies  $3\mathcal{L}^N(E_t) < \omega_N r_0^N$ . So by the estimate 3.197 and the coarea formula we see that

$$\begin{aligned}
 \int_{\Gamma \cap J_u} u^+ + u^- d\mathcal{H}^{N-1} &= \int_0^\infty \mathcal{H}^{N-1}(\{x \in \Gamma \cap J_u \mid u^+(x) + u^-(x) > t\}) dt \\
 &\leq \int_0^\infty \mathcal{H}^{N-1}((E_t)^{\frac{1}{2}} \cap \Gamma) dt \\
 &\leq t_0 \mathcal{H}^{N-1}(\Gamma) + \frac{3\gamma_1 \xi \Lambda}{\omega_N} \int_{t_0}^\infty P(E_t, \Omega) dt \\
 &\leq \left( \frac{6\mathcal{H}^{N-1}(\Gamma)}{\omega_N r_0^N} + \frac{3\gamma_1 \xi \Lambda}{\omega_N} \right) \|u\|_{BV(\Omega)}.
 \end{aligned} \tag{3.200}$$

Noting that the functions  $u^{x,r}(y) = u(x + ry)$  converge in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$  to  $\tilde{u}(x)$  for every  $x \in \Omega \setminus S_u$  and that

$$\{x \in \Omega \setminus S_u \mid \tilde{u}(x) > t\} \subset \left\{u > \frac{t}{2}\right\}^{\frac{1}{2}}, \tag{3.201}$$

again by the estimate 3.197 and the coarea formula we see that

$$\begin{aligned} \int_{\Gamma \setminus S_u} \tilde{u}(x) d\mathcal{H}^{N-1} &= \int_0^\infty \mathcal{H}^{N-1}(\{x \in \Omega \setminus S_u \mid \tilde{u}(x) > t\}) dt \\ &\leq \left( \frac{6\mathcal{H}^{N-1}(\Gamma)}{\omega_N r_0^N} + \frac{3\gamma_1 \xi \Lambda}{\omega_N} \right) \|u\|_{BV(\Omega)}. \end{aligned} \quad (3.202)$$

So for  $C = 2 \left( \frac{6\mathcal{H}^{N-1}(\Gamma)}{\omega_N r_0^N} + \frac{3\gamma_1 \xi \Lambda}{\omega_N} \right)$  we conclude 3.193.  $\square$

Using Theorem 3.64 and Theorem 3.66 we are now in a place where we can state and prove the *boundary trace theorem* for open sets with bounded Lipschitz boundary.

**Theorem 3.67** *Let  $\Omega \subset \mathbb{R}^N$  be an open set with with bounded Lipschitz boundary and  $u \in BV(\Omega)$ . Then, for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial\Omega$  there exists  $u^\Omega(x) \in \mathbb{R}$  such that*

$$\lim_{r \rightarrow 0^+} r^{-N} \int_{\Omega \cap B(x,r)} |u(y) - u^\Omega(x)| dy = 0. \quad (3.203)$$

Furthermore, we have the following

- (i)  $\|u^\Omega\|_{L^1(\partial\Omega, \mathcal{L}^N)} \leq C \|u\|_{BV(\Omega)}$  for some constant  $C > 0$  only depending on  $\Omega$ .
- (ii) The extension  $u_0$  of  $u$  to 0 outside of  $\Omega$  is in  $BV(\mathbb{R}^N)$  and if we see  $Du$  as a measure on  $\mathbb{R}^N$  concentrated on  $\Omega$ , we obtain the following formula for  $Du_0$ :

$$Du_0 = Du + (u^\Omega \otimes \nu_\Omega) \mathcal{H}^{N-1} \llcorner \partial\Omega. \quad (3.204)$$

**Proof** We find by Propositions 3.18 and 3.47 that  $\Omega$  is an extension domain and a set of finite perimeter satisfying  $\mathcal{H}^{N-1}(\partial\Omega \setminus \partial^*\Omega) = 0$ , as  $\|D\chi_\Omega\| = \mathcal{H}^{N-1} \llcorner \partial\Omega$ . Since  $\Omega$  has Lipschitz boundary, we can choose  $r_0 > 0$  such that for every  $x \in \partial\Omega$  we see that  $\partial\Omega \cap B(x, r_0)$  is the graph of some Lipschitz function  $\varphi_x$ . Then, by the compactness of  $\partial\Omega$  we can choose  $L = \sup_{x \in \partial\Omega} \text{Lip}(\varphi_x)$  and set  $\Lambda = \Omega_{N-1} \sqrt{1 + L^2}$ . Then we see that for these choices of  $r_0$  and  $\Lambda$  the inequality 3.192 is satisfied. Moreover, the constant  $r_0$  and  $\Lambda$ . Now, let  $w \in BV(\mathbb{R}^N)$  be an extension of  $u$  satisfying  $\|w\|_{BV(\mathbb{R}^N)} \leq C_1 \|u\|_{BV(\Omega)}$ , where  $C_1 > 0$  only depends on  $\Omega$ <sup>7</sup>. By Remark 3.65 we have that that  $w_{\partial^*\Omega}^+$  is defined by

$$\begin{aligned} \lim_{r \rightarrow 0^+} \int_{\Omega \cap B(x,r)} |u(y) - w_{\partial^*\Omega}^+(x)| dy &= \lim_{r \rightarrow 0^+} \int_{\Omega \cap B(x,r)} |w(y) - w_{\partial^*\Omega}^+(x)| dy \\ &= 0, \end{aligned} \quad (3.205)$$

<sup>7</sup>Such a extension exists, since  $\Omega$  is an extension domain.

whenever  $x \in \partial^* E$  and  $w_{\partial^* \Omega}^\pm$  are both defined. Now  $u^\Omega = w_{\partial^* \Omega}^+$  satisfies 3.203. By Theorem 3.66 we find a constant  $C_2$  only depending on  $r_0$  and  $\Lambda$  such that;

$$\begin{aligned} \int_{\partial \Omega} |u^\Omega| d\mathcal{H}^{N-1} &= \int_{\partial^* \Omega} |w_{\partial^* \Omega}^+| d\mathcal{H}^{N-1} \\ &= C_2 \|w\|_{BV(\mathbb{R}^N)} \\ &\leq C_1 C_2 \|u\|_{BV(\Omega)}. \end{aligned} \quad (3.206)$$

This proves (i). Now for (ii) we can apply Theorem 3.64 in  $\mathbb{R}^N$  to  $v = 0$  and  $E = \Omega$  to obtain that the function  $u_0 = w\chi_\Omega$  belongs to  $BV(\mathbb{R}^N)$  and its distributional derivative is given by, noting that  $\Omega^1 = \Omega$  and  $w|_\Omega = u$ :

$$\begin{aligned} Du_0 &= Dw \llcorner \Omega^1 + w_{\partial^* \Omega}^+ \otimes \nu_\Omega \mathcal{H}^{N-1} \llcorner \partial^* \Omega \\ &= Du + u^\Omega \otimes \nu_\Omega \mathcal{H}^{N-1} \llcorner \partial \Omega. \end{aligned} \quad (3.207)$$

□

The trace  $u^\Omega$  from Theorem 3.67 actually defines a bounded linear operator.

**Theorem 3.68** *Let  $\Omega \subset \mathbb{R}^N$  be an open set with bounded Lipschitz boundary. The operator  $u \mapsto u^\Omega$  assigning to each  $u \in BV(\Omega)$  its **trace on  $\partial \Omega$**  defines a bounded linear operator from  $BV(\Omega)$  to  $L^1(\partial \Omega, \mathcal{H}^{N-1} \llcorner \partial \Omega)$ , where  $BV(\Omega)$  is endowed with topology induced by the convergence of  $u_h$  to  $u$  in  $BV(\Omega)$  if  $u_h$  converges to  $u$  in  $L^1(\Omega, \mathcal{L}^N)$  and  $\|Du_h\|$  converges to  $\|Du\|$ .*

**Proof** The proof of this result is given for Theorem 3.88 in [3]. There, it is argued locally, where  $\partial \Omega$  is a graph of a Lipschitz function. □

We also have the following glueing result for the traces.

**Corollary 3.69** *Let  $\Omega \subset \mathbb{R}^N$  be an open set with bounded Lipschitz boundary,  $u \in BV(\Omega)$  and  $v \in BV(\mathbb{R}^N \setminus \overline{\Omega})$ . Then the function  $w$  on  $\mathbb{R}^N$  defined by*

$$w(x) := \begin{cases} u(x), & x \in \Omega \\ v(x), & x \in \mathbb{R}^N \setminus \overline{\Omega}, \end{cases} \quad (3.208)$$

*is an element of  $BV(\mathbb{R}^N)$ , and by viewing  $Du$  resp.  $Dv$  as measures on  $\mathbb{R}^N$  concentrated on  $\Omega$  resp.  $\mathbb{R}^N \setminus \overline{\Omega}$  we have the following formula for  $Dw$ :*

$$Dw = Du + (u^\Omega - v^{\mathbb{R}^N \setminus \overline{\Omega}}) \otimes \nu_\Omega \mathcal{H}^{N-1} \llcorner \partial \Omega + Dv. \quad (3.209)$$

**Proof** Noting that  $\nu_{\mathbb{R}^N \setminus \overline{\Omega}} = -\nu_\Omega$ , and that  $u, v$  are integrable by Theorem 3.67 on  $\partial \Omega$  and extended by 0 can be viewed as functions  $u_0, v_0 \in BV(\mathbb{R}^N)$ .

Hence we may write  $w = u_0 + v_0$ , which clearly is in  $BV(\mathbb{R}^N)$  and satisfies by 3.204:

$$\begin{aligned} Dw &= D(u_0 + v_0) \\ &= Du + u^\Omega \otimes \nu_\Omega \mathcal{H}^{N-1} \llcorner \partial\Omega + Dv + v^{\mathbb{R}^N \setminus \bar{\Omega}} \otimes \nu_{\mathbb{R}^N \setminus \bar{\Omega}} \mathcal{H}^{N-1} \llcorner \partial\Omega \quad (3.210) \\ &= Du + (u^\Omega - v^{\mathbb{R}^N \setminus \bar{\Omega}}) \otimes \nu_\Omega \mathcal{H}^{N-1} \llcorner \partial\Omega + Dv. \end{aligned}$$

□

We now give the main application of Theorem 3.67, which is of importance to us.

**Example 3.70** Let  $u \in BV(\Omega)$  and  $x \in \Omega$ . By slight abuse of notation, we may consider  $u$  as a function in  $BV(\mathbb{R}^N)$  by extension by 0 outside of  $\Omega$ . For every  $r \in (0, d(x, \partial\Omega))$ , consider the function  $v_r := u\chi_{B(x,r)} \in BV(B(x,r))$ . By the formula 3.204 we have for every  $r \in (0, d(x, \partial\Omega))$ :

$$Dv_r = Du \llcorner B(x,r) - u^{B(x,r)} \otimes \nu_{B(x,r)} \mathcal{H}^{N-1} \llcorner \partial B(x,r). \quad (3.211)$$

By Theorem 3.204 for  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial B(x,r)$  we have

$$\lim_{s \rightarrow 0^+} s^{-N} \int_{B(x,r) \cap B(y,s)} |u(z) - u^{B(x,r)}(y)| dz = 0. \quad (3.212)$$

Moreover,  $\mathcal{L}^N$ -a.e.  $y \in \Omega$  is a Lebesgue point for  $u$ , i.e.

$$\lim_{s \rightarrow 0^+} s^{-N} \int_{B(y,s)} |u(z) - u(y)| dz = 0. \quad (3.213)$$

By the coarea formula for any real-valued Lipschitz function  $f$  it holds for  $\mathcal{L}^1$ -a.e.  $t \in (0, d(x, \partial\Omega))$  that:

$$\|\nabla f\| \mathcal{L}^N = \mathcal{H}^{N-1} \llcorner \{f = t\} \otimes \mathcal{L}^1. \quad (3.214)$$

Hence, for  $f = |\cdot - x|$  we find that for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  we have the following:

$$\mathcal{L}^N = \mathcal{H}^{N-1} \llcorner \partial B(x,r) \otimes \mathcal{L}^1. \quad (3.215)$$

From 3.213 and 3.215 we find for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  and  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial B(x,r)$  we find:

$$\lim_{s \rightarrow 0^+} s^{-N} \int_{B(x,r) \cap B(y,s)} |u(z) - u(y)| dz = 0. \quad (3.216)$$

Since 3.212 characterizes the trace  $u^{B(x,r)}$  uniquely, we find by 3.216 that  $\mathcal{H}^{N-1}$ -a.e. on  $\partial B(x,r)$ :

$$u^{B(x,r)} = u|_{\partial B(x,r)}, \quad \text{for } \mathcal{L}^1\text{-a.e. } r \in (0, d(x, \partial\Omega)). \quad (3.217)$$

Moreover, by 3.211 for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  we then have:

$$Dv_r = Du \llcorner B(x, r) - u \otimes \nu_{B(x, r)} \mathcal{H}^{N-1} \llcorner \partial B(x, r). \quad (3.218)$$

Consider now  $E \subset \mathbb{R}^N$  to be a set of finite perimeter in  $\Omega$ . Then in particular, we can apply the above to  $u = \chi_E$ , which implies the following:

(i) By 3.217, for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  the **trace of  $E$  on  $B(x, r)$**  is given by

$$\chi_E^{B(x, r)} = \chi_{E \cap \partial B(x, r)}. \quad (3.219)$$

(ii) By 3.218, for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  it holds that:

$$D\chi_{E \cap B(x, r)} = D\chi_E \llcorner B(x, r) - \chi_E \otimes \nu_{B(x, r)} \mathcal{H}^{N-1} \llcorner \partial B(x, r). \quad (3.220)$$

(iii) By 3.220 we infer that for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$  it holds that:

$$P(E \cap B(x, r), \Omega) = P(E, B(x, r)) + \mathcal{H}^{N-1}(E \cap \partial B(x, r)). \quad (3.221)$$

This concludes our discussion of functions of bounded variation. In the next chapter, we will turn our attention to *minimal perimeter sets* and the *Plateau problem* related to that.



## Chapter 4

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# Minimal variation and perimeter

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In this chapter, we are going to consider *sets of minimal perimeter* and *functions of minimal variation*, for which we will define some notion of minimality. We will take the theory developed in Chapter 2 and use its full force to show existence and regularity results. This chapter follows the exposition given in [9], [10], [5] and [11].

**Definition 4.1** Let  $\Omega \subset \mathbb{R}^N$  be an open set, and  $u_0 \in BV_{\text{loc}}(\mathbb{R}^N)$ . We define the following quantities

$$\gamma(u_0, \Omega) := \inf \left\{ \|Du\|(\Omega) \mid u \in BV_{\text{loc}}(\mathbb{R}^N), \text{supp}(u - u_0) \subset \Omega \right\}. \quad (4.1)$$

If  $E_0 \subset \mathbb{R}^N$  is a set of locally finite perimeter in  $\mathbb{R}^N$ , then we denote

$$\gamma(E_0, \Omega) := \gamma(\chi_{E_0}, \Omega). \quad (4.2)$$

We say that a function  $u : \mathbb{R}^N \rightarrow \mathbb{R}$  is a function of *minimal variation in  $\Omega$  with respect to  $u_0$*  if

$$\|Du\|(\Omega) = \gamma(u_0, \Omega). \quad (4.3)$$

We say that a set  $E \subset \mathbb{R}^N$  has *minimal perimeter in  $\Omega$  with respect to  $E_0$*  if

$$P(E, \Omega) = \gamma(E_0, \Omega). \quad (4.4)$$

### 4.1 Existence and compactness results

We dive right into and prove the following existence result.

**Theorem 4.2** Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $u_0 \in BV_{\text{loc}}(\mathbb{R}^N)$ . Then, there exists a function  $u$  of minimal variation in  $\Omega$  with respect to  $u_0$ .

**Proof** For any  $u \in BV_{\text{loc}}(\mathbb{R}^N)$  such that  $\text{supp}(u - u_0) \subset \Omega$  we have that

$$\|Du\|(B(0, R)) = \|Du\|(\Omega) + \|Du\|(B(0, R) \setminus \Omega). \quad (4.5)$$

Now, since  $\text{supp}(u - u_0) \subset \Omega$ , we have  $u = u_0$  outside of  $\Omega$ . Then, we in particular have  $u^\Omega = u_0^{\mathbb{R}^N \setminus \bar{\Omega}}$  and so by Corollary 3.69 it follows that  $\|Du\|(B(0, R) \setminus \Omega) = \|Du_0\|(B(0, R) \setminus \Omega)$ . So, it suffices to find a minimizer of  $\gamma(u_0, B(0, R))$ . We note that  $\gamma(u_0, B(0, R)) \leq \|Du_0\|(B(0, R)) < \infty$ , since  $u_0 \in BV_{\text{loc}}(\mathbb{R}^N)$  and  $\text{supp}(u_0 - u_0) = \emptyset \subset \Omega$ . Then, we can consider a minimizing sequence  $(u_h)_{h \in \mathbb{N}} \subset BV_{\text{loc}}(\mathbb{R}^N)$ , i.e.

$$\text{supp}(u_h - u_0) \subset B(0, R) \quad (4.6)$$

$$\|Du_h\|(B(0, R)) \leq \|Du_0\|(B(0, R)) \quad (4.7)$$

$$\lim_{h \rightarrow \infty} \|Du_h\|(B(0, R)) = \gamma(u_0, B(0, R)). \quad (4.8)$$

By 4.7, the sequence  $(u_h)_{h \in \mathbb{N}}$  is bounded in  $BV(B(0, R))$ . By Theorem 3.19 there exists a subsequence  $(u_{h_k})_{k \in \mathbb{N}}$  converging to some  $u$  in  $L^1(B(0, R), \mathcal{L}^N)$ . Since  $u_{h_k} = u_0$  outside of  $B(0, R)$  for all  $k \in \mathbb{N}$ , we have  $\text{supp}(u - u_0) \subset B(0, R)$ . Moreover, by Theorem 3.19 we know that  $u \in BV_{\text{loc}}(\mathbb{R}^N)$  and by lower semicontinuity of variation we find that

$$\begin{aligned} \gamma(u_0, B(0, R)) &\leq \|Du\|(B(0, R)) \\ &\leq \liminf_{h \rightarrow \infty} \|Du_h\|(\|DB(0, R)\|) \\ &= \gamma(u_0, B(0, R)). \end{aligned} \quad (4.9)$$

□

As a corollary, we get the following existence result for minimal perimeter.

**Corollary 4.3** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set and  $E_0 \subset \mathbb{R}^N$  is a set of locally finite perimeter in  $\mathbb{R}^N$ . Then, there exists a set  $E$  of minimal perimeter in  $\Omega$  with respect to  $E_0$ .*

**Proof** Applying Theorem 4.2 to  $u_0 = \chi_{E_0} \in BV_{\text{loc}}(\mathbb{R}^N)$  and using Theorem 3.25 instead of Theorem 3.19, yields the result. □

Furthermore, we have the following compactness result for functions of minimal variation.

**Theorem 4.4** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $u_0 \in BV_{\text{loc}}(\mathbb{R}^N)$ . Suppose  $(u_h)_{h \in \mathbb{N}} \subset BV_{\text{loc}}(\mathbb{R}^N)$  is a sequence of functions of minimal variation in  $\Omega$  with respect to  $u_0$  and suppose  $u_h \rightarrow u$  in  $L^1_{\text{loc}}(\mathbb{R}^N, \mathcal{L}^N)$ . Then,  $u \in BV_{\text{loc}}(\mathbb{R}^N)$  is a function of minimal variation in  $\Omega$  with respect to  $u_0$ . Moreover, we have for any open ball  $B(x, r) \subset \subset \Omega$  that:*

$$\lim_{h \rightarrow \infty} \|Du_h\|(B(x, r)) = \|Du\|(B(x, r)). \quad (4.10)$$

**Proof** Let  $A \subset\subset \Omega$  be open and  $\varphi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  a Lipschitz function such that  $\varphi(A) \subset A$  and  $\varphi(x) = x$  outside of  $A$ . Then, we have  $v = \varphi \circ u$ . Define now for any  $h \in \mathbb{N}$

$$v_h = v\chi_A + u_h\chi_{\mathbb{R}^N \setminus \bar{A}}. \quad (4.11)$$

Clearly, we have  $\text{supp}(v_h - u_0) \subset \Omega$  and so by the minimality of  $u_h$  we find the following:

$$\|Du_h\|(\Omega) \leq \|Dv_h\|(\Omega), \quad \forall h \in \mathbb{N}. \quad (4.12)$$

We also note that

$$\|Dv_h\|(\Omega) = \|Dv\|(A) + \int_{\partial A} |u^A - u_h^{\mathbb{R}^N \setminus \bar{A}}| d\mathcal{H}^{N-1} + \|Du_h\|(\mathbb{R}^N \setminus \bar{A}). \quad (4.13)$$

By  $L^1_{\text{loc}}$ -convergence of  $u_h \rightarrow u$  and continuity of trace, we have

$$\int_{\partial A} |u^A - u_h^{\mathbb{R}^N \setminus \bar{A}}| d\mathcal{H}^{N-1} \xrightarrow{h \rightarrow \infty} 0. \quad (4.14)$$

Now, by taking the limit  $h \rightarrow \infty$  we find by the lower semicontinuity of variation that

$$\|Du\|(A) \leq \liminf_{h \rightarrow \infty} \|Du_h\|(A) \leq \|Dv\|(A). \quad (4.15)$$

So by taking the limit  $A \uparrow \Omega$  we have then  $\|Du\|(\Omega) = \gamma(u_0, \Omega)$ . Now by considering  $\varphi = \text{id}_{\mathbb{R}^N}$  and  $A = B(x, r)$  we obtain that

$$\begin{aligned} \|Du\|(B(x, r)) &\leq \liminf_{h \rightarrow \infty} \|Du_h\|(B(x, r)) \\ &\leq \limsup_{h \rightarrow \infty} \|Du_h\|(B(x, r)) \\ &\leq \|Du\|(\bar{B}(x, r)). \end{aligned} \quad (4.16)$$

By 3.70 we know that  $\|Du\|(\partial B(x, r))$  for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$ . So by 4.16 we find that  $\|Du\|(\bar{B}(x, r)) = \|Du\|(B(x, r))$  and 4.10 follow.  $\square$

As a corollary to Theorem 4.4 we get a compactness result for sets of minimal perimeter.

**Corollary 4.5** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  a set of locally finite perimeter in  $\mathbb{R}^N$ . Suppose  $(E_h)_{h \in \mathbb{N}}$  is a sequence of sets of minimal perimeter in  $\Omega$  with respect to  $E_0$  and suppose  $E_h \rightarrow E$  locally in  $\mathbb{R}^N$ . Then,  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$ . Moreover, we have for any open ball  $B(x, r) \subset\subset \Omega$  that:*

$$\lim_{h \rightarrow \infty} P(E_h, B(x, r)) = P(E, B(x, r)). \quad (4.17)$$

## 4.2 Uniform density estimates

For sets of minimal perimeter, we get even better density estimates than the ones from the isoperimetric inequalities. This will also yield an improvement on the compactness result. We begin with the uniform density estimates for sets of minimal perimeter.

**Theorem 4.6** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Suppose  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $x \in \partial E \cap \Omega$ . Then, there is  $r_0 \in (0, d(x, \partial\Omega))$  such that for all  $r \in (0, r_0)$  it holds that:*

$$\mathcal{L}^N(E \cap B(x, r)), \mathcal{L}^N(B(x, r) \setminus E) \geq cr^N \quad (4.18)$$

$$c_1 r^{N-1} \leq P(E, B(x, r)) \leq C_1 r^{N-1} \quad (4.19)$$

$$\mathcal{H}^{N-1}(\Omega \cap (\partial E \setminus \partial^* E)) = 0, \quad (4.20)$$

where  $c, c_1, C_1 > 0$  are only depending on  $N$ .

**Proof** We consider  $\ell(r) = \mathcal{L}^N(E \cap B(x, r))$  for  $r \in (0, d(x, \partial\Omega))$ . Note that  $F_r = E \setminus B(x, r)$  is a competitor for perimeter. Indeed, we have  $F_r$  is a set of locally finite perimeter in  $\mathbb{R}^N$  and  $F_r \setminus \Omega = E_0 \setminus \Omega$ . Then, we have for  $\mathcal{L}^1$ -a.e.  $s \in (r, d(x, \partial\Omega))$  by minimality of perimeter of  $E$  in  $\Omega$

$$\begin{aligned} P(E, B(x, s)) &\leq P(E, B(x, s)) \\ &= P(E \setminus B(x, r), B(x, s)) \\ &= \mathcal{H}^{N-1}(E \cap \partial B(x, r)) + P(E, B(x, s) \setminus \overline{B}(x, r)). \end{aligned} \quad (4.21)$$

Hence, we have as  $s \rightarrow r^+$  we get that

$$P(E, B(x, r)) \leq \mathcal{H}^{N-1}(E \cap \partial B(x, r)). \quad (4.22)$$

Also note that we have for  $r \in (0, d(x, \partial\Omega))$ :

$$\ell(r) = \mathcal{L}^N(E \cap B(x, r)) = \int_0^r \mathcal{H}^{N-1}(E \cap \partial B(x, s)) ds. \quad (4.23)$$

Then, 4.23 implies that for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$   $\ell'(r) = \mathcal{H}^{N-1}(E \cap \partial B(x, r))$ . By Theorem 3.30 we find a dimensional  $\gamma_2 > 0$  such that for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$ :

$$\begin{aligned} \ell(r)^{\frac{N-1}{N}} &\leq \gamma_2 P(E \cap B(x, r), \mathbb{R}^N) \\ &\leq \gamma_2 (P(E, B(x, r)) + \mathcal{H}^{N-1}(E \cap \partial B(x, r))) \\ &\leq 2\gamma_2 \mathcal{H}^{N-1}(E \cap \partial B(x, r)) \\ &= 2\gamma_2 \ell'(r). \end{aligned} \quad (4.24)$$

Inequality 4.24 yields for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$

$$\begin{aligned} \left(\ell^{\frac{1}{N}}\right)'(r) &= \frac{1}{N} \ell'(r) \ell(r)^{\frac{1-N}{N}} \\ &\geq \frac{1}{2N\gamma_2}. \end{aligned} \quad (4.25)$$

So for  $c = \left(\frac{1}{2N\gamma_2}\right)^N$  and integration from 0 to  $r$  yields for every  $r \in (0, d(x, \partial\Omega))$ :

$$\begin{aligned} \mathcal{L}^N(E \cap B(x, r)) &= \ell(r) \\ &\geq cr^N \end{aligned} \quad (4.26)$$

Applying the same argument to  $\mathbb{R}^N \setminus E$  gives the result 4.18. By Theorem 3.34 (iii) there is a dimensional constant  $\gamma_5 > 0$  such that for all  $r \in (0, d(x, \partial\Omega))$ :

$$\begin{aligned} P(E, B(x, r)) &\geq \frac{1}{\gamma_5} (cr^N)^{\frac{N-1}{N}} \\ &= \frac{c^{N-1}}{\gamma_5} r^{N-1}. \end{aligned} \quad (4.27)$$

So  $c_1 = \frac{c^{N-1}}{\gamma_5}$  yields the lower bound in 4.19. The upper bound we get by Lemma `lem::bddpervol`, where there is some  $r_0 \in (0, d(x, \partial\Omega))$  and a dimensional constant  $C_1 = \alpha > 0$  such that for all  $r \in (0, r_0)$  it holds that:

$$P(E, B(x, r)) \leq C_1 r^{N-1}. \quad (4.28)$$

4.20 is a consequence of Theorem 3.45 (ii).  $\square$

This gives us the following improvement on the compactness of minimal perimeter.

**Proposition 4.7** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Suppose  $(E_h)_{h \in \mathbb{N}}$  be a sequence of sets of minimal perimeter in  $\Omega$  with respect to  $E_0$  and suppose that  $(E_h)_{h \in \mathbb{N}}$  converges to some set  $E$  in  $L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ . For any  $x \in \partial E$ , there is some  $r_0 \in (0, d(x, \partial\Omega))$  such that for all  $\varepsilon > 0$  there is  $h_0 \in \mathbb{N}$  such that for all  $h \geq h_0$ :*

$$\partial E_h \cap B\left(x, \frac{r_0}{2}\right) \subset \left(\partial E \cap B\left(x, \frac{r_0}{2}\right)\right)_\varepsilon, \quad (4.29)$$

where  $U_\varepsilon = \bigcup_{y \in U} \{z \in \mathbb{R}^N \mid \|z - y\| < \varepsilon\}$ .

**Proof** Suppose that this is not true. Then there is some  $\varepsilon_0 > 0$  such that there exists a sequence  $(z_h)_{h \in \mathbb{N}}$  of points  $z_h \in \partial E_h \cap B(x, \frac{r_0}{2})$  such that  $\|z_h - y\| \geq \varepsilon$  for all  $y \in \partial E \cap B(x, \frac{r_0}{2})$ . By Theorem 4.6 we then find for all  $h \in \mathbb{N}$ :

$$\begin{aligned} \int_{B(x, \frac{r_0}{2})} |\chi_{E_h} - \chi_E| \, dy &\geq \min \left\{ \mathcal{L}^N(E_h \cap B(z_h, \varepsilon_0)), \mathcal{L}^N(B(z_h, \varepsilon_0) \setminus E_h) \right\} \\ &\geq c\varepsilon_0^N. \end{aligned} \quad (4.30)$$

But this is a contradiction to the convergence  $\chi_{E_h} \rightarrow \chi_E$  in  $L^1_{\text{loc}}(\Omega, \mathcal{L}^N)$ .  $\square$

This ends this short section on uniform density estimates.

### 4.3 Monotonicity formula

In this section we would like to show the monotonicity of the following function

$$\Phi(E, x)(\cdot) : r \in (0, d(x, \partial\Omega)) \mapsto \frac{\|D\chi_E\|(B(x, r))}{\omega_{N-1}r^{N-1}} \quad (4.31)$$

for every  $x \in \partial E$  and  $E \subset \mathbb{R}^N$  a set of minimal perimeter in open set  $\Omega$ . But first we note the following scaling property of  $\Phi$ .

**Lemma 4.8** *Let  $E \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Then for any  $x \in \partial E$  and  $r > 0$  we find that:*

$$\Phi(E, x)(r) = \Phi\left(\frac{E-x}{r}, 0\right)(1). \quad (4.32)$$

**Proof** We consider any  $\varphi \in C^1_c(B(0, 1))$  with  $\|\varphi\|_\infty \leq 1$  and set  $\psi(x + ry) = \varphi(y)$ . Then  $\psi \in C^1_c(B(x, r))$  with  $\|\psi\|_\infty \leq 1$  and we find by chain rule and substitution:

$$\begin{aligned} \frac{1}{\omega_{N-1}} \int_{B(0, 1)} \chi_{\frac{E-x}{r}}(y) \operatorname{div}(\varphi)(y) \, dy &= r \int_{B(0, 1)} \chi_E(x + ry) \operatorname{div}(\psi)(x + ry) \, dy \\ &= \frac{1}{\omega_{N-1}r^{N-1}} \int_{B(x, r)} \chi_E(y) \operatorname{div}(\psi)(y) \, dy. \end{aligned} \quad (4.33)$$

By taking the supremum over all such  $\varphi$  and noting that

$$\varphi \in C^1_c(B(0, 1)) \mapsto (x + ry \mapsto \varphi(y)) \in C^1_c(B(x, r)) \quad (4.34)$$

defines a bijection, we obtain that  $\Phi(E, x)(r) = \Phi\left(\frac{E-x}{r}, 0\right)(1)$ .  $\square$

For cones, we get the following preliminary result. For the definition of a cone see Definition 5.17.

**Lemma 4.9** *Let  $\mathcal{C} \subset \mathbb{R}^N$  be a cone of locally finite perimeter in  $\mathbb{R}^N$  centered at some  $x \in \partial^* \mathcal{C}$ . Then the map  $\Phi(\mathcal{C}, x)(\cdot)$  is constant.*

**Proof** Since  $\mathcal{C}$  is a cone centered at  $x$  we have that  $\mathcal{C} - x$  is a cone centered at the origin. Therefore,  $\frac{\mathcal{C} - x}{r} = \mathcal{C} - x$  for all  $r > 0$ . By Lemma 4.8 it follows for all  $r > 0$  that:

$$\begin{aligned} \Phi(\mathcal{C}, x)(r) &= \Phi\left(\frac{\mathcal{C} - x}{r}, 0\right)(1) \\ &= \Phi(\mathcal{C} - x, 0)(1). \end{aligned} \quad (4.35)$$

Since the last expression is independent of  $r$ , it follows that  $\Phi(\mathcal{C}, x)(\cdot)$  is a constant function.  $\square$

Now we can state the monotonicity result for minimal perimeter.

**Theorem 4.10** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Suppose  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $x \in \partial^* E \cap \Omega$ . Then the function  $\Phi(E, x)(\cdot)$  is a non-decreasing function on  $(0, d(x, \partial\Omega))$ . Moreover,  $\Phi(E, x)(\cdot)$  is constant if and only if  $E^1$  is a cone centered at  $x$ .*

#### 4.3.1 Cones of locally finite perimeter

To prove Theorem 4.10, we need to first prove an alternative characterisation of cones of locally finite perimeter.

**Proposition 4.11** *Let  $\mathcal{C} \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . Then  $\mathcal{C}^1$  is a cone if and only if for  $\mathcal{H}^{N-1}$ -a.e.  $x \in \partial^* \mathcal{C}$  it holds that  $x \cdot \nu_E(x) = 0$ .*

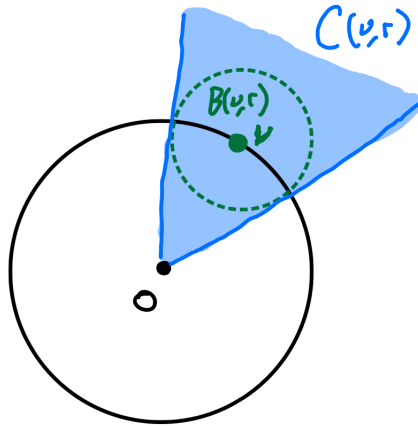


Figure 4.1: Open cone  $\mathcal{C}(v, r)$ .

**Proof** For any  $v \in S^{N-1}$  and  $r \in (0, 1)$  we consider the open cone

$$\mathcal{C}(v, r) := \{\lambda z \mid \lambda > 0, z \in B(v, r)\}. \quad (4.36)$$

Since  $\mathcal{C}(v, r)$  has Lipschitz boundary, it is a set of locally finite perimeter and we have by the generalized Gauss-Green formula 3.46 that for any  $\varepsilon > 0$  and  $x \in \partial\mathcal{C}(v, r)$

$$\begin{aligned} \int_{\partial\mathcal{C}(v, r) \cap B(x, \varepsilon)} y \cdot \nu_{\mathcal{C}(v, r)}(y) d\mathcal{H}^{N-1}(y) &= - \int_{\mathcal{C}(v, r) \cap B(x, \varepsilon)} \operatorname{div}(y) dy \\ &\leq N\omega_N \varepsilon^N. \end{aligned} \quad (4.37)$$

By this computation and taking limit  $\varepsilon \rightarrow 0$ , we must have that for all  $x \in \partial\mathcal{C}(v, r)$  we have

$$x \cdot \nu_{\mathcal{C}(v, r)}(x) = 0. \quad (4.38)$$

If now  $\mathcal{C}$  is a set of locally finite perimeter in  $\mathbb{R}^N$ , then for any  $r, s > 0$  and  $v \in S^{N-1}$  the set  $\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)$  is a set of finite perimeter and by Theorem 3.41:

$$\begin{aligned} D\chi_{\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)} &= \nu_E \mathcal{H}^{N-1} \llcorner (\partial^* \mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)) \\ &\quad + \nu_{\mathcal{C}(v, r)} \mathcal{H}^{N-1} \llcorner (\mathcal{C}^1 \cap \partial\mathcal{C}(v, r) \cap B(0, s)) \\ &\quad + \frac{y}{\|y\|} \mathcal{H}^{N-1} \llcorner (\mathcal{C}^1 \cap \mathcal{C}(v, r) \cap \partial B(0, s)), \end{aligned} \quad (4.39)$$

where we used that  $\nu_{B(0, s)} = \frac{y}{\|y\|}$  the inner unit normal for an open ball. Since  $x \cdot \nu_{\mathcal{C}(v, r)}(x) = 0$  for  $x \in \partial\mathcal{C}(v, r)$  we obtain by 3.46 that

$$\begin{aligned} N\mathcal{L}^N(\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)) &= \int_{\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)} \operatorname{div}(y) dy \\ &= - \int_{\partial^* \mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)} y \cdot \nu_{\mathcal{C}}(y) d\mathcal{H}^{N-1}(y) \\ &\quad - s \mathcal{H}^{N-1}(\mathcal{C}^1 \cap \mathcal{C}(v, r) \cap \partial B(0, s)). \end{aligned} \quad (4.40)$$

Now set  $u(s) := \mathcal{L}^N(\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s))$ . By the coarea formula we have that for all  $s > 0$ :

$$u(s) = \int_0^s \mathcal{H}^{N-1}(\mathcal{C} \cap \mathcal{C}(v, r) \cap \partial B(0, t)) dt. \quad (4.41)$$

So for  $\mathcal{L}^1$ -a.e.  $s > 0$  we find by differentiation that:

$$\begin{aligned} \left( \frac{u(s)}{s^N} \right)' &= \frac{u'(s)}{s^N} - \frac{Nu(s)}{s^{N+1}} \\ &\stackrel{4.40}{=} \frac{\int_{\partial^* \mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)} y \cdot \nu_{\mathcal{C}}(y) d\mathcal{H}^{N-1}(y)}{s^{N+1}}. \end{aligned} \quad (4.42)$$

We introduce now the following notation for  $y \in \mathbb{R}^N \setminus \{0\}$ ,  $v = \frac{y}{\|y\|}$ ,  $r \in (0, 1)$  and  $s = \|y\|$ :

$$A(y, r) = (\mathcal{C}(v, r) \cap B(0, s(1+r))) \setminus B(0, s(1-r)). \quad (4.43)$$

By boundedness, for any  $y \in \mathbb{R}^N$  we can find constants  $\alpha, \beta, r_0 > 0$  only

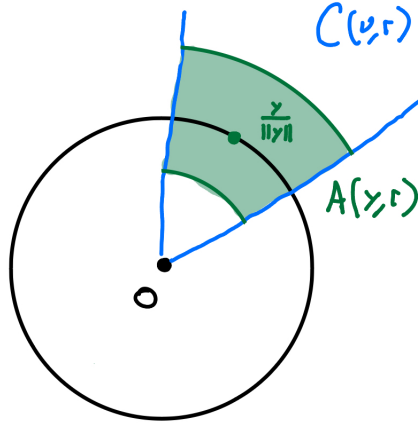


Figure 4.2: Annulus sector  $A(y, r)$ .

depending on  $y$  such that:

$$B(y, \alpha r) \subset A(y, r) \subset B(y, \beta r), \quad \forall r \in (0, r_0). \quad (4.44)$$

Now we show the two implications of Proposition 4.11. Assume first that  $\mathcal{C}^1$  is a cone, then also  $\mathcal{C}^1 \cap \mathcal{C}(v, r)$  is a cone. Then,  $\frac{u(s)}{s^N}$  is constant in  $s$ . Indeed, for all  $s > 0$  we find:

$$\begin{aligned} \frac{u(s)}{s^N} &= \mathcal{L}^N \left( \frac{\mathcal{C} \cap \mathcal{C}(v, r)}{s} \cap B(0, 1) \right) \\ &= \mathcal{L}^N (\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, 1)). \end{aligned} \quad (4.45)$$

Hence, by 4.42 we must have for all  $s > 0, r \in (0, 1), v \in S^{N-1}$ :

$$\int_{\partial^* \mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s)} y \cdot \nu_{\mathcal{C}}(y) d\mathcal{H}^{N-1}(y) = 0. \quad (4.46)$$

This implies for all  $y \in \mathbb{R}^N \setminus \{0\}$  and  $r \in (0, 1)$  we must have:

$$\int_{\partial^* \mathcal{C} \cap A(y, r)} x \cdot \nu_{\mathcal{C}}(x) d\mathcal{H}^{N-1}(x) = 0. \quad (4.47)$$

By Corollary 2.45  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial^* \mathcal{C}$  is a Lebesgue point of the function  $x \cdot \nu_{\mathcal{C}}(x)$  with respect to the Radon measure  $\mathcal{H}^{N-1} \llcorner \partial^* \mathcal{C}$  and therefore:

$$\lim_{r \rightarrow 0^+} \int_{\partial^* \mathcal{C} \cap B(y, r)} |x \cdot \nu_{\mathcal{C}}(x) - y \cdot \nu_{\mathcal{C}}(y)| \, d\mathcal{H}^{N-1} = 0. \quad (4.48)$$

In particular, by the inclusions 4.44 we see for  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial^* \mathcal{C} \setminus \{0\}$ :

$$\lim_{r \rightarrow 0^+} \int_{\partial^* \mathcal{C} \cap A(y, r)} |x \cdot \nu_{\mathcal{C}}(x) - y \cdot \nu_{\mathcal{C}}(y)| \, d\mathcal{H}^{N-1}(x) = 0. \quad (4.49)$$

Together with 4.47, we find that  $y \cdot \nu_{\mathcal{C}}(y) = 0$  for  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial^* \mathcal{C}$ . Conversely, assume that  $y \cdot \nu_{\mathcal{C}}(y) = 0$  for  $\mathcal{H}^{N-1}$ -a.e.  $y \in \partial^* \mathcal{C}$ . Now, let  $y \in \mathcal{C}^1 \setminus \{0\}$  and  $\lambda > 0$ . To show is  $\lambda y \in \mathcal{C}^1$ . By assumption we find that by 4.42,  $\frac{u(s)}{s^N}$  is constant in  $s$ . This implies that for any  $r \in (0, 1)$

$$\begin{aligned} \mathcal{L}^N(\mathcal{C} \cap A(y, r)) &= \mathcal{L}^N(\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s(1+r))) \\ &\quad - \mathcal{L}^N(\mathcal{C} \cap \mathcal{C}(v, r) \cap B(0, s(1-r))) \\ &= \frac{u(\lambda s(1+r))}{\lambda^N} - \frac{u(\lambda s(1-r))}{\lambda^N} \\ &= \frac{\mathcal{L}^N(\mathcal{C} \cap A(\lambda y, r))}{\lambda^N}. \end{aligned} \quad (4.50)$$

Since  $A(\lambda y, r) = \lambda A(y, r)$  we must have by 4.50:

$$\begin{aligned} \frac{\mathcal{L}^N(\mathcal{C} \cap A(\lambda y, r))}{\mathcal{L}^N(A(\lambda y, r))} &= \frac{\mathcal{L}^N(\mathcal{C} \cap A(\lambda y, r))}{\mathcal{L}^N(\lambda A(y, r))} \\ &= \frac{\mathcal{L}^N(\mathcal{C} \cap A(\lambda y, r))}{\lambda^N \mathcal{L}^N(A(y, r))} \\ &\stackrel{4.50}{=} \frac{\mathcal{L}^N(\mathcal{C} \cap A(y, r))}{\mathcal{L}^N(A(y, r))}. \end{aligned} \quad (4.51)$$

Since  $y \in E^1$  we have by Remark 3.44 and the inclusions 4.44 that:

$$\begin{aligned} \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(\mathcal{C} \cap A(\lambda y, r))}{\mathcal{L}^N(A(\lambda y, r))} &= \lim_{r \rightarrow 0^+} \frac{\mathcal{L}^N(\mathcal{C} \cap A(y, r))}{\mathcal{L}^N(A(y, r))} \\ &= 1. \end{aligned} \quad (4.52)$$

So by Remark 3.44 we can conclude  $\lambda y \in E^1$  and hence  $E^1$  is a cone.  $\square$

**Proof (Theorem 4.10)** Up to translation and scaling we may assume that  $x = 0$  and  $d(x, \partial\Omega) = 1$ . Now by Theorem 3.12 we can choose a sequence  $(u_h)_{h \in \mathbb{N}} \subset C^\infty(B(0, 1))$  such that  $\|u_h - \chi_E\|_{BV(B(0, 1))} \rightarrow 0$  as  $h \rightarrow \infty$ . We

consider for every  $h \in \mathbb{N}$  and  $r \in (0, 1)$  the following quantities:

$$a_h(r) := \left| \int_{B(0,r)} \|\nabla u_h\| \, dy - \|D\chi_E\|(B(0,1)) \right| \quad (4.53)$$

$$b_h(r) := \int_{\partial B(0,r)} \left| u_h^{B(0,r)} - \chi_E^{B(0,1)} \right| \, d\mathcal{H}^{N-1}. \quad (4.54)$$

We note that by minimality of perimeter of  $E$ , we find for any function  $v$  with  $v^{B(0,r)} = u_h^{B(0,r)}$  for  $r \in (0, 1)$  extended to be  $\chi_E$  outside of  $B(0, r)$  that using the coarea formula:

$$\begin{aligned} \|D\chi_E\|(B(0,1)) &\leq \int_0^1 P(\{v \geq t\}, B(0,1)) \, dt \\ &\leq \|Dv\|(B(0,1)). \end{aligned} \quad (4.55)$$

This yields, in particular, for any  $r \in (0, 1)$  using Example 3.70

$$\|D\chi_E\|(B(0,r)) \leq \|Dv\|(B(0,r)) + b_h(r), \quad (4.56)$$

and therefore

$$\int_{B(0,r)} \|\nabla u_h\| \, dy \leq \|Dv\|(B(0,r)) + a_h(r) + b_h(r). \quad (4.57)$$

Now set  $J_h(r) := \int_{B(0,r)} \|\nabla u_h\| \, dy$  and consider the radial extension  $v$  of  $u_h$  given by  $v(y) = u_h\left(\frac{y}{\|y\|}\right)$ . We note that we have the following identity:

$$(N-1) \|Dv\|(B(0,1)) = \int_{\partial B(0,1)} \|\nabla_T u_h\| \, dy. \quad (4.58)$$

So, it follows that:

$$J'_h(1) = (N-1) \|Dv\|(B(0,1)) + \left( \int_{\partial B(0,1)} \|\nabla u_h\| - \|\nabla_T u_h\| \, dy \right). \quad (4.59)$$

Since  $v$  is exactly a function as in the preceding discussion, we find that:

$$J'_h(1) \geq (N-1)J_h(r) + \left( \int_{\partial B(0,1)} \|\nabla u_h\| - \|\nabla_T u_h\| \, dy \right) - C_1(a_h(1) + b_h(1)), \quad (4.60)$$

where  $C_1 > 0$  is some dimensional constant. Now, since  $\|\nabla u_h\|^2(y) = \|\nabla_T u_h\|^2(y) + (\nabla u_h(y) \cdot y)^2$  and  $1 - \sqrt{1-t} \geq \frac{t}{2}$  for  $t \in (0, 1)$  it follows that:

$$J'_h(1) - (N-1)J_h(r) \geq \frac{1}{2} \int_{\partial B(0,1)} \frac{(\nabla u_h(y) \cdot y)^2}{\|\nabla u_h(y)\|} \, dy - C(a_h(1) + b_h(1)). \quad (4.61)$$

Note now that if  $\Phi_h(r) = \frac{J_h(r)}{\omega_{N-1}r^{N-1}}$  we find that the left side in 4.61 corresponds to  $\Phi'_h(1)$ . Now, considering the rescaling  $\tilde{\Phi}_h(t) = \Phi_h(tr)$  we find that for  $r \in (0, 1)$

$$\begin{aligned}\Phi'_h(r) &= \frac{1}{r} \tilde{\Phi}'_h(1) \\ &\geq \frac{1}{2} \int_{\partial B(0,r)} \frac{(\nabla u_h(y) \cdot y)^2}{\|\nabla u_h(y)\| r^{N+1}} dy - C_2(a_h(r) + b_h(r)),\end{aligned}\tag{4.62}$$

for some constant  $C_2 > 0$  only depending on  $r$  and  $N$ . Now, integration on an interval  $(r_1, r_2) \subset (0, 1)$  and application of the Cauchy-Schwarz inequality yields:

$$\Phi_h(r_2) - \Phi_h(r_1) \geq \frac{\left( \int_{B(0,r_2) \setminus B(0,r_1)} \frac{\nabla u_h \cdot y}{\|y\|^N} dy \right)^2}{\int_{B(0,r_2) \setminus B(0,r_1)} \frac{\|\nabla u_h\|}{\|y\|^{N-1}} dy} - C_3 \int_{r_1}^{r_2} a_h(r) + b_h(r) dr.\tag{4.63}$$

Now, letting  $h \rightarrow \infty$  and using the convergence  $\int_{B(0,r)} \|\nabla u_h\| dy \rightarrow \|D\chi_E\|(B(0,r))$  for  $\mathcal{L}^1$ -a.e.  $r \in (0, 1)$  and so we must have  $a_h(r) \rightarrow 0$  and  $b_h(r) \rightarrow 0$ . Hence, by taking limits, we find that:

$$\Phi(E, 0)(r_2) - \Phi(E, 0)(r_1) \geq \frac{\left( \int_{B(0,r_2) \setminus B(0,r_1)} \frac{D\chi_E \cdot y}{\|y\|^N} \right)^2}{\int_{B(0,r_2) \setminus B(0,r_1)} \frac{\|D\chi_E\|}{\|y\|^{N-1}}} \geq 0\tag{4.64}$$

for  $\mathcal{L}^1$ -a.e.  $r_1 \leq r_2$  in  $(0, 1)$ . In fact, for any such pair in  $(0, 1)$  as for any sequence  $r_h \rightarrow r^+$  in  $(0, 1)$  we have  $\lim_{h \rightarrow \infty} \|D\chi_E\|(B(0, r_h)) = \|D\chi_E\|(B(0, r))$ . We note that we have equality in 4.64 for all  $r_1 \leq r_2$  if and only if  $D\chi_E \cdot y = 0$  for almost all  $y$  and by Proposition 4.11 this is only the case for a cone  $E^1$  centered at the origin.  $\square$

In Chapter 5 we will see another proof of the monotonicity in Theorem 4.10, which deals with this problem in a different way.

#### 4.4 Singular sets and blowup limits of sets of minimal perimeter

We introduce here the notion of *singular sets* and consider blow-up limits at points of singular sets.

**Definition 4.12** Let  $E \subset \mathbb{R}^N$  be a  $\mathcal{L}^N$ -measurable set and  $\Omega \subset \mathbb{R}^N$  be an open set. We define the **singular set of  $E$  in  $\Omega$**  as the following set:

$$\Sigma(E, \Omega) := \Omega \cap (\partial E \setminus \partial^* E).\tag{4.65}$$

We call  $x \in \Sigma(E, \Omega)$  a **singular point of  $E$  in  $\Omega$** .

**Remark 4.13** If  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  we have already shown in Theorem 4.6, that  $\mathcal{H}^{N-1}(\Sigma(E, \Omega)) = 0$ .

We have the following compactness result for singular points.

**Theorem 4.14** Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . If  $(E_h)_{h \in \mathbb{N}}$  and  $E \subset \mathbb{R}^N$  are a sequence sets and a set of minimal perimeter in  $\Omega$  with respect to  $E_0$  and

- (i)  $(E_h)_{h \in \mathbb{N}}$  converges to  $E$  locally in  $\mathbb{R}^N$ ,
- (ii) for any  $h \in \mathbb{N}$  we have  $x_h \in \Sigma(E_h, \Omega)$ , and
- (iii)  $x \in \Omega \cap \partial E$  such that  $\lim_{h \rightarrow \infty} x_h = x$ ,

then  $x \in \Sigma(E, \Omega)$ . Furthermore, for any  $\varepsilon > 0$  and  $K \subset \Omega$  compact, there exists  $h_0 \in \mathbb{N}$  such that

$$\Sigma(E_h, \Omega) \cap K \subset (\Sigma(E, \Omega) \cap K)_\varepsilon, \quad \forall h \geq h_0. \quad (4.66)$$

**Proof** Suppose for the sake of contradiction that  $x \notin \Sigma(E, \Omega)$ . In particular, we would have  $x \in \partial^* E$ . By Corollary 4.5 we can choose  $h \in \mathbb{N}$  large enough such that  $x_h \in \partial^* E_h$ , which is a contradiction.

Again for the sake of contradiction suppose that there are  $\varepsilon_0 > 0$ , a compact subset  $K_0 \subset \Omega$  and a sequence  $h_k \xrightarrow{k \rightarrow \infty} \infty$ , elements  $y_k \in \Sigma(E_{h_k}, \Omega) \cap K_0$  such that

$$d(y_k, \Sigma(E, \Omega) \cap K_0) \geq \varepsilon_0, \quad \forall k \in \mathbb{N}. \quad (4.67)$$

As  $(y_k)_{k \in \mathbb{N}} \subset K_0$  and  $K_0$  is compact, we can choose a subsequence  $(y_{k_\ell})_{\ell \in \mathbb{N}}$  such that  $y = \lim_{\ell \rightarrow \infty} y_{k_\ell} \in K_0$ . Since  $(E_{k_\ell})_{\ell \in \mathbb{N}}$  still converges locally to  $E$ ,  $y_{k_\ell} \in \Sigma(E_{k_\ell}, \Omega)$ , by the first part of this theorem we know that  $y \in \Sigma(E, \Omega) \cap K_0$ . In particular, there is some  $\ell_0 \in \mathbb{N}$  such that  $y_{k_\ell} \in B(y, \varepsilon) \subset (\Sigma(E, \Omega) \cap K)_\varepsilon$  for all  $\ell \geq \ell_0$ , which contradicts the supposition 4.67.  $\square$

Now we are ready to state and prove the following theorem concerned with blowups of sets of minimal perimeter.

**Theorem 4.15** Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . If  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $x \in \Sigma(E, \Omega)$ , then there exist a sequence  $(r_h)_{h \in \mathbb{N}}$  with  $r_h \xrightarrow{h \rightarrow \infty} 0$  and a cone  $\mathcal{C}$  of minimal perimeter in  $\Omega$  with respect to  $E_0$  with  $0 \in \Sigma(\mathcal{C}, \Omega)$  such that  $\frac{E-x}{r_h} \xrightarrow{h \rightarrow \infty} \mathcal{C}$  locally in  $\mathbb{R}^N$ .

**Proof** By Lemma 4.8 we infer that for all  $r > 0$  it holds:

$$\Phi\left(\frac{E-x}{r}, 0\right)(t) = \Phi(E, x)(tr), \quad \forall t > 0. \quad (4.68)$$

Now by Theorem 4.10 the sequence of perimeters  $P\left(\frac{E-x}{r}, B(0, s)\right)$  is uniformly bounded for all  $s \in (0, d(x, \partial\Omega))$ . By Theorems 3.25 and 4.5, we can extract a subsequence  $(r_h)_{h \in \mathbb{N}} \subset (0, \infty)$  with  $\lim_{h \rightarrow \infty} r_h = 0$  such that  $\left(\frac{E-x}{r_h}\right)_{h \in \mathbb{N}}$  converges to a set  $\mathcal{C}$  with minimal perimeter in  $\Omega$  with respect to  $E_0$ . By Theorem 4.14 since 0 is a singular point of  $\frac{E-x}{r_h}$  for all  $h \in \mathbb{N}$ , we obtain  $0 \in \Sigma(\mathcal{C}, \Omega)$ .

By Theorem 4.5 we have that

$$\begin{aligned} \Phi(\mathcal{C}, 0)(t) &= \lim_{h \rightarrow \infty} \Phi\left(\frac{E-x}{r_h}, 0\right)(t) \\ &= \lim_{h \rightarrow \infty} \Phi(E, x)(tr_h) \\ &= \Phi(E, x)(0^+). \end{aligned} \tag{4.69}$$

The last term in 4.69 is independent of  $t$  and therefore  $\Phi(\mathcal{C}, 0)(\cdot)$  is constant, which implies that  $\mathcal{C}$  is a cone by Theorem 4.10.  $\square$

## 4.5 Partial regularity of sets of minimal perimeter

In this section, we will prove de Giorgi's regularity result for flat sets of minimal perimeter. We show the following result.

**Theorem 4.16** *Let  $E \subset \mathbb{R}^N$  and  $x \in \partial E$ . Assume that  $E$  has minimal perimeter in  $\Omega$  with respect to  $E_0$ . There exists  $\varepsilon > 0$  only depending on  $N$  such that if*

$$\partial E \cap B(x, r) \subset \{|y_N| \leq \varepsilon\} \tag{4.70}$$

*for some  $r \in (0, d(x, \partial\Omega))$ , then there is some unit vector  $v \in S^{N-1}$ , then  $\partial E \cap B(0, \frac{r}{2})$  is the graph of a  $C^{1,\alpha}$  function.*

To prove this, we need the following lemma, the so-called *improvement of flatness*. We will follow [12], [10] and [13].

**Lemma 4.17** *Let  $E \subset \mathbb{R}^N$  and  $x \in \partial E$ . Assume that  $E$  has minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $r \in (0, d(x, \partial\Omega))$ . There exists  $\eta, \rho \in (0, 1)$  and  $\varepsilon_0 > 0$  such that, if*

$$\partial E \cap B(x, r) \subset \{|y_N| \leq \varepsilon\} \tag{4.71}$$

*for some  $\varepsilon \in (0, \varepsilon_0)$ , then*

$$\partial E \cap B(x, \rho r) \subset \{|y \cdot v| \leq \eta \rho \varepsilon\}, \tag{4.72}$$

*for some unit vector  $v \in S^{N-1}$ .*

**Proof** For a proof of Lemma 4.17 see Lemma 6.4 in [13].  $\square$

We use now Lemma 4.17 to sketch a proof of Theorem 4.16.

**Proof (Theorem 4.16)** Up to scaling everything by  $\frac{1}{r}$  and translating by  $-x$ , we can assume that  $x = 0$  and  $r = 1$ . Assume first that  $v = e_N$ . By Lemma 4.17 we know that there exist  $\eta, \rho \in (0, 1)$  and  $\varepsilon_0 > 0$  such that we have the following implication for some  $\varepsilon \in (0, \varepsilon_0)$ :

$$\partial E \cap B(0, 1) \subset \{|y_N| \leq \varepsilon\} \implies \partial E \cap B(0, \rho) \subset \{|y_N| \leq \eta \rho \varepsilon\}. \quad (4.73)$$

Consider now the rescaled set  $E_1 := \frac{E}{\rho}$ . Then we have, by the assumption of the theorem, that  $\partial E_1 \cap B(0, 1) \subset \{|y_N| \leq \eta \varepsilon\}$ . We have  $\rho \varepsilon \in (0, \varepsilon_0)$ , so we may apply Lemma 4.17 to  $E_1$  and obtain the following result:

$$\partial E_1 \cap B(0, \rho) \subset \{|y_N| \leq \eta^2, \rho \varepsilon\}. \quad (4.74)$$

In terms of  $E$ , the inclusion 4.74 then becomes

$$\partial E \cap B(0, \rho^2) \subset \{|y_N| \leq \eta^2, \rho^2 \varepsilon\}. \quad (4.75)$$

By iterating this process, we find for any  $k \in \mathbb{N}$ , that

$$\partial E \cap B(0, \rho^k) \subset \{|y_N| \leq \eta^k \rho^k \varepsilon\} = \{|y_N| \leq \rho^{(1+\alpha)k} \varepsilon\}, \quad (4.76)$$

Where we have chose  $\alpha > 0$  such that  $\rho^\alpha = \eta$ . Now, given any  $s \in (0, 1)$ , there exists  $k \in \mathbb{N}$  such that  $\rho^k \leq s \leq \rho^{k-1}$  and, therefore,

$$\begin{aligned} \partial E \cap B(0, s) &\subset \partial E \cap B(0, \rho^{k-1}) \\ &\subset \{|y_N| \leq \rho^{(1+\alpha)(k-1)} \varepsilon\} \\ &\subset \{|y_N| \leq \rho^{-(1+\alpha)} s^{1+\alpha} \varepsilon\}. \end{aligned} \quad (4.77)$$

Thus, we conclude that  $\partial E \cap B(0, s) \subset \{|y_N| \leq C \varepsilon s^{1+\alpha}\}$  for all  $s \in (0, 1]$  with  $C = \rho^{-(1+\alpha)}$ .

In general, we do not have  $v = e_N$ . In general, our iteration 4.76 will be of the form

$$\partial E \cap B(0, \rho^k) \subset \{|y \cdot v_k| \leq \rho^{(1+\alpha)k} \varepsilon\}, \quad (4.78)$$

for some sequence  $(v_k)_{k \in \mathbb{N}} \subset S^{N-1}$ . However, the inner products  $v_k \cdot v_{k+1}$  at each iteration step can be too far from 1 (see Figure 4.3), implying that

$$\|v_k - v_{k+1}\|^2 = 2(1 - v_k \cdot v_{k+1}) \quad (4.79)$$

becomes small as  $k \rightarrow \infty$ . So, the sequence  $(v_k)_{k \in \mathbb{N}}$  converges to some unit vector  $v \in S^{N-1}$ . Hence, the general bound inclusion we can obtain is:

$$\partial E \cap B(0, s) \subset \{|y \cdot v| \leq C \varepsilon s^{1+\alpha}\}, \quad s \in (0, 1]. \quad (4.80)$$

Let now  $z \in \partial E \cap B(0, \frac{1}{2})$ . Since  $B(z, \frac{1}{2}) \subset B(0, 1)$ , we must have that

$$\partial E \cap B\left(z, \frac{1}{2}\right) \subset \{|y_N| \leq \varepsilon\}. \quad (4.81)$$

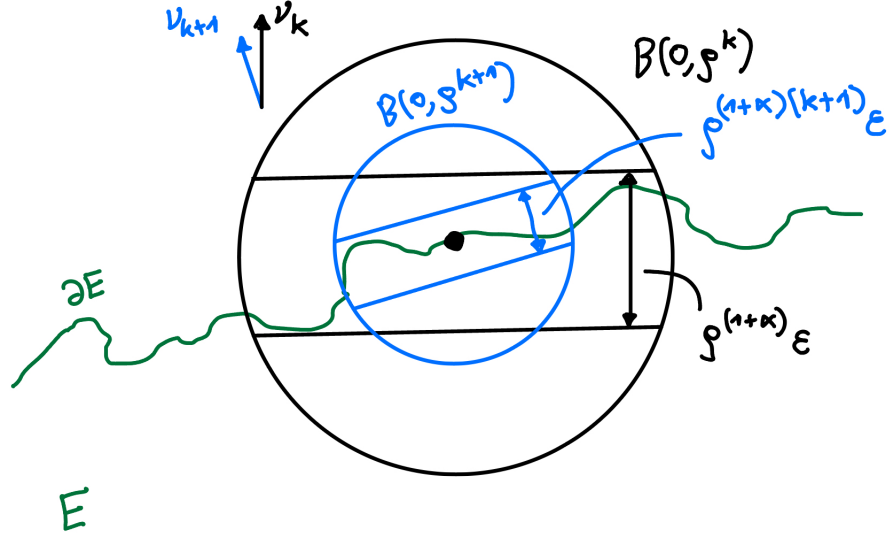


Figure 4.3:  $\nu_k \cdot \nu_{k+1}$  is not too large.

Suppose now  $\epsilon \leq \frac{\epsilon_0}{2}$  and set  $E_z := 2(E - z)$ . Then we find that by 4.81 we find that

$$\partial E_z \cap B(0, 1) \subset \{|y_N| \leq 2\epsilon\}. \quad (4.82)$$

So by the argument above applied to  $E_z$  in place of  $E$ , we obtain for any  $z \in \partial E \cap B(0, \frac{1}{2})$  that there is some unit vector  $\nu_z \in S^{N-1}$  such that:

$$\partial E \cap B(z, s) \subset \{|y - z \cdot \nu_z| \leq 2C\epsilon s^{1+\alpha}\}, \quad s \in \left(0, \frac{1}{2}\right). \quad (4.83)$$

From this we obtain, that  $\partial E \cap B(0, \frac{1}{2})$  is the graph of a  $C^{1,\alpha}$  function. Indeed,

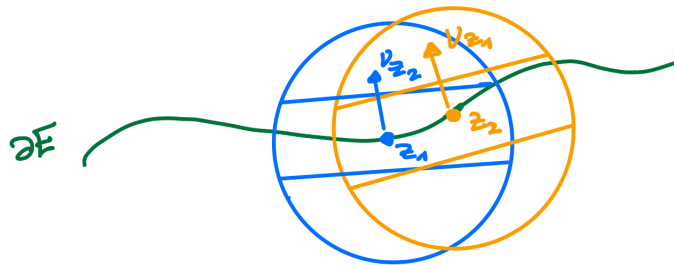


Figure 4.4:  $\partial E \cap B(0, \frac{1}{2})$  is a  $C^{1,\alpha}$  graph.

in Figure 4.4, it is evident, that if we take  $z_1, z_2 \in B(0, \frac{1}{2})$ , which are very close too eachother, that their normals  $\nu_{z_1}$  and  $\nu_{z_2}$  to  $\partial E$ , are in fact very close too eachother, which can only be the case if  $\partial E \cap B(0, \frac{1}{2})$  is the graph of a  $C^{1,\alpha}$  function.  $\square$

Now we have the following result due to Simon [14] classifying cones of minimal perimeter.

**Theorem 4.18** *Let  $N \geq 2$  and  $\Omega \subset \mathbb{R}^N$  be a bounded open set and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . If there is a cone  $\mathcal{C} \subset \mathbb{R}^N$  of minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $\Sigma(\mathcal{C}, \Omega) \neq \emptyset$ , then  $N \geq 8$ .*

**Remark 4.19** *Theorem 4.18 together with Remark ?? in fact implies that if  $\mathcal{C} \subset \mathbb{R}^N$ , for  $2 \leq N \leq 7$ , is a cone of minimal perimeter in  $\Omega$ , then its boundary must be a  $C^{1,\alpha}$  hypersurface, as locally by Remark ??  $\partial \mathcal{C}$  is the graph of a function  $C^{1,\alpha}$ , and since  $\mathcal{C}$  is scaling invariant, its boundary must globally be  $C^{1,\alpha}$ . This is in fact consistent with the result for soap-films by Taylor in [1], if we have a minimal soap-film, which is the interface between two sides, i.e. modeled by a set of minimal perimeter.*

## 4.6 Quasi-minimizer of perimeter

We now generalize our understanding of sets of minimal perimeter a bit with the concept of *quasi-minimizers of perimeter*. We will also discuss some properties, like density estimates and compactness, that we have seen in the preceding sections. This section will follow [11].

**Definition 4.20** *Let  $\xi : (0, \infty) \rightarrow (0, \infty)$  be an increasing continuous function with  $\lim_{r \rightarrow 0} \xi(r) = 0$  and  $\Omega \subset \mathbb{R}^N$  be an open set. We say that a  $\mathcal{L}^N$ -measurable set  $E \subset \mathbb{R}^N$  is a **quasi minimizer of perimeter in  $\Omega$  for gauge function  $\xi$**  if*

$$P(E, B(x, r)) \leq (1 + \xi(r))P(F, B(x, r)), \quad (4.84)$$

*whenever  $x \in \partial E \cap \Omega$  and  $E \Delta F \subset \subset B(x, r) \cap \Omega$ . We denote by  $\mathbf{M}_\xi(\Omega)$  the set of quasi-minimizers for the perimeter in  $\Omega$  for the gauge function  $\xi$ .*

We have already discussed the prototypical example of quasi-minizers of perimeter in the preceding sections. We call them sets of minimal perimeter. This is the content of the following proposition.

**Proposition 4.21** *Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set, and  $E_0 \subset \mathbb{R}^N$  be a set of locally finite perimeter in  $\mathbb{R}^N$ . If  $E \subset \mathbb{R}^N$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$ , then we have  $E \in \mathbf{M}_0(\Omega)$ .*

**Proof** Fix  $x \in \partial E \cap \Omega$ ,  $F \subset \mathbb{R}^N$  and  $r > 0$  such that  $E \Delta F \subset\subset B(x, r) \cap \Omega$ . We note that we have:

$$\begin{aligned} \text{supp}(\chi_F - \chi_{E_0}) &= \text{supp}((\chi_F - \chi_E) + (\chi_E - \chi_{E_0})) \\ &\subset \text{supp}(\chi_F - \chi_E) \cup \text{supp}(\chi_E - \chi_{E_0}) \\ &= (E \Delta F) \cup \text{supp}(\chi_E - \chi_{E_0}) \\ &\subset \Omega. \end{aligned} \tag{4.85}$$

This implies that  $F$  is a competitor for perimeter for  $E$  in  $\Omega$ . Since  $E$  has minimal perimeter in  $\Omega$ , it follows that  $P(E, \Omega) \leq P(F, \Omega)$ . Now, we can observe that:

$$\begin{aligned} P(E, B(x, r)) - P(F, B(x, r)) &= P(E \Delta F, B(x, r)) \\ &= P(E \Delta F, \Omega) \\ &= P(E, \Omega) - P(F, \Omega) \\ &\leq 0. \end{aligned} \tag{4.86}$$

This shows that  $E \in M_0(\Omega)$ .  $\square$

The next proposition establishes the rescaling and translation properties of quasi-minimizers, which will be crucial, when considering blowups.

**Proposition 4.22** *Let  $\Omega \subset \mathbb{R}^N$  be an open set,  $E \in M_{\xi}(\Omega)$ ,  $\lambda > 0$ , and  $x \in \mathbb{R}^N$ . Then we have  $\frac{E-x}{\lambda} \in M_{\xi'}(\frac{\Omega-x}{\lambda})$ , where  $\xi'(r) := \xi(\lambda r)$ .*

**Proof** Set  $E' = \frac{E-x}{\lambda}$  and  $\Omega' = \frac{\Omega-x}{\lambda}$ . Now choose  $y' \in \partial E'$ ,  $r > 0$ , and  $F' \subset \mathbb{R}^N$  such that  $E' \Delta F' \subset\subset B(y', r) \cap \tilde{\Omega}$ . Set  $F = \lambda F' + x$  and  $y = \lambda y' + x$ . Then we note that:

$$\begin{aligned} E \Delta F &= \lambda(E' \Delta F') + x \\ &\subset\subset \lambda(B(y', r) \cap \Omega') + x \\ &= B(y, \lambda r) \cap \Omega. \end{aligned} \tag{4.87}$$

Then it follows that:

$$\begin{aligned} P(E', B(y', r)) &= P(E, B(y, \lambda r)) \\ &\leq (1 + \xi(\lambda r))P(F, B(y, \lambda r)) \\ &= (1 + \xi'(r))P(F', B(y', r)). \end{aligned} \tag{4.88}$$

This shows that  $E' \in M_{\xi'}(\Omega')$ .  $\square$

Like sets of minimal perimeter, quasiminimizer also have an upper density estimate.

**Proposition 4.23** *Let  $\Omega \subset \mathbb{R}^N$  be an open set. If  $E \in M_{\xi}(\Omega)$ ,  $x \in \Omega$  and  $r \in (0, d(x, \partial\Omega))$ , then we have:*

$$P(E, B(x, r)) \leq c(1 + \xi(2r))r^{N-1}, \tag{4.89}$$

where  $c > 0$  only depends on  $N$ .

**Proof** We may assume that  $P(E, B(x, r)) \neq 0$ , otherwise the statment is trivially true. Fix now  $y \in \partial E \cap B(x, r)$  and  $s < r$ . Then we have:

$$\begin{aligned}
 P(E, B(x, r)) &\leq P(E, B(y, 2r)) \\
 &\stackrel{(*)}{\leq} (1 + \xi(2r))P(E \setminus B(y, 2s), B(y, 2r)) \\
 &\stackrel{(**)}{=} (1 + \xi(2r))(\mathcal{H}^{N-1}(E \cap \partial B(y, 2s)) + P(E, B(y, 2r) \setminus \bar{B}(y, 2s))) \\
 &\leq (1 + \xi(2r))(N\omega_N(2s)^{N-1} + P(E, B(y, 2r) \setminus \bar{B}(y, 2s))),
 \end{aligned} \tag{4.90}$$

where we used at  $(*)$  that  $E \in \mathbf{M}_\xi(\Omega)$ , and Example 3.70 at  $(**)$ . The result follows by taking the limit  $s \rightarrow r$ .  $\square$

We also have the following compactness result for quasi-minimizers.

**Proposition 4.24** *Let  $\Omega \subset \mathbb{R}^N$  be an open set and  $(E_h)_{h \in \mathbb{N}}$  be a sequence of quasi-minimizers  $E_h \in \mathbf{M}_{\xi_h}(\Omega)$ , such that  $(\xi_h)_{h \in \mathbb{N}}$  converges pointwise to an increasing function  $\xi$  with  $\lim_{r \rightarrow 0} \xi(r) = 0$ . Then, there exists a subsequence  $(E_{h_k})_{k \in \mathbb{N}}$  converging locally in  $\mathbb{R}^N$  to a set  $E \in \mathbf{M}_\xi(\Omega)$ .*

**Proof** By Proposition 4.23, we find for every  $h \in \mathbb{N}$  and every  $B(x, r) \subset\subset \Omega$ :

$$P(E, B(x, r)) \leq N\omega_N 2^{N-1} (1 + \xi_h(2r)) r^{N-1}. \tag{4.91}$$

Since  $\xi_h \rightarrow \xi$  pointwise, we must have that  $P(E_h, B(x, r))$  is uniformly bounded for every  $h$ . By a covering argument, we also find that  $P(E_h, A)$  is uniformly bouded for every  $h$  and every open subset  $A \subset\subset \Omega$ . By Theorem 3.25, we must have that there is a subsequence  $(E_{h_k})_{k \in \mathbb{N}}$  converging locally in  $\Omega$  to some set  $E$ . Now, fix  $x \in \partial E$ ,  $r > 0$  and  $F \subset \mathbb{R}^N$  such that  $E \Delta F \subset\subset B(x, r) \cap \Omega$ . By local convergence of  $E_{h_k}$  to  $E$  we may choose  $k_0 > 0$  such that  $E_{h_k} \Delta F \subset\subset B(x, r) \cap \Omega$  for all  $k \geq k_0$ . By lower semicontinuity of perimeter we find that

$$\begin{aligned}
 P(E, B(x, r)) &\leq \liminf_{k \rightarrow \infty} P(E_{h_k}, B(x, r)) \\
 &\leq \liminf_{k \rightarrow \infty} (1 + \xi_{h_k}(r)) P(F, B(x, r)) \\
 &= (1 + \xi(r)) P(F, B(x, r)).
 \end{aligned} \tag{4.92}$$

This shows that  $E \in \mathbf{M}_\xi(\Omega)$ .  $\square$

There are also similar bounds for the Lebesgue measure of quasi-minimizers of perimeter as in Theorem 4.6 for sets of minimal perimeter.

**Proposition 4.25** *Let  $\Omega \subset \mathbb{R}^N$  be an open set,  $E \in \mathbf{M}_\xi(\Omega)$ ,  $x \in \partial E \cap \Omega$ . Then for all  $r \in (0, d(x, \partial\Omega))$  it holds that:*

$$\frac{cr^N}{2 + \xi(r)} \leq \mathcal{L}^N(E \cap B(x, r)) \leq \left(1 - \frac{1}{(2 + \xi(r))^N}\right) Cr^N, \tag{4.93}$$

where  $c, C > 0$  only depend on  $N$ .

**Proof** Let  $\ell(r) := \mathcal{L}^N(E \cap B(x, r))$ . Then by the coarea formula it follows for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$ :

$$\max\{P(E \cap B(x, r), \partial B(x, r)), P(E \setminus B(x, r), \partial B(x, r))\} \leq \ell'(r). \quad (4.94)$$

By the Poincaré inequality (Theorem 3.34 (iii)) we know that:

$$\begin{aligned} \frac{1}{\gamma_5} \ell(r)^{\frac{N-1}{N}} &\leq P(E \cap B(x, r), B(x, r)) \\ &= P(E, B(x, r)) + P(E \cap B(x, r), \partial B(x, r)). \end{aligned} \quad (4.95)$$

Consider now  $F = E \setminus B(x, r)$ . Then we have  $E \Delta F \subset\subset B(x, s) \cap \Omega$  for all  $s > r$ . Since  $E$  is a quasi minimizer for perimeter in  $\Omega$  for the gauge function  $\xi$  we have that:

$$P(E, B(x, s)) \leq (1 + \xi(s))P(E \setminus B(x, r), B(x, s)). \quad (4.96)$$

By continuity of  $\xi$  we get for  $s \rightarrow r$  that

$$\begin{aligned} P(E, \bar{B}(x, r)) &\leq (1 + \xi(r))P(E \setminus B(x, r), \bar{B}(x, r)) \\ &= (1 + \xi(r))P(E \setminus B(x, r), \partial B(x, r)). \end{aligned} \quad (4.97)$$

By 4.95 and 4.97 we find for  $\mathcal{L}^1$ -a.e.  $r \in (0, d(x, \partial\Omega))$ :

$$\begin{aligned} \frac{1}{\gamma_5} \ell(r)^{\frac{N-1}{N}} &\leq (2 + \xi(r))P(E \setminus B(x, r), \partial B(x, r)) \\ &\leq (2 + \xi(r))\ell'(r). \end{aligned} \quad (4.98)$$

This gives us for some  $c > 0$  only depending on  $N$ :

$$\left(\ell^{\frac{1}{N}}\right)'(r) \geq \frac{c}{2 + \xi(r)}. \quad (4.99)$$

By integrating 4.99 from 0 to  $r$ , we get the lower bound of the theorem. By replacing  $E$  by  $\mathbb{R}^N \setminus E$  and arguing in the same way, we find the upper bound.  $\square$

Similarly to Theorem 4.6, we find a lower density bound for quasi-minimizers of perimeter.

**Proposition 4.26** *There exists a constant  $r_0 > 0$  only depending on the gauge function  $\xi$ , such that if  $E \in \mathbf{M}_\xi(\Omega)$ ,  $x \in \partial E \cap \Omega$  and  $r < r_0$ , then*

$$P(E, B(x, r)) \geq \frac{\omega_{N-1}}{2} r^{N-1}. \quad (4.100)$$

**Proof** Suppose for the sake of contradiction that there exist a sequence  $(E_h)_{h \in \mathbb{N}} \subset \mathbf{M}_\zeta(\Omega)$ , a sequence of points  $x_h \in \partial E_h \cap \Omega$  and a sequence of positive real numbers  $r_h$  with  $\lim_{h \rightarrow \infty} r_h = 0$  such that

$$P(E_h, B(x_h, r_h)) < \frac{\omega_{N-1}}{2} r_h^{N-1}. \quad (4.101)$$

By Proposition 4.22 we know that  $F_h := \frac{E - x_h}{r_h} \in \mathbf{M}_{\zeta_h}\left(\frac{\Omega - x_h}{r_h}\right)$  for  $\zeta_h(r) := \zeta(r_h r)$ . Note that  $\zeta_h \rightarrow 0$  as  $h \rightarrow \infty$  and so by Proposition 4.24, up to restricting to a subsequence, we can assume that  $F_h$  converges locally in  $\mathbb{R}^N$  to a set  $F \in \mathbf{M}_0(\mathbb{R}^N)$ . By lower semicontinuity of perimeter we find

$$\begin{aligned} P(F, B(0, 1)) &\leq \liminf_{h \rightarrow \infty} P(F_h, B(0, 1)) \\ &= \liminf_{h \rightarrow \infty} \frac{1}{r_h^{N-1}} P(E_h, B(x_h, r_h)) \\ &\leq \frac{\omega_{N-1}}{2}. \end{aligned} \quad (4.102)$$

By Proposition 4.25 there are constants  $c, C > 0$  depending only on  $N$  and  $r$  such that  $cr^n \leq \mathcal{L}^N(F \cap B(0, r)) \leq Cr^N$ . Since  $\lim_{h \rightarrow \infty} \mathcal{L}^N(F_h \cap B(0, r)) = \mathcal{L}^N(F \cap B(0, r))$ , it must follow that  $0 \in \partial F$ . Since  $F$  is a set of minimal perimeter, it follows that  $P(F, B(0, 1)) \geq \omega_{N-1}$ , which is in contradiction to 4.102.  $\square$

Finally, we have the following result for the reduced boundary of quasi-minimizers of perimeter in  $\mathbb{R}^N$ .

**Corollary 4.27** *Let  $\Omega \subset \mathbb{R}^N$  be an open set. If  $E \in \mathbf{M}_\zeta(\Omega)$ , then*

$$\mathcal{H}^{N-1}(\Omega \cap (\partial E \setminus \partial^* E)) = 0. \quad (4.103)$$

*In particular, we have that  $\|D\chi_E\| = \mathcal{H}^{N-1} \llcorner \partial E \cap \Omega$  and in Proposition 4.24 we have convergence of  $\partial E_h$  to  $\partial E$  in the following sense:*

- (i) *If  $(x_h)_{h \in \mathbb{N}}$  is a convergent sequence of points  $x_h \in \partial E_h \cap \Omega$  with limit  $x \in \Omega$ , then  $x \in \partial E \cap \Omega$ .*
- (ii) *If  $x \in \partial E \cap \Omega$ , then there exists a sequence  $(x_h)_{h \in \mathbb{N}}$  of points  $x_h \in \partial E_h \cap \Omega$  converging to  $x$ .*

**Proof** By Proposition 4.26, we find that the measure  $\mathcal{H}^{N-1} \llcorner \partial^* E = P(E, \cdot)$  has a positive  $(N-1)$ -dimensional density on the set  $\partial E$  and so in particular on the set  $\partial E \setminus \partial^* E$ . By Theorem 2.54, we then know that for any Borel set  $A \subset \mathbb{R}^N$  with  $\mathcal{H}^{N-1}(A) < \infty$ ,  $\mathcal{H}^{N-1} \llcorner A$  has 0  $(N-1)$ -dimensional density in  $\mathcal{H}^{N-1}$ -a.e. points of  $\mathbb{R}^N \setminus A$ . Applying this to  $A = \partial E \cap \Omega$ , we obtain  $\mathcal{H}^{N-1}(\Omega \cap (\partial E \setminus \partial^* E)) = 0$ .

If now  $E_h$  converges to  $E$  and satisfies all the assumptions of Proposition 4.24,

any point of  $\partial E$  can be approximated by points of  $\partial E_h$ , by the convergence of  $D\chi_{E_h}$  to  $D\chi_E$ , which proves (ii). By Proposition 4.26 and the convergence of  $\|D\chi_{E_h}\|$  to  $\|D\chi_E\|$ , it follows that every limit point of  $\partial E_h$  belongs to  $\partial E$ , proving (i).  $\square$

To end this final discussion of perimeter minimizer, we will state an extended partial regularity result for quasi-minimizers, which is even more along the lines of Taylor's results for soap-film regularity. The proof of this result can be looked up in [15].

**Theorem 4.28** *Let  $\Omega \subset \mathbb{R}^N$  be an open set and suppose that  $\alpha \in (0, \frac{1}{2}]$  and  $\xi(r) = cr^{2\alpha}$ . There exists a constant  $C > 1$ , depending only on  $N$  and  $\xi$ , such that if  $E \in M_\xi(\Omega)$ ,  $x \in \partial E \cap \Omega$  and  $r \in (0, 1)$ , then there are  $y \in \partial E$ ,  $t \geq \frac{r}{C}$  with  $B(y, t) \subset B(x, r)$  and such that  $\partial E \cap B(y, t)$  is the graph of  $C^{1,\alpha}$  function.*

**Remark 4.29** *This is very much along the lines of Taylor's results in [1]. There they consider exactly Almgren minimal sets as in Definition 5.3 with gauge functions exactly of the form as in Theorem 4.28. In essence, Theorem 4.28 states that locally a soap-film-like minimal surface consisting only of one sheet, i.e. the perimeter of a set of minimal perimeter, is a  $C^{1,\alpha}$  hypersurface.*

This concludes our discussion of quasi-minimizers of perimeter and this chapter on minimal variation and minimal perimeter.

## Chapter 5

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# Almost minimal sets and the Plateau problem

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In this, we discuss minimal sets in the sense of Almgren and Taylor in [16] and [1]. Moreover, we delve further into the theory of almost-minimality in the sense discussed in [2].

### 5.1 Plateau's problem for spanning sets

This section is concerned with the modeling of soap films by so-called *spanning sets* and the related Plateau problem. The goal is to understand what it means for a set to be spanning with respect to some contour, modeling the physical case in three-dimensional Euclidean space of a soap-film spanning a wire. This section will follow the exposition given in [17]. This section mainly establishes some kind of existence of *minimal* soap-films, which are of interest to us.

**Definition 5.1** Let  $N \geq 2$ ,  $\Gamma \subset \mathbb{R}^{N+1}$  be a closed set and set  $\Omega := \mathbb{R}^{N+1} \setminus \Gamma$ . A **spanning class** for  $\Gamma$  is collection  $\mathcal{C}$  of smooth embeddings  $S^1 \rightarrow \Omega$  closed under homotopy, i.e. if  $\{\varphi_t\}_{t \in [0,1]}$  is a continuous one-parameter family of such that  $\varphi_0 \in \mathcal{C}$ , then  $\varphi_t \in \mathcal{C}$  for all  $t \in (0,1]$ . Given a spanning class  $\mathcal{C}$  for  $\Gamma$ , we call a relatively closed subset  $E \subset \Omega$   **$\mathcal{C}$ -spanning** if

$$E \cap \gamma(S^1), \text{ for all } \gamma \in \mathcal{C}. \quad (5.1)$$

Denote by  $\mathcal{F}(\Gamma, \mathcal{C})$  the family of all  $\mathcal{C}$ -spannings set of  $\Gamma$ .

The next theorem is part of a theorem in [17] establishing the existence of a solution to the Plateau problem for spanning sets.

**Theorem 5.2** Let  $N \geq 2$ ,  $\Gamma \subset \mathbb{R}^{N+1}$  be a closed set and set  $\Omega := \mathbb{R}^{N+1} \setminus \Gamma$  and let  $\mathcal{C}$  be a spanning class for  $\Gamma$ . Assume that

$$\inf\{\mathcal{H}^N(E) \mid E \in \mathcal{F}(\Gamma, \mathcal{C})\} < \infty. \quad (5.2)$$

Then the following holds:

- (i) There is a minimizing sequence of  $\mathcal{H}^N$ -rectifiable sets  $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{F}(\Gamma, \mathcal{C})$  for 5.2. Moreover, the sequence of restricted Hausdorff measure  $\mathcal{H}^n \llcorner E_j$  locally weakly\* converges on  $\Omega$  to a Radon measure  $\mu = \theta \mathcal{H}^N \llcorner E$ , where  $\theta \geq 1$  and  $E = \text{supp}(\mu) \setminus \Gamma$  is a rectifiable minimizer for 5.2, i.e.  $E \in \mathcal{F}(\Gamma, \mathcal{C})$ .
- (ii) The set  $E$  from (i) is a  $(M, 0, \infty)$ -minimal set in  $\Omega$  in the sense of Almgren.

### 5.1.1 Almgren minimal sets

To finish this short section, we explain what (ii) in Theorem 5.2 actually means. To this end, we introduce the concept of Almgren minimal sets as described in [1].

**Definition 5.3** Let  $N \geq 2$ ,  $1 \leq k \leq N$ ,  $\Gamma \subset \mathbb{R}^{N+1}$  be a closed set and set  $\Omega := \mathbb{R}^{N+1} \setminus \Gamma$ . A relatively closed and  $\mathcal{H}^k$ -rectifiable subset  $E \subset \Omega$  is called  **$(M, \xi, \delta)$ -minimal in  $\Omega$  of dimension  $m$**  for non-decreasing continuous function  $\xi : (0, \infty) \rightarrow [0, \infty)$  with  $\lim_{r \rightarrow \infty} \xi(r) = 0$  and  $\delta > 0$ , if

- (i) For any Lipschitz continuous  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  such that

$$d(W \cup \varphi(W), \Gamma) > 0 \quad (5.3)$$

$$r := \text{diam}(W \cup \varphi(W)) < \delta \quad (5.4)$$

where  $W := \{x \in \mathbb{R}^{N+1} \mid \varphi(x) \neq 0\}$ , it holds that:

$$\mathcal{H}^k(E \cap W) \leq (1 + \xi(r)) \mathcal{H}^k(\varphi(E \cap W)). \quad (5.5)$$

- (ii)  $E = \text{supp}(\mathcal{H}^k \llcorner E) \setminus \Gamma$ .

We say that  $E$  is  **$(M, 0, \infty)$ -minimal in  $\Omega$  of dimension  $k$**  if  $E$  is  $(M, 0, \delta)$ -minimal in  $\Omega$  of dimension  $k$  for all  $\delta > 0$ .

## 5.2 Minimal sets

In the following we introduce a new kind of minimality, which is less restrictive than the Almgren minimality from Definition 5.3. This will enable us to prove a monotonicity result, i.e. that a function

$$r \rightarrow \Theta_k(E, x)(r) = \frac{1}{\omega_k r^k} \mathcal{H}^k(E \cap B(x, r))$$

is non-decreasing for minimal set  $E \subset \mathbb{R}^{N+1}$  in this new sense. This section is closely based on the treatment in [2].

As promised, we will start with the definition of a broader sense of minimality of sets in open subsets in  $\mathbb{R}^{N+1}$ .

**Definition 5.4** Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset. Fix a relatively closed and  $\mathcal{H}^k$ -rectifiable subset  $E \subset \Omega$ . Then we define:

- (i) A **competitor for  $E$  in a compact set  $K \subset \Omega$**  is a set  $F = \varphi(E)$ , where  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  is a Lipschitz continuous with
  - $\varphi(x) = x$  for all  $x \in \mathbb{R}^{N+1} \setminus K$ , and
  - $\varphi(K) \subset K$ .
- (ii) A **gauge function** is a non-decreasing continuous function  $\xi : (0, \infty) \rightarrow [0, \infty)$  such that  $\lim_{r \rightarrow \infty} \xi(r) = 0$ .
- (iii)  $E$  is called **almost minimal in  $\Omega$  of dimension  $k$  for the gauge function  $\xi$**  if for any compact set  $K \subset \Omega$  with  $r := \text{diam}(K)$  and any competitor for  $E$  in  $K$  we have the following:

$$\mathcal{H}^k(E \cap K) \leq \mathcal{H}^k(\varphi(E \cap K)) + r^m \xi(r). \quad (5.6)$$

- (iv)  $E$  is called **minimal in  $\Omega$  of dimension  $k$**  if  $E$  is almost minimal in  $\Omega$  of dimension  $k$  for gauge function  $\xi = 0$ .
- (v) We say  $E$  is **reduced** if  $\mathcal{H}^k(E \cap B(x, r)) > 0$  for all  $x \in E$  and  $r > 0$ .

First, we make the observation that small parts of an almost minimal set are still almost minimal.

**Proposition 5.5** Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset and  $E \subset \Omega$  be an almost minimal set in  $\Omega$  of dimension  $m$  for gauge function  $\xi$ . Then it holds for any  $x \in \Omega$  and  $r > 0$  such that  $B(x, r) \subset \Omega$  that  $E \cap B(x, r)$  is almost minimal in  $B(x, r)$  of dimension  $k$  for the gauge function  $\xi$ .

**Proof** Fix  $x \in \Omega$  and  $r > 0$  so that  $B(x, r) \subset \Omega$ . Let  $K \subset B(x, r)$  be compact and  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  be Lipschitz continuous such that  $\varphi(y) = y$  for all  $y \in \mathbb{R}^{N+1} \setminus K$  and  $\varphi(K) \subset K$ . Then  $\varphi(E)$  is a competitor for  $E$  in  $K$  and so by almost minimality of  $E$  it follows by 5.6:

$$\mathcal{H}^k(E \cap K) \leq \mathcal{H}^k(\varphi(E \cap K)) + r^k \xi(r), \quad (5.7)$$

where  $r = \text{diam}(K)$ . Since  $K \subset B(x, r)$  we find for  $\tilde{E} = E \cap B(x, r)$  that:

$$\begin{aligned} \mathcal{H}^k(\tilde{E} \cap K) &= \mathcal{H}^k(E \cap B(x, r) \cap K) \\ &\stackrel{K \subset B(x, r)}{=} \mathcal{H}^m(E \cap K) \\ &\stackrel{5.7}{\leq} \mathcal{H}^k(\varphi(E \cap K)) + r^k \xi(r) \\ &\stackrel{K \subset B(x, r)}{=} \mathcal{H}^k(\varphi(E \cap B(x, r) \cap K)) + r^k \xi(r) \\ &= \mathcal{H}^k(\varphi(\tilde{E} \cap K)) + \xi(r) r^k. \end{aligned} \quad (5.8)$$

Since  $\varphi(\tilde{E})$  is a competitor of  $\tilde{E}$  in  $K$ , 5.8 shows that  $\tilde{E} = E \cap B(x, r)$  is almost minimal in  $B(x, r)$  for the gauge function  $\xi$ .  $\square$

We now show a useful equivalent form of the inequality 5.6.

**Proposition 5.6** *Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset. Let  $E \subset \Omega$  be relatively closed and  $\mathcal{H}^k$ -rectifiable. Then  $E$  is almost minimal in  $\Omega$  of dimension  $k$  for gauge function  $\xi$  if and only if for every compact set  $K \subset \Omega$  with diameter  $\text{diam}(K) = r$  and every competitor  $F$  for  $E$  in  $K$  it holds that:*

$$\mathcal{H}^k(E \setminus F) \leq \mathcal{H}^k(F \setminus E) + \xi(r)r^k. \quad (5.9)$$

In order to prove this proposition, we will need one lemma from point-set topology, one from classical analysis, and one from measure theory.

**Lemma 5.7** *Let  $N \geq 1$  and  $U \subset \mathbb{R}^N$  be an open subset. If  $C \subset U$  is compact then there exists a compact subset  $D \subset U$  such that  $C \subset D^\circ \subset D$ .*

**Proof** For every  $x \in C$ , since  $x \in U$  is open we can choose closed ball  $\bar{B}(x, r_x) \subset U$  for some radius  $r_x > 0$  (just choose a open ball around  $x$  contained in  $U$  and then a closed ball of smaller radius). Then  $\{B(x, r_x)\}_{x \in C}$  is an open cover of  $C$ . By the compactness of  $C$  there is a finite subset  $S \subset C$  such that  $C \subset \bigcup_{x \in S} B(x, r_x)$ . Now set  $D := \bigcup_{x \in S} \bar{B}(x, r_x)$ . Then, by construction,  $D \subset U$  is compact with  $C \subset D^\circ \subset D$ .  $\square$

**Lemma 5.8** *Let  $U \subset \mathbb{R}^N$  be an open set and  $C \subset U$  be a compact set. Then there exists a smooth function  $\psi : \mathbb{R}^N \rightarrow \mathbb{R}$  such that*

- (i)  $0 \leq \psi(x) \leq 1$  for all  $x \in \mathbb{R}^N$ .
- (ii)  $\psi(x) = 1$  for all  $x \in C$ .
- (iii)  $\psi(x) = 0$  for all  $x \in \mathbb{R}^N \setminus U$ .

**Lemma 5.9** *Let  $\mu$  be a Radon measure of  $\mathbb{R}^N$ ,  $U \subset \mathbb{R}^N$  be an open set, and  $C \subset U$  compact. For every  $\varepsilon > 0$  there exists a compact set  $D_\varepsilon \subset U$  such that  $C \subset D_\varepsilon^\circ \subset D_\varepsilon$  and  $\mu(D_\varepsilon \setminus C) < \varepsilon$ .*

**Proof** For every  $t > 0$  define

$$D_t := \{x \in \mathbb{R}^N \mid d(x, C) \leq t\}. \quad (5.10)$$

By the compactness of  $C$  we have that  $d(\cdot, C)$  is a continuous function on  $\mathbb{R}^N$  and so  $D_t$  is compact for all  $t > 0$ . Clearly, we have  $C = \{x \in \mathbb{R}^N \mid d(x, C) = 0\} \subset D_t^\circ$  for all  $t > 0$  and  $C = \bigcap_{t>0} D_t$ . So it follows that:

$$\lim_{t \rightarrow 0} \mu(D_t) = \mu(C). \quad (5.11)$$

In particular, we can choose  $t_0 > 0$  such that  $\mu(D_{t_0} \setminus C) < \varepsilon$ . Now set  $D_\varepsilon := D_{t_0} \cap U$ . Then by the above, we have  $D_\varepsilon \subset U$  compact with  $C \subset D_\varepsilon^\circ \subset D_\varepsilon$  and  $\mu(D_\varepsilon \setminus C) < \varepsilon$ .  $\square$

This enables us now to prove the proposition 5.6.

**Proof (Proposition 5.6)** Fix  $K \subset \Omega$  be a compact set with  $\text{diam}(K) = r$  and let  $F = \varphi(E)$  be a competitor for  $E$  in  $\Omega$ , that is  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  is a Lipschitz continuous function with Lipschitz constant  $C > 0$  such that

$$\varphi(x) = x, \quad \forall x \in \mathbb{R}^{N+1} \setminus K \quad (5.12)$$

$$\varphi(K) \subset K. \quad (5.13)$$

Also, define  $W := \{x \in \mathbb{R}^{N+1} \mid \varphi(x) \neq x\}$ . Then, by the definition of the competitor, we have  $W \subset K$ .

Assume now first that competitor  $F = \varphi(E)$  for  $E$  in  $K$  satisfies 5.9. Set now first  $E_1 := E \cap K$  and  $E_2 := E \setminus K$ . Then we have by assumption:

$$\begin{aligned} \mathcal{H}^k(E_1) &= \mathcal{H}^k(E_1 \cap F) + \mathcal{H}^k(E_1 \setminus F) \\ &\stackrel{E_1 \subset E}{\leq} \mathcal{H}^k(E_1 \cap F) + \mathcal{H}^k(E \setminus F) \\ &\stackrel{5.9}{\leq} \mathcal{H}^k(E_1 \cap F) + \mathcal{H}^k(F \setminus E) + \zeta(r)r^k. \end{aligned} \quad (5.14)$$

Since  $W \subset K$ , we have  $E \cap W \subset E_1$ . This implies that on  $E_2$  we have  $\varphi(x) = x$  and thus  $E_2 = \varphi(E_2) \subset F$ . Now observe that:

$$F = \varphi(E) = \varphi(E_1) \cup \varphi(E_2), \quad (5.15)$$

and consequently

$$\begin{aligned} E_1 \cap F &= E_1 \cap [\varphi(E_1) \cup \varphi(E_2)] \\ &= [E_1 \cap \varphi(E_1)] \cup [E_1 \cap E_2] \\ &= E_1 \cap \varphi(E_1). \end{aligned} \quad (5.16)$$

Similarly, we find that:

$$\begin{aligned} F \setminus E &= [\varphi(E_1) \cup \varphi(E_2)] \setminus E \\ &= [\varphi(E_1) \setminus E] \cup [E_2 \setminus E] \\ &= \varphi(E_1) \setminus E. \end{aligned} \quad (5.17)$$

Putting 5.14, 5.16 and 5.17 together we find:

$$\begin{aligned} \mathcal{H}^k(E \cap K) &= \mathcal{H}^k(E_1) \\ &\stackrel{5.14}{\leq} \mathcal{H}^k(E_1 \cap F) + \mathcal{H}^k(F \setminus E) + \zeta(r)r^k \\ &\stackrel{5.16 \text{ and } 5.17}{\leq} \mathcal{H}^k(E_1 \cap \varphi(E_1)) + \mathcal{H}^k(\varphi(E_1) \setminus E) + \zeta(r)r^k \\ &\stackrel{E_1 \subset E}{\leq} \mathcal{H}^k(E_1 \cap \varphi(E_1)) + \mathcal{H}^k(\varphi(E_1) \setminus E_1) + \zeta(r)r^k \\ &= \mathcal{H}^k(\varphi(E_1)) + \zeta(r)r^k \\ &= \mathcal{H}^k(\varphi(E \cap K)) + \zeta(r)r^k. \end{aligned} \quad (5.18)$$

This shows that if 5.9 holds for all competitors of  $E$ , then  $E$  is almost minimal in  $\Omega$  of dimension  $k$  for the gauge function  $\xi$ .

Conversely, suppose that  $E$  is almost minimal in  $\Omega$  of dimension  $k$  for the gauge function  $\xi$ . By Lemma 5.7 we can choose a compact subset  $B \subset \Omega$  such that  $W \subset K \subset B^\circ \subset B$  with  $\text{diam}(B) = \delta$ . By Lemma 5.8 we can choose a smooth function  $\psi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}$  such that

$$0 \leq \psi(x) \leq 1, \quad \forall x \in \mathbb{R}^{N+1} \quad (5.19)$$

$$\psi(x) = 1, \quad \forall x \in K \quad (5.20)$$

$$\psi(x) = 0, \quad \forall x \in \mathbb{R}^{N+1} \setminus B. \quad (5.21)$$

Now for every  $h \in \mathbb{N}$  define the Lipschitz continuous function  $\varphi_h : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by

$$\varphi_h(x) := \varphi(x) + \varepsilon_h \psi(\varphi(x))v, \quad (5.22)$$

where  $(\varepsilon_h)_{h \in \mathbb{N}}$  is a monotone decreasing sequence of positive real numbers with  $\lim_{h \rightarrow \infty} \varepsilon_h = 0$ , and  $v \in \partial B(0, 1)$  a unit vector. Note that the Lipschitz continuous functions  $\varphi_h$  have the following properties:

- For every  $x \in \mathbb{R}^{N+1} \setminus B$  we have  $\varphi_h(x) = x$  as  $\varphi(x) = x$  outside of  $K$  and  $K \subset B$  and  $\psi(x) = 0$  outside of  $B$ .
- For all  $x \in \mathbb{R}^{N+1}$  we find that:

$$\|\varphi_h(x) - \varphi(x)\| = \|\varepsilon_h \psi(\varphi(x))v\| \leq \varepsilon_h. \quad (5.23)$$

- We can choose  $h_0 \in \mathbb{N}$  such that for all  $h \geq h_0$  it holds that  $\varphi_h(B) \subset B$ . Indeed, let  $x \in B$ . Since  $\varphi(K) \subset K$  and  $\varphi(x) = x$  outside of  $K$  we have that  $\varphi(x) \in B$ . By continuity we have if  $\varphi(x) \in \partial B$  that  $\psi(\varphi(x)) = 0$  and so  $\varphi_h(x) = \varphi(x) \in B$ . By Lemma 5.7 and compactness of  $\varphi(B) \subset B$  there is open set  $V \subset B$  such that  $\varphi(B) \subset V$ . Choose now open balls  $B(\varphi(x), r_x) \subset V$  about  $\varphi(x)$  for every  $x \in B$ . By compactness of  $\varphi(B)$  there are finitely many elements  $x_1, \dots, x_\ell \in B$  such that  $\varphi(B) \subset \bigcup_{i=1}^{\ell} B(\varphi(x_i), r_{x_i})$ . Since  $\varepsilon_h \xrightarrow{h \rightarrow \infty} 0$  and  $V \subset B$  is open, for every  $i = 1, \dots, \ell$  we can choose  $h_i \in \mathbb{N}$  such that  $B(\varphi(x_i), r_{x_i}) \subset \varphi_h(B(\varphi(x_i), r_{x_i})) \subset V$  for all  $h \geq h_i$ . Now set  $h_0 = \max_{i=1, \dots, \ell} h_i$ . Then as claimed for every  $h \geq h_0$  we have  $\varphi_h(B) \subset B$ .

This shows that for every  $h \geq h_0$ , we have that  $\varphi_h(E)$  is a competitor for  $E$  in  $B$  and so we have:

$$\mathcal{H}^k(E \cap B) \leq \mathcal{H}^k(\varphi_h(E \cap B)) + \xi(\delta)\delta^k. \quad (5.24)$$

Now, in analogy to  $W$  define  $W_h := \{x \in \mathbb{R}^{N+1} \mid \varphi_h(x) \neq x\}$  for every  $h \in \mathbb{N}$ . Also set  $A = B \setminus K$  and  $E_1 = E \cap K$  and  $E_2 = E \setminus B$ . We now make the following observations:

- For all  $x \in E_2$  we have  $\varphi(x) = x$  as  $x \notin K$  and  $\psi(\varphi(x)) = 0$  as  $x \notin B$ . Therefore,  $\varphi_h(x) = x$ .
- It is  $\varphi(E_1) \subset K$ , as  $E_1 \subset K$  and  $\varphi(K) \subset K$ .
- Consequently, we must have  $\varphi_h(E_1) \subset B$  for all  $h \geq h_0$ , as  $K \subset B$ .
- In particular,  $\varphi_h(E_1)$  and  $E_2 = \varphi_h(E_2)$  are disjoint.
- Furthermore,  $W_h$  and  $E_2$  are disjoint by definition of  $W_h$ .
- For all  $h \geq h_0$  and  $x \in E_1$  we then find:

$$\begin{aligned} \varphi_h(x) - \varphi(x) &= \varepsilon_h \psi(\varphi(x))v \\ &\stackrel{\varphi(x) \in K}{=} \varepsilon_h v. \end{aligned} \quad (5.25)$$

This shows that  $\varphi_h - \varphi$  is constant on  $E_1$  for all  $h \geq h_0$ .

Now for any  $h \in \mathbb{N}$  we claim that

$$E_1 \setminus W_h \subset Z_h := \{x \in E_1 \mid 0 < \|\varphi(x) - x\| \leq \varepsilon_h\}. \quad (5.26)$$

Indeed, let  $x \in E_1 \setminus Z_h$ . If  $\varphi(x) = x$ , then as  $\varphi(x) \in K$ , it follows that  $\varphi_k(x) = x + \varepsilon_h v \neq x$ , that is,  $x \in W_h$ . On the other hand if  $\|\varphi(x) - x\| > \varepsilon_h$ , then by the reverse triangle inequality we find

$$\begin{aligned} \|\varphi_h(x) - x\| &\geq \|x - \varphi(x)\| - \|\varphi_h(x) - \varphi(x)\| \\ &\stackrel{5.23}{>} \varepsilon_h - \varepsilon_h = 0. \end{aligned} \quad (5.27)$$

Again, we have  $x \in W_h$ . This shows that  $W_k \subset E_1 \setminus Z_k$ . Therefore,  $E_1 \setminus W_k \subset Z_k$ , showing 5.26.

Using 5.24, 5.25 and 5.26 we can now show for every  $h \geq h_0$ :

$$\begin{aligned} \mathcal{H}^k(E_1) &= \mathcal{H}^k(E_1 \cap W_h) + \mathcal{H}^k(E_1 \setminus W_h) \\ &\stackrel{5.26}{\leq} \mathcal{H}^k(E \cap B) + \mathcal{H}^k(Z_h) \\ &\stackrel{5.24}{\leq} \mathcal{H}^k(\varphi_h(E \cap B)) + \mathcal{H}^k(Z_h) + \zeta(r)r^k \\ &= \mathcal{H}^k(\varphi_h(E \cap (K \cup A))) + \mathcal{H}^k(Z_h) + \zeta(r)r^k \\ &\leq \mathcal{H}^k(\varphi_h(E_1)) + \mathcal{H}^k(\varphi_h(E \cap A)) + \mathcal{H}^k(Z_h) + \zeta(\delta)\delta^k \\ &\stackrel{5.25}{=} \mathcal{H}^k(\varphi(E_1)) + \mathcal{H}^k(\varphi_h(E \cap A)) + \mathcal{H}^k(Z_h) + \zeta(\delta)\delta^k. \end{aligned} \quad (5.28)$$

Observe now that since  $E_2 = E \setminus B \subset E$  and  $\varphi(E_2) = E_2$  we have  $E_2 \subset \varphi(E)$ , we have if  $x \in E \setminus \varphi(E)$ , then in particular  $x \notin E_2$  and so  $x \in B$ . This shows that  $E \setminus \varphi(E) = (E \cap B) \setminus \varphi(E)$  and so:

$$\mathcal{H}^k(E \setminus \varphi(E)) = \mathcal{H}^k(E \cap B) - \mathcal{H}^k(E \cap \varphi(E) \cap B). \quad (5.29)$$

Similarly, we have  $\varphi(E) \setminus E \supset (\varphi(E) \cap B) \setminus E$  and therefore:

$$\mathcal{H}^k(\varphi(E) \setminus E) \geq \mathcal{H}^k(\varphi(E) \cap B) - \mathcal{H}^k(E \cap \varphi(E) \cap B). \quad (5.30)$$

Recall now that  $F = \varphi(E)$  and set:

$$\Delta := \mathcal{H}^k(E \setminus F) - \mathcal{H}^k(F \setminus E). \quad (5.31)$$

With 5.29 and 5.30 we have just shown that:

$$\begin{aligned} \Delta &\leq \mathcal{H}^k(E \cap B) - \mathcal{H}^k(E \cap \varphi(E) \cap B) - \mathcal{H}^k(\varphi(E) \cap B) + \mathcal{H}^k(E \cap \varphi(E) \cap B) \\ &= \mathcal{H}^k(E \cap B) + \mathcal{H}^k(\varphi(E) \cap B). \end{aligned} \quad (5.32)$$

Because of 5.28 and since  $\varphi(E_1) \subset \varphi(E) \cap B$ :

$$\begin{aligned} \Delta &\stackrel{5.31}{\leq} \mathcal{H}^k(E \cap B) + \mathcal{H}^k(\varphi(E) \cap B) \\ &\leq \mathcal{H}^k(E \cap B) - \mathcal{H}^k(\varphi(E_1)) \\ &= \mathcal{H}^k(E_1 \cup (E \cap A)) - \mathcal{H}^k(\varphi(E_1)) \\ &= \mathcal{H}^k(E_1) + \mathcal{H}^k(E \cap A) - \mathcal{H}^k(\varphi(E_1)) \\ &\stackrel{5.28}{\leq} \mathcal{H}^k(E \cap A) + \mathcal{H}^k(\varphi_h(E \cap A)) + \mathcal{H}^k(Z_h) + \xi(\delta)\delta^k. \end{aligned} \quad (5.33)$$

Note that for every  $h \in \mathbb{N}$  we can write  $\varphi_h = L_h \circ \varphi$ , where  $L_h : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  is the function defined by:

$$L_h(x) := x + \varepsilon_h \psi(x)v. \quad (5.34)$$

Then  $L_h$  is Lipschitz. Indeed, since  $\psi$  is smooth, for any  $x, y \in \mathbb{R}^{N+1}$ , by the mean value theorem, we can find  $t \in [0, 1]$  such that for  $z = (1-t)x + ty$  it holds that:

$$\|\psi(x) - \psi(y)\| = \|\nabla \psi(z)\| \|x - y\|. \quad (5.35)$$

Since  $\psi$  is compactly supported, its gradient vector field  $\nabla \psi$  is bounded (as  $\nabla \psi$  is continuous on the support of  $\psi$ ), so we find for all  $x, y \in \mathbb{R}^{N+1}$ :

$$\|\psi(x) - \psi(y)\| \leq \left( \sup_{z \in \mathbb{R}^{N+1}} \|\nabla \psi(z)\| \right) \|x - y\|. \quad (5.36)$$

Since  $\lim_{h \rightarrow \infty} \varepsilon_h = 0$  we can choose  $\ell_0 \in \mathbb{N}$  such that for all  $h \geq \ell_0$  it holds that:

$$\varepsilon_h \left( \sup_{z \in \mathbb{R}^{N+1}} \|\nabla \psi(z)\| \right) \leq 1. \quad (5.37)$$

Then for all  $h \geq \ell_0$  we have that:

$$\begin{aligned} \|L_h(x) - L_h(y)\| &\leq \|x - y\| + \varepsilon_h \|\psi(x) - \psi(y)\| \\ &\leq \|x - y\| + \varepsilon_h \left( \sup_{z \in \mathbb{R}^{N+1}} \|\nabla \psi(z)\| \right) \|x - y\| \\ &\leq 2 \|x - y\|. \end{aligned} \quad (5.38)$$

So for all  $h \geq \ell_0$ ,  $L_h$  is Lipschitz continuous with Lipschitz constant 2. Let now  $C_\varphi = 2C^m$ .  $C_\varphi$  is a constant only depending on  $\varphi$  and  $m$  such that for all  $k \geq \ell_0$  by Proposition 2.51 (iv) we have the following:

$$\mathcal{H}^k(\varphi_h(E \cap A)) \leq C_\varphi \mathcal{H}^k(E \cap A). \quad (5.39)$$

For all  $h \geq \max\{h_0, \ell_0\}$ , it then follows by 5.33 and 5.39 that:

$$\Delta \leq (1 + C_\varphi) \mathcal{H}^k(E \cap A) + \mathcal{H}^k(Z_h) + \zeta(\delta) \delta^k. \quad (5.40)$$

Since  $\varepsilon_k$  is a monotone decreasing sequence converging to 0 we have that  $\lim_{h \rightarrow \infty} \mathcal{H}^k(Z_h) = 0$ . So let  $\eta > 0$ . Then we can choose by Lemma 5.9 a compact subset  $B_\eta \subset \Omega$  such that  $K \subset B_\eta^\circ$ ,  $\text{diam}(B_\eta) \leq \text{diam}(K) + \eta = r + \eta$  and  $\mu(B_\eta \setminus K) < \frac{\eta}{2(1+C_\varphi)}$ . By choosing in the above  $B = B_\eta$ , i.e.  $A = B_\eta \setminus K$ , and choosing  $n_0 \in \mathbb{N}$  such that for all  $h \geq n_0$  we have  $\mathcal{H}^k(Z_h) < \frac{\eta}{2}$ , and noting that  $\zeta$  is non-decreasing, we get by 5.40 that for all  $h \geq \max\{h_0, \ell_0, n_0\}$ :

$$\begin{aligned} \Delta &\leq (1 + C_\varphi) \mathcal{H}^k(E \cap A) + \mathcal{H}^k(Z_h) + \zeta(r + \eta)(r + \eta)^k \\ &\leq (1 + C_\varphi) \frac{\eta}{2(1 + C_\varphi)} + \frac{\eta}{2} + \zeta(r + \eta)(r + \eta)^k \\ &= \eta + \zeta(r + \eta)(r + \eta)^k. \end{aligned} \quad (5.41)$$

By continuity of the gauge function  $\zeta$  we find if we take  $\eta \rightarrow 0$  in 5.41 that  $\Delta \leq \zeta(r)r^k$ , which proves that:

$$\mathcal{H}^k(E \setminus F) \leq \mathcal{H}^k(F \setminus E) + \zeta(r)r^k. \quad (5.42)$$

Next, we will prove that the  $(M, 0, \infty)$ -minimal sets are actually minimal and reduced in the sense of Definition 5.4. The proof will be quite similar to the proof of Proposition 5.6.

**Proposition 5.10** *Let  $N \geq 2$ ,  $1 \leq k \leq N$ ,  $\Gamma \subset \mathbb{R}^{N+1}$  be closed,  $\Omega = \mathbb{R}^{N+1} \setminus \Gamma$ . If a relatively closed and  $\mathcal{H}^k$ -rectifiable subset  $E \subset \Omega$  is  $(M, 0, \infty)$ -minimal in  $\Omega$  of dimension  $k$ , then  $E$  is reduced minimal in  $\Omega$  of dimension  $k$ , i.e., reduced almost minimal in  $\Omega$  of dimension  $k$  for the gauge function  $\zeta = 0$ .*

**Proof** Let  $E \subset \Omega$  be relatively closed and  $\mathcal{H}^k$ -rectifiable and assume that it is  $(M, 0, \infty)$ -minimal in  $\Omega$  of dimension  $k$ . As  $E$  is  $(M, 0, \infty)$ -minimal in  $\Omega$  of dimension  $k$ , it follows that  $E = \text{supp}(\mathcal{H}^k \llcorner E) \setminus \Gamma$ . Let  $x \in E$  and  $r > 0$ . Then  $x \in \text{supp}(\mathcal{H}^k \llcorner E)$  and as  $B(x, r)$  is an open neighborhood of  $x$ , it follows that  $\mathcal{H}^k(B(x, r)) > 0$ . Since  $x \in E$  and  $r > 0$  were arbitrary, we find that  $E$  is a reduced set.

Now we show that any competitor for  $E$  satisfies 5.9. Then by Proposition 5.6, we have  $E$  reduced minimal in  $\Omega$  of dimension  $k$  as in Definition 5.4. Let  $K \subset \Omega$  be compact with  $\text{diam}(K) = r$  and let  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  be Lipschitz continuous such that:

$$\varphi(x) = x, \quad \forall x \in \mathbb{R}^{N+1} \setminus K \quad (5.43)$$

$$\varphi(K) \subset K. \quad (5.44)$$

Then  $F = \varphi(E)$  is a competitor for  $E$  in  $K$ . Let  $\eta > 0$ . Then in Lemma 5.9 we can choose a compact set  $B \subset \Omega$  such that  $K \subset B^\circ \subset B$  and  $\mathcal{H}^k(E \cap A) < \eta$ , where  $A = B \setminus K$ . By Lemma 5.8 we can choose a smooth function  $\psi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}$  such that  $\psi|_K = 1$  and  $\psi|_{\mathbb{R}^{N+1} \setminus B} = 0$ . Now for monotone decreasing sequence  $(\varepsilon_h)_{h \in \mathbb{N}}$  with  $\lim_{h \rightarrow \infty} \varepsilon_h = 0$  and  $v \in \partial B(0, 1)$  define the Lipschitz continuous functions  $\varphi_k : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  as in 5.22 in the proof of Proposition 5.6. Define now  $W_h = \{x \in \mathbb{R}^{N+1} \mid \varphi_h(x) \neq x\}$ . Choose now some  $\delta > \text{diam}(B)$ . Note that by the same logic as in the proof of Proposition 5.6, we can choose  $h_0 \in \mathbb{N}$  such that for all  $h \geq h_0$  we have  $\varphi_k(B) \subset B$ . As  $W_h \subset B$  we have  $W_h \cup \varphi_h(W_h) \subset B$  for all  $h \geq h_0$ . That shows by the compactness of  $B \subset \Omega$ , that  $d(W_h \cup \varphi_h(W_h), \Gamma) > 0$  and  $\text{diam}(W_h \cup \varphi_h(W_h)) < \delta$ . That is, for all  $h \geq h_0$  the Lipschitz continuous function  $\varphi_h$  satisfies the conditions set forth by Definition 5.3 of  $(M, 0, \delta)$ -minimality in  $\Omega$ . So we know that  $\varphi_h$  satisfies 5.5 for the gauge function  $\xi = 0$  and all  $h \geq h_0$ :

$$\mathcal{H}^k(E \cap W_h) \leq \mathcal{H}^k(\varphi_h(E \cap W_h)). \quad (5.45)$$

Define now  $E_1 = E \cap K$  and  $E_2 = E \setminus B$ . We recall, that as in proof of Proposition 5.6, that the following properties are satisfied by  $\varphi_h$  for all  $h \geq h_0$ :

$$\varphi_h(E_2) = E_2 \quad (5.46)$$

$$\varphi(E_1) \subset K \quad (5.47)$$

$$\varphi_h(E_1) \subset K \subset B \quad (5.48)$$

$$\varphi_h(E_1) \cap E_2 = \emptyset \quad (5.49)$$

$$W_h \cap E_2 = \emptyset. \quad (5.50)$$

In particular, by 5.47 and 5.48, it follows that  $\varphi_h - \varphi$  is constant on  $E_1$  for all

$h \geq h_0$  and therefore for all  $h \geq h_0$ :

$$\begin{aligned} \mathcal{H}^k(\varphi_h(E \cap W_h)) &\stackrel{W_h \subseteq B}{=} \mathcal{H}^k(\varphi_h(E \cap B \cap W_h)) \\ &= \mathcal{H}^k(\varphi_h(E_1 \cap W) \cup \varphi_h(E \cap A \cap W_h)) \\ &\leq \mathcal{H}^k(\varphi_h(E_1)) + \mathcal{H}^k(\varphi_h(E \cap A)) \end{aligned} \quad (5.51)$$

Defining  $Z_h := \{x \in E_1 \mid 0 < \|\varphi(x) - x\| \leq \varepsilon_h\}$  we have (as shown in proof of Proposition 5.6) that  $E_1 \setminus W_h \subset Z_h$  and so we find that for all  $h \geq h_0$ :

$$\begin{aligned} \mathcal{H}^k(E_1) &= \mathcal{H}^k(E_1 \cap W_h) + \mathcal{H}^k(E_1 \setminus W_h) \\ &\leq \mathcal{H}^k(E_1 \cap W_h) + \mathcal{H}^k(Z_h) \\ &\stackrel{5.51}{\leq} \mathcal{H}^k(\varphi_h(E_1)) + \mathcal{H}^k(\varphi_h(E \cap A)) + \mathcal{H}^k(Z_h). \end{aligned} \quad (5.52)$$

Now, we note as in proof of Proposition 5.6 that  $E \setminus \varphi(E) = (E \cap B) \setminus \varphi(E)$  and  $\varphi(E) \setminus E \supset (\varphi(E) \cap B) \setminus B$  and hence:

$$\mathcal{H}^k(E \setminus \varphi(E)) = \mathcal{H}^k(E \cap B) - \mathcal{H}^k(E \cap B \cap \varphi(E)) \quad (5.53)$$

$$\mathcal{H}^k(\varphi(E) \setminus E) \geq \mathcal{H}^k(\varphi(E) \cap B) - \mathcal{H}^k(E \cap B \cap \varphi(E)). \quad (5.54)$$

For  $F = \varphi(E)$  consider  $\Delta = \mathcal{H}^k(E \setminus F) - \mathcal{H}^k(F \setminus E)$ . With 5.53 and 5.54 we have just shown that:

$$\Delta \leq \mathcal{H}^k(E \cap B) - \mathcal{H}^k(\varphi(E) \cap B). \quad (5.55)$$

Note that since  $\varphi(E_1) \subset K$  we have  $\varphi(E_1) \subset \varphi(E) \cap B$ . Using 5.52 and 5.55 we find that for all  $h \geq h_0$ :

$$\begin{aligned} \Delta &\stackrel{5.55}{\leq} \mathcal{H}^k(E \cap B) - \mathcal{H}^k(\varphi(E) \cap B) \\ &\leq \mathcal{H}^k(E_1) + \mathcal{H}^k(E \cap A) - \mathcal{H}^k(\varphi(E_1)) \\ &\stackrel{5.52}{\leq} \mathcal{H}^k(E \cap A) + \mathcal{H}^k(\varphi_h(E \cap A)) + \mathcal{H}^k(Z_h). \end{aligned} \quad (5.56)$$

Now, as in proof of Proposition 5.6 we can find a constant  $C_\varphi > 0$  only depending on  $\varphi$  and  $k$  and some  $\ell_0 \in \mathbb{N}$  such that  $\mathcal{H}^k(Z_h) < \eta$  and  $\mathcal{H}^k(\varphi_h(E \cap A)) \leq C_\varphi \mathcal{H}^m(E \cap A)$  for all  $h \geq \ell_0$ . Then by choice of the compact set  $B$  we have for all  $h \geq \max\{h_0, \ell_0\}$  it holds that:

$$\begin{aligned} \Delta &\leq (1 + C_\varphi) \mathcal{H}^k(E \cap A) + \mathcal{H}^k(Z_h) \\ &\leq (1 + C_\varphi) \eta + \eta \\ &= (2 + C_\varphi) \eta. \end{aligned} \quad (5.57)$$

Since  $C_\varphi$  depends only on  $\varphi$  and  $m$  we can take in 5.57 the limit  $\eta \rightarrow 0$  and obtain  $\Delta \leq 0$ , which shows that competitor  $F = \varphi(E)$  of  $E$  in  $K$  satisfies 5.9. This proves the proposition.  $\square$

Next, we will also show that the reduced boundary of a set of minimal perimeter in the sense of Definition 4.1 is also minimal in the sense of Definition 5.4.

**Proposition 5.11** *Let  $\Omega \subset \mathbb{R}^{N+1}$  be a bounded open set, and  $E_0 \subset \mathbb{R}^{N+1}$  be a set of locally finite perimeter in  $\mathbb{R}^{N+1}$ . Suppose  $E \subset \mathbb{R}^{N+1}$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$ . Then  $\partial^* E$  is minimal in  $\Omega$  of dimension  $N$ , that is,  $\partial^* E \cap \Omega$  is almost minimal of dimension  $N$  with gauge function  $\xi = 0$ .*

**Proof** Set  $\tilde{E} = \partial^* E \cap \Omega$ . By Theorem 3.41  $\tilde{E}$  is  $\mathcal{H}^N$ -rectifiable and clearly is closed in  $\Omega$  as  $\partial^* E$  is defined by a closed condition. Consider now  $K \subset \Omega$  be a compact subset and  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  be a Lipschitz function such that  $\varphi(K) \subset K$  and  $\varphi(x) = x$  for all  $x \in \mathbb{R}^{N+1} \setminus K$ . Then,  $\tilde{F} = \varphi(\tilde{E})$  is a competitor for  $\tilde{E}$  in  $K$ . Set also  $F = \varphi(E)$ . By Proposition 5.6, it suffices to show that:

$$\mathcal{H}^N(\tilde{E} \setminus \tilde{F}) \leq \mathcal{H}^N(\tilde{F} \setminus \tilde{E}). \quad (5.58)$$

We note that since  $\varphi(x) = x$  outside of  $K \subset \Omega$  we have

$$F \setminus \Omega = E \setminus \Omega = E_0 \setminus \Omega. \quad (5.59)$$

So we have that  $F$  is a competitor for perimeter in  $\Omega$  of  $E$ . That is, by Theorem 3.41 and since  $\tilde{E} \setminus \tilde{F} \subset \tilde{E}$  we have the following:

$$\begin{aligned} \mathcal{H}^N(\tilde{F}) &= P(F, \Omega) \\ &\geq P(E, \Omega) \\ &= \mathcal{H}^N(\tilde{E}) \\ &\geq \mathcal{H}^N(\tilde{E} \setminus \tilde{F}). \end{aligned} \quad (5.60)$$

Now consider  $G = F \setminus \tilde{E}$  and note that

$$G \setminus \Omega = F \setminus \Omega = E_0 \setminus \Omega. \quad (5.61)$$

So  $G$  is a competitor for perimeter in  $\Omega$  of  $E$ . Then we find:

$$\begin{aligned} \mathcal{H}^N(\tilde{E} \setminus \tilde{F}) &\leq \mathcal{H}^N(\tilde{E}) \\ &= P(E, \Omega) \\ &\leq P(G, \Omega) \\ &= \mathcal{H}^N(\tilde{F} \setminus \tilde{E}). \end{aligned} \quad (5.62)$$

This shows 5.58 and therefore  $\tilde{E}$  is minimal in  $\Omega$  of dimension  $N$ .  $\square$

### 5.3 Monotonicity of density

In this section, we shall show that the function

$$r \in (0, d(x, \mathbb{R}^{N+1} \setminus \Omega)) \mapsto \frac{1}{\omega_k r^k} \mathcal{H}^k(E \cap B(x, r))$$

is nondecreasing for  $x \in E$  and  $E \subset \Omega$  a minimal set. First, we show an integral version of this fact for almost minimal sets, where we will follow [2].

**Lemma 5.12** *Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset. Let  $E \subset \Omega$  be a reduced almost minimal set in  $\Omega$  of dimension  $k$  for the gauge function  $\xi$ . For any  $x \in E$  and  $0 < \rho_1 < \rho_2 < d(x, \mathbb{R}^{N+1} \setminus \Omega)$  it holds that:*

$$\begin{aligned} \mathcal{H}^k(E \cap B(x, \rho_1)) &\leq \xi(2\rho_2)(2\rho_2)^k \\ &\quad + \frac{1}{\rho_2 - \rho_1} \int_{E \cap A(x, \rho_1, \rho_2)} \frac{r(y) \cos \theta(y)}{k} d\mathcal{H}^k(y), \end{aligned} \quad (5.63)$$

where  $A(x, \rho_1, \rho_2) = B(x, \rho_2) \setminus B(x, \rho_1)$ ,  $r(y) = \|y - x\|$  and  $\theta(y) \in [0, \pi/2]$  is the smallest angle between the line segment connecting  $x$  and  $y$  and the approximate  $m$ -tangent space of  $E$  at  $y \in E$  (which by Proposition 2.62 exists almost everywhere, as  $E$  is  $\mathcal{H}^k$ -rectifiable).

To prove this, we first show that some radially defined functions are Lipschitz continuous.

**Lemma 5.13** *Let  $f : [0, \infty) \rightarrow \mathbb{R}$  with  $f(0) = 0$  be a Lipschitz function with Lipschitz constant  $C \geq 0$ . Then for  $k \geq 1$  the function  $g : \mathbb{R}^N \rightarrow \mathbb{R}^N$  defined by*

$$g(x) = \begin{cases} f(\|x\|) \frac{x}{\|x\|}, & x \neq 0, \\ 0, & x = 0, \end{cases} \quad (5.64)$$

*is Lipschitz continuous.*

**Proof** Let  $x, y \in \mathbb{R}^N$ . If  $x = 0$  or  $y = 0$ , then

$$\|g(x) - g(y)\| \leq C \|x - y\|. \quad (5.65)$$

So we can assume  $x, y \neq 0$ . Without loss of generality, we assume that

$\|x\| \leq \|y\|$ . Then it follows that:

$$\begin{aligned}
 \|g(x) - g(y)\| &= \left\| f(\|x\|) \frac{x}{\|x\|} - f(\|x\|) \frac{y}{\|y\|} + f(\|x\|) \frac{y}{\|y\|} - f(\|y\|) \frac{y}{\|y\|} \right\| \\
 &\leq |f(\|x\|)| \left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\| + |f(\|x\|) - f(\|y\|)| \\
 &\leq C \|x\| \left\| \frac{\|y\| x - \|x\| x + \|x\| x - \|x\| y}{\|x\| \|y\|} \right\| + C \|x - y\| \\
 &\leq \frac{C}{\|y\|} [\|x\| |\|x\| - \|y\|| + \|x\| \|x - y\|] + C \|x - y\| \\
 &\leq C \left[ \frac{2\|x\|}{\|y\|} + 1 \right] \|x - y\| \\
 &\leq 3C \|x - y\|.
 \end{aligned} \tag{5.66}$$

This shows that  $g$  is Lipschitz with Lipschitz constant  $3C$ .  $\square$

**Proof (Lemma 5.12)** First, note that we may assume that  $x = 0$  up to replacing  $E$  by  $E - x$ . The idea is now to construct a competitor for  $E$  in a compact subset of  $\Omega$ . Fix  $0 < r_1 < r_2$  such that  $\bar{B}(0, r_2) \subset \Omega$ . Now, define the function  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by  $\varphi(y) = \psi(\|y\|) \frac{y}{\|y\|}$  for  $y \neq 0$  and  $\varphi(0) = 0$ . Here,  $\psi : [0, \infty) \rightarrow \mathbb{R}$  is the piecewise linear function defined by:

$$\psi(r) = \begin{cases} 0, & 0 \leq r \leq r_1, \\ \frac{r_2(r-r_1)}{r_2-r_1}, & r_1 \leq r \leq r_2, \\ r, & r \geq r_2. \end{cases} \tag{5.67}$$

Note that we have for any  $r, s \geq 0$ , that:

$$|\psi(s) - \psi(r)| \leq \max \left\{ \frac{r_2}{r_2 - r_1}, 1 \right\} |s - r|. \tag{5.68}$$

Hence,  $\psi$  is Lipschitz continuous and, by Lemma 5.13, the function  $\varphi$  is also Lipschitz continuous. We note that  $\varphi(y) = y$  for all  $y \in \mathbb{R}^{N+1} \setminus \bar{B}(0, r_2)$ . Consider now  $y \in \bar{B}(0, r_2)$ . If  $\|y\| \leq r_1$ , then  $\varphi(y) = 0$ . Otherwise we have  $r_1 \leq \|y\| \leq r_2$  and so:

$$\begin{aligned}
 \|\varphi(y)\| &= \left\| \psi(\|y\|) \frac{y}{\|y\|} \right\| \\
 &= |\psi(\|y\|)| \\
 &= \frac{r_2(\|y\| - r_1)}{r_2 - r_1} \\
 &\leq r_2.
 \end{aligned} \tag{5.69}$$

This shows that  $\varphi(\overline{B}(0, r_2)) \subset \overline{B}(0, r_2)$ . So in fact  $F = \varphi(E)$  is a competitor for  $E$  in the compact subset  $\overline{B}(0, r_2) \subset \Omega$ . So by almost minimality of  $E$  for gauge function  $\xi$  it follows that:

$$\mathcal{H}^k(E \cap B(0, r_2)) \leq \mathcal{H}^k(\varphi(E \cap B(0, r_2))) + \xi(\delta)\delta^k, \quad (5.70)$$

for any  $\delta > 2r_2$ .

Now the goal is to evaluate  $\mathcal{H}^k(\varphi(E \cap B(0, r_2)))$ . Since  $B(0, r_1)$  is mapped to the origin, it suffices to evaluate  $\mathcal{H}^k(\varphi(E \cap A))$  for  $A = B(0, r_2) \setminus B(0, r_1)$ , i.e., we have

$$\mathcal{H}^k(\varphi(E \cap B(0, r_2))) = \mathcal{H}^k(\varphi(E \cap A)). \quad (5.71)$$

Now we will use the rectifiability of  $E$ . Choose  $y \in E \cap A$  such that there exists the approximate tangent space  $\pi_y$  of  $E$  at  $y$ . Consider now  $r(y) = \|y\|$ . We want to compute the tangential derivative  $d^E \varphi(y)$  at  $y$ . That is for any  $w = y + v \in y + \pi_x$  we want to compute  $d^E \varphi(y)w$ .

- First we consider vectors, that are parallel to  $y$ , i.e. in radial direction. For this, we consider  $y_0 = \frac{y}{\|y\|}$ . For this we note that by definition of  $\varphi$  we know that for  $s > 0$  such that  $sy_0 \in A$  it holds that:

$$\varphi(sy_0) = \psi(s)y_0 = \frac{r_2(s - r_1)}{r_2 - r_1}y_0. \quad (5.72)$$

Hence,  $d^E \varphi(y)$  acts as follows on  $y_0$  (and by linearity on any other vector in radial direction):

$$d^E \varphi(y)(y_0) = \frac{r_2}{r_2 - r_1}y_0. \quad (5.73)$$

- Now consider a unit vector  $v \in P_y$  orthogonal to  $y$ . Define now the following quantities for  $h \geq 0$ :

$$z(h) = hv + r(y)y_0 \quad (5.74)$$

$$s(h) = \sqrt{h^2 + r(y)^2} \quad (5.75)$$

$$z_0(h) = \frac{z(h)}{s(h)}. \quad (5.76)$$

Now we can compute using Definition 2.63 and the continuity of  $\varphi$ :

$$\begin{aligned}
 d^E \varphi(y)(v) &= \lim_{h \rightarrow 0} \frac{\varphi(r(y)y_0 + hv) - \varphi(r(y)y_0)}{h} \\
 &\stackrel{5.76}{=} \lim_{h \rightarrow 0} \frac{\varphi(s(h)z_0(h)) - \varphi(r(y)y_0)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\psi(s(h))z_0(h) - \psi(r(y))y_0}{h} \\
 &\stackrel{5.76}{=} \lim_{h \rightarrow 0} \frac{1}{h} \left[ \psi(s(h)) \frac{1}{s(h)} (hv + r(y)y_0) - \psi(r(y))y_0 \right] \\
 &= \lim_{h \rightarrow 0} \frac{\psi(s(h))}{s(h)} v - \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\psi(s(h))}{s(h)} r(y) - \psi(r(y)) \right] y_0 \\
 &= \frac{\psi(r(y))}{r(y)} v.
 \end{aligned} \tag{5.77}$$

In the last line we used that  $s(h) \rightarrow r(y)$  as  $h \rightarrow 0$ . So, on vectors orthogonal to  $y$  we have that:

$$d^E \varphi(y)(v) = \frac{\psi(r(y))}{r(y)} v. \tag{5.78}$$

These computations show that if  $\pi_y$  is orthogonal to the line segment from the origin to  $y$ , then  $d^E \varphi(y)$  acts on  $y + \pi_y$  by multiplication with  $\frac{\psi(r(y))}{r(y)}$ . Otherwise, we can consider the orthogonal projection  $p \in P_y$  of the origin onto  $\pi_y$ . Then using the unit vector  $w = \frac{y-p}{\|y-p\|} \in y + \pi_y$ , by definition of  $\theta(y)$ ,  $\theta(y)$  is the angle between the line segment from the origin to  $y$  and  $w$ , so that we can write

$$w = \cos \theta(y) y_0 + \sin \theta(y) v, \tag{5.79}$$

for some unit vector  $v$  orthogonal to  $y$ . By our computations above and the linearity of  $d^E \varphi(y)$  we find that:

$$d^E \varphi(y)(w) = \frac{r_2}{r_2 - r_1} \cos \theta(y) y_0 + \frac{\psi(r(y))}{r(y)} \sin \theta(y) v. \tag{5.80}$$

Now also introduce the quantity:

$$\begin{aligned}
 \beta(y) &:= \left\| d^E \varphi(y)(w) \right\| \\
 &= \left\{ \left[ \frac{r_2}{r_2 - r_1} \right]^2 \cos^2 \theta(y) + \left[ \frac{\psi(r(y))}{r(y)} \right]^2 \sin^2 \theta(y) \right\}^{\frac{1}{2}}.
 \end{aligned} \tag{5.81}$$

The next step is to compute the tangential Jacobian  $J_k d^E \varphi(y)$  of  $\varphi$  at  $y$ . For this, using the Gram-Schmidt orthonormalization procedure, we can construct an orthonormal basis  $w_1, \dots, w_k$  of  $y + \pi_y$  such that  $w_1 = w$ . We

know that  $w$  is the linear combination of orthogonal unit vectors  $y_0$  and  $v$ , so it must follow that  $w_i$  is in particular orthogonal to  $y_0$  for all  $i = 2, \dots, k$ . So we can compute in this basis the matrix representation of  $d^E \varphi(y)$  by:

$$d^E \varphi(y)(w_i) = \begin{cases} d^E \varphi(y)(w), & i = 1, \\ \frac{\psi(r(y))}{r(y)} w_i, & i = 2, \dots, k. \end{cases} \quad (5.82)$$

This yields that  $d^E \varphi(y)$  can be identified by the following matrix:

$$d^E \varphi(y) = \begin{bmatrix} d^E \varphi(y)(w) & \frac{\psi(r(y))}{r(y)} w_2 & \dots & \frac{\psi(r(y))}{r(y)} w_m \end{bmatrix}. \quad (5.83)$$

This allows to compute:

$$\begin{aligned} d^E \varphi(y)^t d^E \varphi(y) &= \begin{bmatrix} d^E \varphi(y)(w)^t \\ \frac{\psi(r(y))}{r(y)} w_2^t \\ \frac{\psi(r(y))}{r(y)} w_m^t \end{bmatrix} \begin{bmatrix} d^E \varphi(y)(w) & \frac{\psi(r(y))}{r(y)} w_2 & \dots & \frac{\psi(r(y))}{r(y)} w_m \end{bmatrix} \\ &= \begin{bmatrix} \beta(y)^2 & 0 & \dots & \dots & 0 \\ 0 & \left[ \frac{\psi(r(y))}{r(y)} \right]^2 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & 0 & \left[ \frac{\psi(r(y))}{r(y)} \right]^2 \end{bmatrix}. \end{aligned} \quad (5.84)$$

This is a  $k \times k$  diagonal matrix with diagonal entries  $\beta(y)^2$  and  $k - 1$  times the entry  $\left[ \frac{\psi(r(y))}{r(y)} \right]^2$ . Using this we can easily compute the Jacobian:

$$\begin{aligned} J_k d^E \varphi(y) &= \sqrt{\det d^E \varphi(y)^* d^E \varphi(y)} \\ &= \sqrt{\beta(y)^2 \left[ \frac{\psi(r(y))}{r(y)} \right]^{2(k-1)}} \\ &= \beta(y) \left[ \frac{\psi(r(y))}{r(y)} \right]^{k-1}. \end{aligned} \quad (5.85)$$

This is even compatible with the case that  $\pi_y$  is orthogonal to  $y$ , as we have then  $\theta(y) = \frac{\pi}{2}$  and therefore  $\beta(y) = \left[ \frac{\psi(r(y))}{r(y)} \right]$ , which exactly yields the expression 5.85. Since  $\varphi$  is bijective on  $A$  and because of our computation we can apply now the area formula 2.100 from Theorem 2.67 for  $E \cap A$ ,

which yields:

$$\begin{aligned}
 \mathcal{H}^k(\varphi(E \cap B(0, r_2))) &= \mathcal{H}^k(\varphi(E \cap A)) \\
 &\stackrel{2.100}{=} \int_{E \cap A} \mathbf{J}_k d^E \varphi(y) d\mathcal{H}^k(y) \\
 &= \int_{E \cap A} \beta(y) \left[ \frac{\psi(r(y))}{r(y)} \right]^{k-1} d\mathcal{H}^k(y).
 \end{aligned} \tag{5.86}$$

With this result in our pocket, we now can tackle the main problem: Fix  $0 < \rho_1 < \rho_2$  such that  $B(0, \rho_2) \subset \Omega$ . Now pick  $\varepsilon > 0$  small enough such that for all  $t \in (\rho_1, \rho_2)$  we have  $0 < t - \varepsilon < t$  and  $\overline{B}(0, t) \subset \Omega$ . By applying inequality 5.70 and equation 5.86 to  $r_1 = t - \varepsilon$  and  $r_2 = t$  we find for all such  $t \in (\rho_1, \rho_2)$  it holds for all  $\delta > 2t$  that:

$$\begin{aligned}
 \mathcal{H}^k(E \cap B(0, t - \varepsilon)) &\leq \mathcal{H}^k(E \cap B(0, t)) \\
 &\stackrel{5.70}{\leq} \mathcal{H}^k(E)(\varphi_t(E \cap B(0, t))) + \zeta(\delta)\delta^k \\
 &\stackrel{5.86}{=} \int_{E \cap A_t} \beta_t(y) \left[ \frac{\psi_t(r(y))}{r(y)} \right]^{k-1} d\mathcal{H}^k(y) + \zeta(\delta)\delta^k.
 \end{aligned} \tag{5.87}$$

Here  $\varphi_t : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$ ,  $\psi_t : [0, \infty) \rightarrow \mathbb{R}$ ,  $\beta_t(y)$  and  $A_t$  are defined in the following obvious way:

$$\varphi_t(y) = \begin{cases} \psi_t(\|y\|) \frac{y}{\|y\|}, & y \neq 0, \\ 0, & y = 0. \end{cases} \tag{5.88}$$

$$\psi_t(r) = \begin{cases} 0, & r \in [0, t - \varepsilon], \\ \frac{t(r - (t - \varepsilon))}{\varepsilon}, & r \in [t - \varepsilon, t], \\ r, & r \in [t, \infty). \end{cases} \tag{5.89}$$

$$A_t = B(0, t) \setminus B(0, t - \varepsilon) \tag{5.90}$$

$$\beta_t(y) = \left\{ \left[ \frac{t}{\varepsilon} \right]^2 \cos^2 \theta(y) + \left[ \frac{\psi_t(r(y))}{r(y)} \right]^2 \sin^2 \theta(y) \right\}^{\frac{1}{2}}. \tag{5.91}$$

Now we can average the inequality 5.87 over  $(\rho_1, \rho_2)$  and get the following inequality:

$$\mathcal{H}^k(E \cap B(0, \rho_1 - \varepsilon)) \leq \frac{1}{\rho_2 - \rho_1} I(\varepsilon) + \zeta(2\rho_2)(2\rho_2)^k, \tag{5.92}$$

where we have denoted:

$$I(\varepsilon) := \int_{\rho_1}^{\rho_2} \int_{y \in E: t - \varepsilon \leq r(y) \leq t} \beta_t(y) \left[ \frac{\psi_t(r(y))}{r(y)} \right]^{k-1} d\mathcal{H}^k(y). \tag{5.93}$$

By applying Fubini's theorem we find:

$$I(\varepsilon) = \int_{y \in E: \rho_1 - \varepsilon < r(y) < \rho_2} F_\varepsilon(y) d\mathcal{H}^k(y), \quad (5.94)$$

where we defined:

$$F_\varepsilon(y) := \int_{t \in (\rho_1, \rho_2) \cap [r(y), r(y) + \varepsilon]} \beta_t(y) \left[ \frac{\psi_t(r(y))}{r(y)} \right]^{k-1} dt. \quad (5.95)$$

Now we bound  $\beta_t(y)$  and  $\frac{\psi_t(r(y))}{r(y)}$ . We get:

- We start with  $\beta_t(y)$ :

$$\begin{aligned} \beta_t(y) &= \left\{ \left[ \frac{t}{\varepsilon} \right]^2 \cos^2 \theta(y) + \left[ \frac{\psi_t(r(y))}{r(y)} \right]^2 \sin^2 \theta(y) \right\}^{\frac{1}{2}} \\ &\stackrel{(*)}{\leq} \left\{ \left[ \frac{t^2}{\varepsilon} \cos^2 \theta(y) + 1 \right] \right\}^{\frac{1}{2}} \\ &\stackrel{(**)}{\leq} \frac{t}{\varepsilon} \cos \theta(y) + 1. \end{aligned} \quad (5.96)$$

At  $(*)$  we used  $\psi_t(r(y)) \leq r(y)$  and  $\sin^2 \theta(y) \leq 1$ , and at  $(**)$  we used that for  $a, b \geq 0$  we have  $a^2 + b^2 \leq (a + b)^2$ . So for  $t \in (\rho_1, \rho_2) \cap [r(y), r(y) + \varepsilon]$  it holds that:

$$\beta_t(y) \leq \frac{1}{\varepsilon} [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon]. \quad (5.97)$$

- Now we bound  $\frac{\psi_t(r(y))}{r(y)}$ . First we may assume that  $\varepsilon \in (0, 1)$  is small enough such that:

$$\frac{\varepsilon}{r(y)} \leq \frac{1}{k}. \quad (5.98)$$

For  $t \in (\rho_1, \rho_2) \cap [r(y), r(y) + \varepsilon]$  we find then that:

$$\begin{aligned} \frac{\psi_t(r(y))}{r(y)} &= \frac{t(r(y) - (t - \varepsilon))}{r(y)\varepsilon} \\ &= \frac{t}{r(y)} \frac{r(y) - t + \varepsilon}{\varepsilon} \\ &\leq \frac{1}{\varepsilon} \frac{r(y) + \varepsilon}{r(y)} (r(y) - t + \varepsilon). \end{aligned} \quad (5.99)$$

Using the bounds 5.97 and 5.99 we can bound  $F_\varepsilon(y)$  as follows since the integrand in  $F_\varepsilon(y)$  is nonnegative:

$$\begin{aligned}
 F_\varepsilon(y) &\leq \int_{r(y)}^{r(y)+\varepsilon} \beta_t(y) \left[ \frac{\psi_t(r(y))}{r(y)} \right]^{k-1} dt \\
 &\leq \int_{r(y)}^{r(y)+\varepsilon} \frac{1}{\varepsilon} [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon] \left[ \frac{1}{\varepsilon} \frac{r(y) + \varepsilon}{r(y)} (r(y) - t + \varepsilon) \right]^{k-1} dt \\
 &= \left( \frac{1}{\varepsilon} \right)^k [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon] \left( \frac{r(y) + \varepsilon}{r(y)} \right)^{k-1} \int_0^\varepsilon s^{k-1} ds \\
 &= \frac{1}{k} \left( \frac{1}{\varepsilon} \right)^k [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon] \left( \frac{r(y) + \varepsilon}{r(y)} \right)^{k-1} \varepsilon^k \\
 &= \frac{1}{k} [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon] \left( 1 + \frac{\varepsilon}{r(y)} \right)^{k-1} \\
 &\leq \frac{1}{k} [(r(y) + \varepsilon) \cos \theta(y) + \varepsilon] \left( 1 + 2(k-1) \frac{\varepsilon}{r(y)} \right) \\
 &\leq \frac{1}{k} r(y) \cos \theta(y) + \left( 4 + \frac{2}{r(y)} \right) \varepsilon.
 \end{aligned} \tag{5.100}$$

We set  $C = C(y) = 4 + \frac{2}{r(y)} \geq 0$ . This constant is independent of  $\varepsilon$  and clearly bounded for  $y \in (\rho_1 - \varepsilon, \rho_2)$ . This yields the following bound on  $I(\varepsilon)$ :

$$\begin{aligned}
 I(\varepsilon) &= \int_{y \in E: \rho_1 - \varepsilon < r(y) < \rho_2} F_\varepsilon(y) d\mathcal{H}^k(y) \\
 &\stackrel{5.100}{\leq} \int_{y \in E: \rho_1 - \varepsilon < r(y) < \rho_2} \frac{1}{k} r(y) \cos \theta(y) + C\varepsilon d\mathcal{H}^k(y).
 \end{aligned} \tag{5.101}$$

This finally enables us to prove the lemma:

$$\begin{aligned}
 \mathcal{H}^k(E \cap B(0, \rho_1 - \varepsilon)) &\leq \frac{1}{\rho_2 - \rho_1} I(\varepsilon) + \xi(2\rho_2)(2\rho_2)^k \\
 &\stackrel{5.101}{\leq} \frac{1}{\rho_2 - \rho_1} \int_{y \in E: \rho_1 - \varepsilon < r(y) < \rho_2} \frac{1}{k} r(y) \cos \theta(y) + C\varepsilon d\mathcal{H}^k(y) \\
 &\quad + \xi(2\rho_2)(2\rho_2)^k.
 \end{aligned} \tag{5.102}$$

So by letting  $\varepsilon$  tend to zero, we find that:

$$\begin{aligned}
 \mathcal{H}^k(E \cap B(0, \rho_1)) &\leq \xi(2\rho_2)(2\rho_2)^k \\
 &\quad + \frac{1}{\rho_2 - \rho_1} \int_{E \cap A(0, \rho_1, \rho_2)} \frac{r(y) \cos \theta(y)}{k} d\mathcal{H}^k(y).
 \end{aligned} \tag{5.103}$$

<sup>1</sup>We have for  $x \leq \frac{1}{h}$  and  $h \in \mathbb{N}$  that  $(1+x)^h \leq 1+2hx$ .

This result enables us to prove the monotonicity result for reduced minimal sets.

**Proposition 5.14** *Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset. Let  $E$  be a reduced minimal set in  $\Omega$  of dimension  $k$  (i.e.  $E$  is a reduced almost minimal set in  $\Omega$  of dimension  $k$  for gauge function  $\xi = 0$ ). Then for every  $x \in \Omega$  the function*

$$\Theta_k(E, x)(\cdot) : r \in (0, d(x, \mathbb{R}^{N+1} \setminus \Omega)) \mapsto \omega_k^{-1} r^{-k} \mathcal{H}^k(E \cap B(x, r)), \quad (5.104)$$

*is nondecreasing.*

For this proof, we will extensively make use of the Lebesgue-Besicovitch differentiation theorem for Radon measure (see Theorem 2.43). But first we show the following small lemma for absolutely continuous measures with respect to the Lebesgue measure on  $\mathbb{R}$ , which can be thought of as a special version of the fundamental theorem of calculus for the Radon-Nikodym derivative.

**Lemma 5.15** *Let  $\mu$  be a measure on  $\mathbb{R}$ , which is absolutely continuous with respect to the Lebesgue measure  $\lambda$  on  $\mathbb{R}$ , with Radon-Nikodym derivative  $g := \frac{d\mu}{d\lambda}$ . Suppose that there are  $a < b$  and a differentiable function  $f : (a, b) \rightarrow \mathbb{R}$  such that  $f'(t) = g(t)$  for  $\lambda$ -a.e.  $t \in (a, b)$ . Then for any smooth function  $\varphi : (a, b) \rightarrow \mathbb{R}$  and  $a < c < d < b$  it holds that:*

$$\varphi(d)f(d) - \varphi(c)f(c) = \int_c^d \varphi(t) d\mu(t) + \int_c^d \varphi'(t)f(t) dt. \quad (5.105)$$

**Proof** Let  $\varphi : (a, b) \rightarrow \mathbb{R}$  be smooth and  $a < c < d < b$  and set

$$I := \int_c^d \int_c^t \varphi'(s) ds d\mu(t). \quad (5.106)$$

By the fundamental theorem of calculus we get that:

$$\begin{aligned} I &= \int_c^d \int_c^t \varphi'(s) ds d\mu(t) \\ &= \int_c^d [\varphi(t) - \varphi(s)] d\mu(t) \\ &= \int_c^d \varphi(t) d\mu(t) + \varphi(c) \int_c^d g(t) d\lambda(t) \\ &= \int_c^d \varphi(t) d\mu(t) + \varphi(c) \int_c^d f'(t) dt \\ &= \int_c^d \varphi(t) d\mu(t) + \varphi(c) [f(d) - f(c)]. \end{aligned} \quad (5.107)$$

On the other hand by an application of Fubini's theorem we find that:

$$\begin{aligned}
 I &= \int_c^d \int_c^t \varphi'(s) ds d\mu(t) \\
 &= \int_c^d \int_s^d \varphi'(s) d\mu(t) ds \\
 &= \int_c^d \varphi'(s) \int_s^d g(t) d\lambda(t) ds \\
 &= \int_c^d \varphi'(s) \int_s^d f'(t) dt ds \\
 &= \int_c^d \varphi'(s) [f(d) - f(s)] ds \\
 &= f(d) [\varphi(d) - \varphi(c)] - \int_c^d \varphi'(s) f(s) ds.
 \end{aligned} \tag{5.108}$$

Putting 5.107 and 5.108 together we find that:

$$\begin{aligned}
 \int_c^d \varphi(t) d\mu(t) + \int_c^d \varphi'(t) f(t) dt &= (I + \varphi(c) [f(d) - f(c)]) \\
 &\quad + (f(d) [\varphi(d) - \varphi(c)] - I) \\
 &= \varphi(c) f(d) - \varphi(c) f(c) \\
 &\quad + \varphi(d) f(d) - \varphi(c) f(d) \\
 &= \varphi(d) f(d) - \varphi(c) f(c).
 \end{aligned} \tag{5.109}$$

This proves the lemma.  $\square$

**Proof (Proposition 5.14)** Let  $x \in \Omega$  and  $R = d(x, \mathbb{R}^{N+1} \setminus \Omega)$ . By Proposition 2.51 (i) and since  $\mathcal{H}^k(E) < \infty$  by rectifiability of  $E$ , it follows that  $\mathcal{H}^k \llcorner E$  defines a Radon measure on  $\mathbb{R}^{N+1}$ . Define now the radial projection  $\pi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}$  by  $\pi(y) := \|x - y\|$ . Clearly,  $\pi$  is continuous. It is even a proper map as: If  $K \subset \mathbb{R}$  is compact, then it is in particular closed (Heine-Borel). So, by continuity, it follows that  $\pi^{-1}(K)$  is closed in  $\mathbb{R}^{N+1}$ . Now, we show that  $\pi^{-1}(K)$  is bounded and therefore compact (Heine-Borel). We note that for any  $y \in \pi^{-1}(K)$ , it follows that  $\|x - y\| \in K$ . Since  $K$  is bounded in  $\mathbb{R}$ , there is a constant  $M > 0$  such that  $|a| \leq M$  for any  $a \in K$ . So we must have  $\|x - y\| \leq M$  for any  $y \in \pi^{-1}(K)$ . This shows that  $\pi^{-1}(K) \subset \overline{B}(x, M)$ . So  $\pi^{-1}(K)$  is bounded in  $\mathbb{R}^{N+1}$ . By Lemma 2.31, it follows that  $\nu := \pi_{\#}(\mathcal{H}^k \llcorner E)$  is a Radon measure on  $\mathbb{R}$ . Since  $[0, R)$  is a Borel set and

$$\nu([0, R)) = \mathcal{H}^k(E \cap B(x, R)) < \infty, \tag{5.110}$$

it follows that  $\mu = \nu \llcorner [0, R)$  is a Radon measure on  $\mathbb{R}$  with  $\mu(A) = 0$  for all  $A \subset \mathbb{R}$  with  $A \cap [0, R) = \emptyset$ . We also note that for  $r \in [0, R)$  and  $h > 0$  such

that  $(r - h, r + h) \subset [0, R)$  we find that:

$$\pi^{-1}((r - h, r + h) \cap [0, R)) = B(x, r + h) \setminus B(x, r - h). \quad (5.111)$$

Now define the function  $f : [0, R) \rightarrow [0, \infty)$  by

$$f(r) := \mathcal{H}^k(E \cap B(x, r)). \quad (5.112)$$

Let  $\lambda = \mathcal{L}^1$  be the Lebesgue measure on  $\mathbb{R}$ . Since  $\lambda$  and  $\mu$  are both Radon measures on  $\mathbb{R}$ , we know by Theorem 2.43, that the density  $g := \frac{d\mu}{d\lambda}$  of  $\mu$  is defined  $\lambda$ -a.e. on  $\mathbb{R}$ . In particular,  $g(r)$  for  $\lambda$ -a.e.  $r \in [0, r)$ . By the observation 5.111 we have for  $\lambda$ -a.e.  $r \in [0, r)$  that in the following all the limits exist:

$$\begin{aligned} g(r) &= \frac{d\mu}{d\lambda}(r) \\ &\stackrel{2.65}{=} \lim_{h \rightarrow 0^+} \frac{\mu((r - h, r + h))}{\lambda((r - h, r + h))} \\ &\stackrel{\lambda((a,b))=b-a}{=} \lim_{h \rightarrow 0^+} \frac{\mu((r - h, r + h))}{2h} \\ &= \lim_{h \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap \pi^{-1}((r - h, r + h) \cap [0, R)))}{2h} \\ &\stackrel{h \text{ small}}{=} \lim_{h \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap \pi^{-1}((r - h, r + h)))}{2h} \\ &\stackrel{5.111}{=} \lim_{h \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap (B(x, r + h) \setminus B(x, r - h)))}{2h} \\ &= \lim_{h \rightarrow 0^+} \frac{\mathcal{H}^k(E \cap B(x, r + h)) - \mathcal{H}^k(E \cap B(x, r - h))}{2h} \\ &= \lim_{h \rightarrow 0^+} \frac{f(r + h) - f(r - h)}{2h}. \end{aligned} \quad (5.113)$$

The calculation 5.113 shows that  $f$  is differentiable  $\lambda$ -almost everywhere on  $(0, R)$  and it holds  $f'(r) = g(r)$  for  $\lambda$ -a.e.  $r \in (0, R)$ . Since  $E$  is reduced minimal, i.e. almost minimal for gauge function  $\xi = 0$ , and  $B(0, r) \subset \Omega$  for all  $r \in (0, R)$ , we can apply Lemma 5.12 to  $0 < \rho_1 = r - h < \rho_2 = r$  for  $h > 0$

small enough to find by bounding  $\cos \theta(y) \leq 1$ :

$$\begin{aligned}
 f(r-t) &= \mathcal{H}^k(E \cap B(x, r-t)) \\
 &\leq \frac{1}{\rho_2 - \rho_1} \int_{E \cap A(x, r-h, r)} \frac{r(y) \cos \theta(y)}{k} d\mathcal{H}^k(y) \\
 &\leq \frac{1}{hk} \int_{E \cap (B(x, r) \setminus B(x, r-h))} r(y) d\mathcal{H}^k(y) \\
 &\stackrel{r(y) \leq r}{=} \frac{r}{hk} \mathcal{H}^k(E \cap (B(x, r) \setminus B(x, r-h))) \\
 &= \frac{r}{k} \frac{\mathcal{H}^k(E \cap (B(x, r))) - \mathcal{H}^k(E \cap B(x, r-h))}{h} \\
 &= \frac{r}{k} \frac{f(r) - f(r-h)}{h}.
 \end{aligned} \tag{5.114}$$

By our computations 5.113 and 5.114 we find by letting  $h$  tend to 0:

$$f(r) \leq \frac{r}{k} f'(r) = \frac{r}{k} g(r). \tag{5.115}$$

By Theorem 2.43 we know that  $g = \frac{d\mu^a}{d\lambda}$ , where  $\mu^a$  is the absolutely continuous part of  $\mu$  with respect to the Lebesgue measure  $\lambda$  on  $\mathbb{R}$ . Since  $f'(t) = g(t)$  for  $\lambda$ -a.e.  $t \in (0, R)$ , we can apply Lemma 5.15 for the smooth function  $\varphi(t) = t^{-k}$  on  $(0, R)$  to find that for all  $0 < a < b < R$ :

$$\begin{aligned}
 \Theta_k(E, x)(b) - \Theta_k(E, x)(a) &= \varphi(b)f(b) - \varphi(a)f(a) \\
 &\stackrel{5.105}{=} \int_a^b \varphi(t) d\mu^a(t) + \int_a^b \varphi'(t)f(t) dt \\
 &= \int_a^b t^{-k} d\mu^a(t) - k \int_a^b t^{-k-1} f(t) dt.
 \end{aligned} \tag{5.116}$$

Now we note that we have  $\lambda$ -a.e.  $f' = g = \frac{d\mu^a}{d\lambda}$  and we recall from 5.115, that by minimality of  $E$  we have that  $kf(t) \leq tf'(t)$  for  $\lambda$ -a.e.  $t \in (0, R)$ , and therefore:

$$\begin{aligned}
 k \int_a^b t^{-k-1} f(t) dt &\leq k \int_a^b t^{-k-1} \frac{t}{k} f'(t) dt \\
 &= \int_a^b t^{-k} f'(t) dt \\
 &= \int_a^b t^{-k} d\mu^a(t).
 \end{aligned} \tag{5.117}$$

Putting 5.116 and 5.117 yields:

$$\Theta_k(E, x)(b) - \Theta_k(E, x)(a) \geq 0. \tag{5.118}$$

That is, for all  $a, b \in (0, R)$  with  $a \leq b$  we have proven  $\Theta_k(E, x)(a) \leq \Theta_k(E, x)(b)$ . So  $\Theta_k(E, x)(\cdot)$  is non-decreasing.  $\square$

As a corollary we get the monotonicity result, which we already have proven in Theorem 4.10.

**Corollary 5.16** *Let  $\Omega \subset \mathbb{R}^{N+1}$  be a bounded open set and  $E_0 \subset \mathbb{R}^{N+1}$  be a set of locally finite perimeter in  $\mathbb{R}^{N+1}$ . Suppose  $E \subset \mathbb{R}^{N+1}$  is a set of minimal perimeter in  $\Omega$  with respect to  $E_0$  and  $x \in \Omega$ . Then the function*

$$\Phi(E, x)(\cdot) : r \in (0, d(x, \partial\Omega)) \mapsto \frac{\|D\chi_E\|(B(x, r))}{\omega_N r^N} \quad (5.119)$$

*is nondecreasing.*

**Proof** By Proposition 5.11 we know that  $\partial^* E \cap \Omega$  is minimal in  $\Omega$  of dimension  $N$ . Since  $E$  has finite perimeter in  $\Omega$ , we find that by Theorem 3.41 for any  $r \in (0, d(x, \partial\Omega))$ :

$$\begin{aligned} \Phi(E, x)(r) &= \frac{\|D\chi_E\|(B(x, r))}{\omega_N r^N} \\ &= \frac{P(E, B(x, r))}{\omega_N r^N} \\ &= \frac{\mathcal{H}^N(\partial^* E \cap \Omega)}{\omega_N r^N}. \end{aligned} \quad (5.120)$$

By Proposition 5.14 we can infer that  $\Phi(E, x)(\cdot)$  is non-decreasing.  $\square$

## 5.4 Minimal Cones

This chapter will introduce *cones* and how they relate to minimal sets. This will boil down to considering minimal sets with constant density. We will follow [5] and [2].

### 5.4.1 Minimal cones and minimal sets of constant density

First, we need to come to an agreement, what we mean by cones.

**Definition 5.17** *Let  $\mathcal{C} \subset \mathbb{R}^N$ . We call  $\mathcal{C}$  a **cone centered at the origin** if  $r\mathcal{C} \subset \mathcal{C}$  for all  $r \geq 0$ . We call  $\mathcal{C}$  a **cone centered at  $x \in \mathbb{R}^N$**  if  $\mathcal{C} - x$  is a cone centered at the origin.*

**Remark 5.18** *Note that if  $\mathcal{C} \subset \mathbb{R}^N$  is a cone centered at the origin, then for any  $y \in \mathcal{C}$  and  $r > 0$  we have  $\frac{y}{r} \in \frac{1}{r}\mathcal{C} \subset \mathcal{C}$  and so  $y = r\frac{y}{r} \in r\mathcal{C}$ . So we have  $\mathcal{C} = r\mathcal{C}$  for all  $r > 0$ .*

We now give a useful characterization of cones with vertex at the origin.

**Lemma 5.19** *Let  $\mathcal{C} \subset \mathbb{R}^N$  and define the radial projection onto the unit sphere  $\pi_S : \mathbb{R}^N \setminus \{0\} \rightarrow \mathbb{R}^N$  by  $\pi_S(y) = \frac{y}{\|y\|}$ . Then  $\mathcal{C}$  is a cone centered at the origin if and only if  $0 \in \mathcal{C}$  and*

$$L_y := \pi_S^{-1}(\pi_S(y)) \subset \mathcal{C}, \quad \forall y \in \mathcal{C} \setminus \{0\}. \quad (5.121)$$

**Proof** Assume first that  $\mathcal{C}$  is a cone centered at the origin. Then for all  $r \geq 0$  we have  $r\mathcal{C} = \mathcal{C}$ . Let  $y \in \mathcal{C} \setminus \{0\}$ . Clearly,  $0 \in \mathcal{C}$  and we make the following observation:

$$\begin{aligned} L_y &= \left\{ z \in \mathbb{R}^N \setminus \{0\} \mid \frac{z}{\|z\|} = \frac{y}{\|y\|} \right\} \\ &= \left\{ z \in \mathbb{R}^N \setminus \{0\} \mid \exists r > 0 : z = ry \right\}. \end{aligned} \quad (5.122)$$

So for any  $z \in L_y$  we find  $r > 0$  such that  $z = ry \in r\mathcal{C} = \mathcal{C}$ . This shows that  $L_y \subset \mathcal{C}$  for all  $y \in \mathcal{C}$ .

Conversely, assume that 5.121 holds. Let  $r > 0$  and  $y \in \mathcal{C}$ . If  $y = 0$ , then  $ry = 0 = y \in \mathcal{C}$ . So, assume that  $y \neq 0$ . Note that by observation 5.122 we have  $ry \in L_y$ . And as  $L_y \subset \mathcal{C}$  by assumption, we have  $ry \in \mathcal{C}$ . This shows that  $r\mathcal{C} \subset \mathcal{C}$  and, thus,  $\mathcal{C}$  is a cone with vertex at the origin.  $\square$

We now make the observation that any cone has constant density.

**Proposition 5.20** *Let  $\mathcal{C} \subset \mathbb{R}^{N+1}$  be a cone centered at  $x \in \mathbb{R}^{N+1}$  and  $1 \leq k \leq N$ . The function  $\Theta_k(E, x)(\cdot) : r \in (0, \infty) \mapsto \frac{1}{\omega_k r^k} \mathcal{H}^k(\mathcal{C} \cap B(x, r))$  is constant.*

**Proof** Note that we have that for all  $r \in (0, \infty)$  it holds:

$$\mathcal{H}^k(\mathcal{C} \cap B(x, r)) = \mathcal{H}^k((\mathcal{C} - x) \cap B(0, r)). \quad (5.123)$$

Then, since  $\mathcal{C} - x$  is a cone centered at the origin, we find for all  $r \in (0, \infty)$  that:

$$\begin{aligned} \Theta_k(E, x)(r) &= \omega_k^{-1} r^{-k} \mathcal{H}^k(\mathcal{C} \cap B(x, r)) \\ &\stackrel{5.123}{=} \omega_k^{-1} r^{-k} \mathcal{H}^k((\mathcal{C} - x) \cap B(0, r)) \\ &= \omega_k^{-1} \mathcal{H}^k(r^{-1}(\mathcal{C} - x) \cap r^{-1}B(0, r)) \\ &= \omega_k^{-1} \mathcal{H}^k((\mathcal{C} - x) \cap B(0, 1)) \\ &= \omega_k^{-1} \mathcal{H}^k(\mathcal{C} \cap B(x, 1)) \\ &= \Theta_k(E, x)(1). \end{aligned} \quad (5.124)$$

This shows that  $\Theta_k(E, x)(r) = \Theta_k(E, x)(1)$  for all  $r \in (0, \infty)$ , that is,  $\Theta_k(E, \cdot)$  is constant.  $\square$

Now we turn our attention to minimal sets with constant 2-dimensional density. By that we mean minimal sets  $E$  in open set  $\Omega$  of dimension  $k$  such that the function for some  $x \in \Omega$  and  $R = d(x, \mathbb{R}^{N+1} \setminus \Omega)$ :

$$\Theta_k(E, x)(\cdot) : r \in (0, R) \mapsto \frac{1}{\omega_k r^k} \mathcal{H}^k(E \cap B(x, r)) \quad (5.125)$$

is constant on some interval  $(a, b) \subset (0, R)$ . We will show that such minimal sets coincide locally with a cone. This will follow the result given in [2].

**Theorem 5.21** *Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset. Let  $E \subset \Omega$  be a reduced minimal set in  $\Omega$  of dimension  $k$ . Let  $x \in \Omega$  and set  $R = d(x, \mathbb{R}^{N+1} \setminus \Omega)$ . Suppose there are  $0 \leq a < b \leq R$  and some  $\vartheta_0 \geq 0$  such that*

$$\Theta_k(E, x)(r) = \vartheta_0, \quad \forall r \in (a, b). \quad (5.126)$$

*Then there exists a reduced minimal cone  $\mathcal{C}$  in  $\mathbb{R}^{N+1}$  of dimension  $k$  centered at  $x^2$  such that:*

$$E \cap [B(x, b) \setminus \overline{B}(x, a)] = \mathcal{C} \cap [B(x, b) \setminus \overline{B}(x, a)]. \quad (5.127)$$

The proof of Theorem 5.21 is based on the following crucial proposition from [2].

**Proposition 5.22** *Let  $N \geq 2$ ,  $1 \leq k \leq N$  and  $\Omega \subset \mathbb{R}^{N+1}$  be an open subset with  $0 \in \Omega$ . Let  $E \subset \Omega$  be a reduced minimal set in  $\Omega$  of dimension  $k$  and let  $\pi_S : \mathbb{R}^{N+1} \setminus \{0\} \rightarrow \mathbb{R}^{N+1}$  denote the radial projection onto the unit sphere. Suppose there are  $0 \leq a < b \leq R$  and some  $\vartheta_0 \geq 0$  such that*

$$\Theta_k(E, 0)(r) = \vartheta_0, \quad \forall r \in (a, b). \quad (5.128)$$

*Define  $A := B(0, b) \setminus \overline{B}(0, a)$  and let  $y \in E \cap A$  be a point such that the approximate  $m$ -tangent space to  $E$  at  $y$  goes through the origin. Then we have  $L = A \cap \pi_S^{-1}(\pi_S(y)) \subset E$ .*

**Proof** For a proof of Proposition 5.22, see the proof of Proposition 6.11 in [2].  $\square$

Now we prove Theorem 5.21.

**Proof (Theorem 5.21)** Without loss of generality, we may assume that  $x = 0\Omega$ . Otherwise, we may replace  $E$  with  $\tilde{E} = E - x$  and  $\tilde{\Omega} = \Omega - x$ . Then  $\tilde{E}$  is minimal in  $\tilde{\Omega}$  of dimension  $m$  and  $0 \in \tilde{\Omega}$  such that for all  $r \in (a, b)$  we have:

$$\begin{aligned} \Theta_k(\tilde{E}, 0)(r) &= \omega_k^{-1} r^{-k} \mathcal{H}^k(\tilde{E} \cap B(0, r)) \\ &= \omega_k^{-1} r^{-k} \mathcal{H}^k(E \cap B(x, r)) \\ &= \Theta_k(E, x)(r) \\ &= \vartheta_0. \end{aligned} \quad (5.129)$$

<sup>2</sup>i.e. a minimal set  $\mathcal{C}$ , that is a cone centered at the  $x$ .

So, in fact, we may assume without loss of generality that  $x = 0 \in \Omega$ . Consider now the continuous function  $\pi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}$  defined by  $\pi(y) = \|y\|$  and set  $A := B(0, b) \setminus \overline{B}(0, a)$ . Then we first show that:

$$\cos \theta(y) = 1, \quad \text{for } \mathcal{H}^k\text{-almost every } y \in E \cap A, \quad (5.130)$$

where  $\theta(y)$  is the smallest angle between the approximate  $m$ -tangent space of  $E$  at  $y$  and the line segment from the origin to  $y$ . Since  $\pi$  is a proper continuous function we know that, by Lemma 2.31, both  $\nu$  and  $\mu$  defined by

$$\nu(D) := \int_{E \cap \pi^{-1}(D)} \cos \theta(y) d\mathcal{H}^k(y) \quad (5.131)$$

$$\mu(D) := \mathcal{H}^k(E \cap \pi^{-1}(D)) \quad (5.132)$$

for  $D \subset (a, b)$ , are Radon measures on  $(a, b)$ . Note that  $\mu$  is obtained from  $\nu$  by replacing  $\cos \theta(y)$  by 1 and so we have, in fact,  $\nu \leq \mu$ . Now note that for all  $a < \alpha < \beta < b$  we find that:

$$\begin{aligned} \mu([\alpha, \beta]) &= \mathcal{H}^k(E \cap \pi^{-1}([\alpha, \beta])) \\ &= \mathcal{H}^k(E \cap (B(0, \beta) \setminus \overline{B}(0, \alpha))) \\ &= \mathcal{H}^k(E \cap B(0, \beta)) - \mathcal{H}^k(E \cap B(0, \alpha)) \\ &\stackrel{5.126}{=} \omega_k(\beta^k - \alpha^k) \vartheta_0 \\ &= \omega_k \vartheta_0 k \int_{\alpha}^{\beta} t^{k-1} d\lambda(t), \end{aligned} \quad (5.133)$$

where  $\lambda$  is the Lebesgue measure on  $(a, b)$ . Then 5.133 shows that  $\mu$  is absolutely continuous with respect to the Lebesgue measure  $\lambda$  on  $(a, b)$ . By definition  $\nu$  is absolutely continuous with respect to the measure  $\mu$ , and so by the transitivity of being absolutely continuous, we find that  $\nu$  is absolutely continuous with respect to the Lebesgue measure  $\lambda$  on  $(a, b)$  also.

Then by Theorem 2.43 for  $\lambda$ -almost every  $r \in (a, b)$  the following limit exists:

$$\begin{aligned} g(r) &:= \lim_{t \rightarrow 0} \frac{1}{t} \nu([r-t, r]) \\ &= \frac{1}{t} \int_{E \cap B(0, r) \setminus \overline{B}(0, r-t)} \cos \theta(y) d\mathcal{H}^k(y). \end{aligned} \quad (5.134)$$

Now apply the integral monotonicity result for almost minimal sets (Lemma 5.12) for  $0 < \rho_1 = r-t < \rho_2 = r \in (a, b)$  for  $t$  small enough to find for  $f(r) := \mathcal{H}^k(E \cap B(0, r))$  that:

$$f(r-t) \leq \frac{r}{kt} \int_{E \cap B(0, r) \setminus \overline{B}(0, r-t)} \cos \theta(y) d\mathcal{H}^k(y). \quad (5.135)$$

So as  $t \rightarrow 0$  we have that for  $\lambda$ -almost every  $r \in (a, b)$  it holds that:

$$f(r) \leq \frac{r}{k} g(r). \quad (5.136)$$

By the proof of monotonicity for minimal sets (Proposition 5.14) we know already that for  $\lambda$ -almost every  $r \in (a, b)$  the derivative  $f'(r)$  exists and is the density of  $\mu$  and so we must have:

$$g(r) \leq f'(r), \quad \text{for } \lambda\text{-a.e. } r \in (a, b), \quad (5.137)$$

since  $\nu \leq \mu$ . Now recall that by assumption 5.126 it holds that  $f(r) = r^m \vartheta_0$  for all  $r \in (a, b)$  and hence  $f'(r) = mr^{m-1} \vartheta_0$ . So in summary it holds for  $\lambda$ -a.e.  $r \in (a, b)$  that:

$$\begin{aligned} f(r) &\leq \frac{r}{k} g(r) \\ &\leq \frac{r}{k} f'(r) \\ &= \frac{r}{k} \omega_k k r^{k-1} \vartheta_0 \\ &= \omega_k r^k \vartheta_0 \\ &= f(r). \end{aligned} \quad (5.138)$$

The calculation 5.138 shows that  $g(r) = f'(r)$  for  $\lambda$ -a.e.  $r \in (a, b)$ .

Now, since  $\nu$  is absolutely continuous with respect to  $\lambda$ , it follows that:

$$\nu((a, b)) = \int_a^b g(r) d\lambda(r) \stackrel{5.138}{=} \int_a^b f'(r) d\lambda(r) = \mu((a, b)). \quad (5.139)$$

Then 5.139 and the definitions 5.131 and 5.132 of  $\mu$  resp.  $\nu$  imply that:

$$0 \stackrel{5.139}{=} \nu((a, b)) - \mu((a, b)) \stackrel{5.131, 5.132}{=} \int_{E \cap A} \cos \theta(y) - 1 d\mathcal{H}^k(y). \quad (5.140)$$

So by 5.140 we have  $\cos \theta(y) = 1$  for  $\mathcal{H}^k$ -a.e.  $y \in E \cap A$  as claimed. Geometrically, this means that for  $\mathcal{H}^k$ -a.e.  $y \in E \cap A$  the approximate tangent space of  $E$  at  $y$  goes through the origin, as the angle  $\theta(y)$  between the approximate tangent space of  $E$  at  $y$  and the segment from the origin to  $y$  is  $\mathcal{H}^k$ -a.e. equal to 0. This is already a strong indication that  $E \cap A$  is conical.

Now consider the radial projection  $\pi_S : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \partial B(0, 1)$  onto the unit sphere defined by  $\pi_S(y) = \frac{y}{\|y\|}$ . Then we claim that

$$\mathcal{H}^k(\pi_S(E \cap A)) = 0. \quad (5.141)$$

Again this would be a strong indication that  $E \cap A$  is conical, as for example in for  $N = m = 2$  this would just mean that  $E \cap A$  is the disjoint union

of two planes, which get mapped onto two arcs on the unit sphere in  $\mathbb{R}^3$ , which are  $\mathcal{H}^2$ -null sets. By Theorem 2.57 and the rectifiability of  $E \cap A$  we can find a sequence  $(\Gamma_h)_{h \in \mathbb{N}}$  of  $k$ -dimensional  $C^1$  submanifolds of  $\mathbb{R}^{N+1}$  such that for  $N := (E \cap A) \setminus (\bigcup_{h \in \mathbb{N}} \Gamma_h)$  it holds that  $\mathcal{H}^k(N) = 0$ . Then by Proposition 2.95 for every  $y \in (E \cap A) \setminus N$  the approximate tangent space  $\pi_y$  of  $E$  at  $y$  coincides with tangent space  $T_y \Gamma_{h(y)}$  for some  $h(y) \in \mathbb{N}$ . Then by the result 5.130 proven before, for  $\mathcal{H}^k$ -a.e.  $y \in (E \cap A) \setminus N$  we have that  $T_y \Gamma_{h(y)}$  goes through the origin. A simple computation shows that for  $\mathcal{H}^k$ -a.e.  $y \in (E \cap A) \setminus N$  and  $v \in T_y \Gamma_{h(y)}$  we have:

$$d\pi_S(y)(v) = \frac{v}{\|y\|} - \frac{y \cdot v}{\|y\|^3} y. \quad (5.142)$$

Since  $\cos \theta(y) = 1$  we find that  $d\pi_S(y)$  has not full rank and so  $y$  is a critical point of  $\pi_S$  on  $\Gamma_{h(y)}$ . So by Sard's theorem<sup>3</sup> we find that  $\mathcal{H}^k(\pi_S(E \cap A \cap \Gamma_h)) = 0$  for all  $k \in \mathbb{N}$ . Then by summing over  $h$  we get the following:

$$\begin{aligned} \mathcal{H}^k(\pi_S(E \cap A)) &\leq \mathcal{H}^k(\pi_S(N)) + \sum_{h \in \mathbb{N}} \mathcal{H}^k(\pi_S(E \cap A \cap \Gamma_h)) \\ &= 0 + 0 = 0. \end{aligned} \quad (5.143)$$

That is, as claimed, we find that  $\mathcal{H}^k(\pi_S(E \cap A)) = 0$ .

We will now use Proposition 5.22 to show that there is a cone centered at the origin that coincides with  $E$  in  $A$ . Indeed, let  $E_1$  be the set of points  $y \in E \cap A$  such that the approximate tangent space of  $E$  at  $y$  goes through the origin. We already know by 5.130 that  $\mathcal{H}^k((E \cap A) \setminus E_1) = 0$ . In particular, we then must have that  $E_1$  is a dense subset of  $E \cap A$ . Define now the cone centered at the origin through  $E_1$  by:

$$\mathcal{C} := \{ry \mid y \in E_1, r \geq 0\} = \bigcup_{y \in E_1} \pi_S^{-1}(\pi_S(y)). \quad (5.144)$$

By definition  $\mathcal{C}$  is a cone centered at the origin. Then on the one hand we have  $E \cap A \subset \overline{E_1} \subset \mathcal{C}$ . On the other hand, by Proposition 5.22 we have that  $\pi_S^{-1}(\pi_S(y)) \cap A \subset E$  for all  $y \in E_1$ . and so  $\mathcal{C} \cap A \subset E$  by closedness of  $E$ . So it follows that  $E \cap A = \mathcal{C} \cap A$ .

Now we show that the cone  $\mathcal{C}$  is a reduced minimal set in  $\mathbb{R}^{N+1}$  of dimension  $k$ . We start by showing that  $\mathcal{C}$  is  $\mathcal{H}^k$ -rectifiable:

Recall that  $\mathcal{C} = \{ry \mid y \in E_1, r \geq 0\}$ . Choose now a dense countable subset  $\{r_h\}_{h \in \mathbb{N}}$  of  $[0, \infty)$  (e.g., all nonnegative rational numbers). Then by the definition of  $\mathcal{C}$  we have:

$$\mathcal{C} = \bigcup_{h \in \mathbb{N}} r_h E_1. \quad (5.145)$$

<sup>3</sup>See e.g. [18], but in essence it states the following: Let  $X, Y$  be finite-dimensional manifolds and  $f : Y \rightarrow Z$  be smooth. If  $Z$  is the set of critical points of  $f$ , i.e. the points  $x \in X$  where  $Df(x)$  has not full rank, then  $f(Z)$  has measure 0 in  $Y$ .

Since  $E_1$  is a dense subset of  $E \cap A$  and  $E \cap A$  is  $\mathcal{H}^k$ -rectifiable we find a sequence  $\{f_\ell : \mathbb{R}^k \rightarrow \mathbb{R}^{N+1}\}$  of Lipschitz continuous functions such that

$$\mathcal{H}^k \left( E_1 \setminus \bigcup_{\ell \in \mathbb{N}} f_\ell(\mathbb{R}^k) \right) = 0. \quad (5.146)$$

Define for every  $\ell, h \in \mathbb{N}$  the Lipschitz continuous function  $g_{\ell,h} : \mathbb{R}^k \rightarrow \mathbb{R}^{N+1}$  by  $g_{\ell,h}(y) = r_h f_\ell(y)$ . Then,  $\{g_{\ell,h}\}_{(\ell,h) \in \mathbb{N}^2}$  is a countable collection of Lipschitz continuous functions such that:

$$\begin{aligned} \mathcal{H}^k \left( \mathcal{C} \setminus \bigcup_{(\ell,h) \in \mathbb{N}^2} g_{\ell,h}(\mathbb{R}^k) \right) &\stackrel{5.145}{=} \mathcal{H}^k \left( \bigcup_{h \in \mathbb{N}} r_h E_1 \setminus \bigcup_{h \in \mathbb{N}} \bigcup_{\ell \in \mathbb{N}} r_h f_\ell(\mathbb{R}^k) \right) \\ &= \mathcal{H}^k \left( \bigcup_{h \in \mathbb{N}} r_h \left( E_1 \setminus \bigcup_{\ell \in \mathbb{N}} f_\ell(\mathbb{R}^k) \right) \right) \\ &\leq \sum_{h \in \mathbb{N}} r_h^k \mathcal{H}^k \left( E_1 \setminus \bigcup_{\ell \in \mathbb{N}} f_\ell(\mathbb{R}^k) \right) \\ &\stackrel{5.146}{=} 0. \end{aligned} \quad (5.147)$$

This shows that  $\mathcal{C}$  is, in fact,  $\mathcal{H}^m$ -rectifiable.

Since  $E$  is reduced,  $\mathcal{C}$  is also reduced. Indeed, fix  $y \in \mathcal{C}$  and  $r > 0$ . Because  $E$  is reduced we have and  $0 \leq a < b$  we have:

$$0 \leq \mathcal{H}^k(E \cap \bar{B}(0, a)) < \mathcal{H}^k(E \cap B(0, b)). \quad (5.148)$$

Then we find that:

$$\begin{aligned} \mathcal{H}^k(\mathcal{C} \cap B(y, r)) &= \mathcal{H}^k(\mathcal{C} \cap B(0, r)) \\ &= \mathcal{H}^k \left( \mathcal{C} \cap B \left( 0, \frac{r}{b} b \right) \right) \\ &= \left( \frac{r}{b} \right)^k \mathcal{H}^k(\mathcal{C} \cap B(0, b)) \\ &\geq \left( \frac{r}{b} \right)^k \mathcal{H}^k(\mathcal{C} \cap (B(0, b) \setminus \bar{B}(0, a))) \\ &\stackrel{5.127}{=} \left( \frac{r}{b} \right)^k \mathcal{H}^k(E \cap (B(0, b) \setminus \bar{B}(0, a))) \\ &= \left( \frac{r}{b} \right)^k (\mathcal{H}^k(E \cap B(0, b)) - \mathcal{H}^k(E \cap \bar{B}(0, a))) \\ &\stackrel{5.148}{>} 0. \end{aligned} \quad (5.149)$$

This shows that  $\mathcal{C}$  is reduced.

Finally, we show that  $\mathcal{C}$  is a minimal set in  $\mathbb{R}^{N+1}$  of dimension  $k$ . For the sake of contradiction, suppose that  $\mathcal{C}$  is not minimal in  $\mathbb{R}^{N+1}$ . Then there is

a compact set  $K \subset \mathbb{R}^{N+1}$  and a Lipschitz function  $\varphi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  with  $\varphi(K) \subset K$  and  $\varphi(y) = y$  for all  $y \in \mathbb{R}^{N+1} \setminus K$ , such that for  $F = \varphi(\mathcal{C})$  we have the following:

$$\mathcal{H}^k(F \setminus \mathcal{C}) < \mathcal{H}^k(\mathcal{C} \setminus F). \quad (5.150)$$

We observe that without loss of generality we may assume that  $K \subset \overline{B}(0, a)$  and this will not impact our supposition 5.150. Indeed, we can construct a competitor for  $\mathcal{C}$  in a compact set contained in  $\overline{B}(0, a)$ , such that 5.150 is still satisfied, as follows: Since  $K$  is bounded in  $\mathbb{R}^{N+1}$  there is  $\rho > 0$  such that  $K \subset \overline{B}(0, \rho)$ . If  $\rho \leq a$ , we are done. If  $\rho > a$ , we consider the scaling function

$$\eta : y \in \mathbb{R}^{N+1} \mapsto \frac{a}{\rho} y \in \mathbb{R}^{N+1}, \quad (5.151)$$

which sends  $\overline{B}(0, \rho)$  to  $\overline{B}(0, a)$  and also  $K$  to the compact set  $\tilde{K} = \eta(K) \subset \overline{B}(0, a)$ . Now we define the Lipschitz function  $\tilde{\varphi} : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by  $\tilde{\varphi} = \eta \circ \varphi \circ \eta^{-1}$ . We observe that:

- $\tilde{\varphi}(\tilde{K}) = \eta(\varphi(K)) \subset \eta(K) = \tilde{K}$ .
- For all  $y \in \mathbb{R}^{N+1} \setminus \tilde{K}$  we have  $\eta^{-1}(y) \in \mathbb{R}^{N+1} \setminus K$  and since  $\varphi(z) = z$  for all  $z \in \mathbb{R}^{N+1} \setminus K$  we find that:

$$\tilde{\varphi}(y) = \eta(\varphi(\eta^{-1}(y))) = \eta(\eta^{-1}(y)) = y. \quad (5.152)$$

This shows that  $\tilde{F} = \tilde{\varphi}(\mathcal{C})$  is a competitor for  $\mathcal{C}$  in  $\tilde{K}$ . Now we note that, since  $\mathcal{C}$  is a cone centered at the origin,  $\eta^{-1}(\mathcal{C}) = \mathcal{C}$  and so  $\tilde{F} = \eta(F)$ . Then we have that  $\mathcal{C} \setminus \tilde{F} = \eta(\mathcal{C} \setminus F)$  and  $\tilde{F} \setminus \mathcal{C} = \eta(F \setminus \mathcal{C})$  and so it follows that by 5.150 that:

$$\begin{aligned} \mathcal{H}^k(\tilde{F} \setminus \mathcal{C}) &= \left(\frac{a}{\rho}\right)^k \mathcal{H}^k(F \setminus \mathcal{C}) \\ &< \left(\frac{a}{\rho}\right)^k \mathcal{H}^k(\mathcal{C} \setminus F) \\ &= \mathcal{H}^k(\mathcal{C} \setminus \tilde{F}). \end{aligned} \quad (5.153)$$

So we may assume that  $K \subset \overline{B}(0, a)$ .

Now pick some  $r_0 \in (a, b)$  and define the Lipschitz function  $h : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by

$$h(y) = \begin{cases} 0, & \|y\| \leq a \\ \frac{r_0(\|y\| - a)}{r_0 - a} \frac{y}{\|y\|}, & a < \|y\| < r_0 \\ y, & \|y\| \geq r_0. \end{cases} \quad (5.154)$$

We note that  $h(E \cap B(0, b)) \subset \mathcal{C} \cap B(0, b)$ . Indeed, let  $y \in E \cap B(0, b)$ . If  $\|y\| \leq a$  we find that  $h(y) = 0 \in \mathcal{C} \cap B(0, b)$ . If  $\|y\| \in (a, r_0)$  we have

$\|h(y)\| \in (0, r_0)$  and  $h(y) = ry$  for  $r = \frac{r_0(\|y\| - a)}{(r_0 - a)\|y\|}$ . As  $E$  coincides with  $\mathcal{C}$  on  $B(0, b) \setminus \overline{B}(0, a)$  and  $\mathcal{C}$  is a cone, we must have  $h(y) \in \mathcal{C} \cap B(0, b)$ . Finally, since  $E$  coincides with  $\mathcal{C}$  in  $B(0, b) \setminus \overline{B}(0, a)$  we find for  $\|y\| \geq r_0$  that  $h(y) = y \in \mathcal{C} \cap B(0, b)$ .

Now, define the Lipschitz continuous function  $\psi : \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by  $\psi := \varphi \circ h$  and consider the compact set  $L := \overline{B}(0, r_0) \subset B(0, b)$ . We claim that  $G = \psi(E \cap B(0, b))$  is a competitor for  $E \cap B(0, b)$  in  $L$ . Indeed, we observe the following:

- Since  $h(L) \subset L$  (by the computations above),  $K \subset \overline{B}(0, a)$ ,  $\varphi(K) \subset K$  and  $\varphi(y) = y$  outside of  $K$ , we find that:

$$\begin{aligned} \psi(L) &= \varphi(h(L)) \\ &\subset \varphi(L) \\ &= \varphi((L \setminus K) \cup K) \\ &= \varphi(L \setminus K) \cup \varphi(K) \\ &\subset (L \setminus K) \cup K \\ &= L. \end{aligned} \tag{5.155}$$

- Since  $h(y) = y$  outside of  $L$ ,  $\varphi(y) = y$  outside of  $K$  and  $K \subset L$  we must have  $\psi(y) = y$  for all  $y \in \mathbb{R}^{N+1} \setminus L$ .

This shows that  $G$  is a competitor for  $\tilde{E} = E \cup B(0, b)$  in  $L$ . Since  $E$  is minimal in  $\Omega$  we know by Proposition 5.5 that  $E \cap B(0, b)$  is minimal in  $B(0, b)$ . Since  $G$  is a competitor for  $\tilde{E}$  we must have that:

$$\mathcal{H}^k(\tilde{E} \setminus G) \leq \mathcal{H}^k(G \setminus \tilde{E}). \tag{5.156}$$

Now note that:

$$\begin{aligned} G &= \psi(\tilde{E}) \\ &= \varphi(\mathcal{C} \cap B(0, b)) \\ &\subset F \cap B(0, b). \end{aligned} \tag{5.157}$$

Thus, by 5.156 it follows that:

$$\begin{aligned} \mathcal{H}^k(\tilde{E}) &= \mathcal{H}^k(\tilde{E} \cap G) + \mathcal{H}^k(\tilde{E} \setminus G) \\ &\leq \mathcal{H}^k(\tilde{E} \cap G) + \mathcal{H}^k(G \setminus \tilde{E}) \\ &= \mathcal{H}^k(G) \\ &\leq \mathcal{H}^k(F \cap B(0, b)). \end{aligned} \tag{5.158}$$

Now recall that for all  $r \in (a, b)$  we have  $\mathcal{H}^k(E \cap B(0, r)) = \omega_k \vartheta_0 r^k$  and that

$$\mathcal{C} \cap (B(0, r) \setminus \overline{B}(0, a)) = E \cap (B(0, r) \setminus \overline{B}(0, a)). \tag{5.159}$$

Now consider the sequence  $r_h = a + \frac{b-a}{h} \in (a, b)$  for  $h \in \mathbb{N}$  with  $r_h \xrightarrow{h \rightarrow \infty} a^+$ . Then we find for all  $r \in (a, b)$ , that:

$$\begin{aligned} \mathcal{H}^k(\mathcal{C} \cap (B(0, r) \setminus \overline{B}(0, a))) &= \mathcal{H}^k(E \cap B(0, r)) - \mathcal{H}^k(E \cap \overline{B}(0, a)) \\ &= \mathcal{H}^k(E \cap B(0, r)) - \lim_{h \rightarrow \infty} \mathcal{H}^k(E \cap B(0, r_h)) \\ &= \omega_k \vartheta_0(r^k - \lim_{h \rightarrow \infty} r_h^k) \\ &= \omega_k \vartheta_0(r^k - a^k). \end{aligned} \tag{5.160}$$

This yields for any  $r \in (a, b)$  that

$$\begin{aligned} \mathcal{H}^k(\mathcal{C} \cap B(0, r)) &= \mathcal{H}^k(\mathcal{C} \cap (B(0, r) \setminus \overline{B}(0, a))) + \mathcal{H}^k(\mathcal{C} \cap \overline{B}(0, a)) \\ &= \omega_k \vartheta_0(r^k - a^k) + \mathcal{H}^k(\mathcal{C} \cap \overline{B}(0, a)). \end{aligned} \tag{5.161}$$

Now by Proposition 5.20 we know that  $\omega_k^{-1} s^k \mathcal{H}^k(\mathcal{C} \cap B(0, s)) = \tau_0$  for all  $s \geq 0$  and some constant  $\tau_0 \geq 0$ . Then it follows by 5.161 that for any  $r \in (a, b)$

$$\begin{aligned} r^k \tau_0 = \vartheta_0(r^k - a^k) + a^k \tau_0 &\iff (r^k - a^k) \tau_0 = \vartheta_0(r^k - a^k) \\ &\iff \tau_0 = \vartheta_0. \end{aligned} \tag{5.162}$$

Now consider the sequence  $s_h = b - \frac{b-a}{h} \in (a, b)$  with  $s_h \xrightarrow{h \rightarrow \infty} b^-$ . We note that:

$$\begin{aligned} \mathcal{H}^k(\mathcal{C} \cap B(0, b)) &= \vartheta_0 \omega_k b^k \\ &= \lim_{h \rightarrow \infty} \mathcal{H}^k(\mathcal{C} \cap B(0, s_h)) \\ &= \lim_{h \rightarrow \infty} \mathcal{H}^k(E \cap B(0, s_h)) \\ &= \mathcal{H}^k(E \cap B(0, b)) \\ &= \mathcal{H}^k(\tilde{E}). \end{aligned} \tag{5.163}$$

Hence, by minimality of  $\tilde{E}$  we find that:

$$\mathcal{H}^k(\mathcal{C} \cap B(0, b)) = \mathcal{H}^k(\tilde{E}) \stackrel{5.158}{\leq} \mathcal{H}^k(F \cap B(0, b)). \tag{5.164}$$

But this is a contradiction to the supposition that  $\mathcal{H}^k(F \setminus \mathcal{C}) < \mathcal{H}^k(\mathcal{C} \setminus F)$ . Indeed, let  $F_1 = F \cap B(0, b)$  and  $F_2 = F \setminus B(0, b)$ . Then we have that:

$$\begin{aligned} \mathcal{H}^k(F_1) &= \mathcal{H}^k(F_1 \cap \mathcal{C}) + \mathcal{H}^k(F_1 \setminus \mathcal{C}) \\ &\leq \mathcal{H}^k(F_1 \cap \mathcal{C}) + \mathcal{H}^k(F \setminus \mathcal{C}) \\ &< \mathcal{H}^k(F_1 \cap \mathcal{C}) + \mathcal{H}^k(\mathcal{C} \setminus F). \end{aligned} \tag{5.165}$$

Now we have  $\mathcal{C} = \varphi^{-1}(F) = \varphi^{-1}(F_1) \cup \varphi^{-1}(F_2)$ . Also, we have  $K \subset B(0, b)$  and therefore  $\varphi^{-1}(F_2) = F_2$  and hence:

$$F_1 \cap \mathcal{C} = F_1 \cap \varphi^{-1}(F_1) \quad (5.166)$$

$$\mathcal{C} \setminus F = \varphi^{-1}(F_1) \setminus F. \quad (5.167)$$

This yields:

$$\begin{aligned} \mathcal{H}^k(F \cap B(0, b)) &\stackrel{5.165}{<} \mathcal{H}^k(F_1 \cap \mathcal{C}) + \mathcal{H}^k(\mathcal{C} \setminus F) \\ &\stackrel{5.166, 5.167}{=} \mathcal{H}^k(F_1 \cap \varphi^{-1}(F_1)) + \mathcal{H}^k(\varphi^{-1}(F_1) \setminus F) \\ &\stackrel{F_1 \subset F}{\leq} \mathcal{H}^k(F_1 \cap \varphi^{-1}(F_1)) + \mathcal{H}^k(\varphi^{-1}(F_1) \setminus F_1) \quad (5.168) \\ &= \mathcal{H}^k(\varphi^{-1}(F_1)) \\ &\stackrel{(*)}{=} \mathcal{H}^k(\mathcal{C} \cap B(0, b)), \end{aligned}$$

where at  $(*)$  we used that  $\varphi^{-1}(F_1) = \mathcal{C} \cap \varphi^{-1}(B(0, b))$  and since  $K \subset \varphi^{-1}(K)$  we have that

$$\begin{aligned} \varphi^{-1}(B(0, b)) &= \varphi^{-1}((B(0, b) \setminus K) \cup K) \\ &= (B(0, b) \setminus K) \cup \varphi^{-1}(K) \quad (5.169) \\ &= B(0, b). \end{aligned}$$

So we have shown that  $\mathcal{H}^k(F \cap B(0, b)) < \mathcal{H}^k(\mathcal{C} \cap B(0, b))$ , which in fact contradicts  $\mathcal{H}^k(\mathcal{C} \cap B(0, b)) \leq \mathcal{H}^k(F \cap B(0, b))$ . So  $\mathcal{C}$  is a reduced minimal cone in  $\mathbb{R}^{N+1}$  of dimension  $k$ .  $\square$

**Remark 5.23** This again is a nice result, since the reduced boundary set of minimal perimeter in  $\Omega$  is locally determined to be a cone, if its density is constant, by Theorem 5.21.



## Chapter 6

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# Final Remarks

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In this master thesis, we have delved deeply into the intricate world of geometric measure theory, specifically focusing on the regularity of sets of minimal perimeter, i.e. solutions to the following Plateau type problem

$$\inf \{P(E, \Omega) \mid E \setminus \Omega = E_0 \setminus \Omega\}, \quad (6.1)$$

where  $E_0 \subset \mathbb{R}^N$  has locally finite perimeter in  $\mathbb{R}^N$ .

The journey began with a comprehensive introduction to the essential tools and concepts in geometric measurement theory, as outlined in Chapter 2. This foundational knowledge was crucial for appreciating the advanced topics discussed in the following chapters. The rigorous exploration of abstract measure theory, Radon measures, Sobolev spaces, and Lipschitz functions provided the necessary backdrop for our investigation into sets of finite perimeter.

Chapter 3 extended this foundation by delving into functions of bounded variation and sets of finite perimeter. This chapter was pivotal in understanding the structure and properties of these sets, laying the groundwork for the main results of this thesis. The detailed analysis of isoperimetric inequalities and the structure of reduced boundaries was instrumental in setting the stage for the comprehensive studies that followed. This chapter is also interesting on its own, as it gives us an idea of the structure and regularity of a broader set of functions and sets arising in many problems in mathematics and physics. Other areas of application would be more general plateau type problems with prescribed mean curvature of the form

$$\inf \left\{ \int_E g(x) dx + \mathcal{H}^{N-1}(\partial E) \mid E \subset \mathbb{R}^N \right\}, \quad (6.2)$$

which has been dealt with for example in [19]. More practical applications in engineering are the so-called *Mumford-Shah image segmentation problem*,

which consists of the following minimization problem:

$$\inf \left\{ J(K, u) \mid K \subset \overline{\Omega}, u \in C^1(\Omega \setminus K) \right\}, \quad (6.3)$$

where  $J$  is the functional

$$J(K, u) := \int_{\Omega \setminus K} \left[ \|\nabla u\|^2 + \alpha(u - g)^2 \right] dx + \beta \mathcal{H}^{N-1}(K \cap \Omega), \quad (6.4)$$

where  $\alpha, \beta > 0$  and  $g \in L^\infty(\Omega, \mathcal{L}^N)$ . This problem is very much related to image analysis and faithful reproduction of contours and gray scales in images; see, e.g., the original paper of Mumford and Shah in [20]. This shows that functions of bounded variation appear in many different contexts other than the problem presented in this thesis.

The core of this thesis, presented in Chapter 4, showcased the existence and regularity results for sets of minimal perimeter. These results are central to our understanding of the geometric properties of such sets, which offers comprehensive insights into their behavior and structure. This part gives us partial regularity of sets of minimal perimeter. There are possible extensions of this result mentioned in the preceding paragraph.

Building on these results, Chapter 5 introduced the concept of almost-minimal sets, as defined by David. This chapter placed the findings from Chapter 4 within a broader theoretical framework, highlighting the connections between minimal perimeter sets and almost minimal sets. This perspective not only broadened the scope of our study, but also added a new dimension to the understanding of minimal surfaces. This part, although only scratching the surface, also leads the way to possible limitations and open problems relating to the solution to these Plateau-type problems. Most notably, that even in dimension 2 there seems to be as of yet no complete list for  $N \geq 4$  of the minimal cones as provided by Jean Taylor in [1] for  $N = 3$ .

In conclusion, this thesis has provided a detailed study of the regularity of minimal surfaces within the framework of sets of minimal perimeter. The results presented herein should show, for a more simplified example, the techniques which are applicable to more general problems or geometric problems arising in various areas of mathematics. I think one of the main takeaways for me and for the reader from this thesis should be the following:

- The powerful tool of the direct method in the calculus of variation is to show the existence of minimizers of functional, which was the main theme in Section 4.1. Essentially, this boils down to the following three steps: Given a functional  $T : V \rightarrow \mathbb{R}$  on some Banach space  $V$  and  $\inf_{u \in V} T(u) > -\infty$  we show the existence of a minimizer by

(i) taking a minimizing sequence  $(u_h)_{h \in \mathbb{N}} \subset V$  of  $T$ ;

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(ii) showing that  $(u_h)_{h \in \mathbb{N}}$  admits a converging subsequence  $(u_{h_k})_{k \in \mathbb{N}}$ ;

(iii) showing that  $T$  is lower-semicontinuous.

(ii) and (iii), which is shown for our problem in Section 3.1 for the Banach space of functions of bounded variation and the variation on this space.

- The methods of showing (partial) regularity results for Plateau-type problems, where we first show a monotonicity formula (if possible), then analyze the local behavior, i.e., analyzing the blow-ups of our set, and then using the techniques seen in the proof of Theorem 4.16 to show smoothness in a small neighborhood.

This shows that the themes and techniques used in this thesis can be applicable to geometric problems that arise in other settings.

Finally, I extend my thanks to my thesis advisor, Prof. Dr. Joaquim Serra, whose input was valuable for the development of this thesis.



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